

Shatter Clustering: Distance Quantile Sensitivity Analysis

Sweep over `mean\_to\_other\_vecs\_dist\_quant` with Default $\alpha_{\text{MAD}} = 0.5$, $\alpha_{\text{accept}} = 0.5$

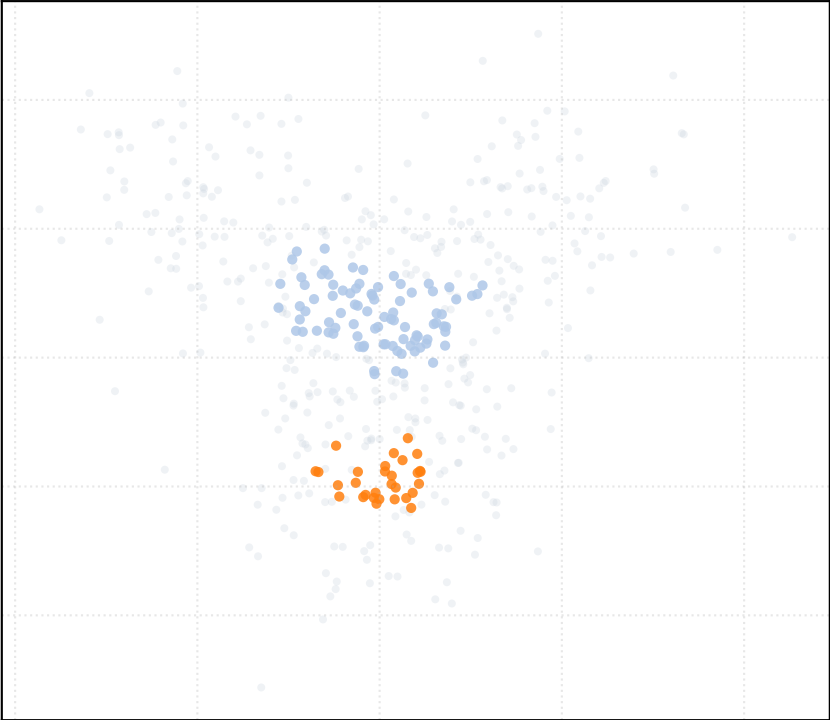
Experimental Configuration

- Total Evaluated Datasets: 30
- Quantile Values Tested: [0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.4, 0.5]
- Total Executions: 240 (8 runs per dataset)
- Fixed Default Parameters: $\alpha_{\text{MAD}} = 0.5$, $\alpha_{\text{accept}} = 0.5$, $t_{\text{percentile}} = 0.15$
- Metric Evaluation: Impact of distance quantile on dynamic search radius r and cluster boundaries

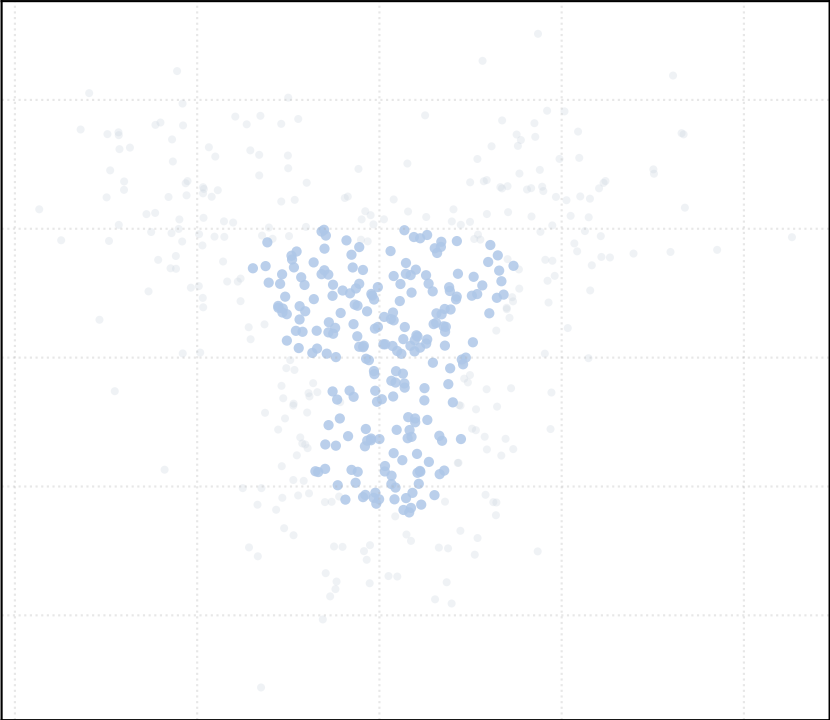
Parameter	Value / Range	Algorithmic Role
mean_to_other_vecs_dist_quant	{0.05, 0.10, 0.15, 0.20, 0.25, 0.30, 0.40, 0.50}	Distance percentile quantile used to calculate search radius
alpha_MAD	0.50 (Default)	MAD multiplier for surface compatibility
alpha_accept	0.50 (Default)	Log2 fold-change acceptance cutoff
t_percentile	0.15 (Default)	Top 15% densest seeds
min_intersection	1 (Default)	Overlapping vectors to trigger merge

Dataset: bifurcation_2way_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

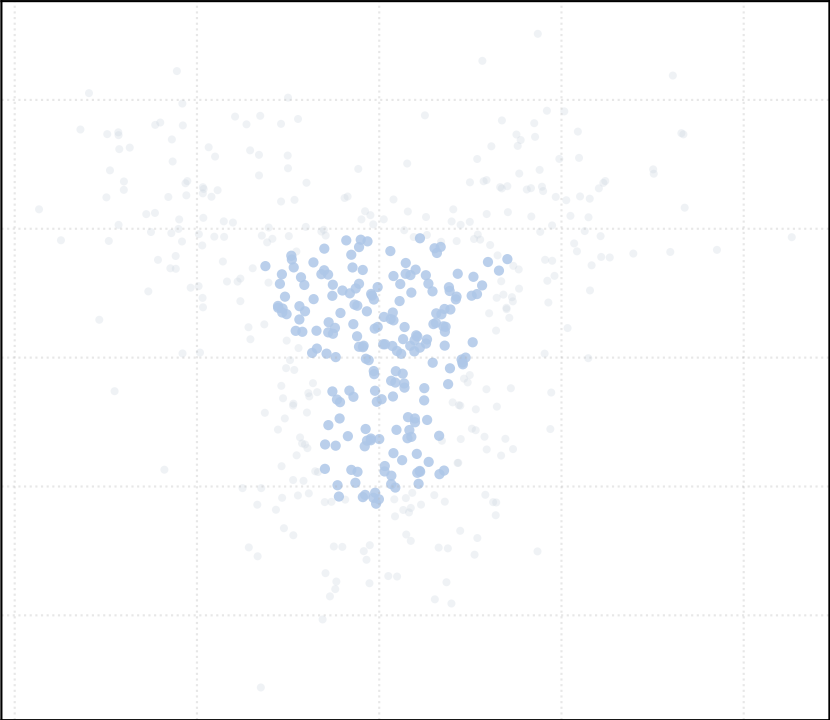
quant = 0.05 [2 Clusters]



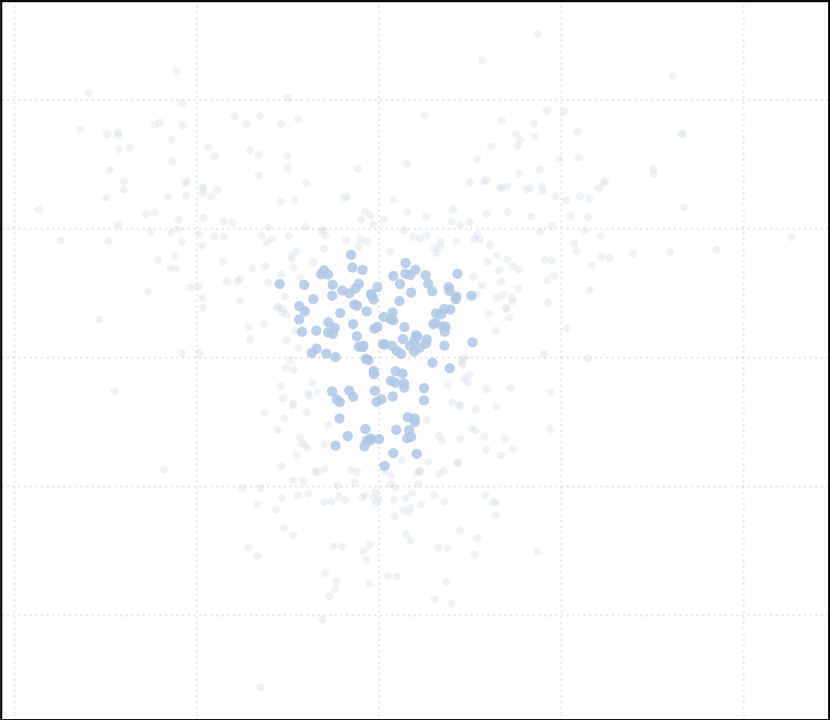
quant = 0.1 [1 Clusters]



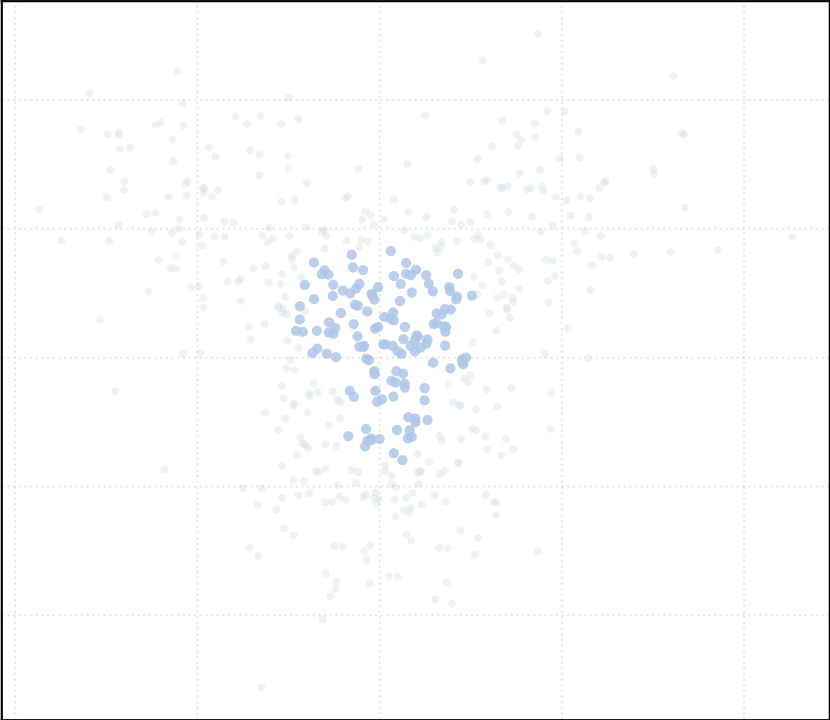
quant = 0.15 (Default) [1 Clusters]



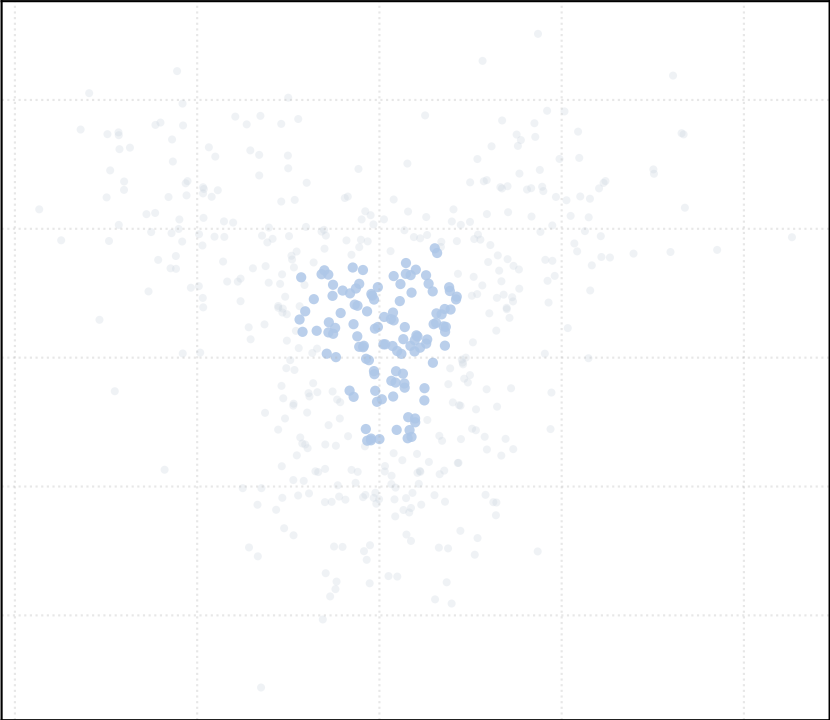
quant = 0.2 [1 Clusters]



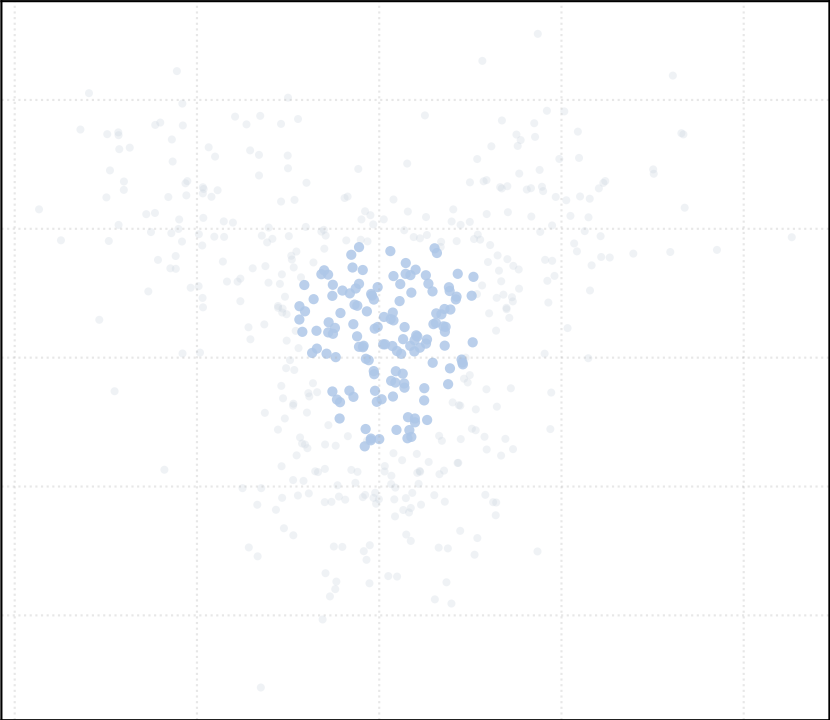
quant = 0.25 [1 Clusters]



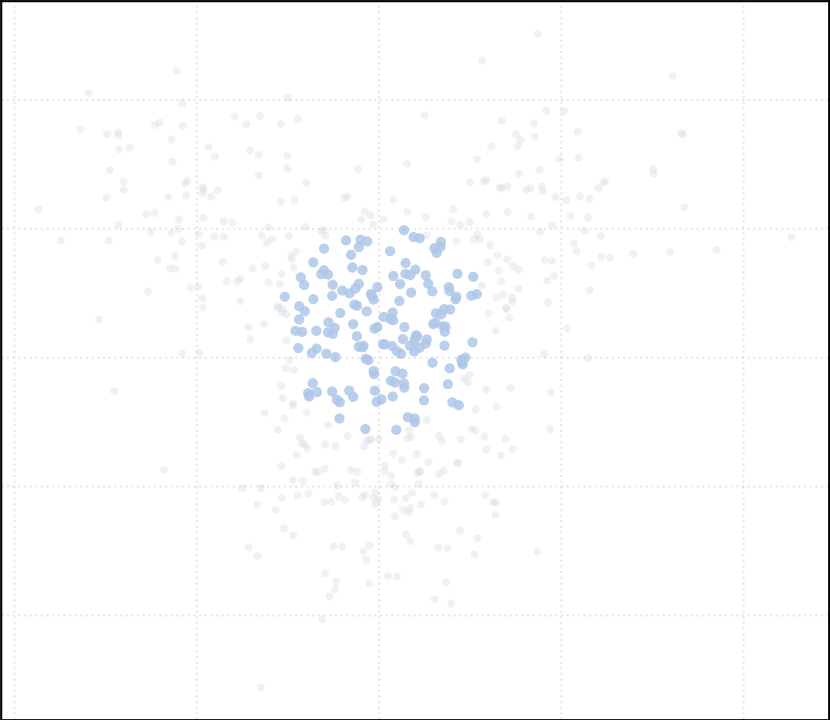
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

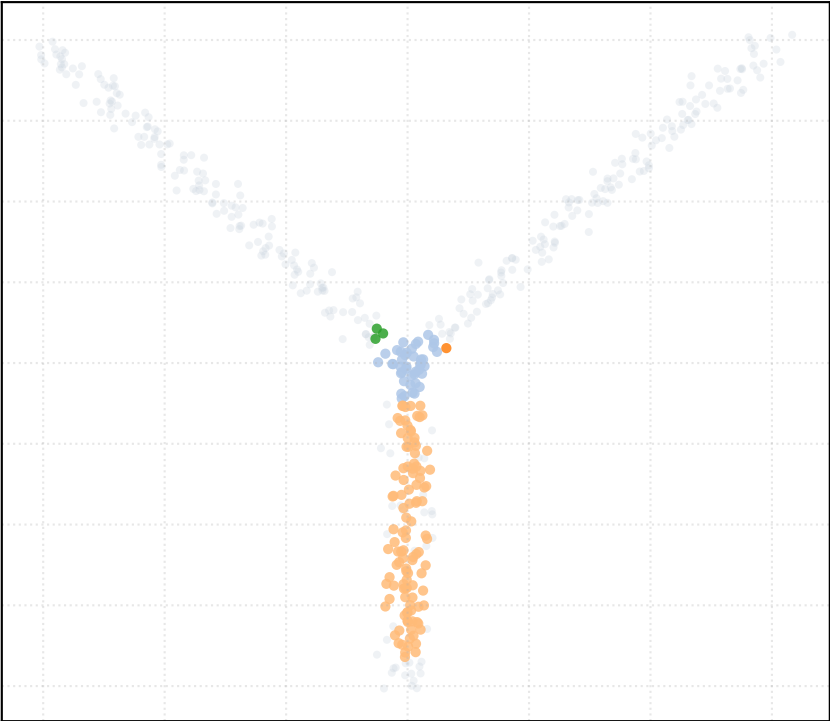


quant = 0.5 [1 Clusters]

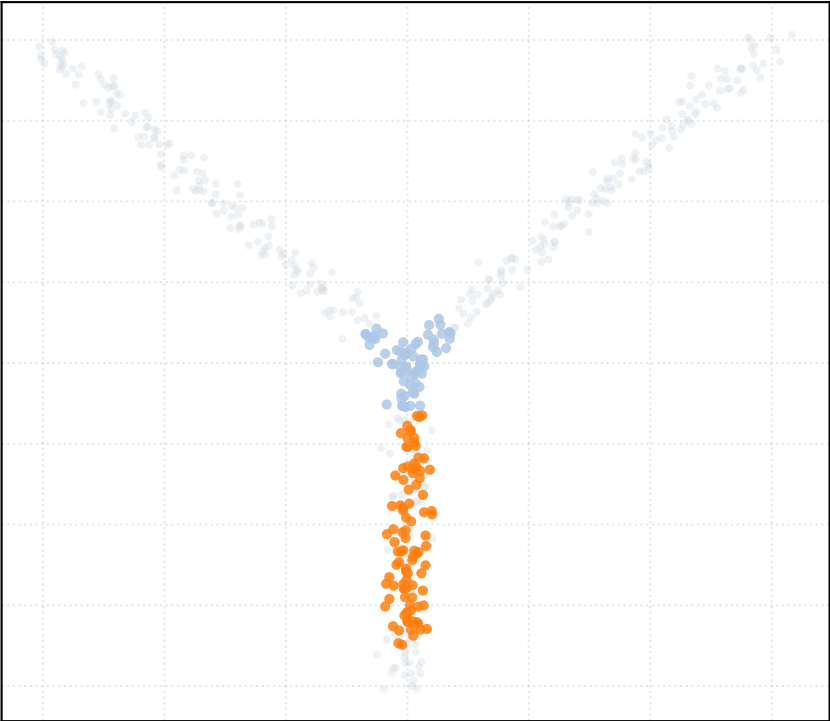


Dataset: bifurcation_2way_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

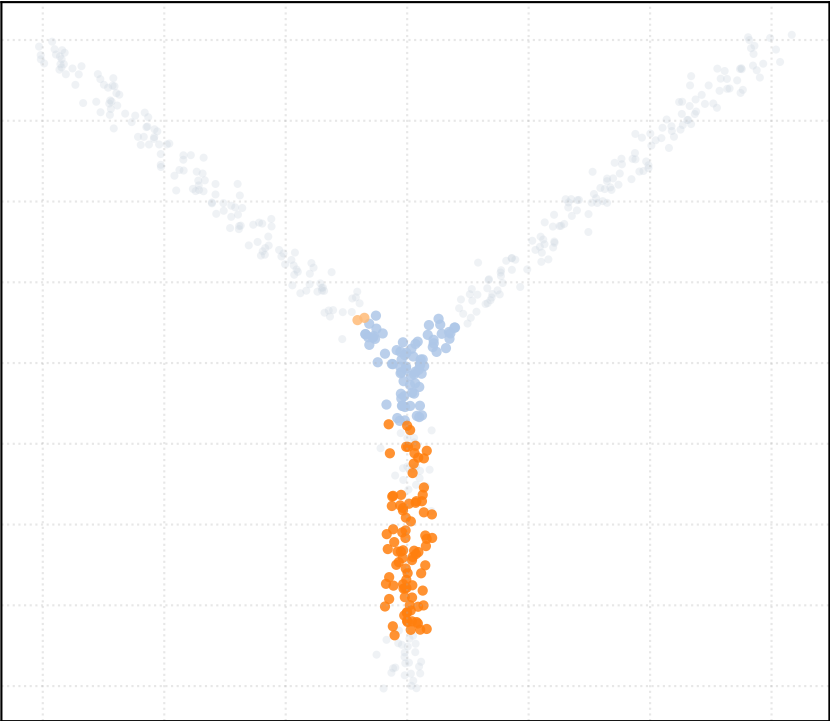
quant = 0.05 [4 Clusters]



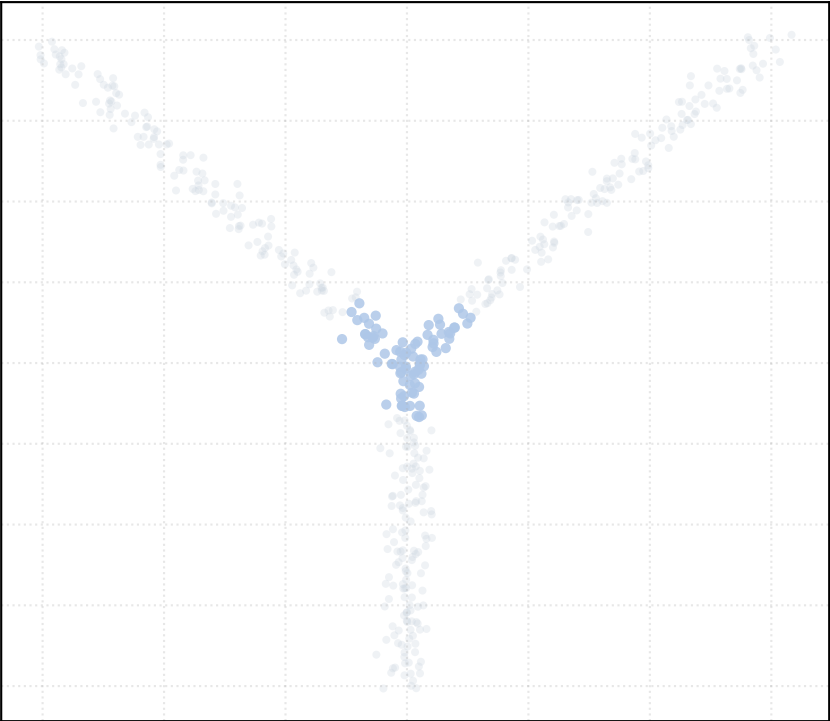
quant = 0.1 [2 Clusters]



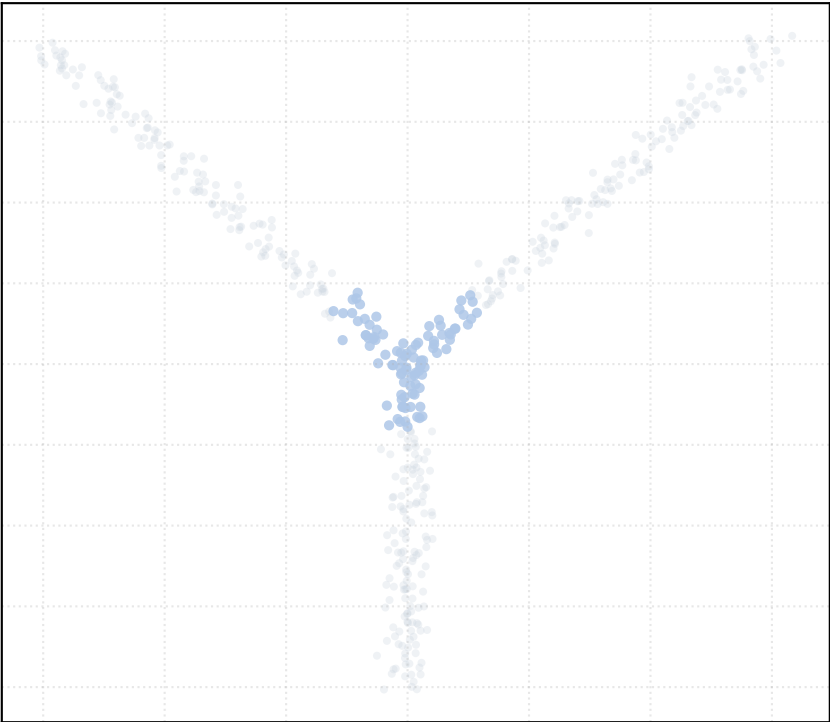
quant = 0.15 (Default) [3 Clusters]



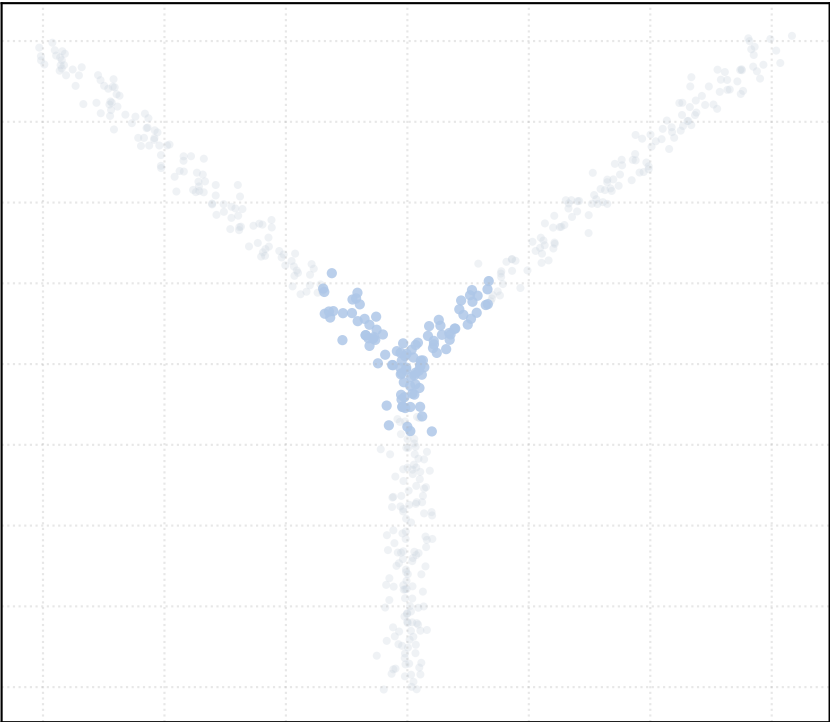
quant = 0.2 [1 Clusters]



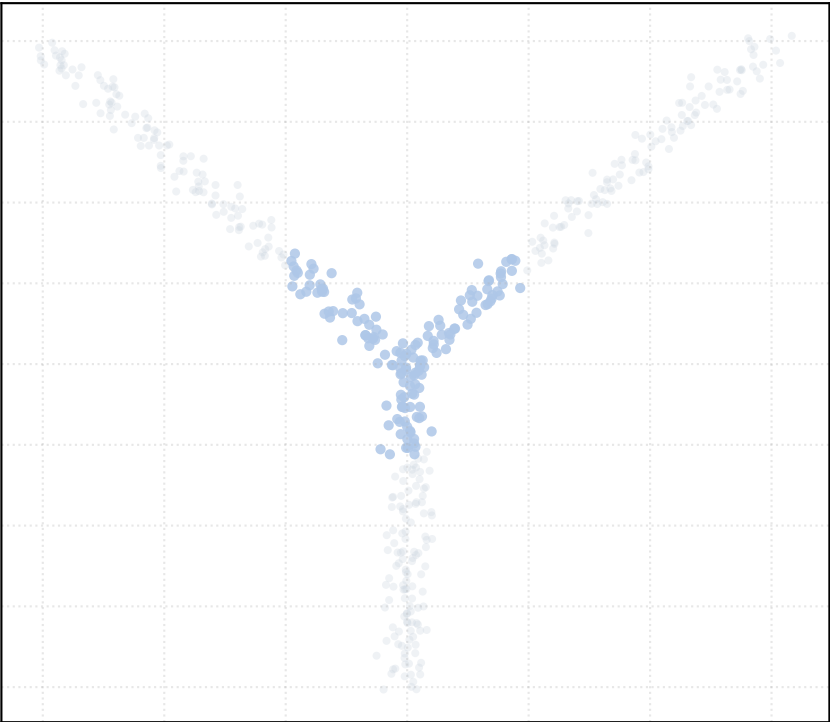
quant = 0.25 [1 Clusters]



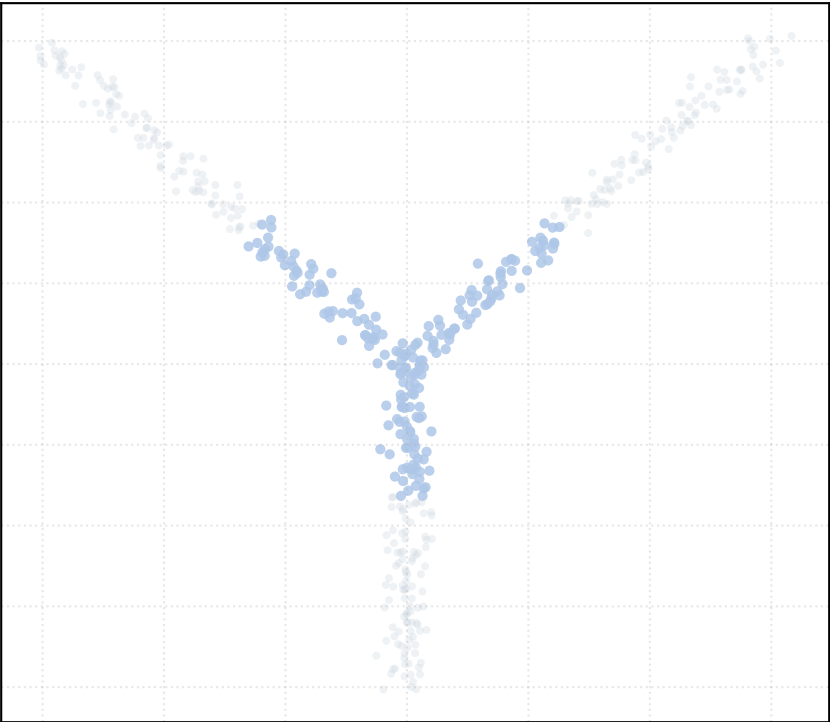
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

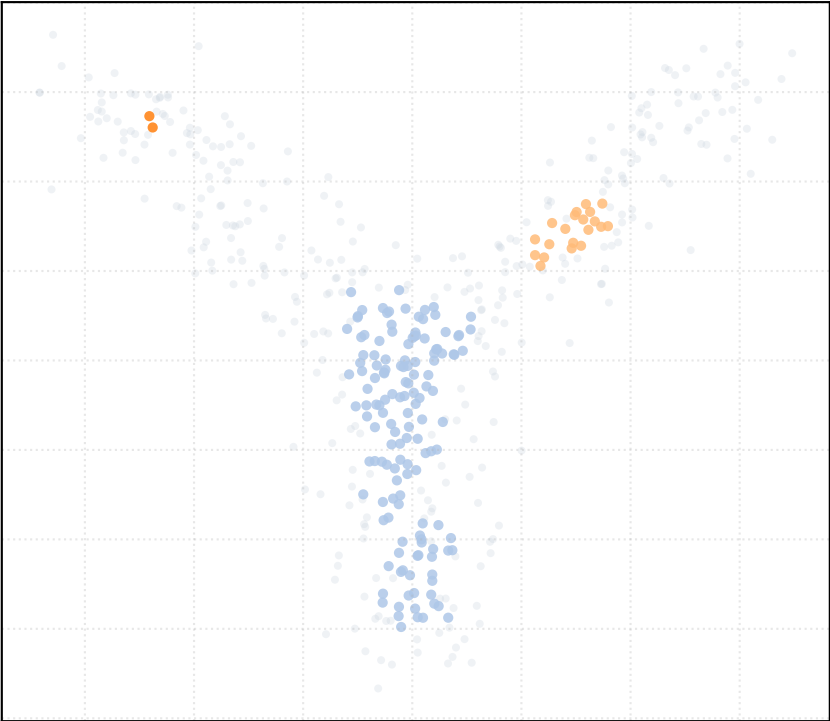


quant = 0.5 [1 Clusters]

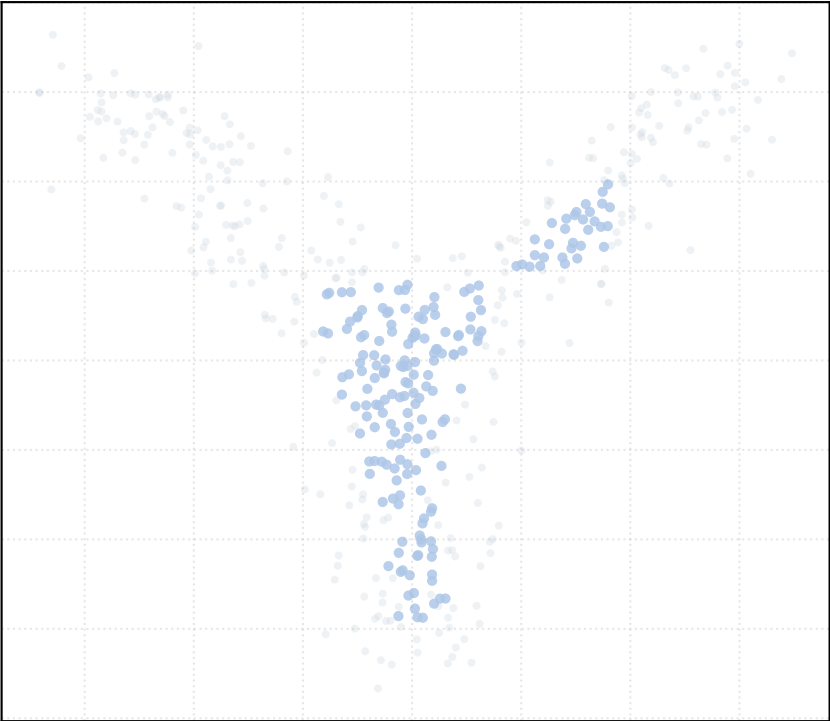


Dataset: bifurcation_2way_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

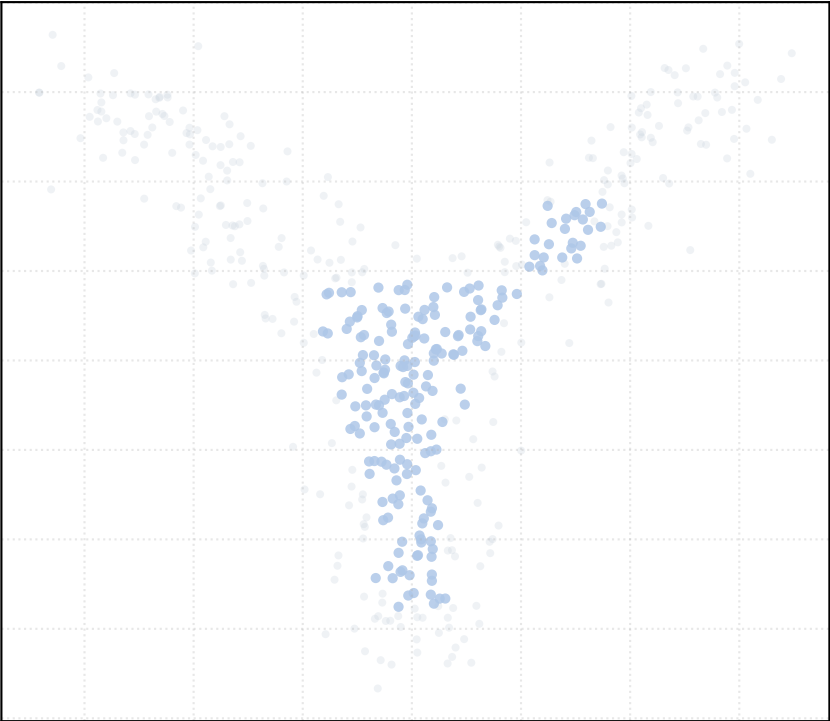
quant = 0.05 [3 Clusters]



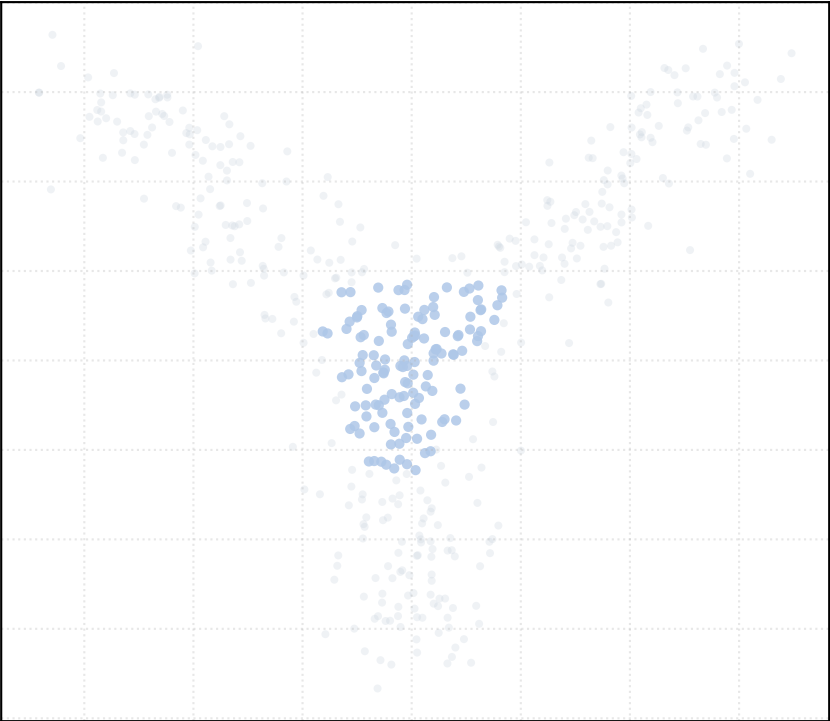
quant = 0.1 [1 Clusters]



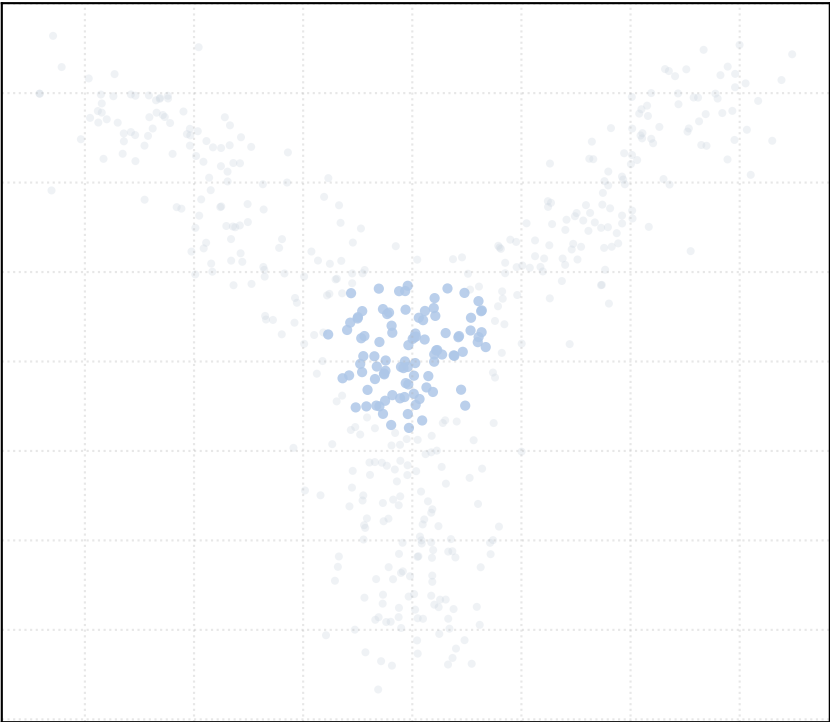
quant = 0.15 (Default) [1 Clusters]



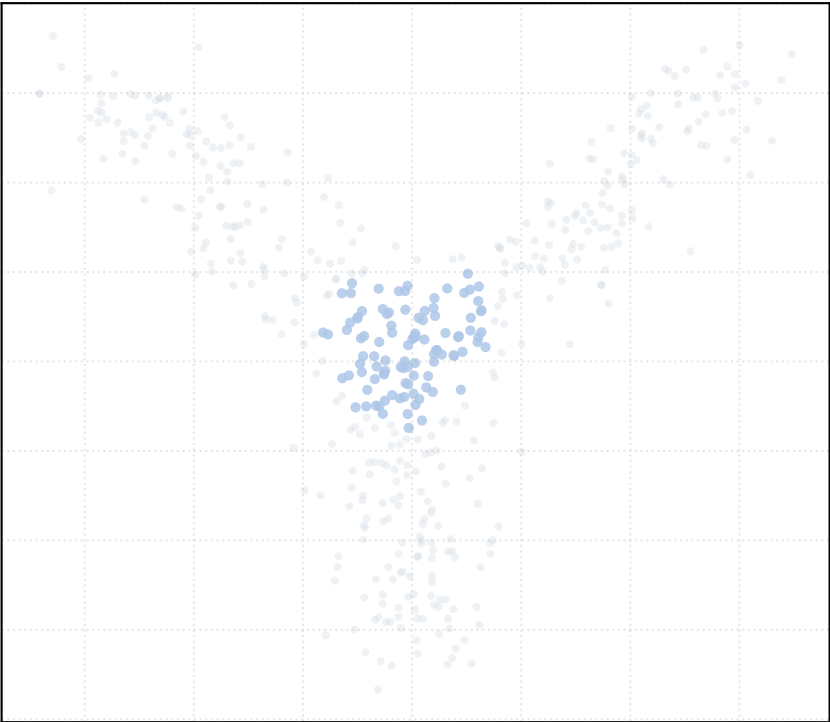
quant = 0.2 [1 Clusters]



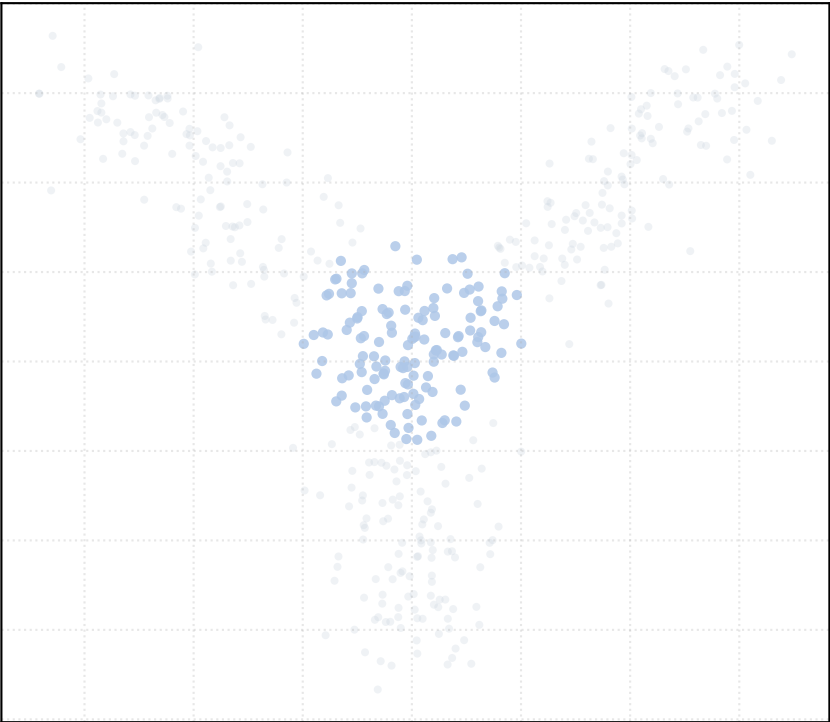
quant = 0.25 [1 Clusters]



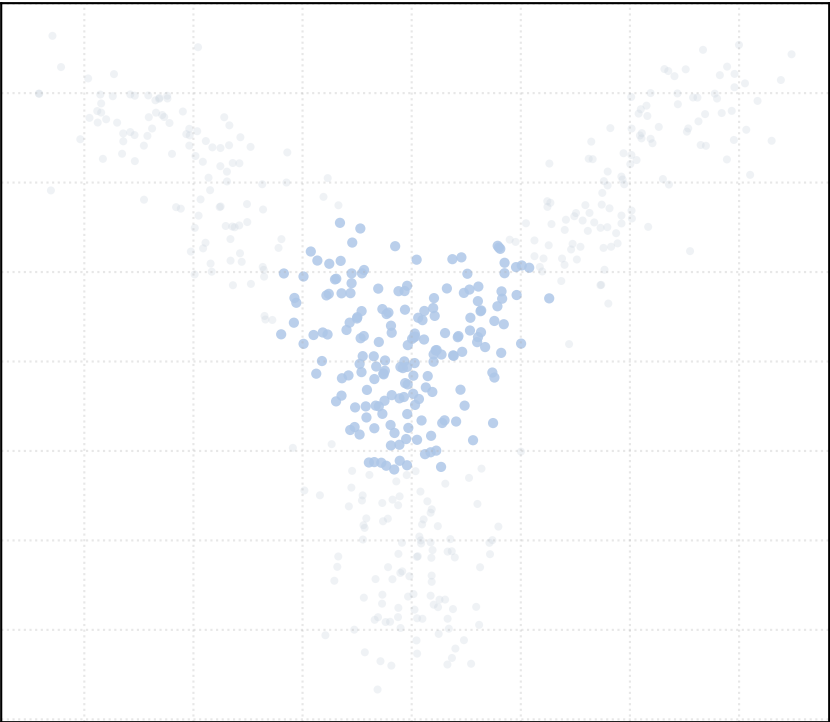
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

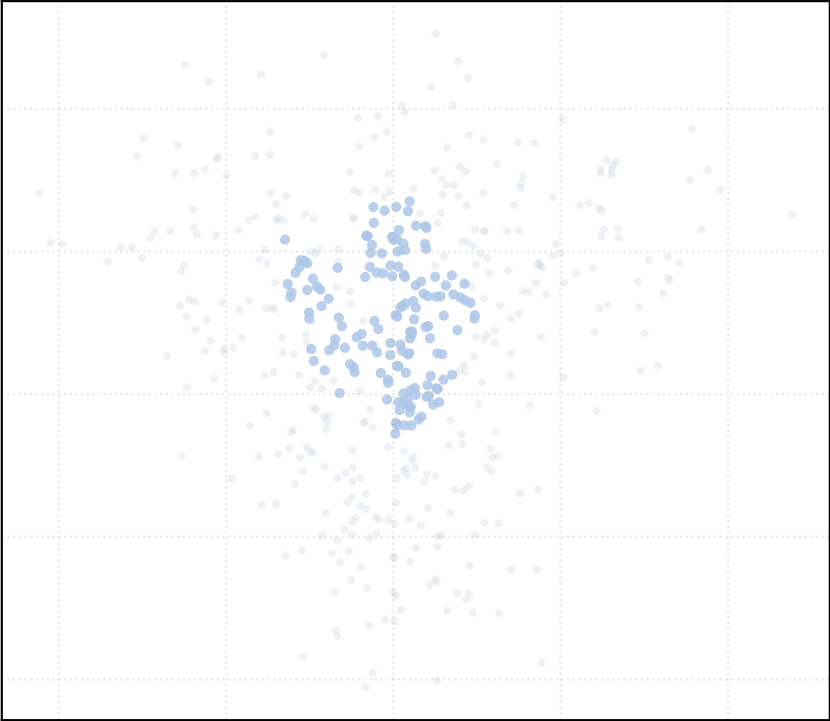


quant = 0.5 [1 Clusters]

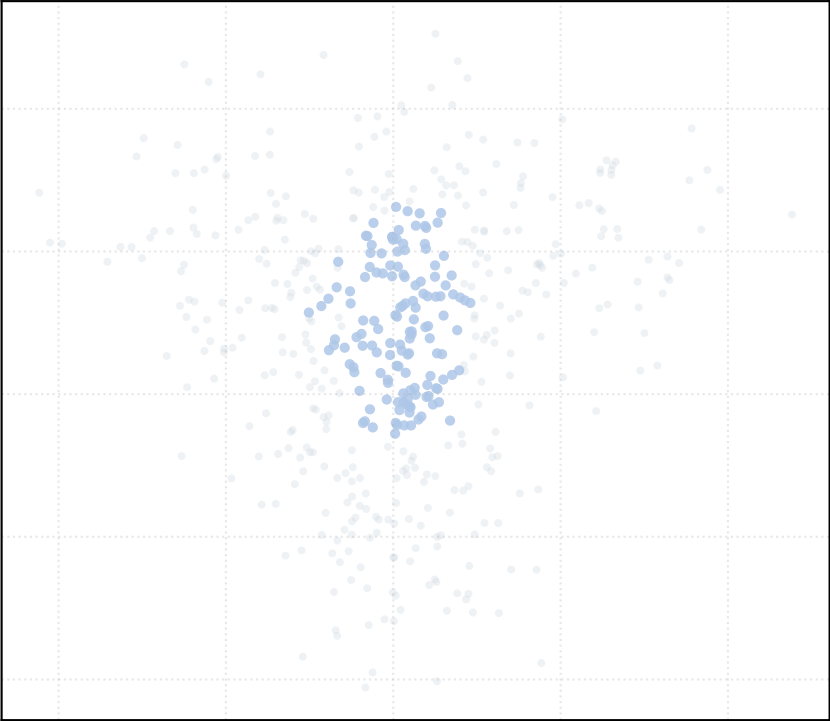


Dataset: bifurcation_3way_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

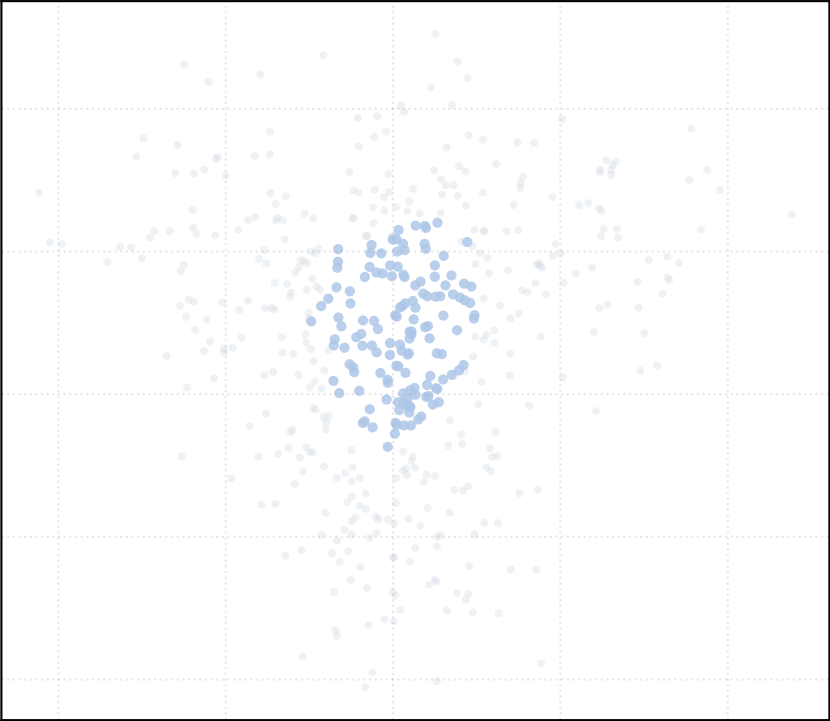
quant = 0.05 [1 Clusters]



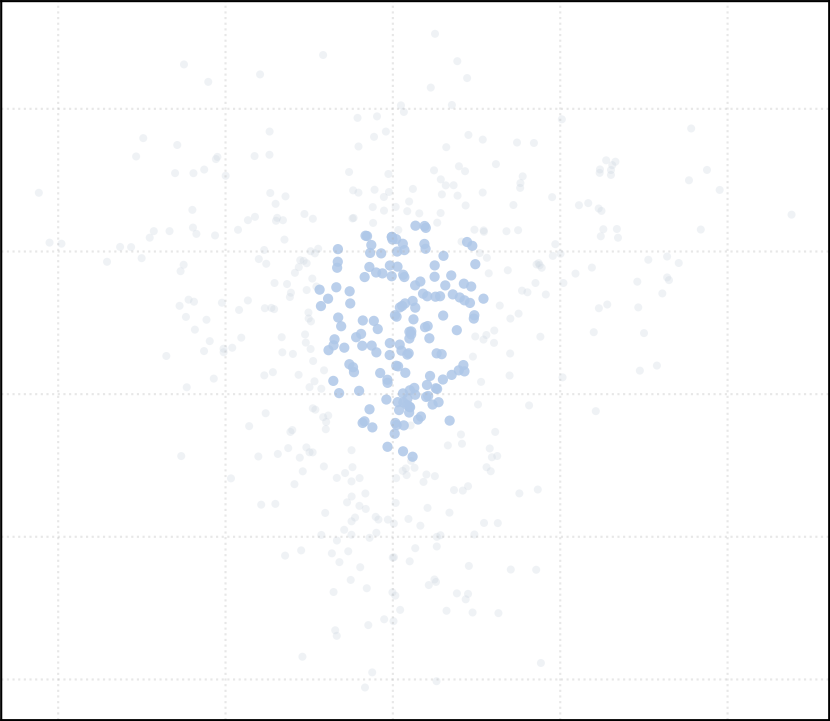
quant = 0.1 [1 Clusters]



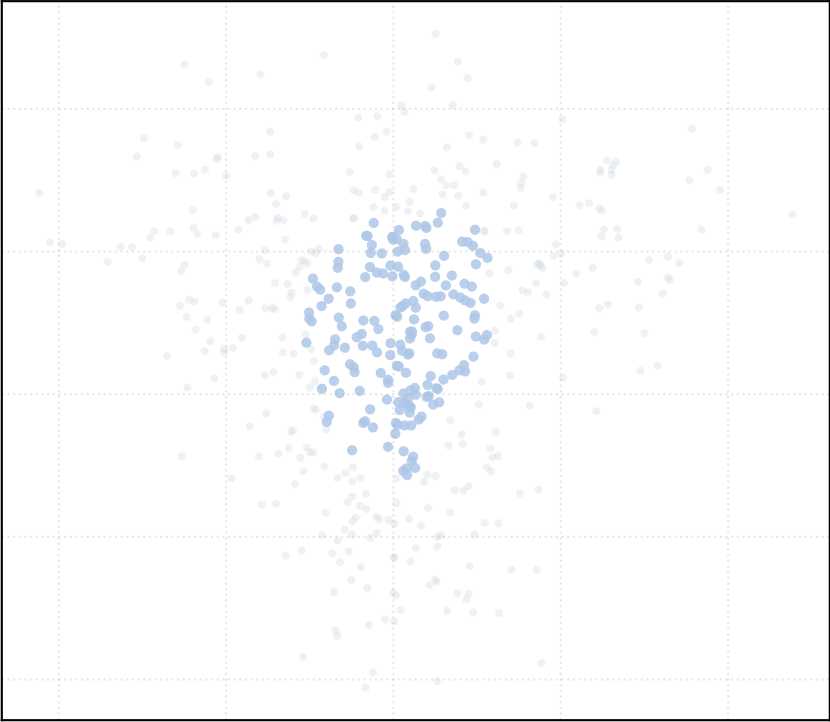
quant = 0.15 (Default) [1 Clusters]



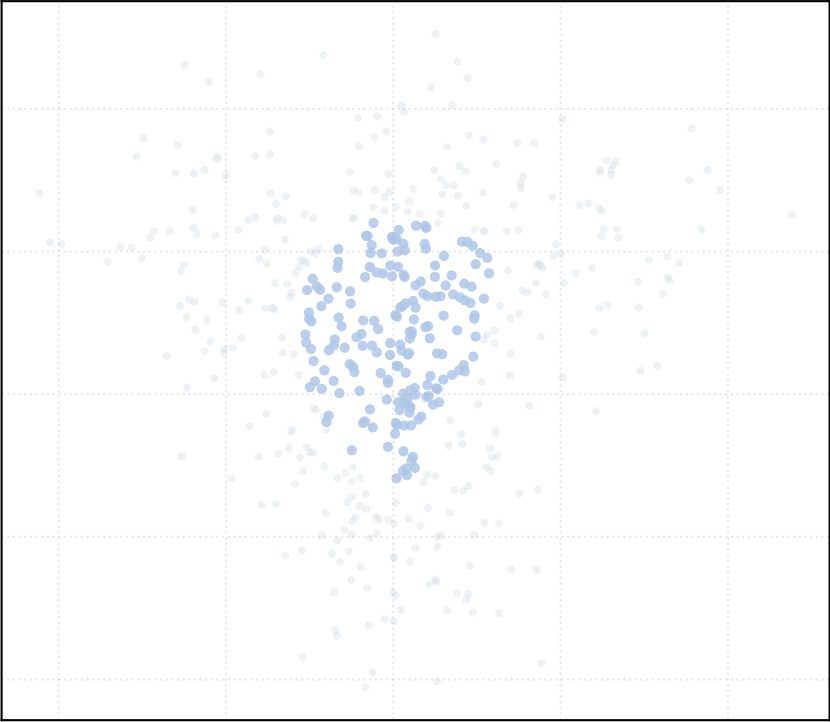
quant = 0.2 [1 Clusters]



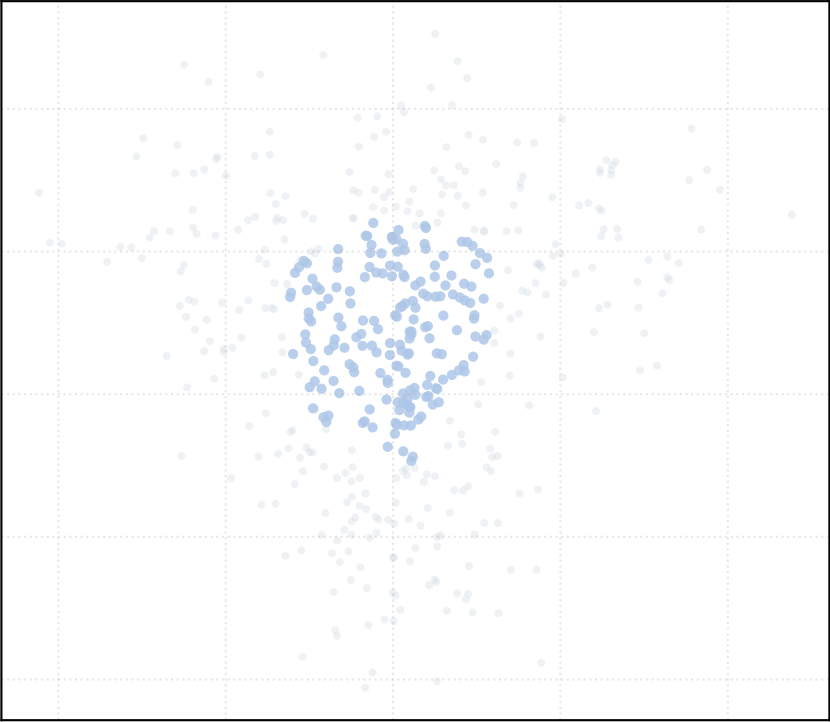
quant = 0.25 [1 Clusters]



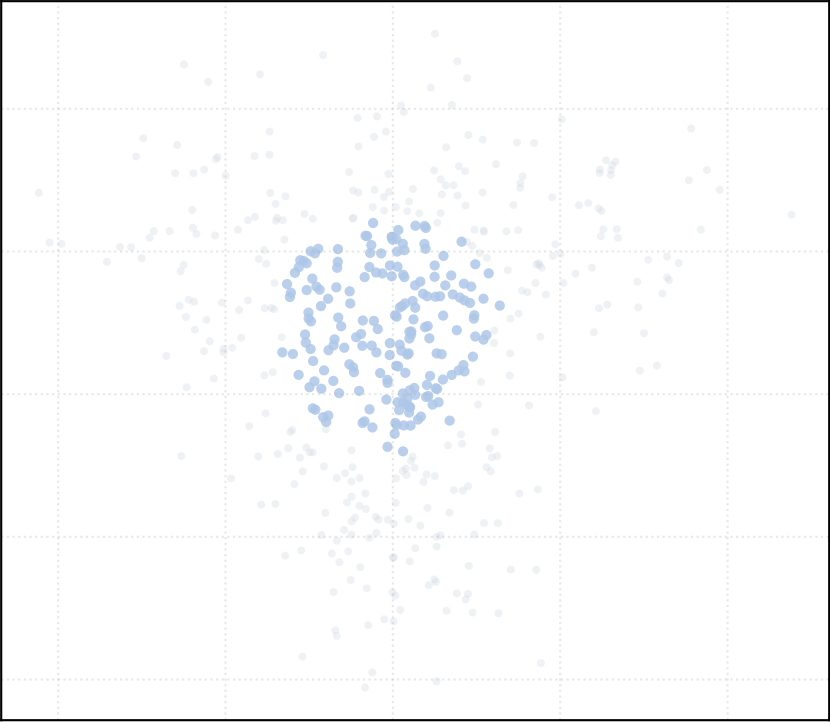
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

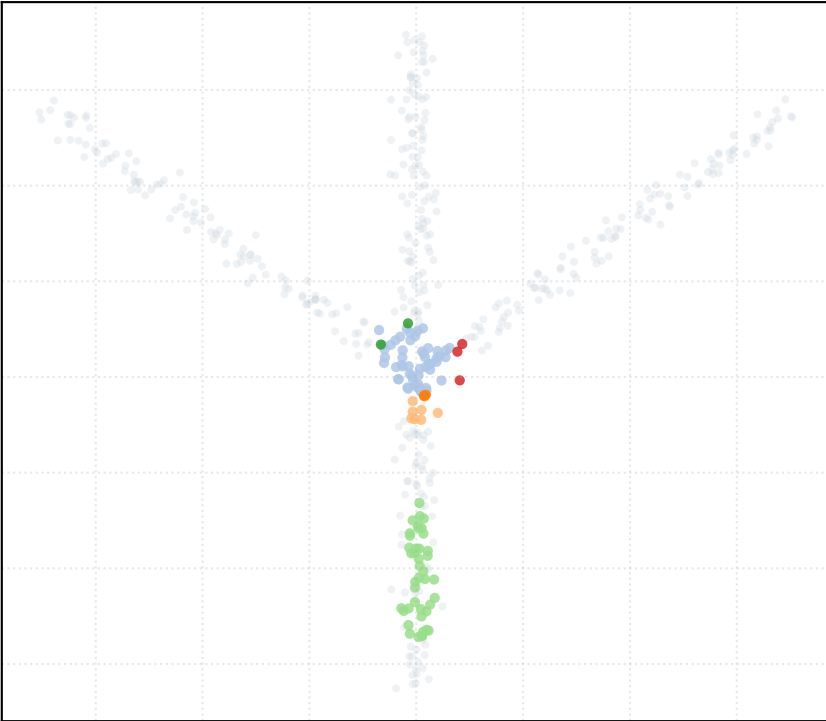


quant = 0.5 [1 Clusters]

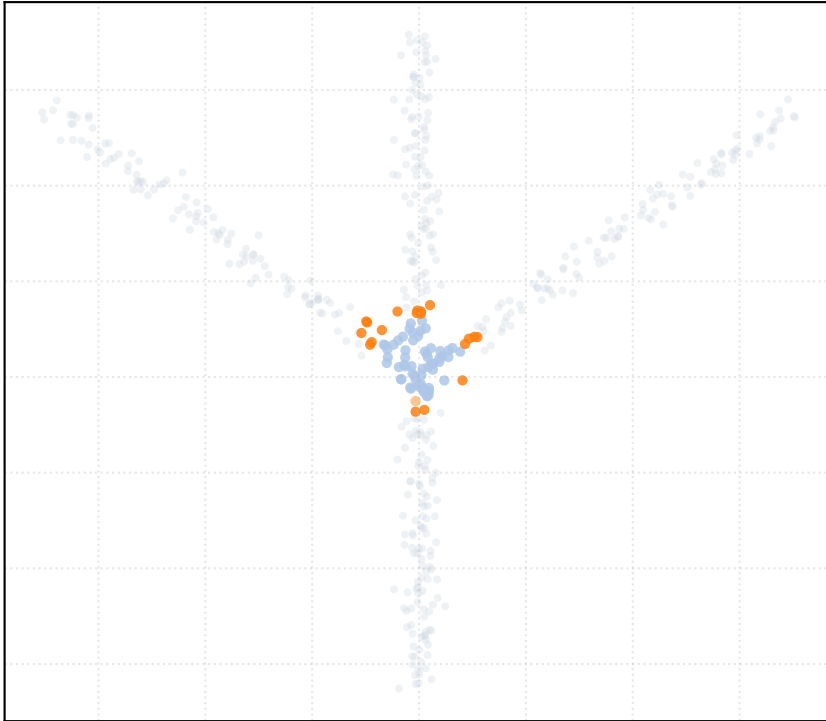


Dataset: bifurcation_3way_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

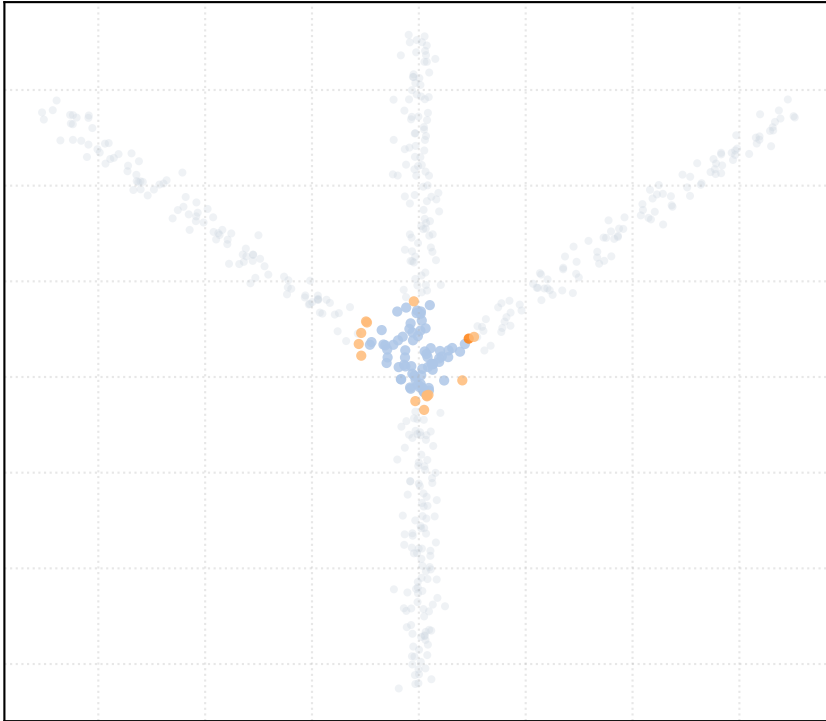
quant = 0.05 [6 Clusters]



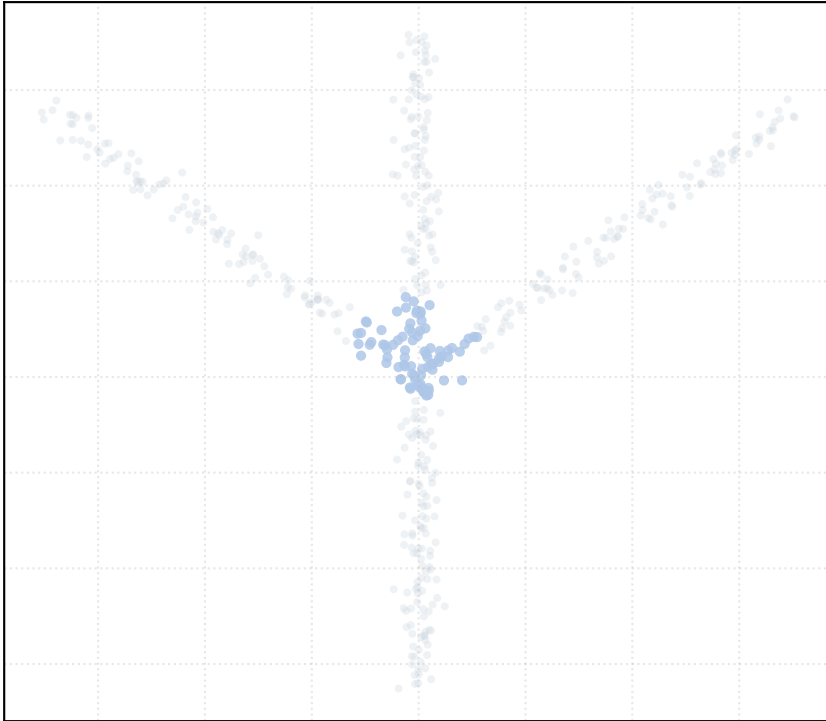
quant = 0.1 [3 Clusters]



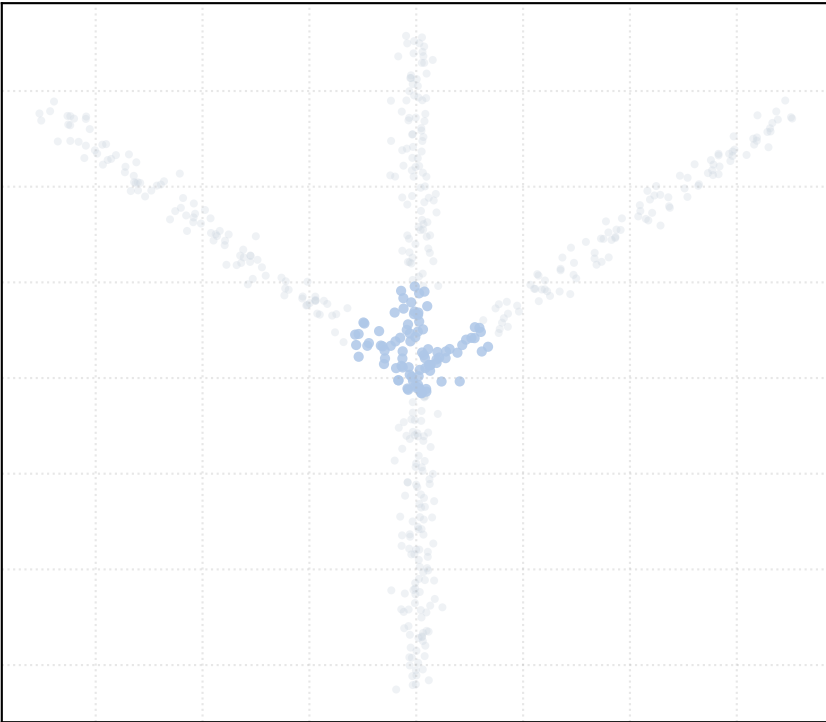
quant = 0.15 (Default) [3 Clusters]



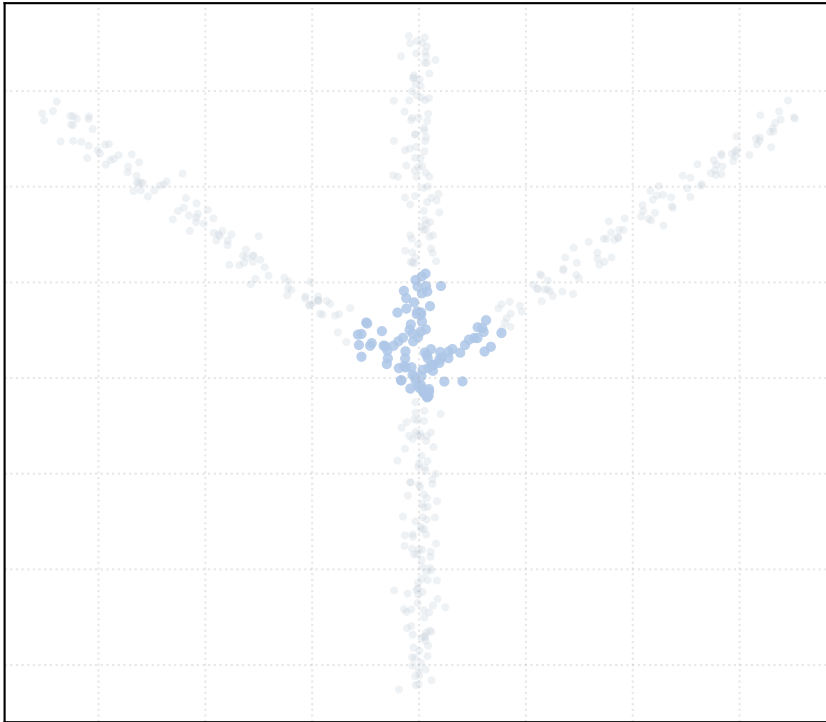
quant = 0.2 [1 Clusters]



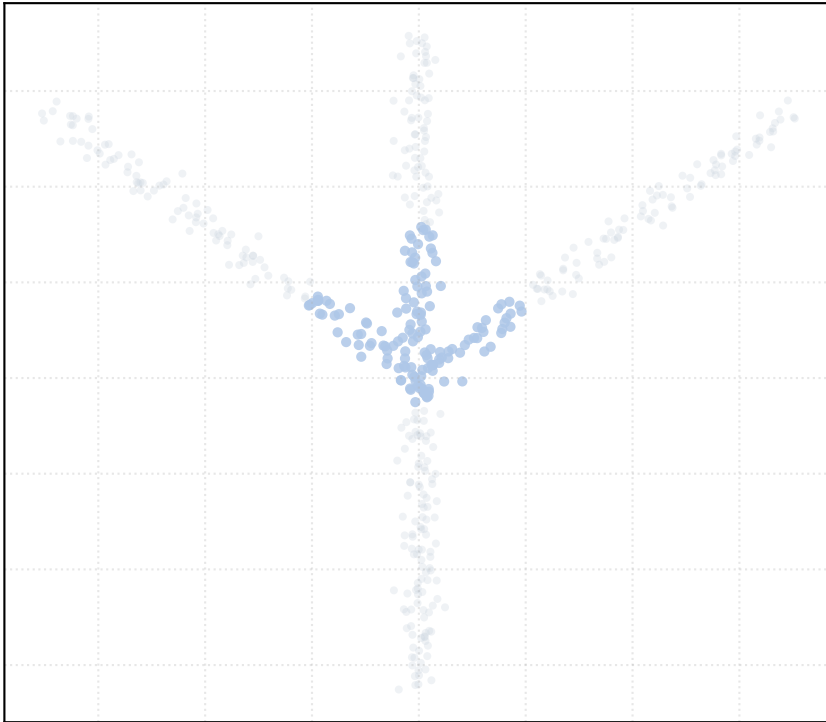
quant = 0.25 [1 Clusters]



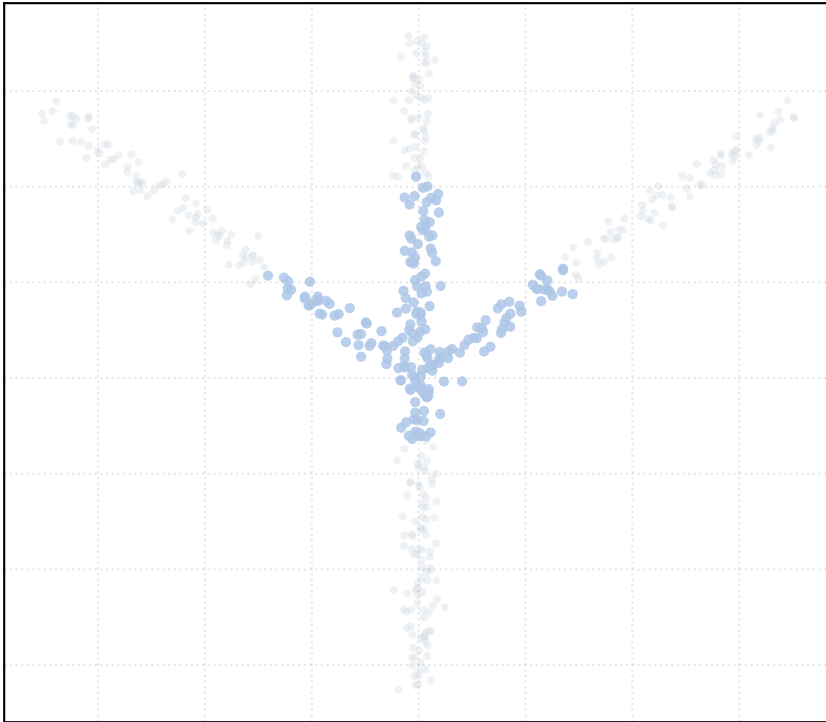
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

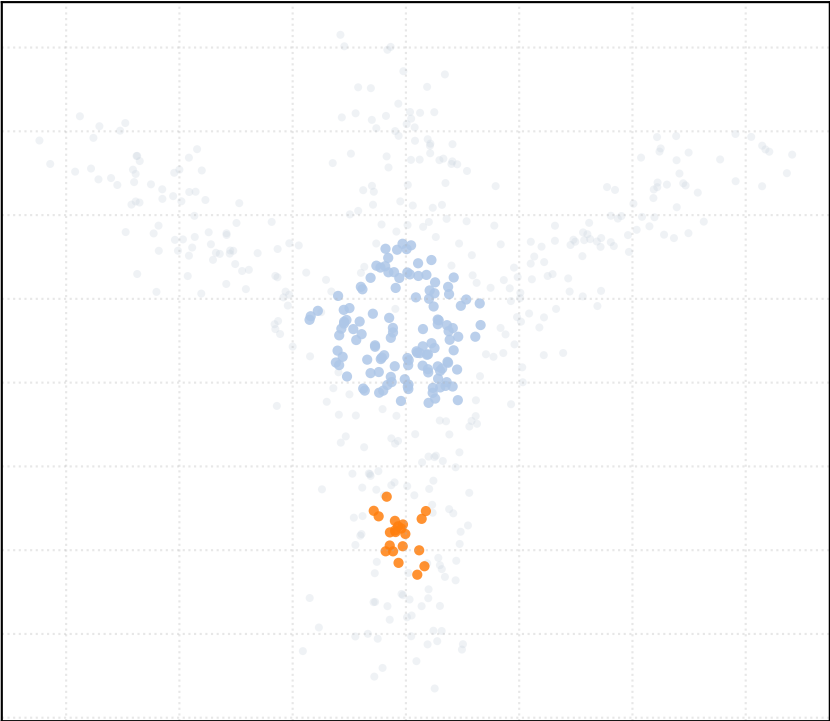


quant = 0.5 [1 Clusters]

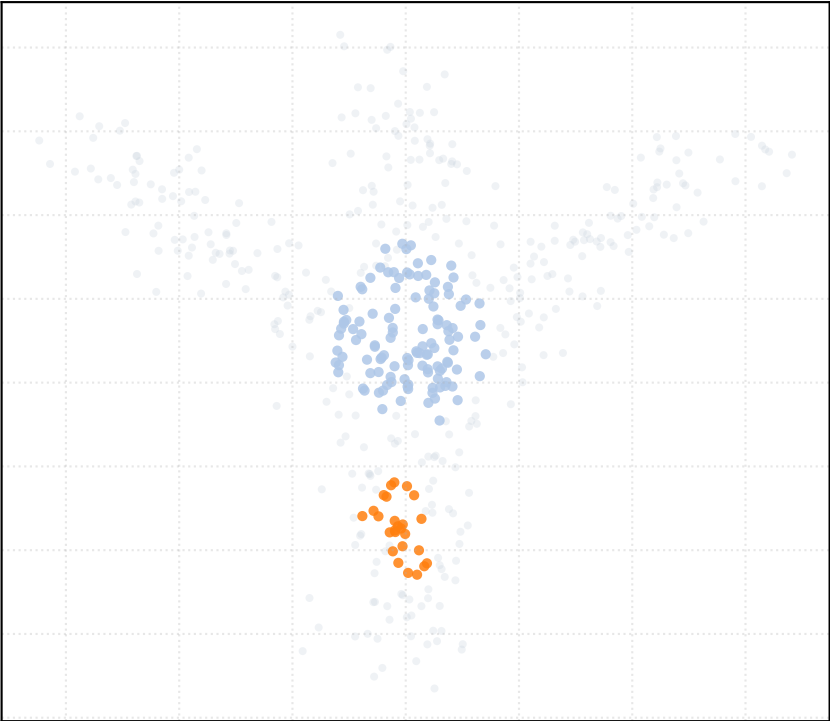


Dataset: bifurcation_3way_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

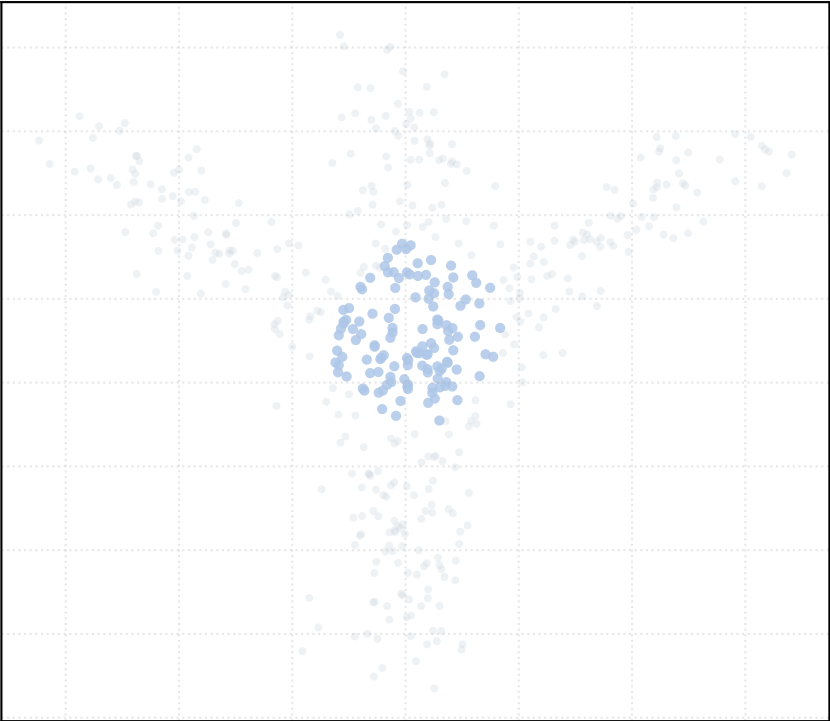
quant = 0.05 [2 Clusters]



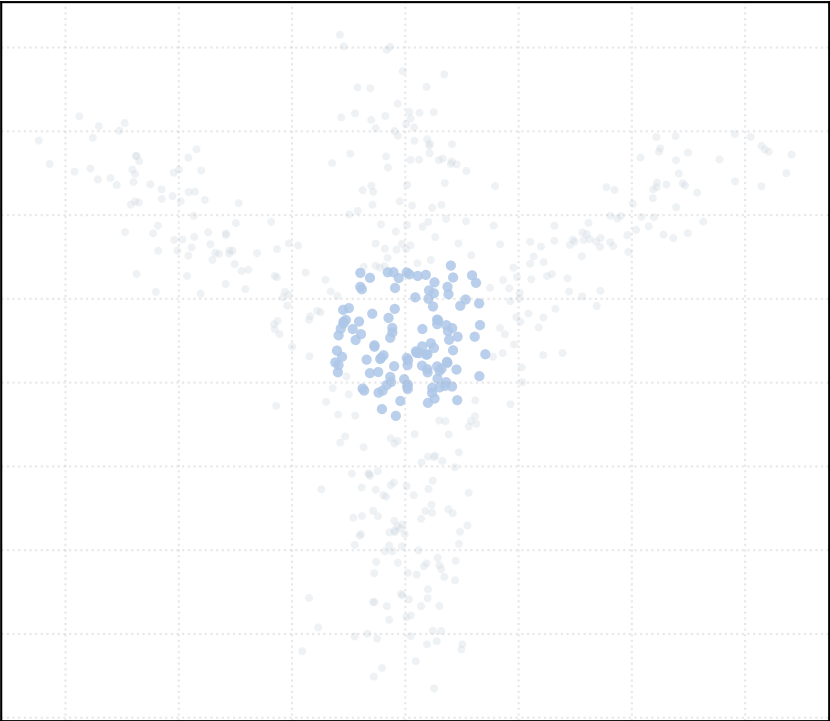
quant = 0.1 [2 Clusters]



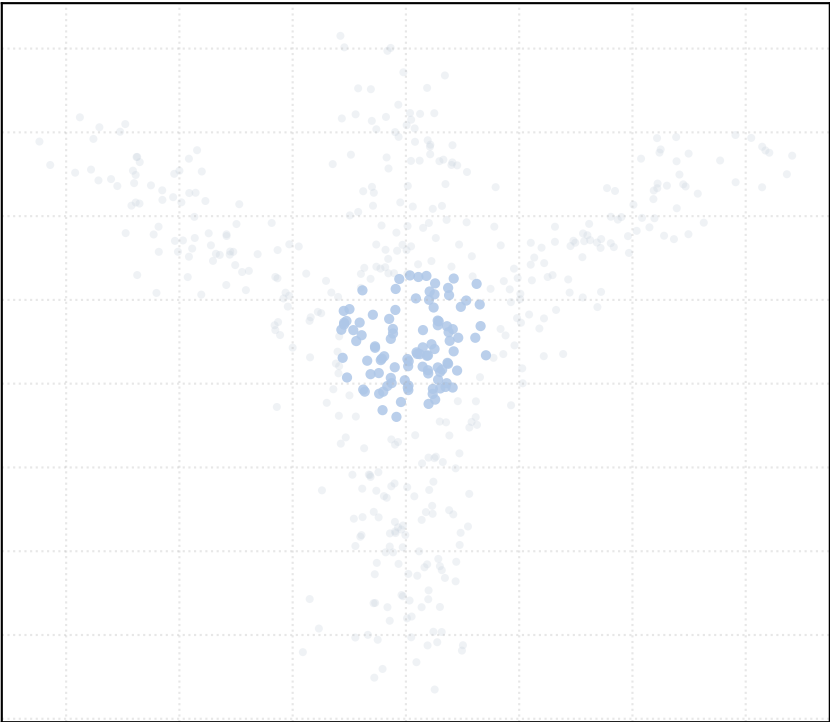
quant = 0.15 (Default) [1 Clusters]



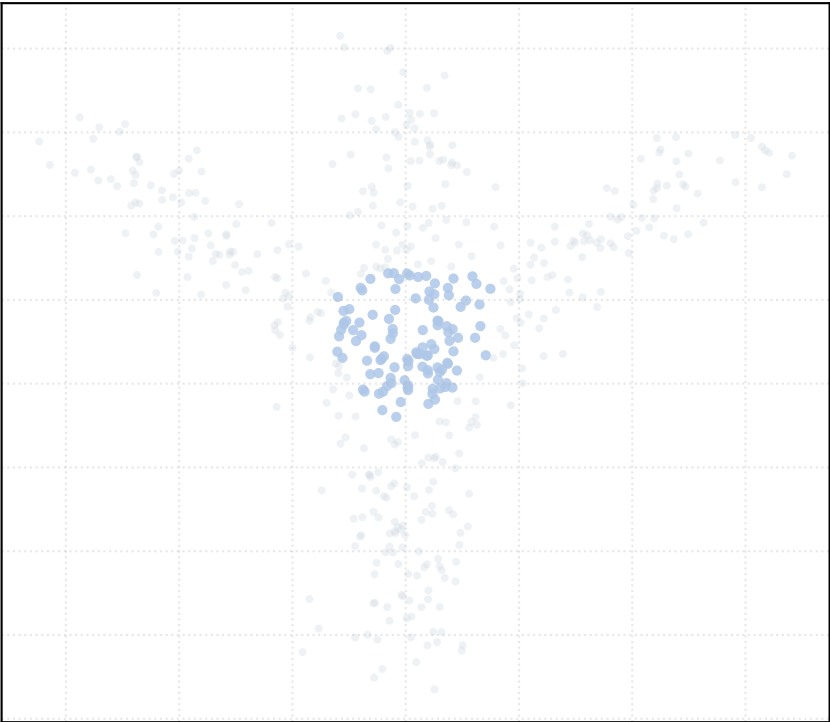
quant = 0.2 [1 Clusters]



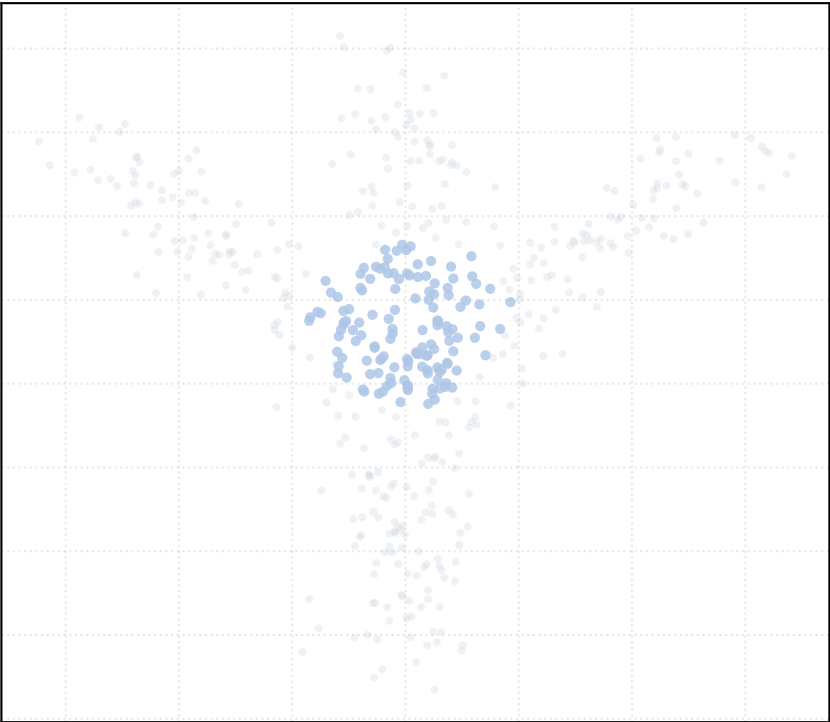
quant = 0.25 [1 Clusters]



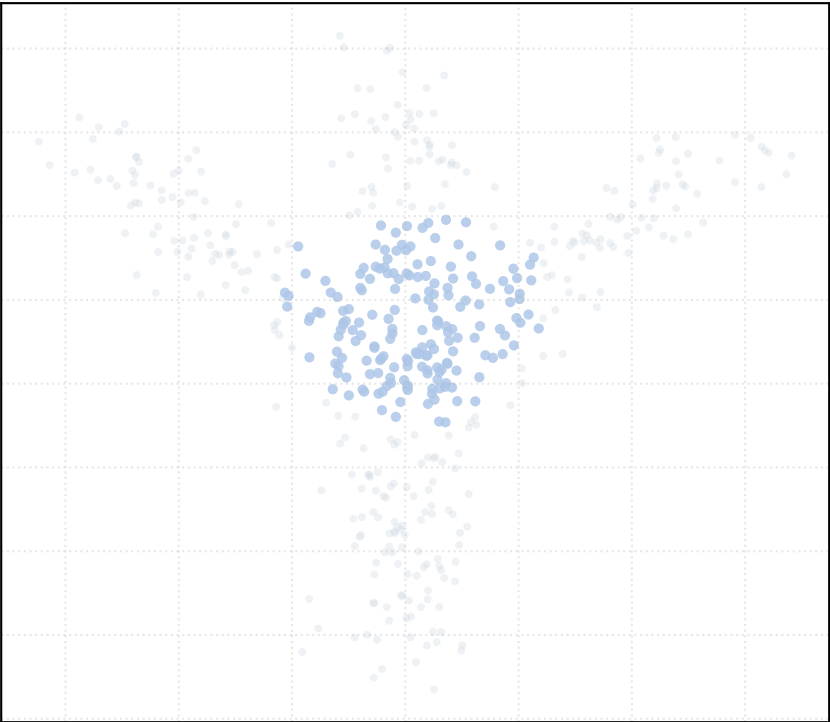
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

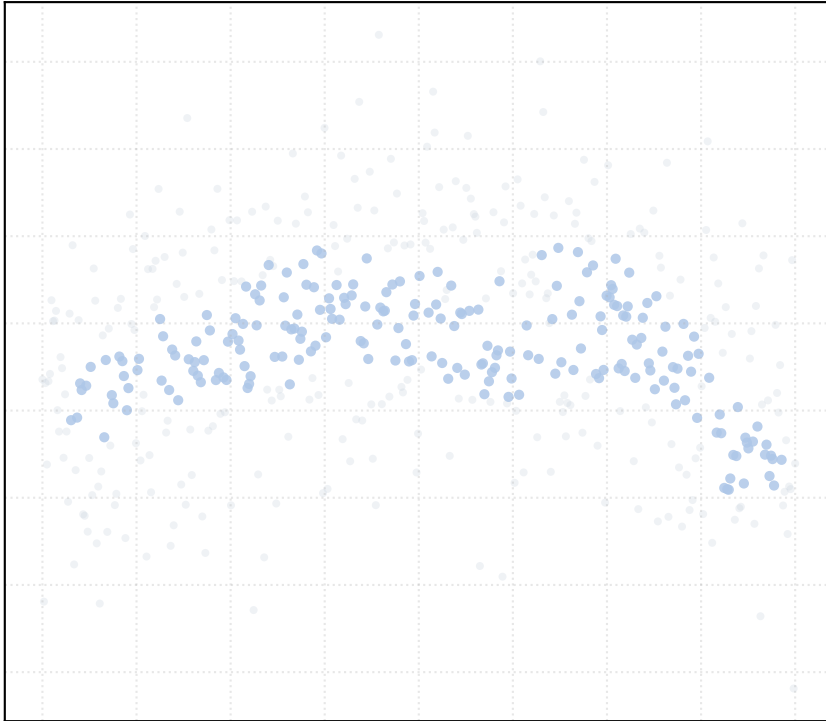


quant = 0.5 [1 Clusters]

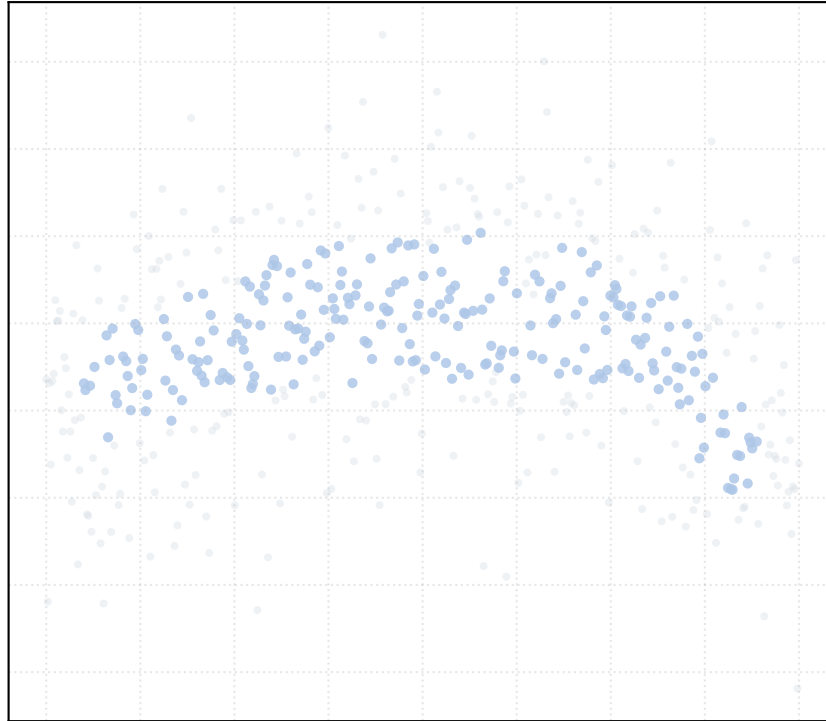


Dataset: circular_arc_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

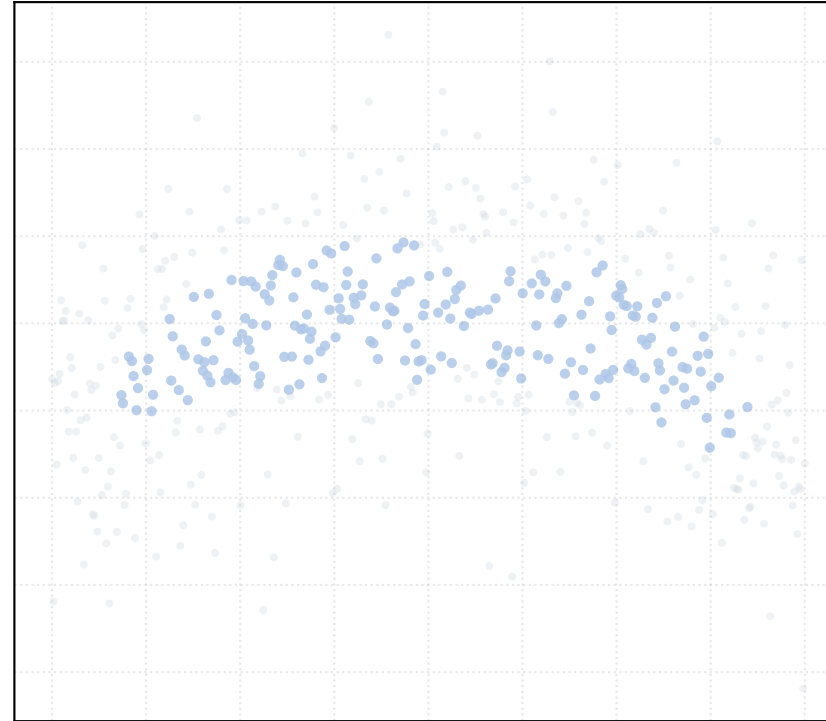
quant = 0.05 [1 Clusters]



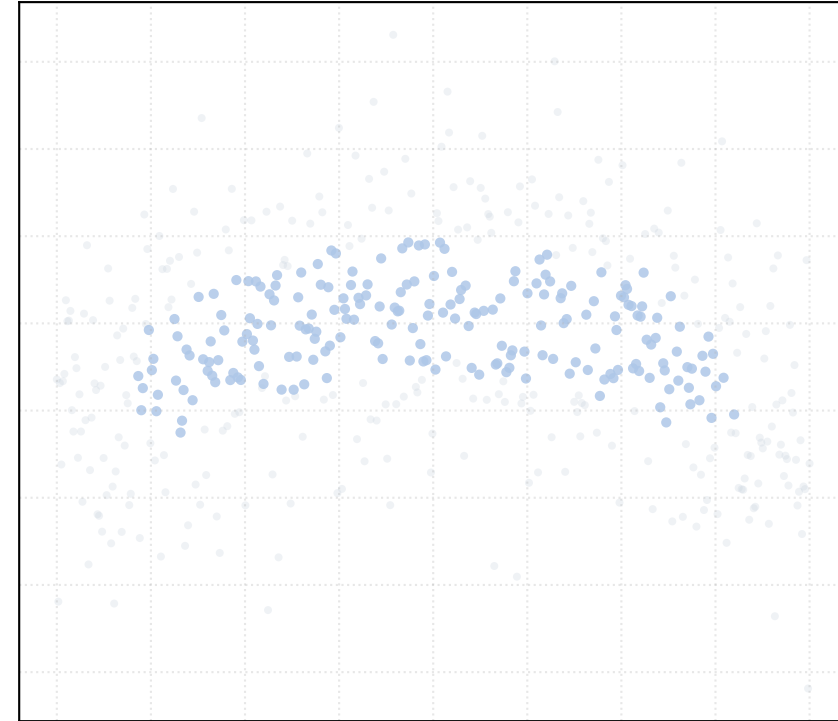
quant = 0.1 [1 Clusters]



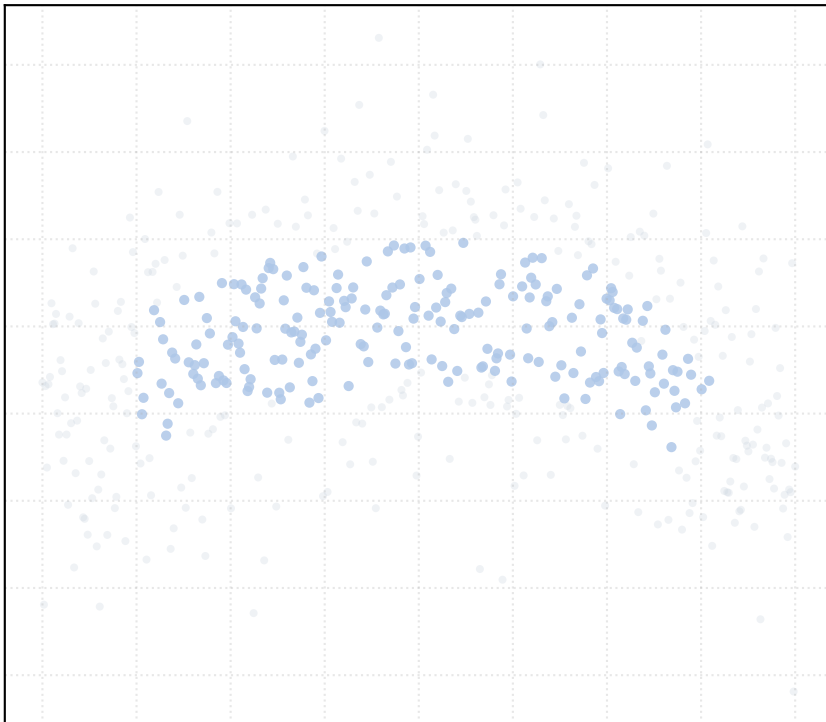
quant = 0.15 (Default) [1 Clusters]



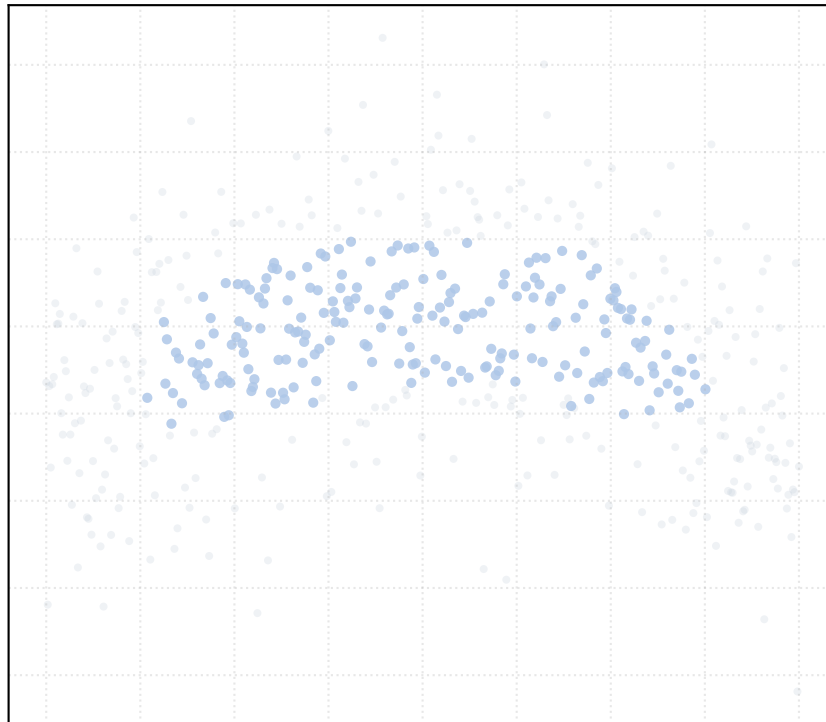
quant = 0.2 [1 Clusters]



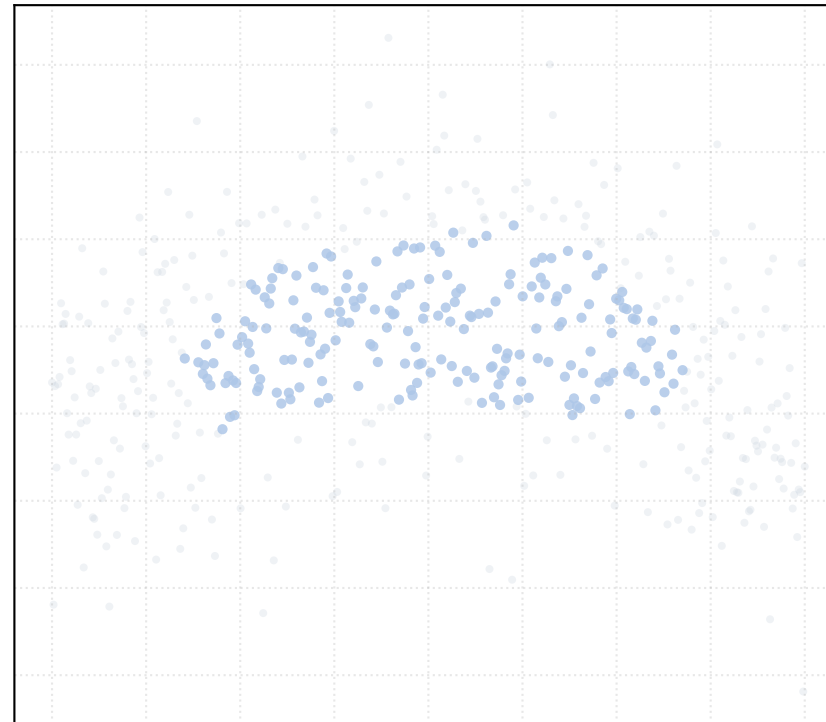
quant = 0.25 [1 Clusters]



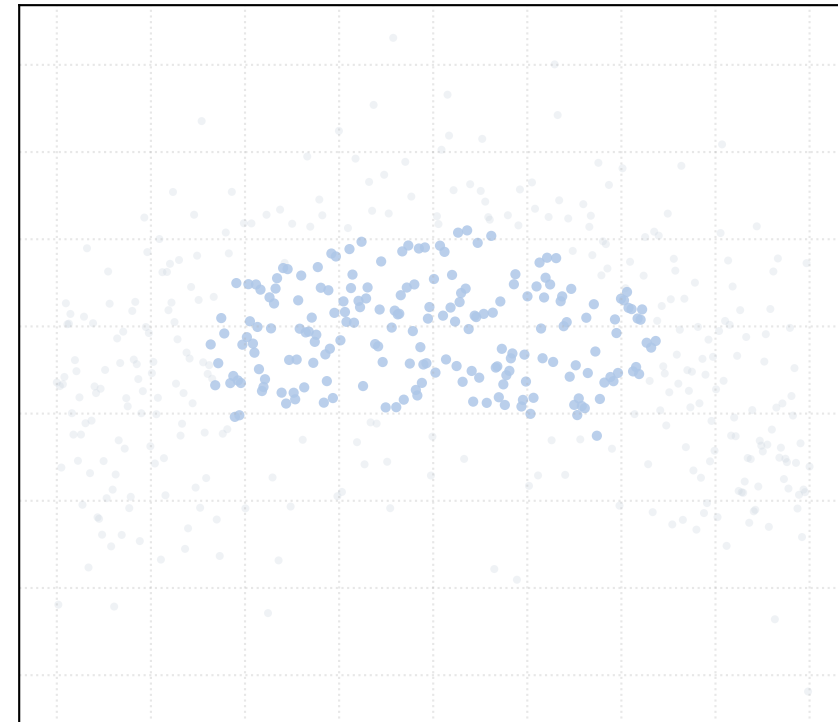
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

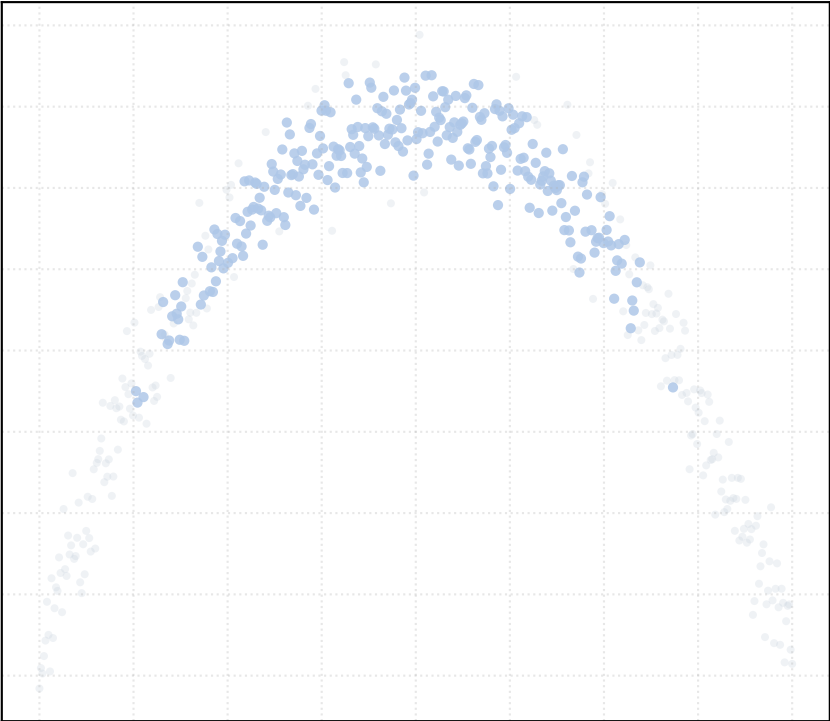


quant = 0.5 [1 Clusters]

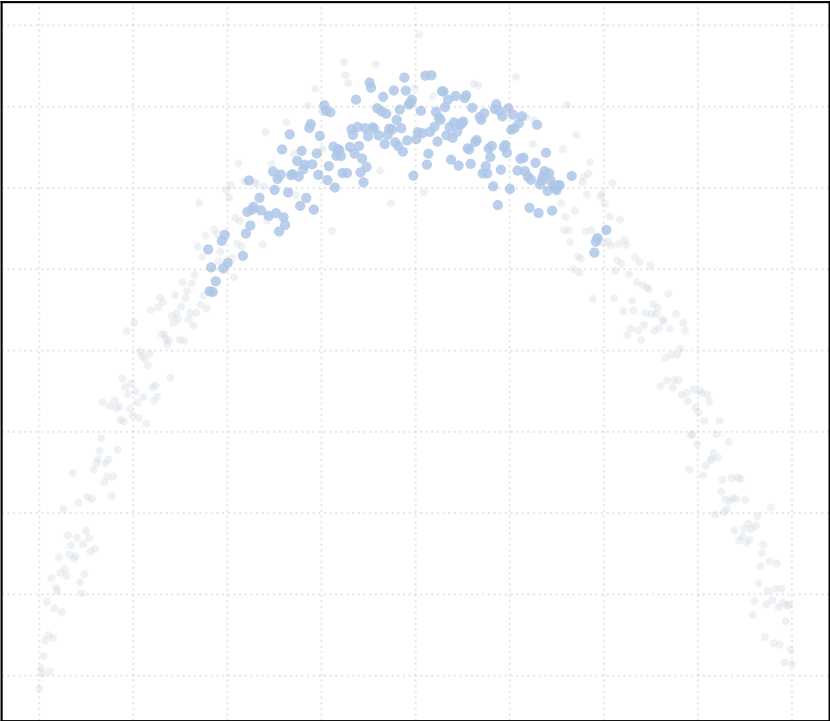


Dataset: circular_arc_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

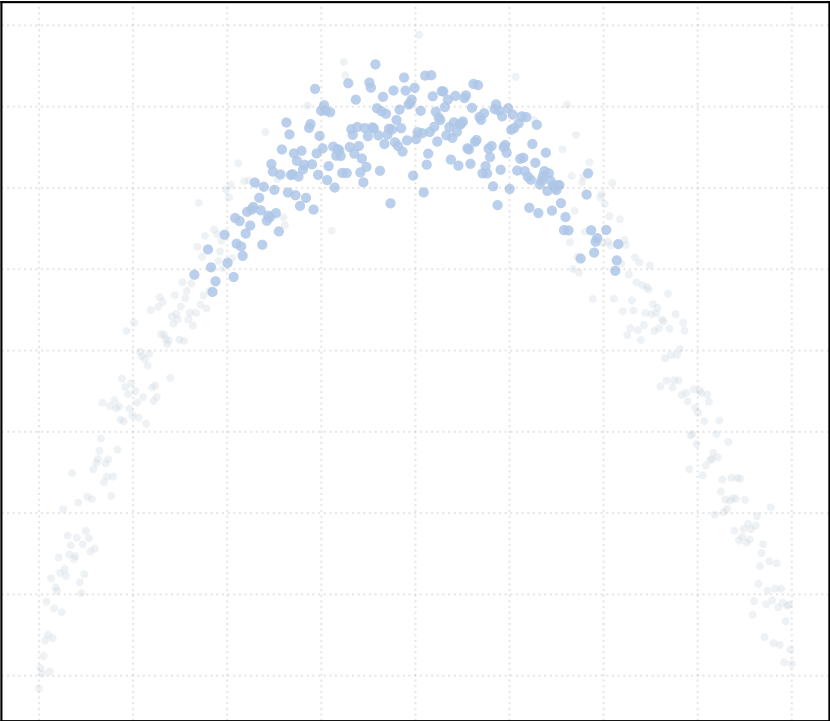
quant = 0.05 [1 Clusters]



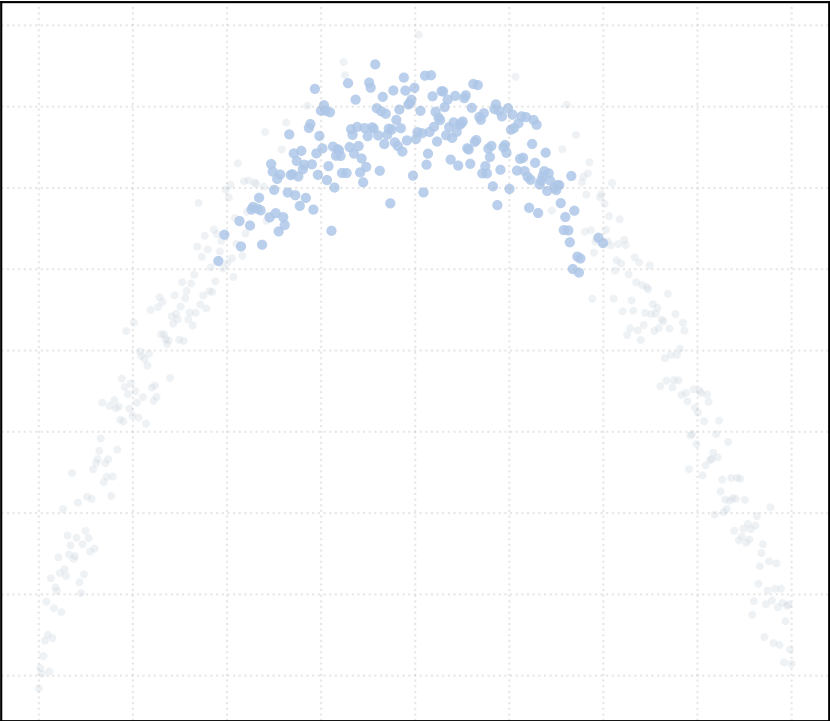
quant = 0.1 [1 Clusters]



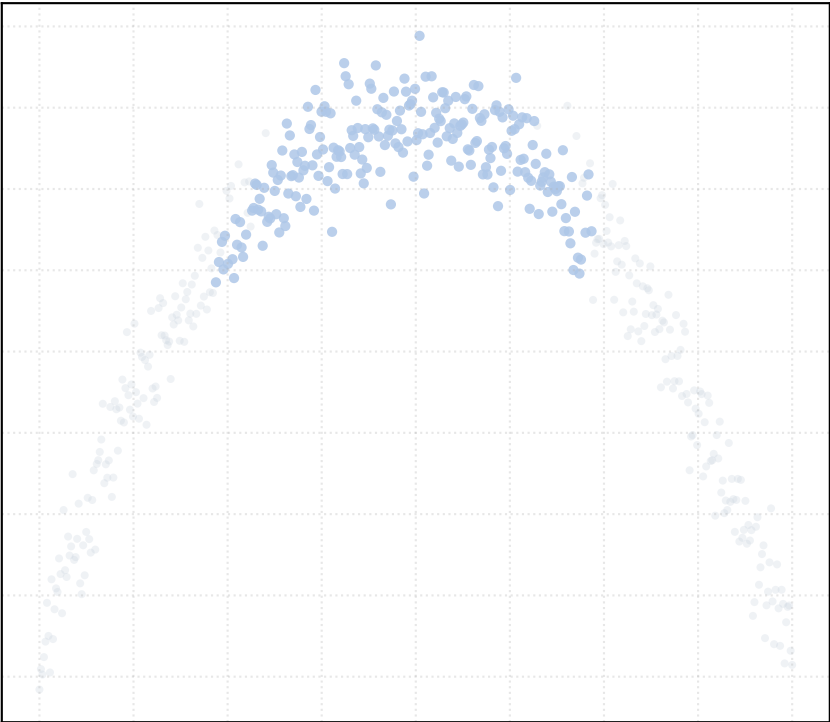
quant = 0.15 (Default) [1 Clusters]



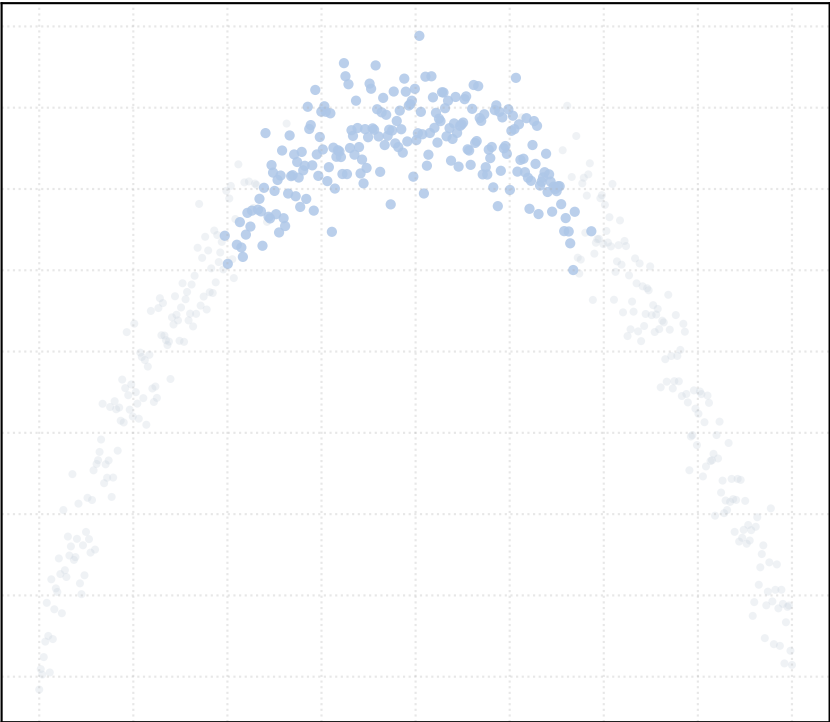
quant = 0.2 [1 Clusters]



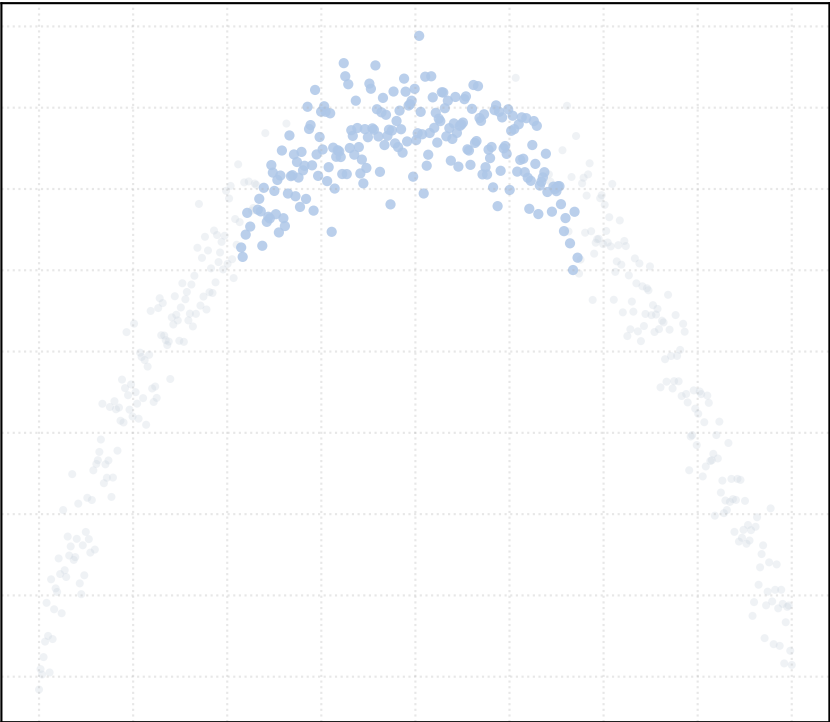
quant = 0.25 [1 Clusters]



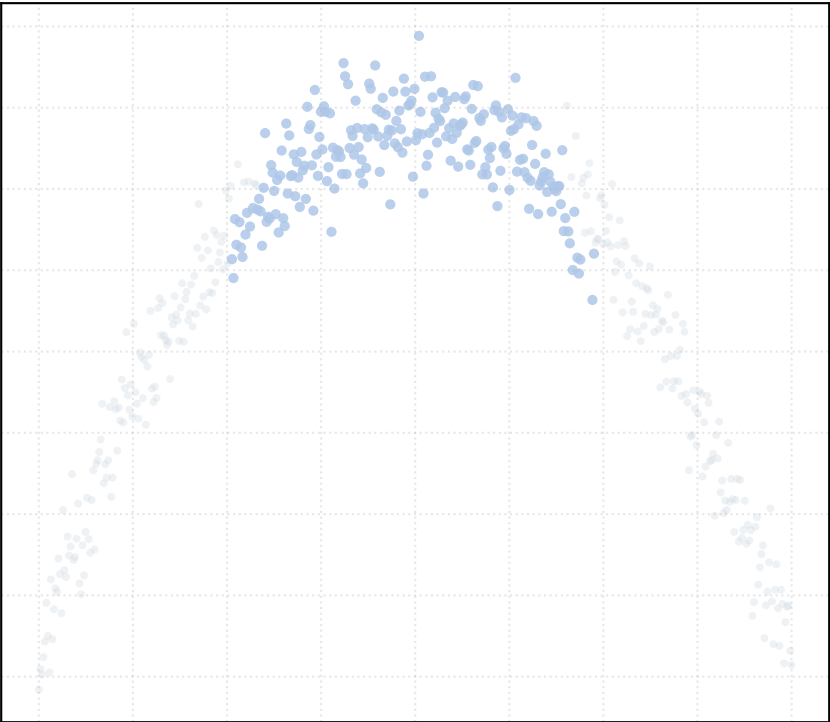
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

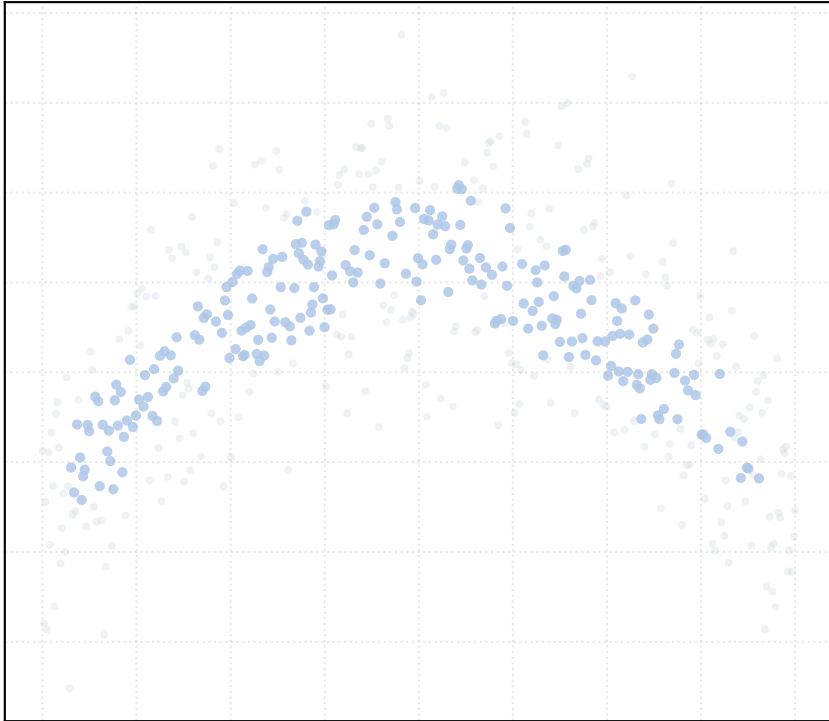


quant = 0.5 [1 Clusters]

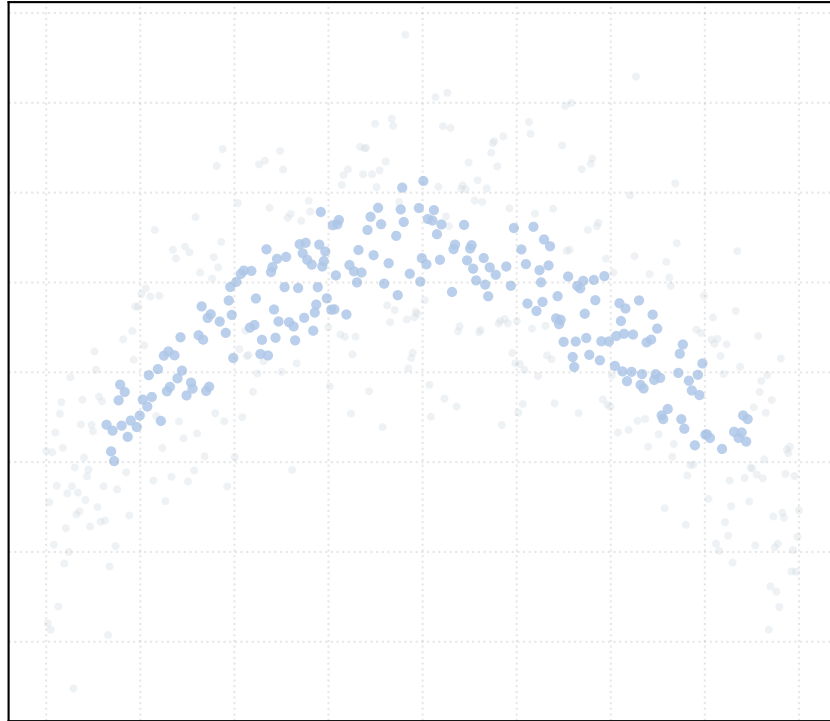


Dataset: circular_arc_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

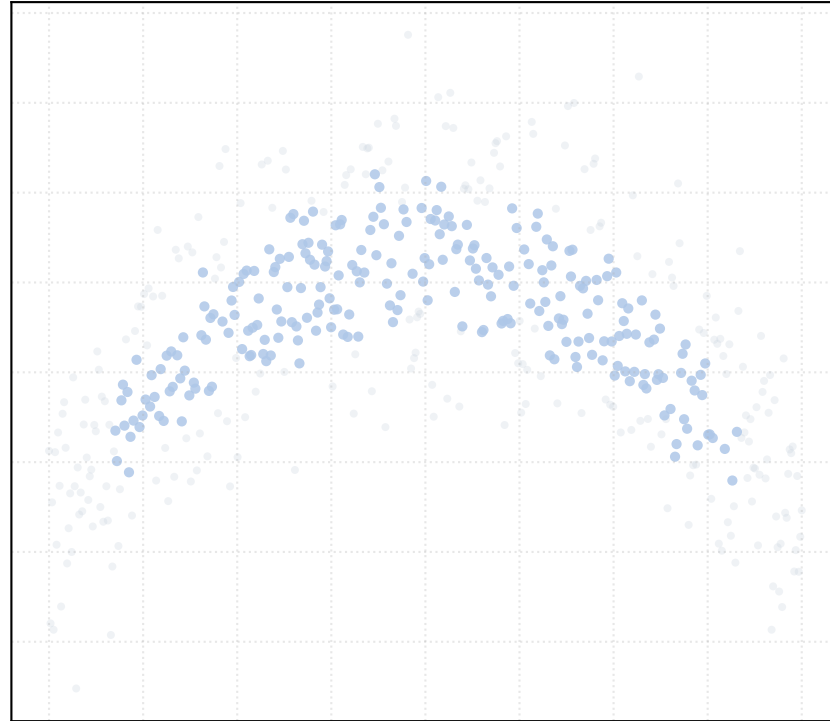
quant = 0.05 [1 Clusters]



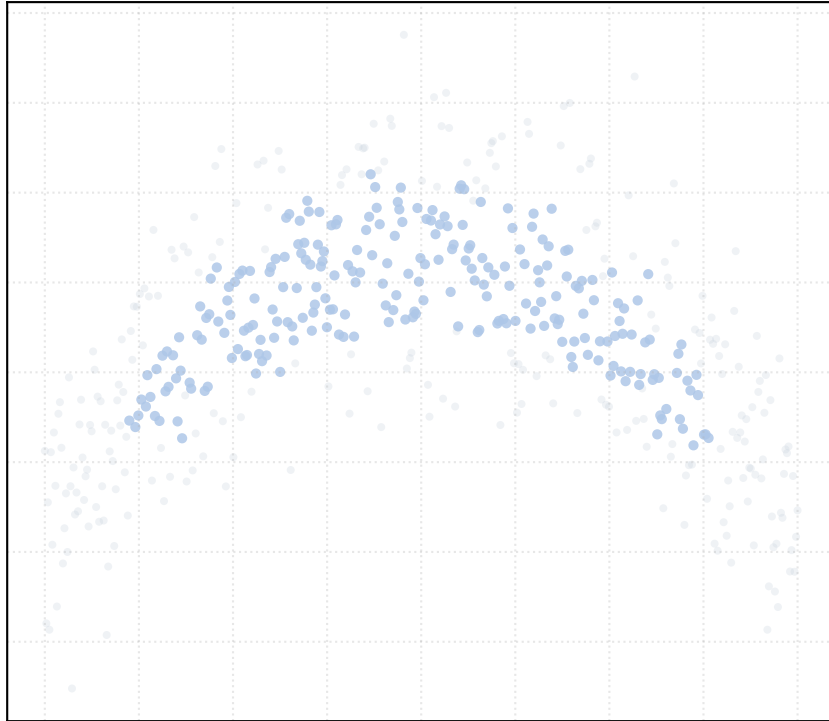
quant = 0.1 [1 Clusters]



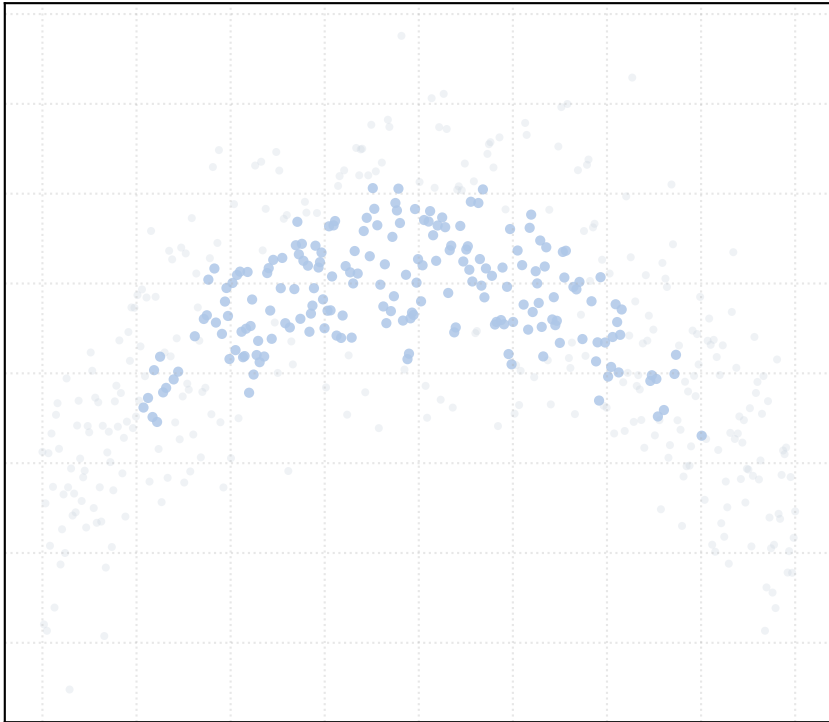
quant = 0.15 (Default) [1 Clusters]



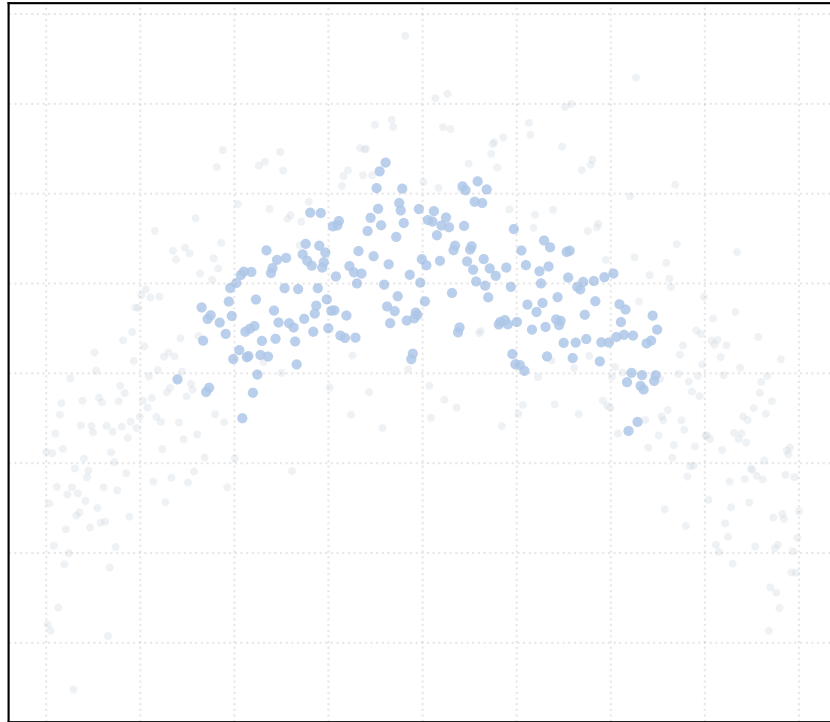
quant = 0.2 [1 Clusters]



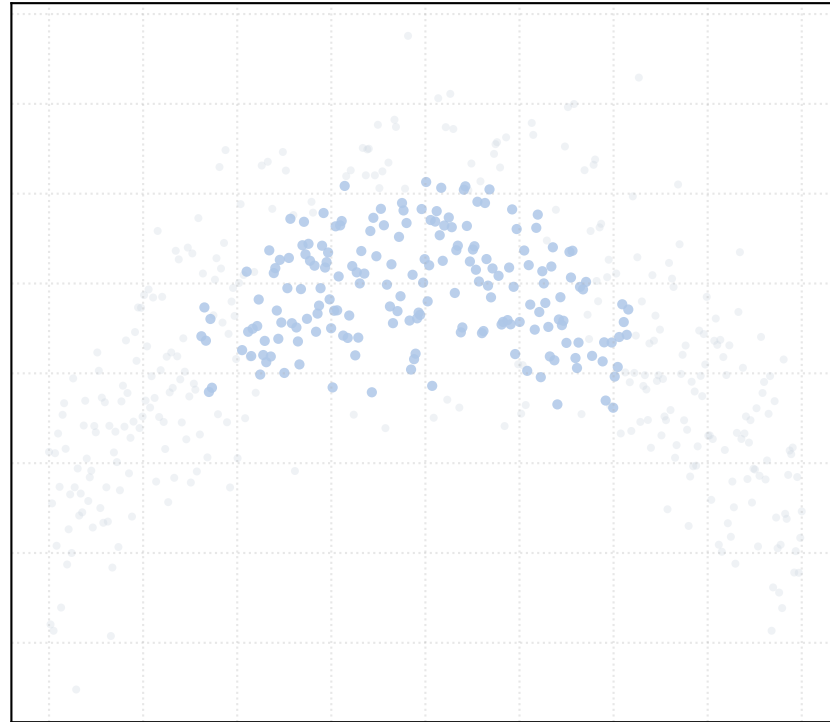
quant = 0.25 [1 Clusters]



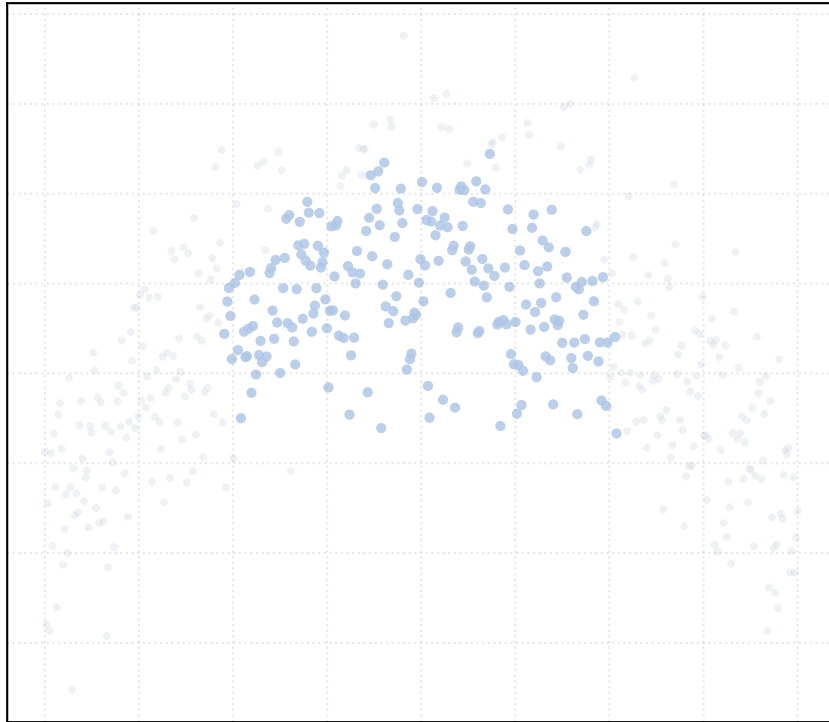
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

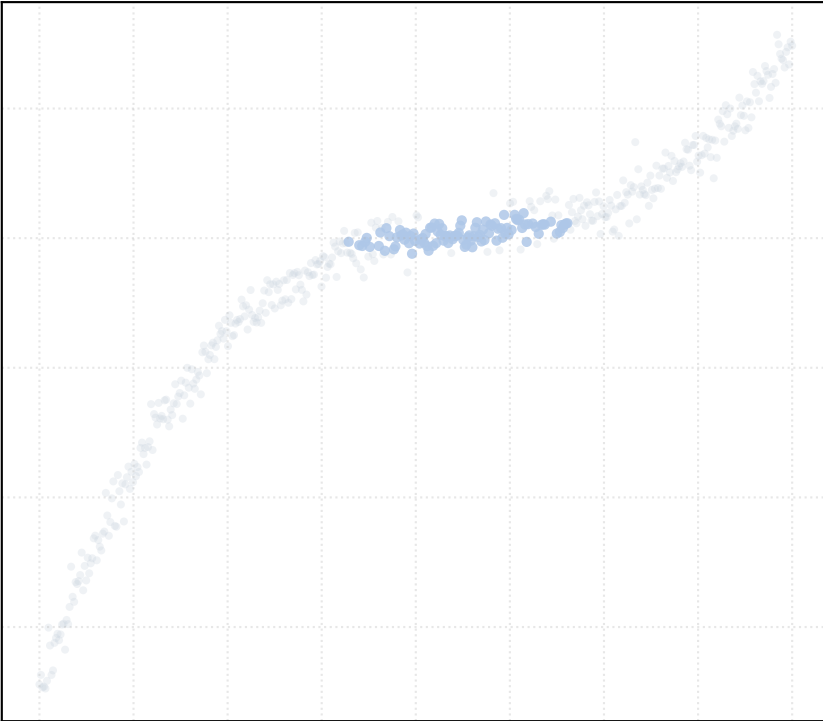


quant = 0.5 [1 Clusters]

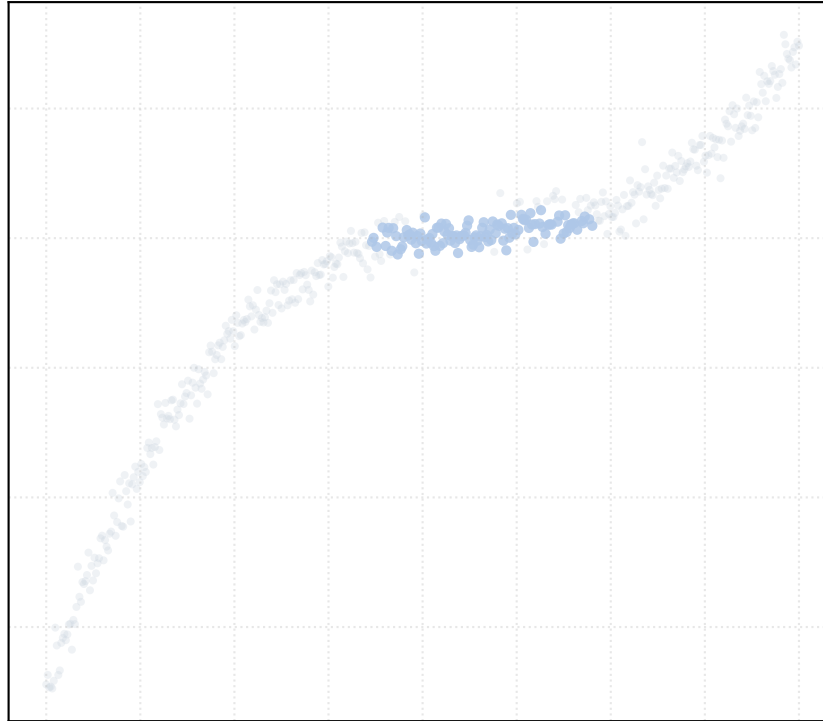


Dataset: cubic_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

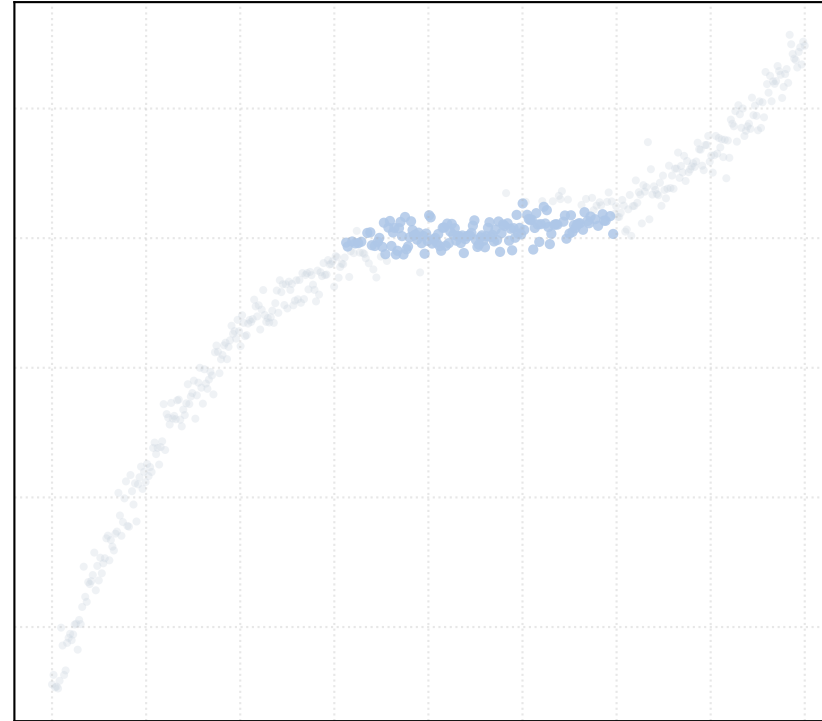
quant = 0.05 [1 Clusters]



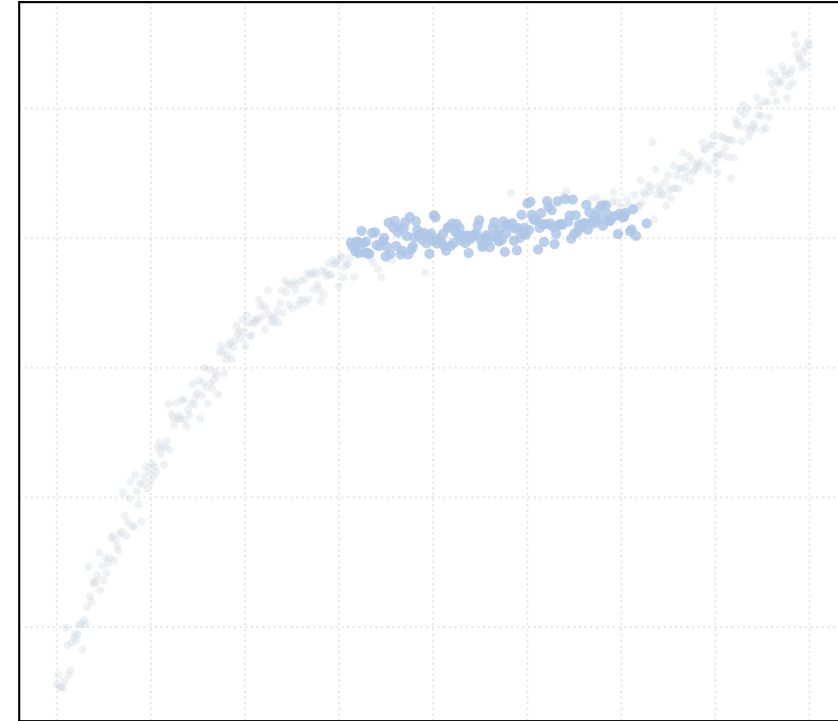
quant = 0.1 [1 Clusters]



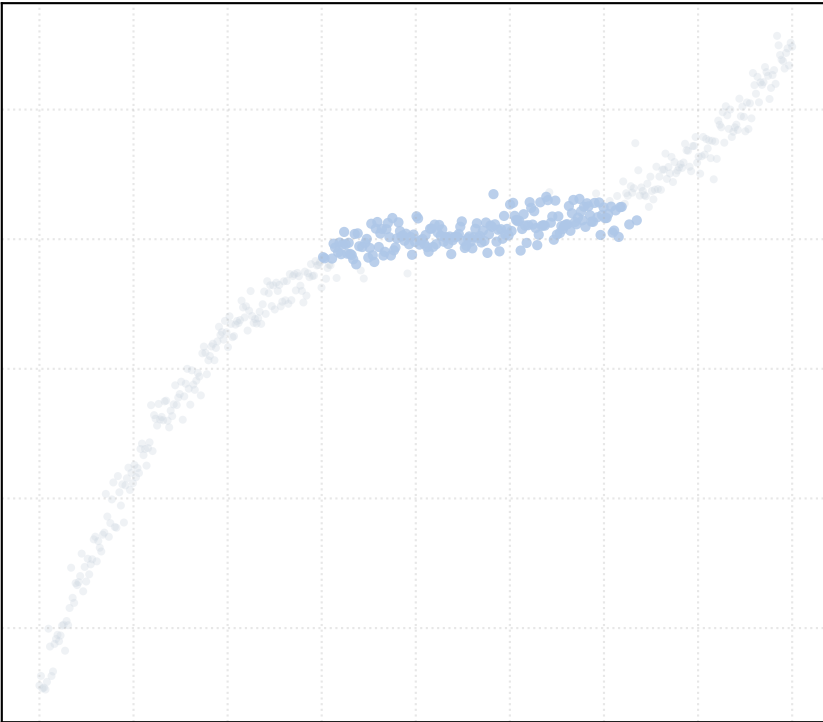
quant = 0.15 (Default) [1 Clusters]



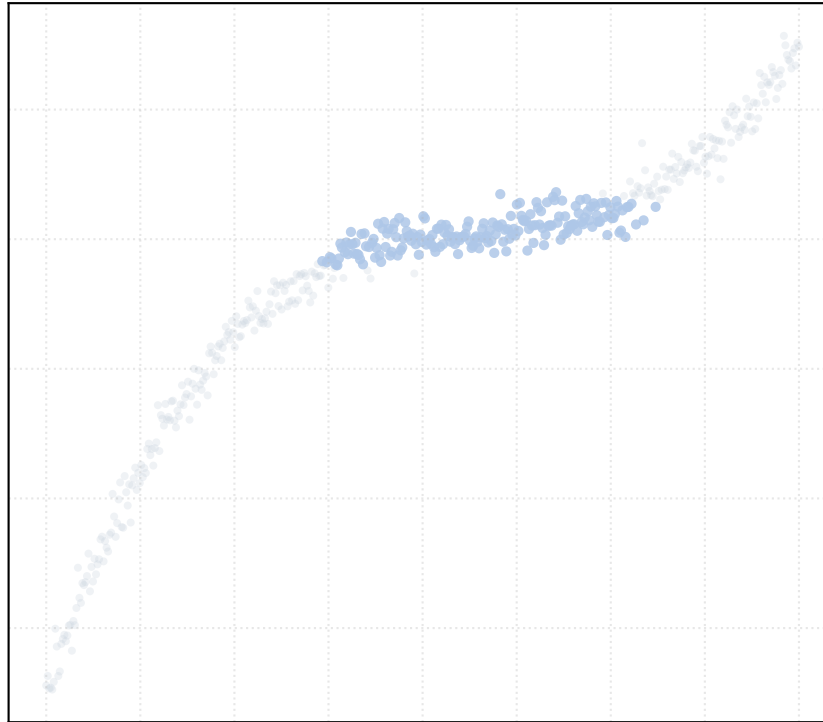
quant = 0.2 [1 Clusters]



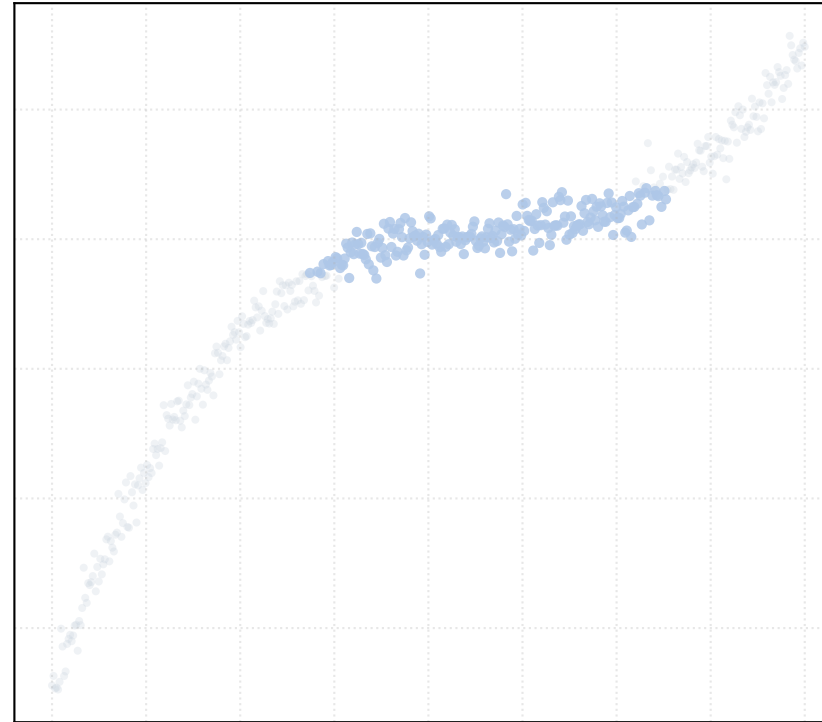
quant = 0.25 [1 Clusters]



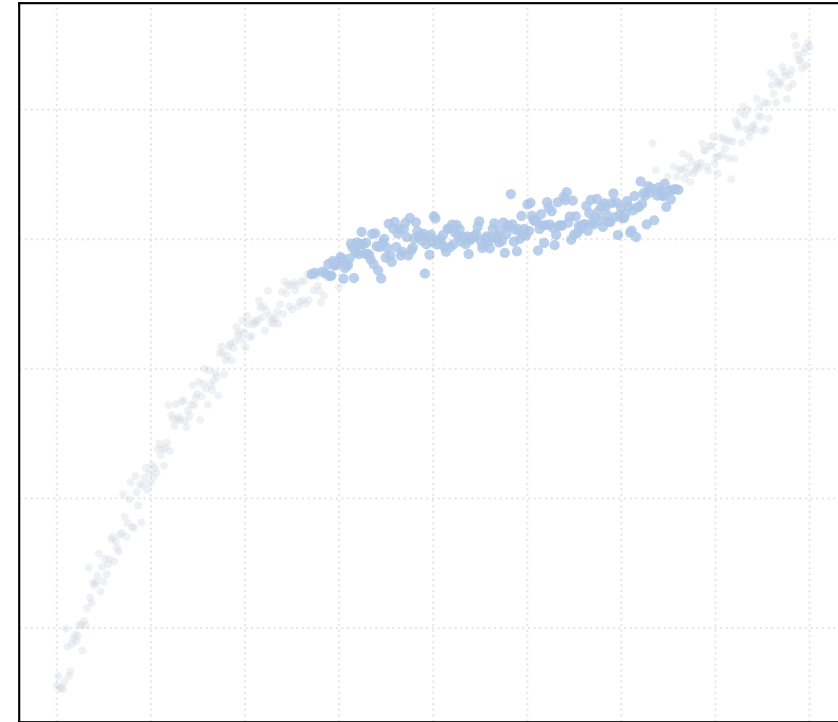
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

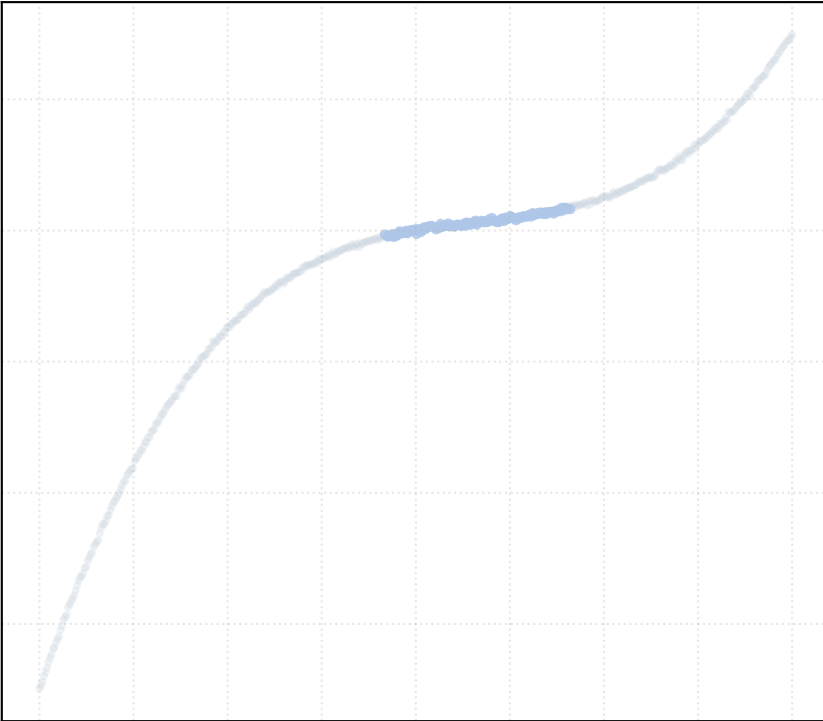


quant = 0.5 [1 Clusters]

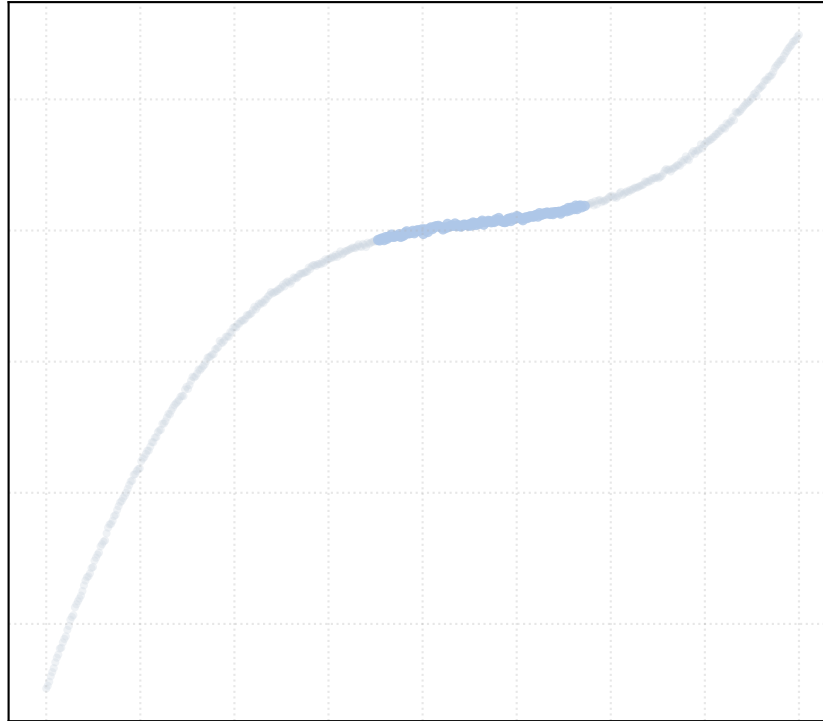


Dataset: cubic_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

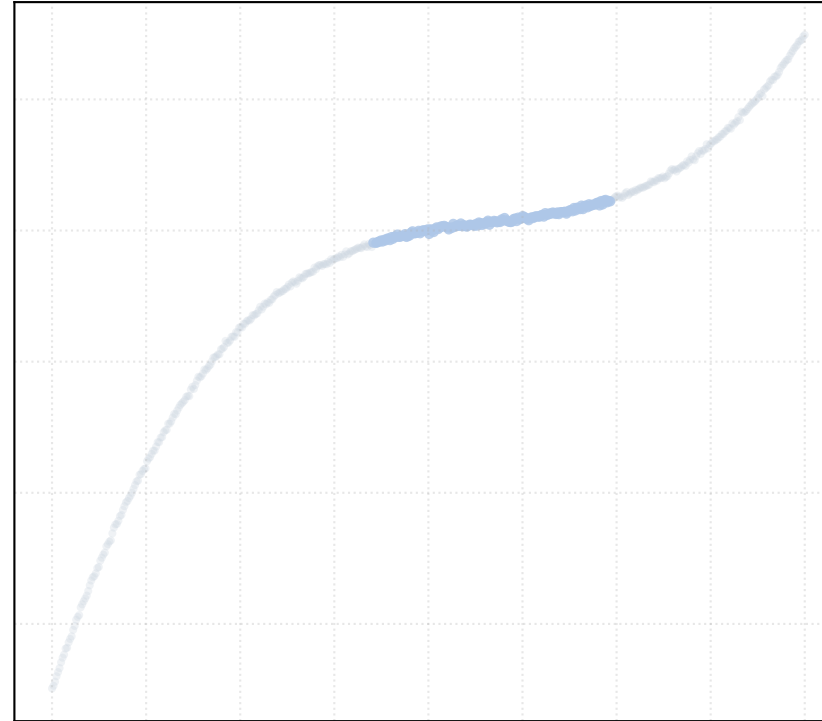
quant = 0.05 [1 Clusters]



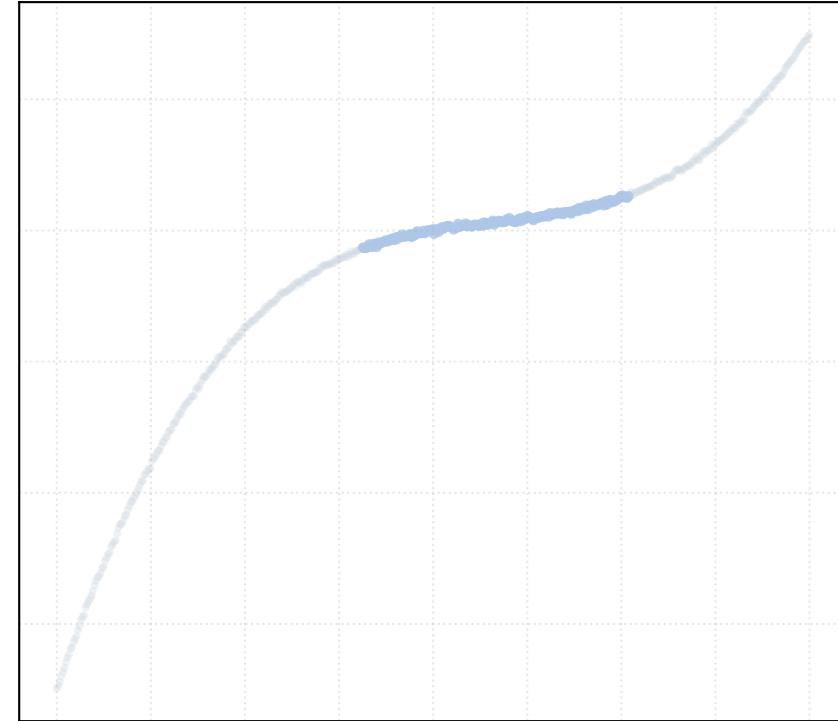
quant = 0.1 [1 Clusters]



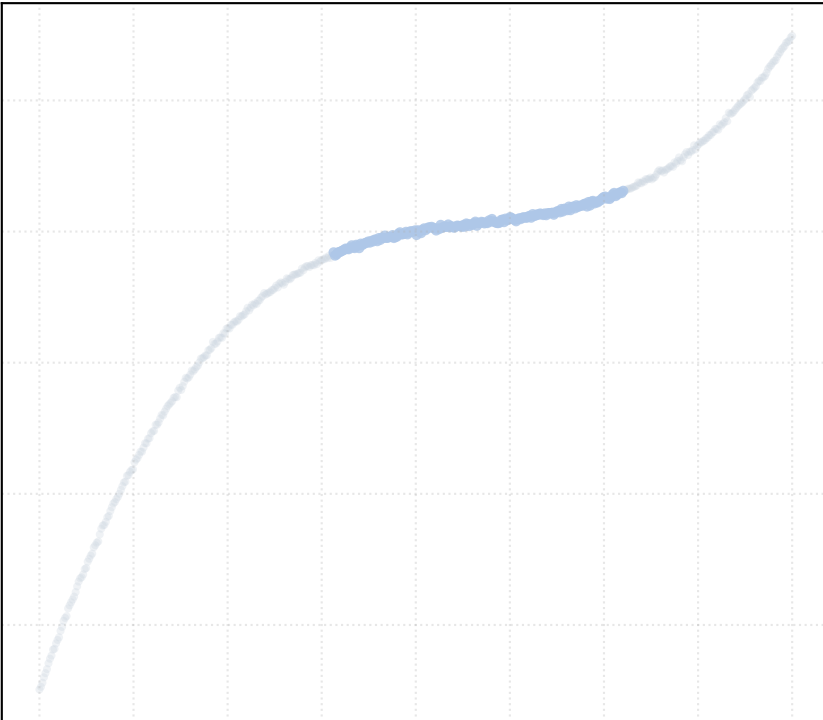
quant = 0.15 (Default) [1 Clusters]



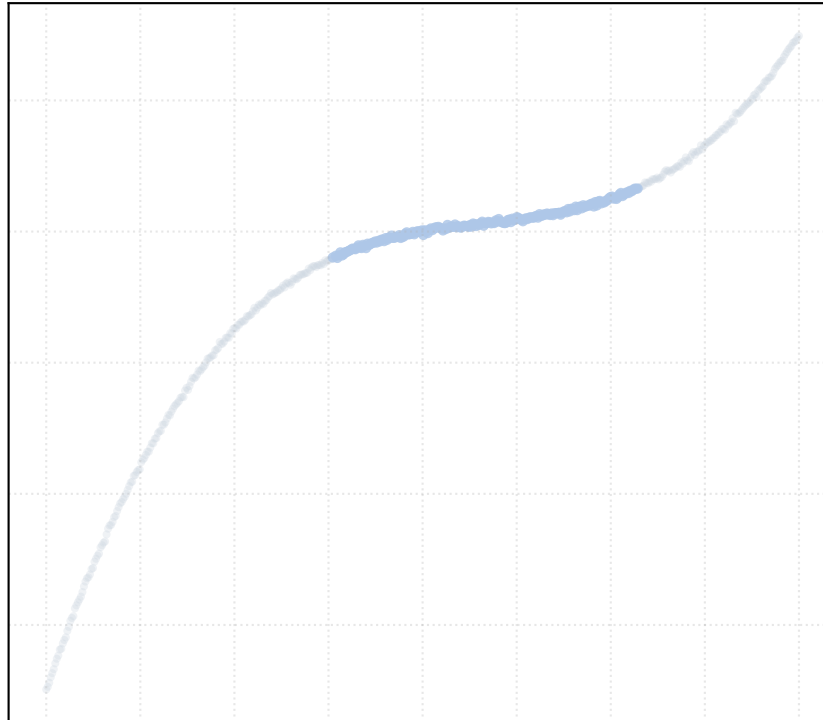
quant = 0.2 [1 Clusters]



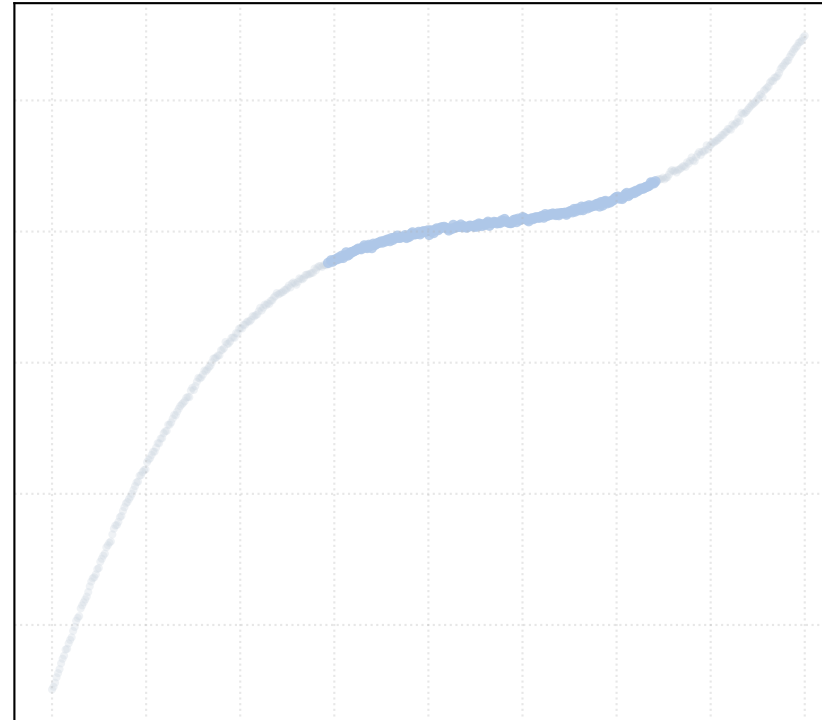
quant = 0.25 [1 Clusters]



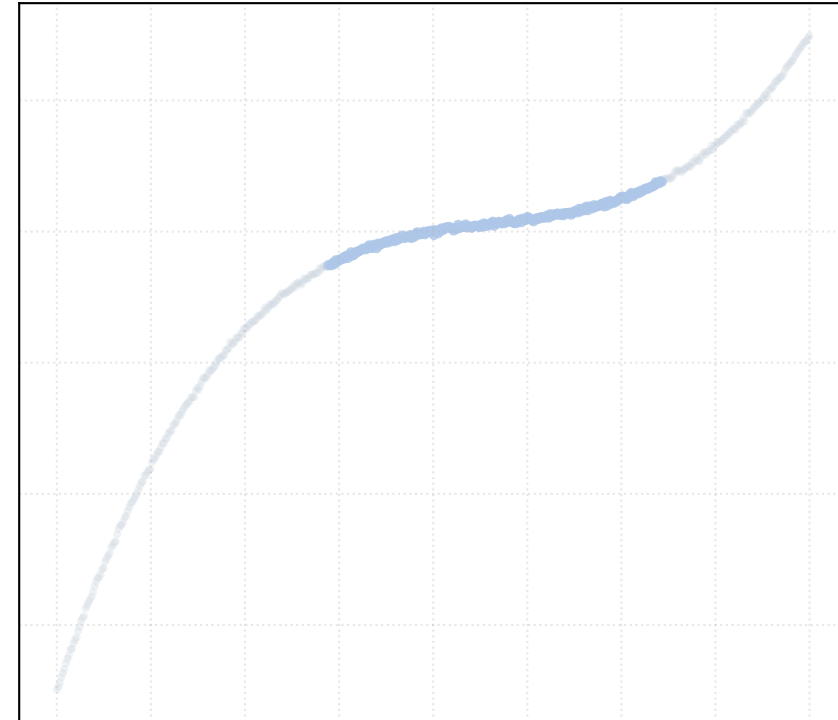
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

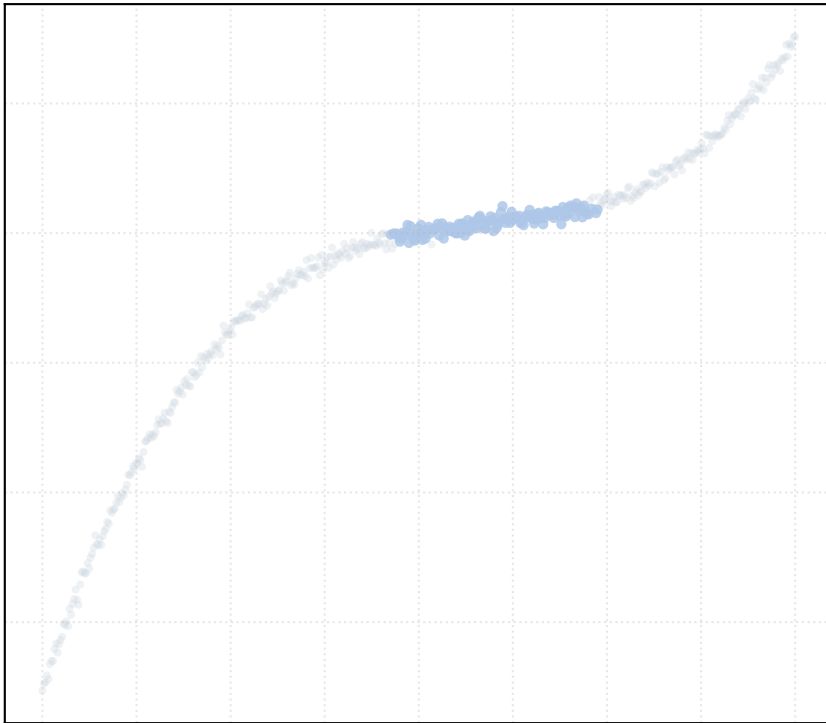


quant = 0.5 [1 Clusters]

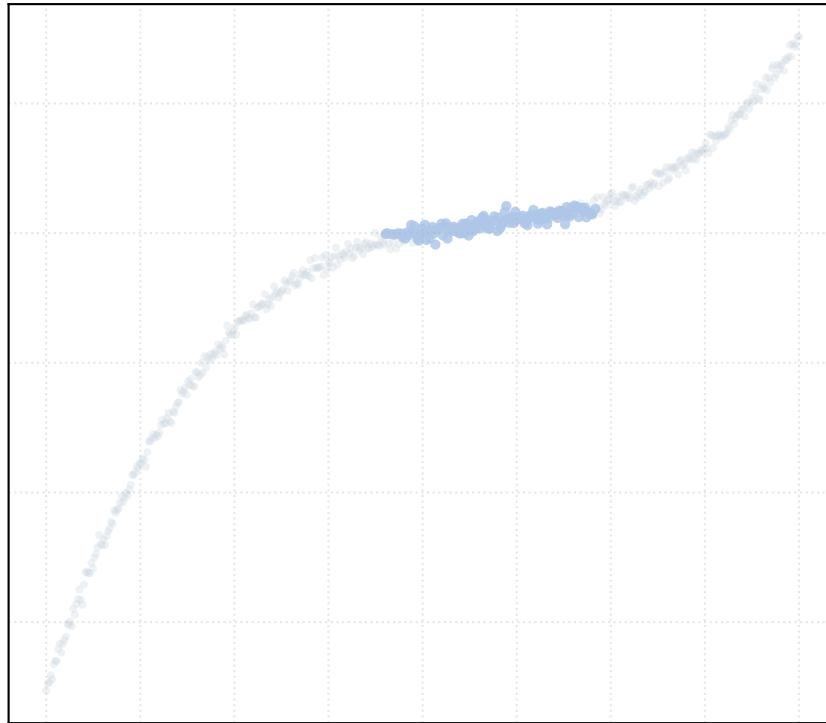


Dataset: cubic_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

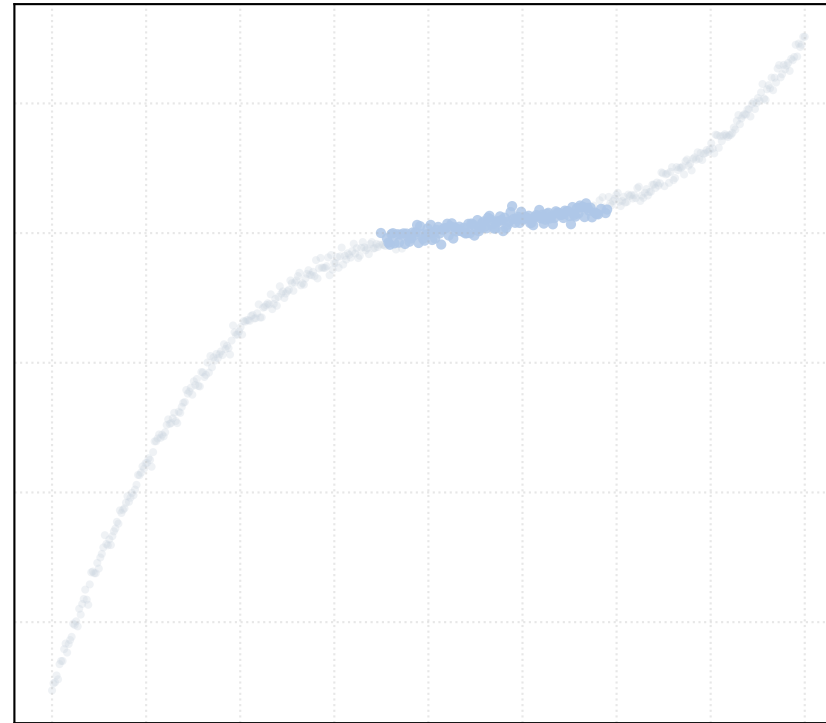
quant = 0.05 [1 Clusters]



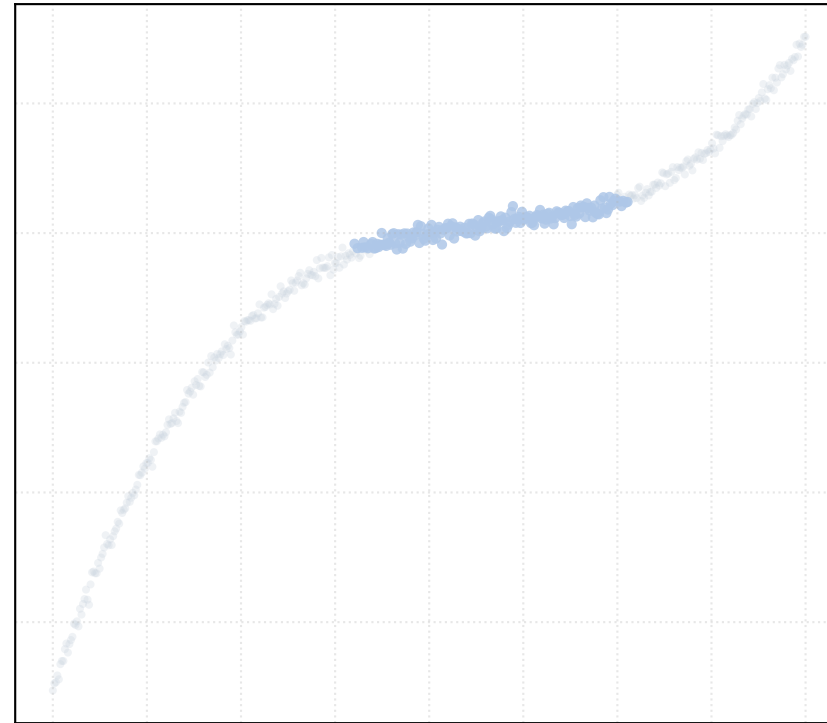
quant = 0.1 [1 Clusters]



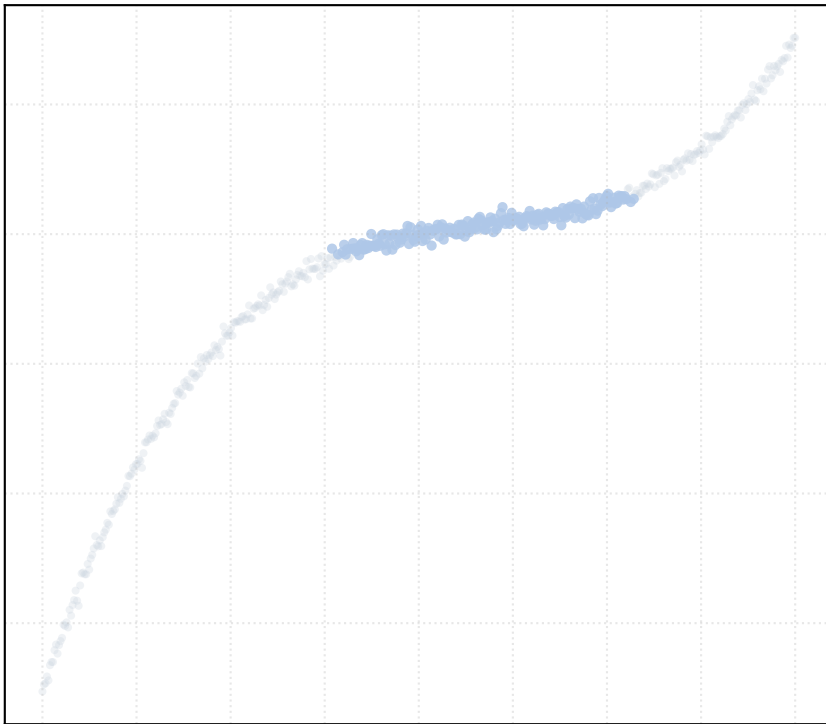
quant = 0.15 (Default) [1 Clusters]



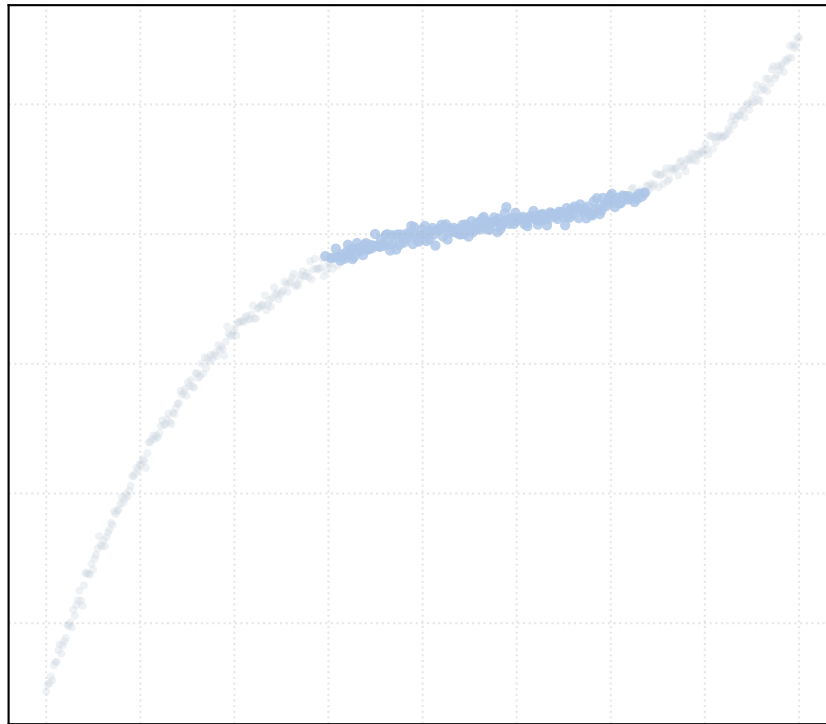
quant = 0.2 [1 Clusters]



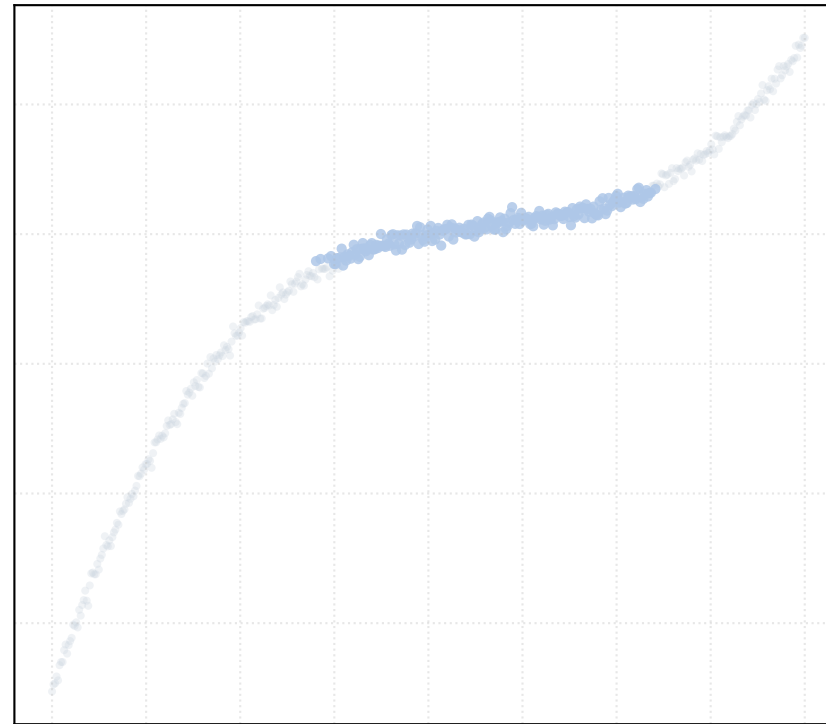
quant = 0.25 [1 Clusters]



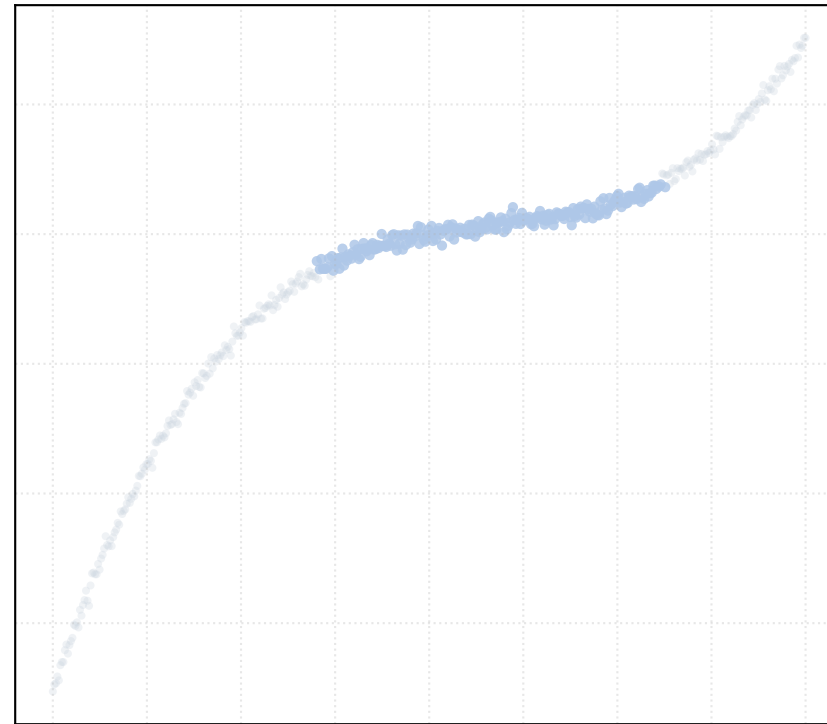
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

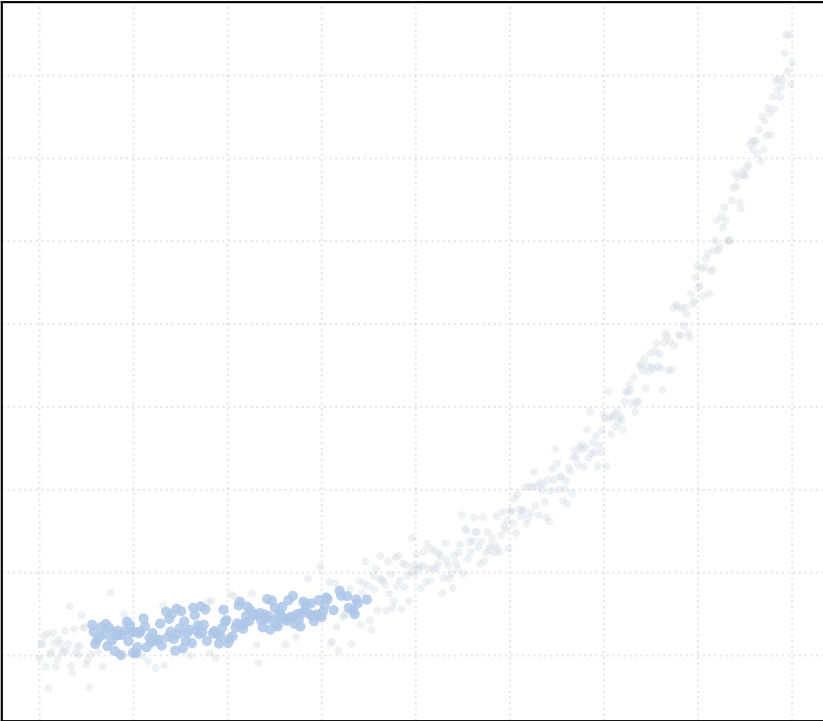


quant = 0.5 [1 Clusters]

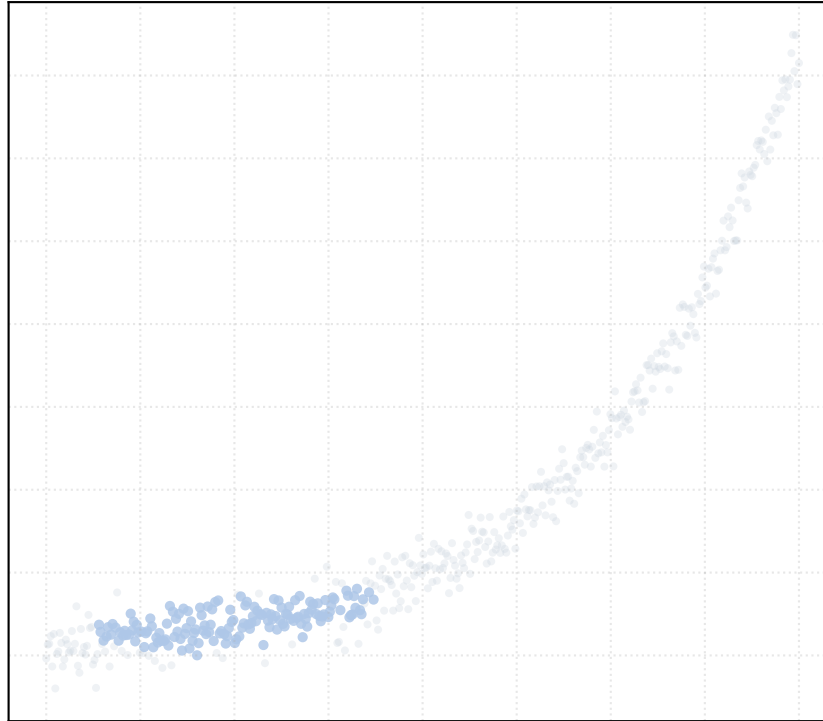


Dataset: exponential_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

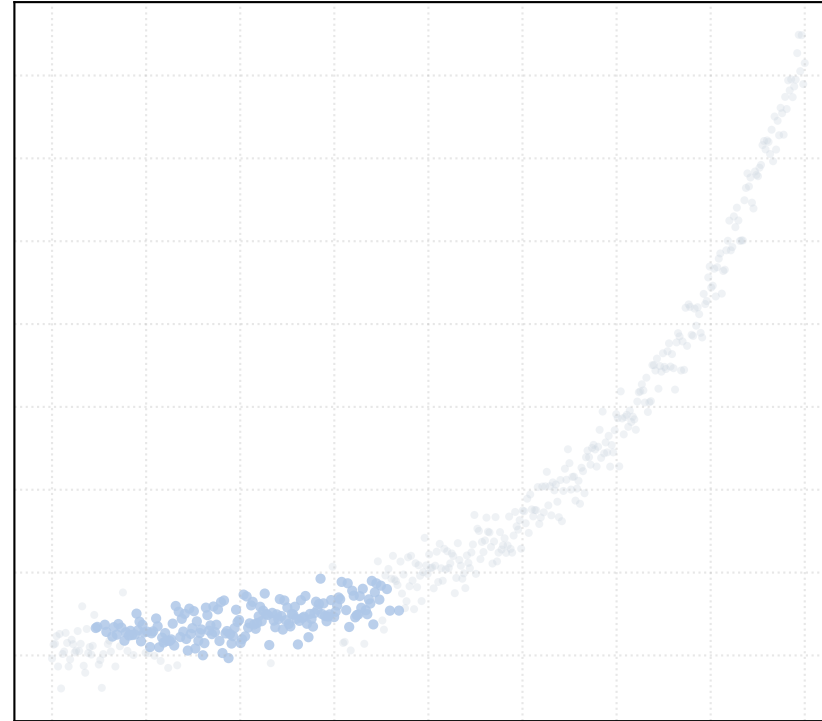
quant = 0.05 [1 Clusters]



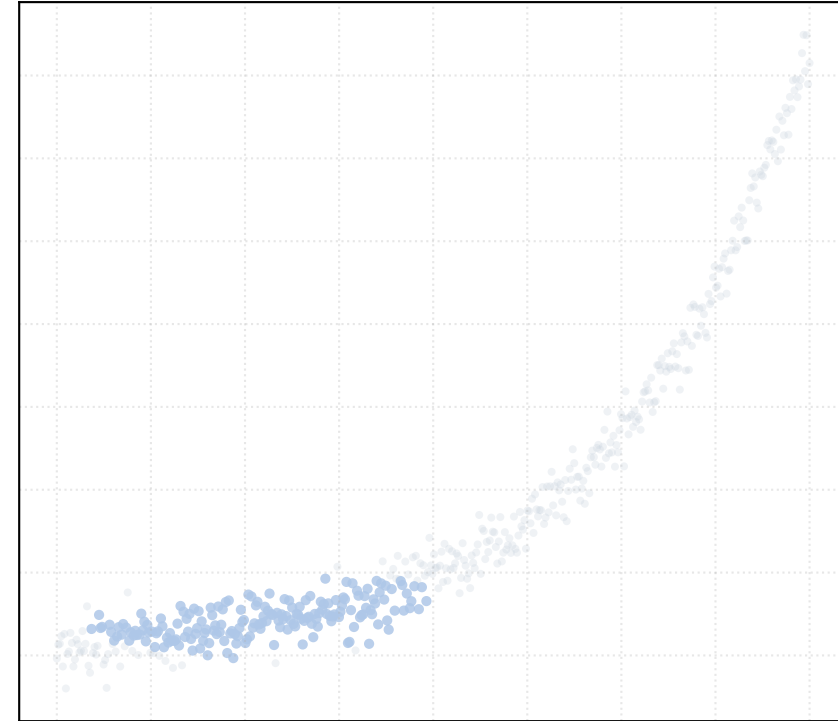
quant = 0.1 [1 Clusters]



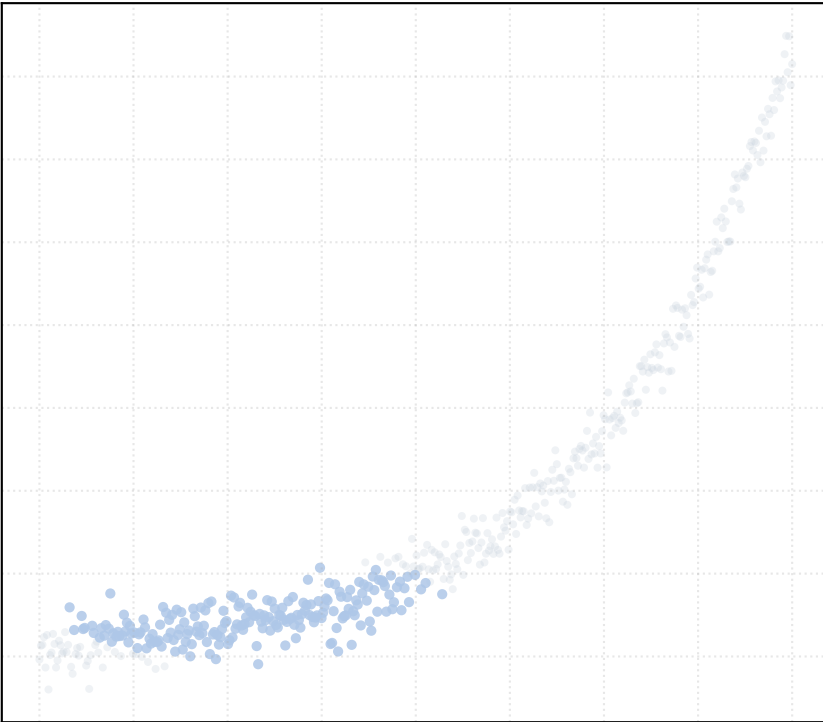
quant = 0.15 (Default) [1 Clusters]



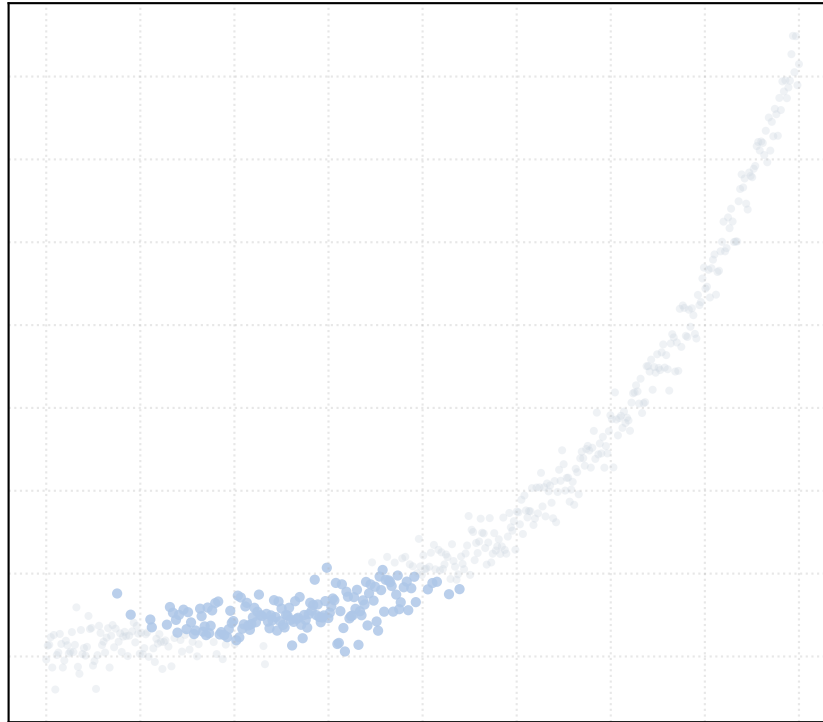
quant = 0.2 [1 Clusters]



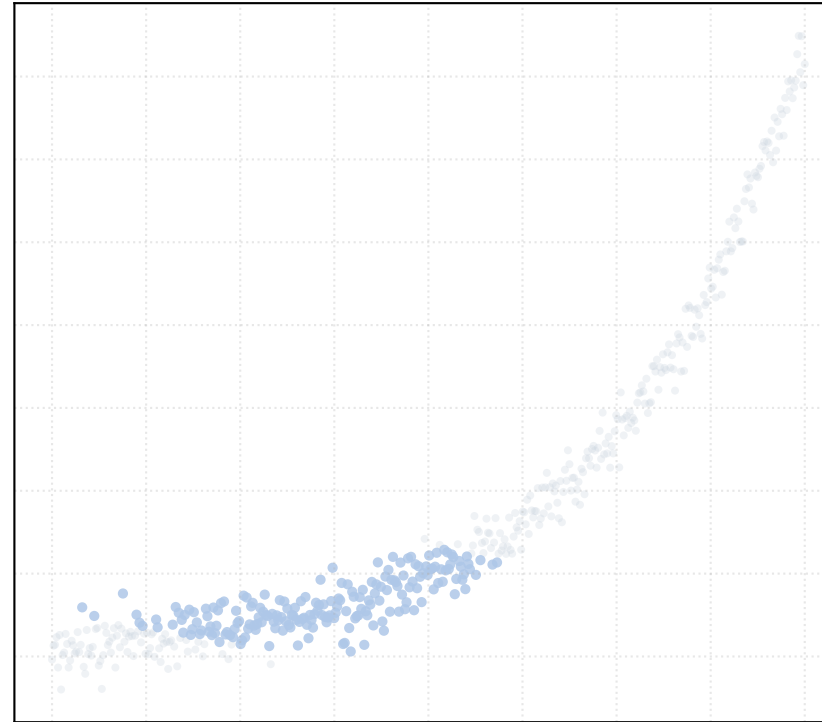
quant = 0.25 [1 Clusters]



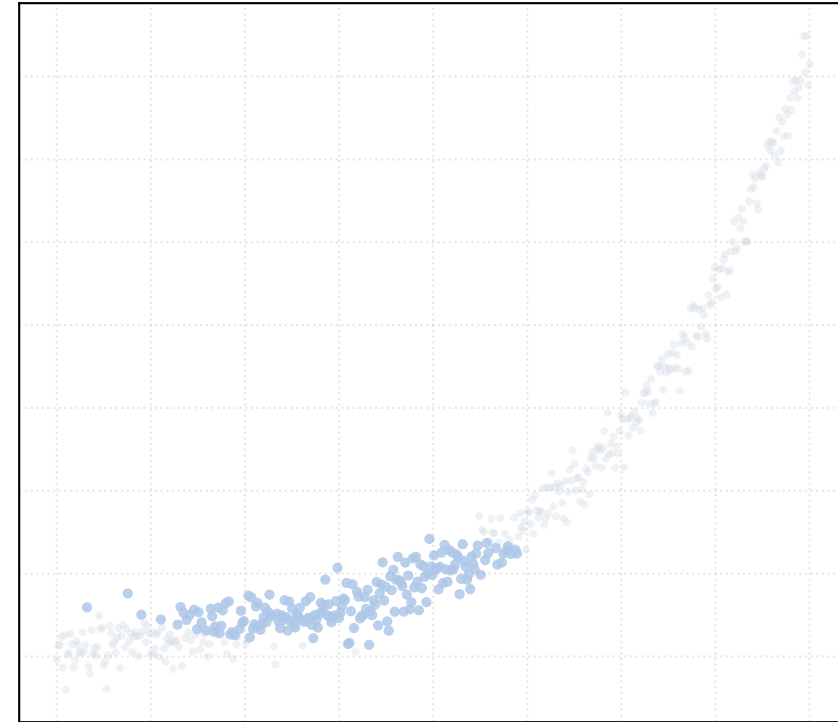
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

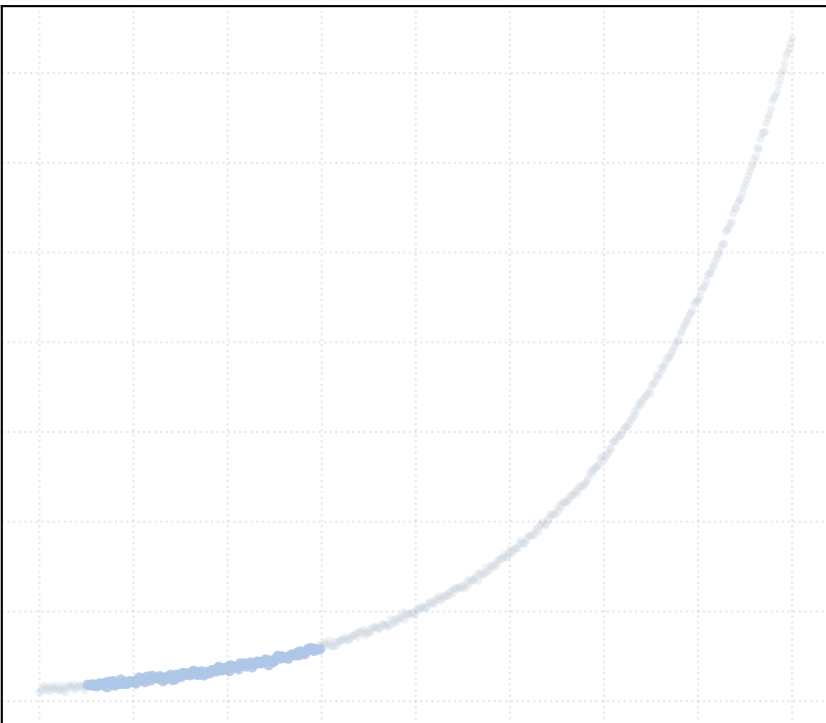


quant = 0.5 [1 Clusters]

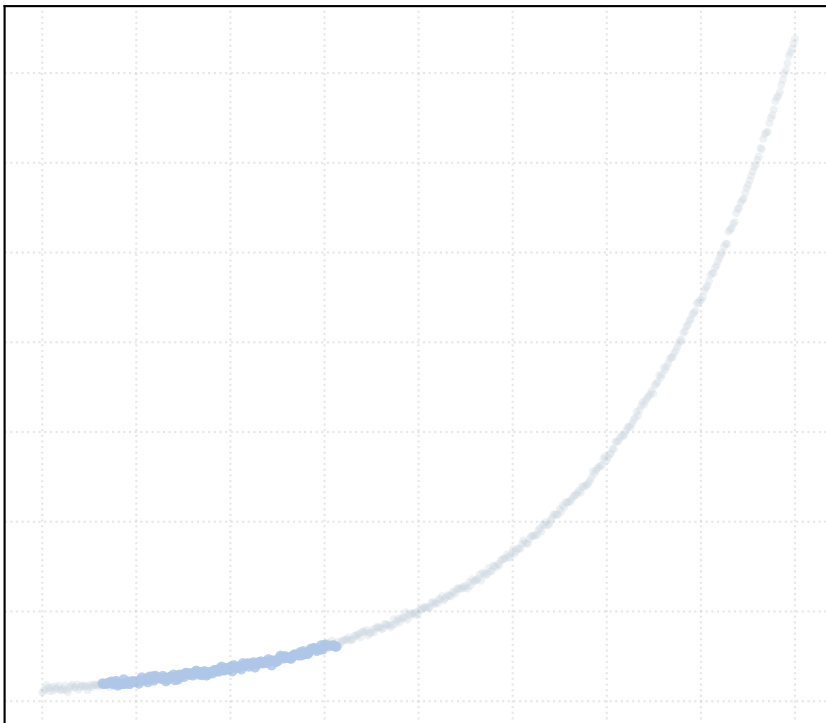


Dataset: exponential_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

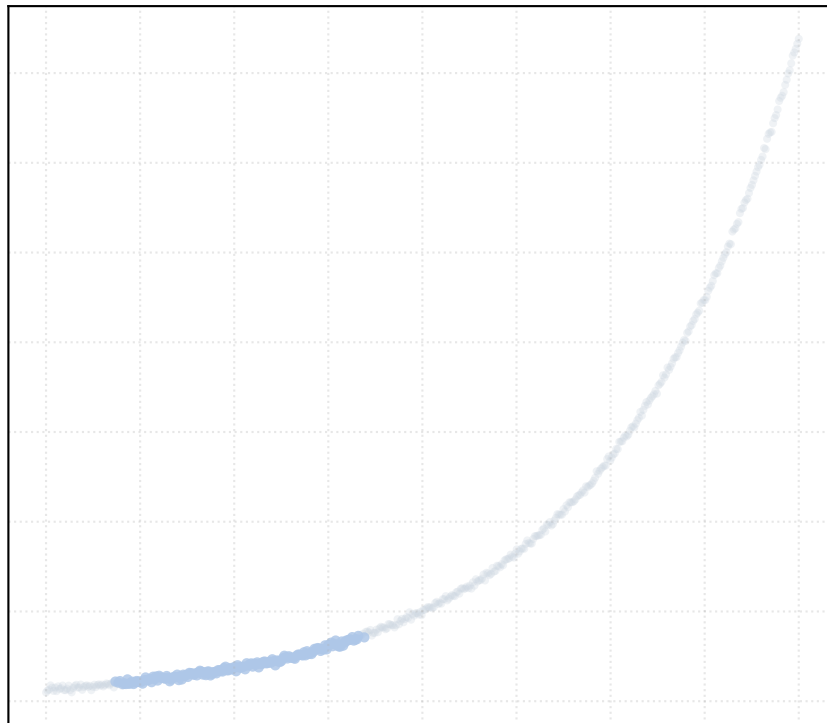
quant = 0.05 [1 Clusters]



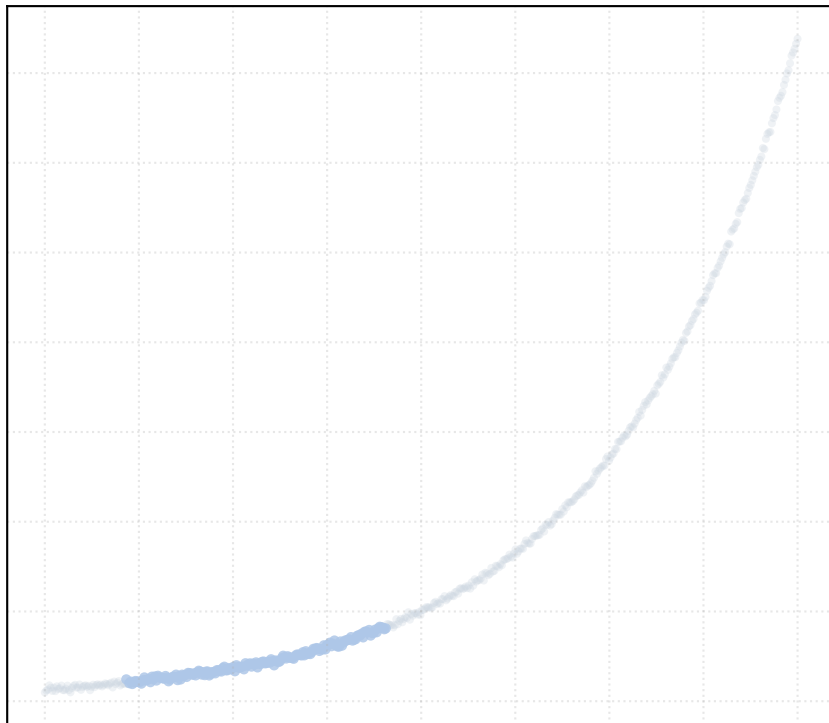
quant = 0.1 [1 Clusters]



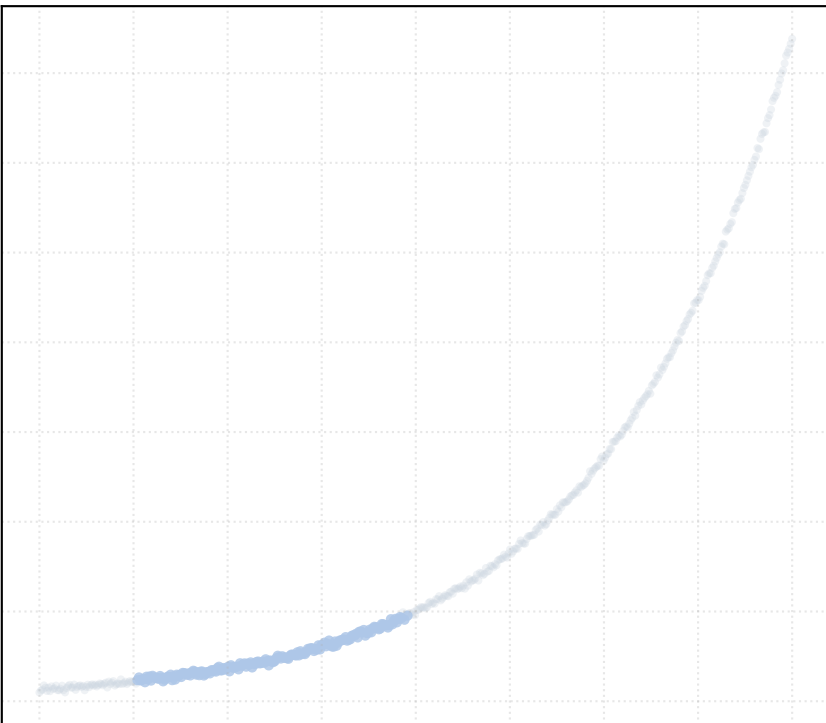
quant = 0.15 (Default) [1 Clusters]



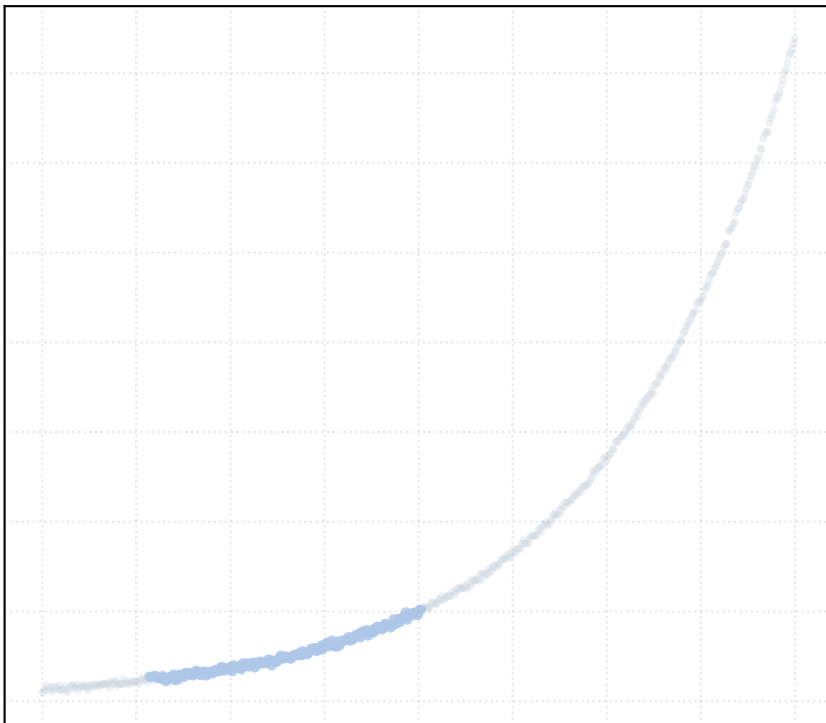
quant = 0.2 [1 Clusters]



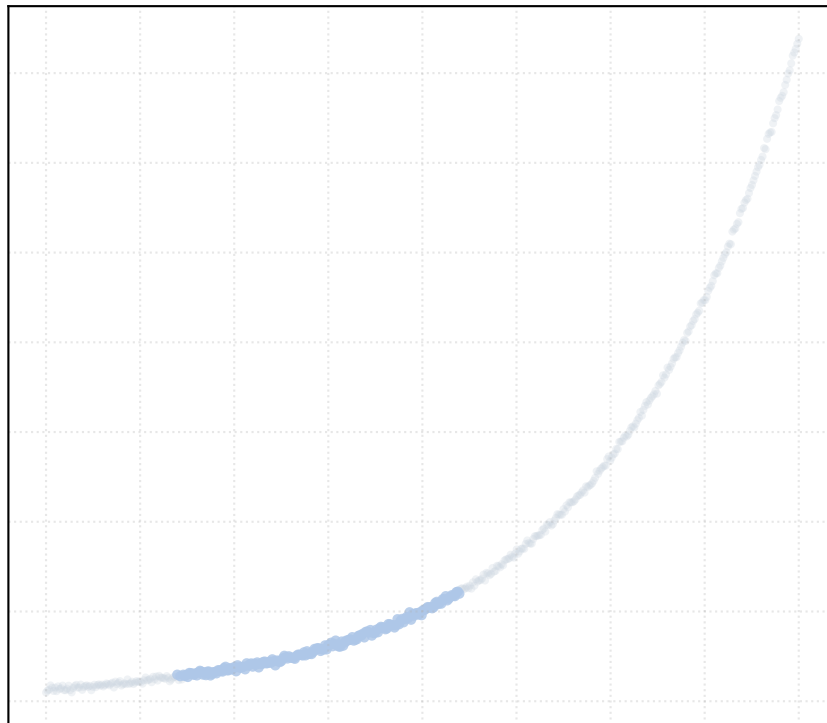
quant = 0.25 [1 Clusters]



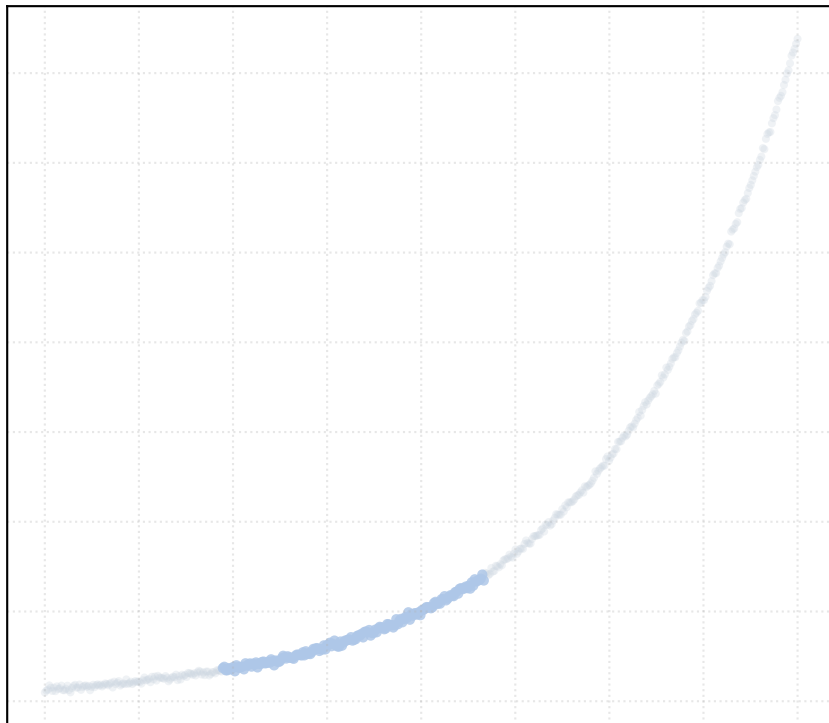
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

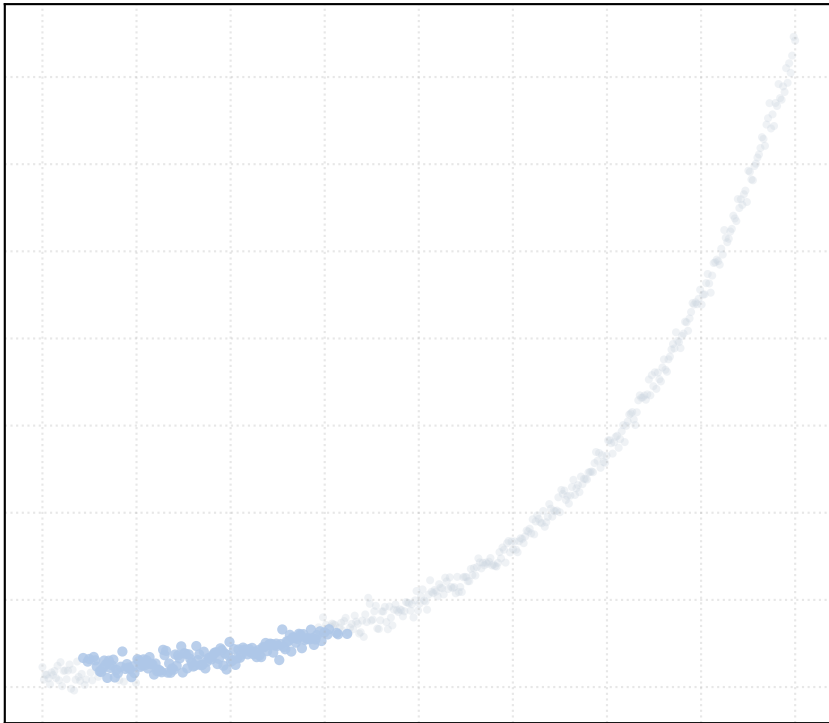


quant = 0.5 [1 Clusters]

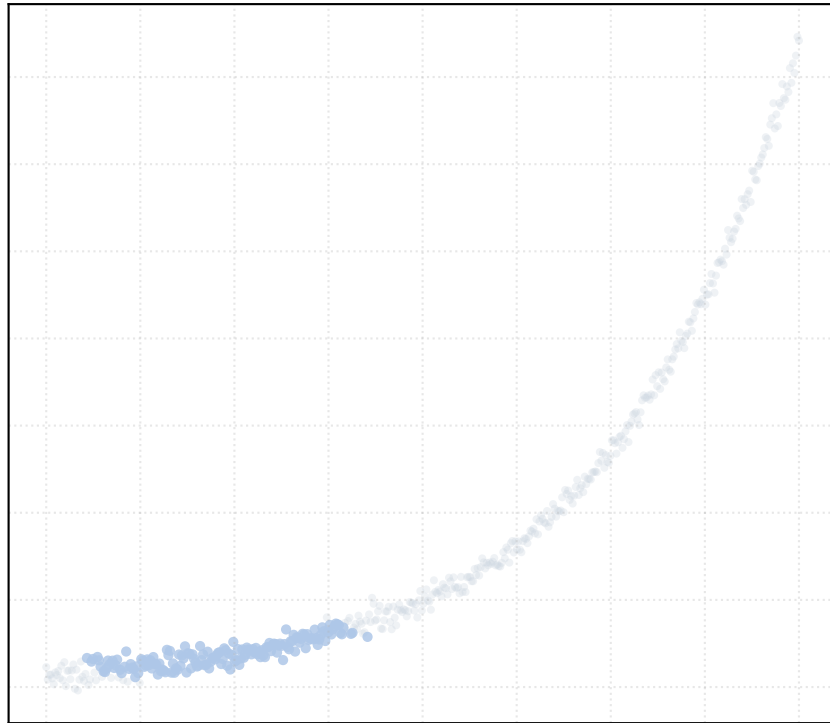


Dataset: exponential_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

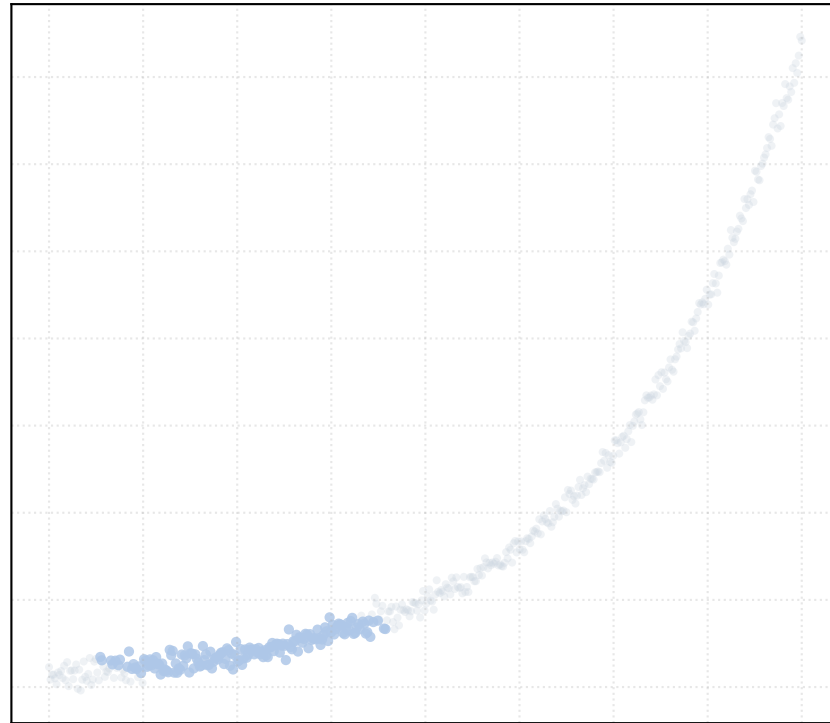
quant = 0.05 [1 Clusters]



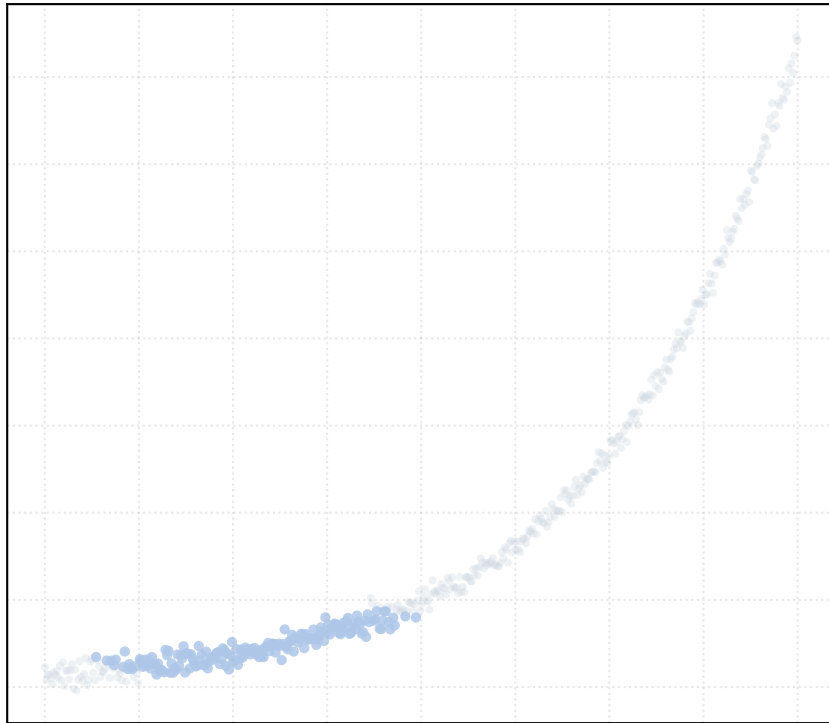
quant = 0.1 [1 Clusters]



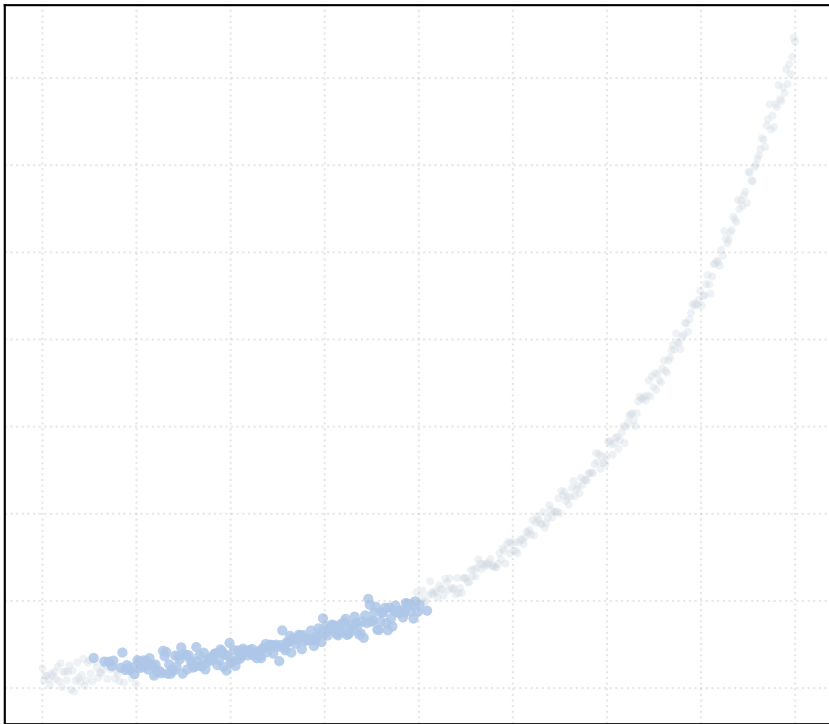
quant = 0.15 (Default) [1 Clusters]



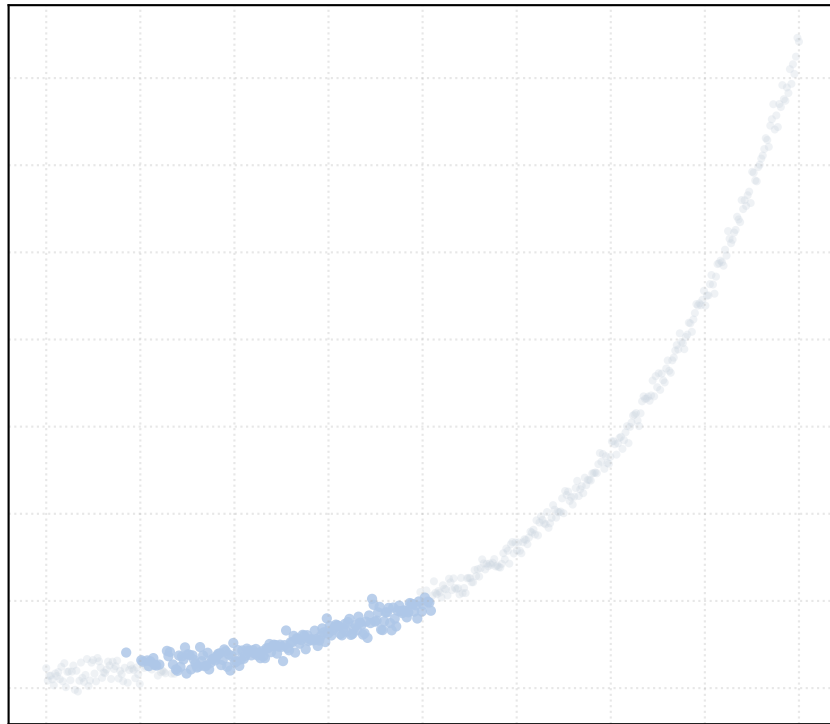
quant = 0.2 [1 Clusters]



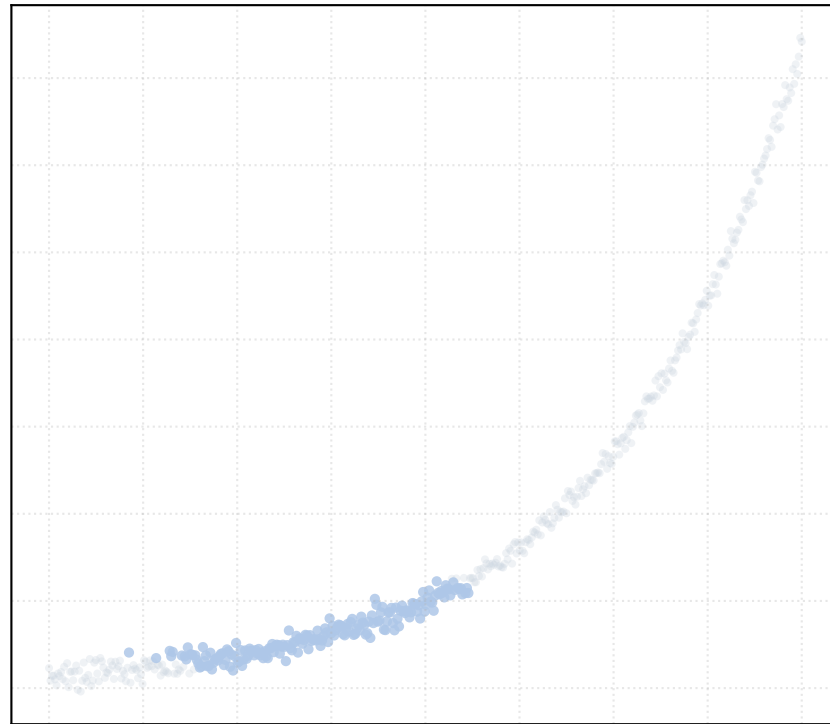
quant = 0.25 [1 Clusters]



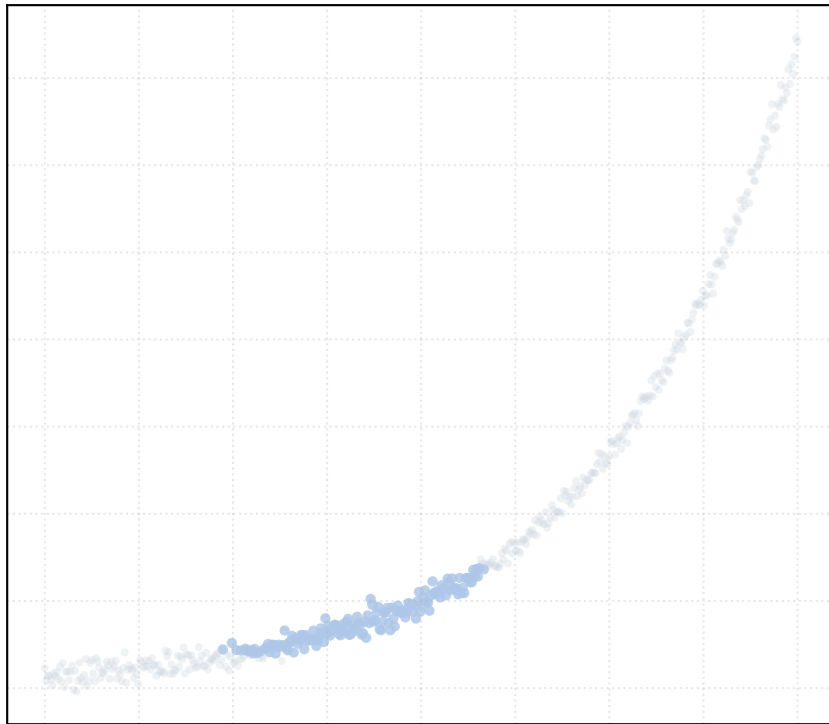
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

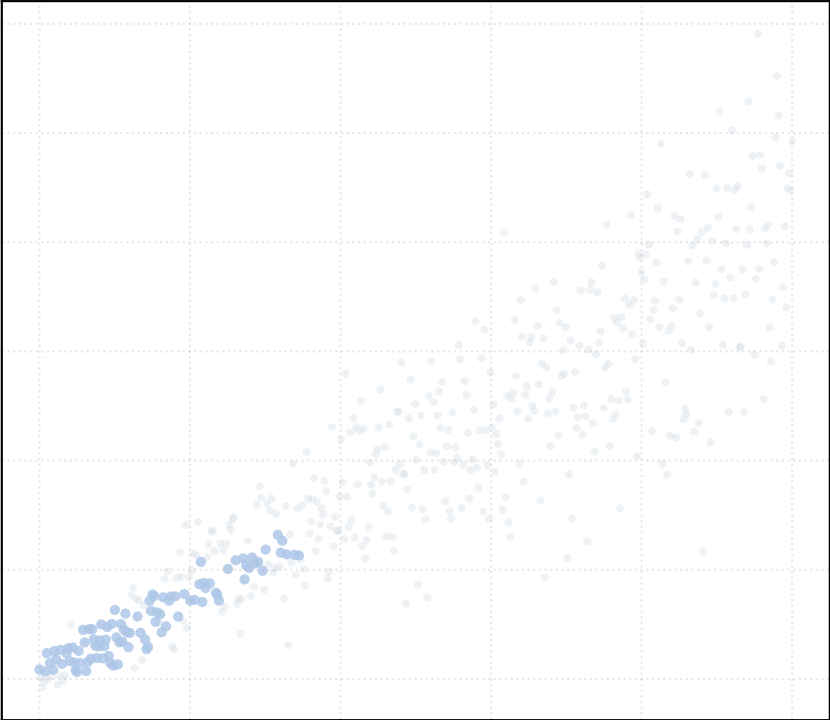


quant = 0.5 [1 Clusters]

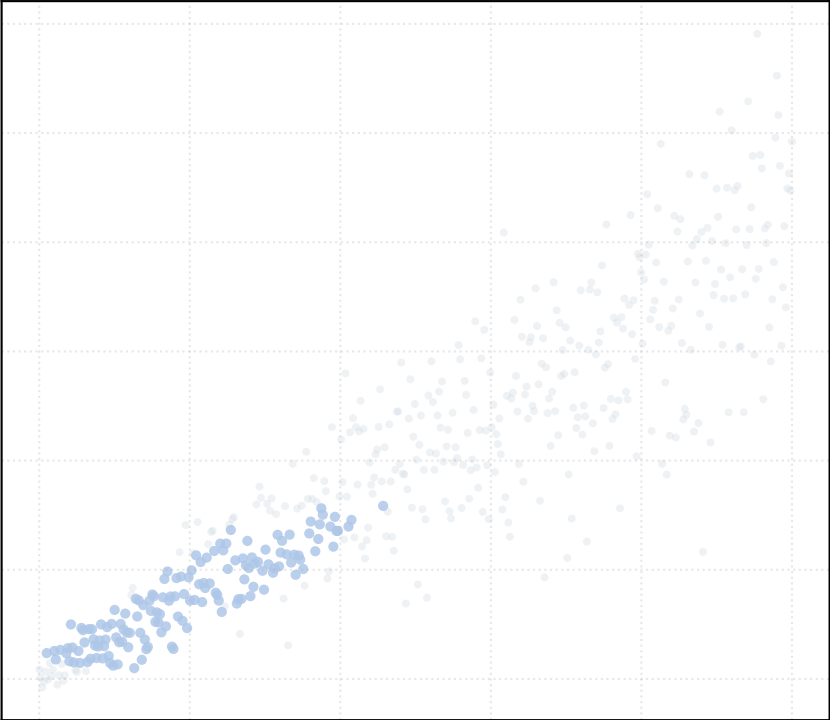


Dataset: heteroscedastic_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

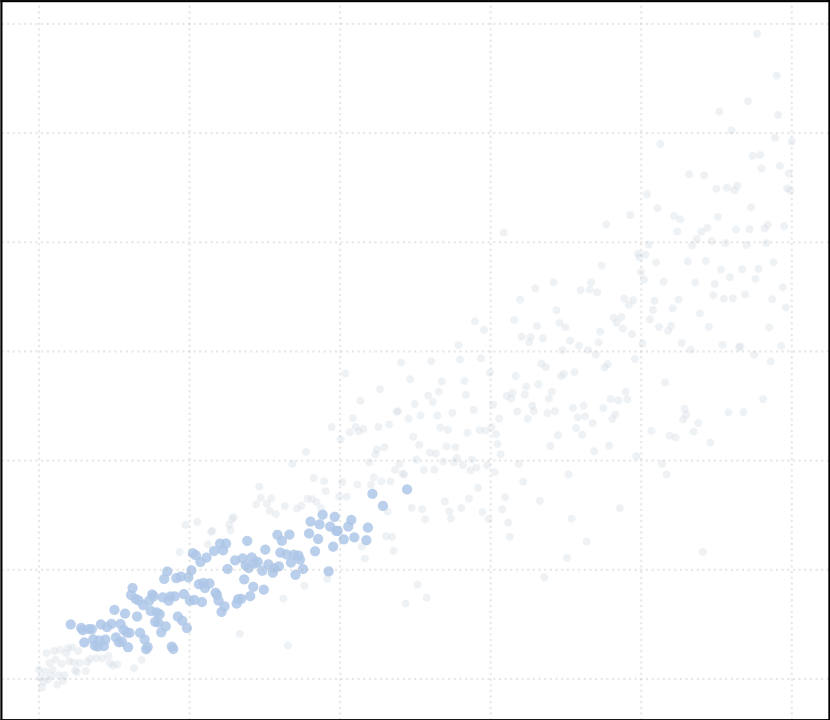
quant = 0.05 [1 Clusters]



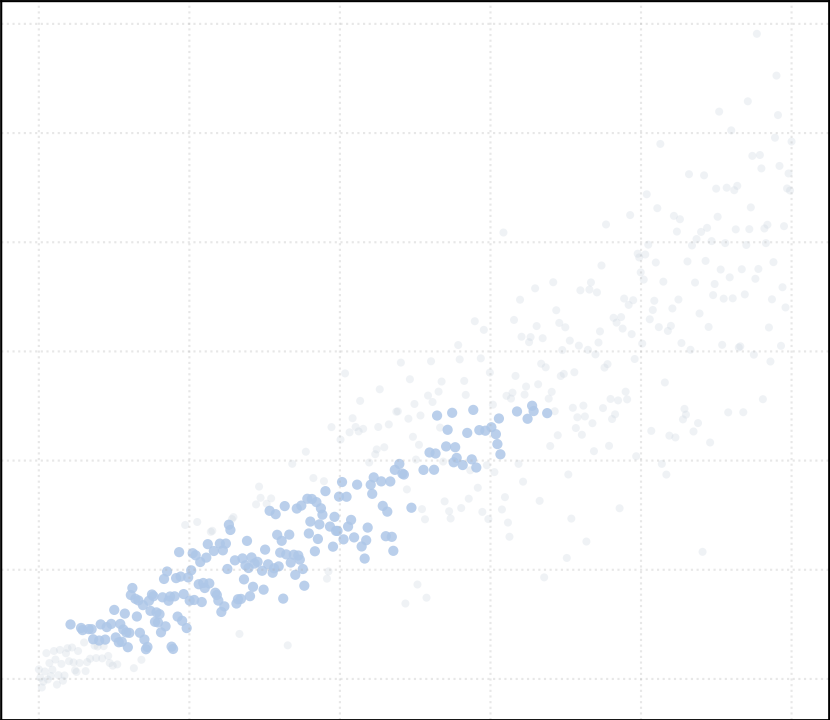
quant = 0.1 [1 Clusters]



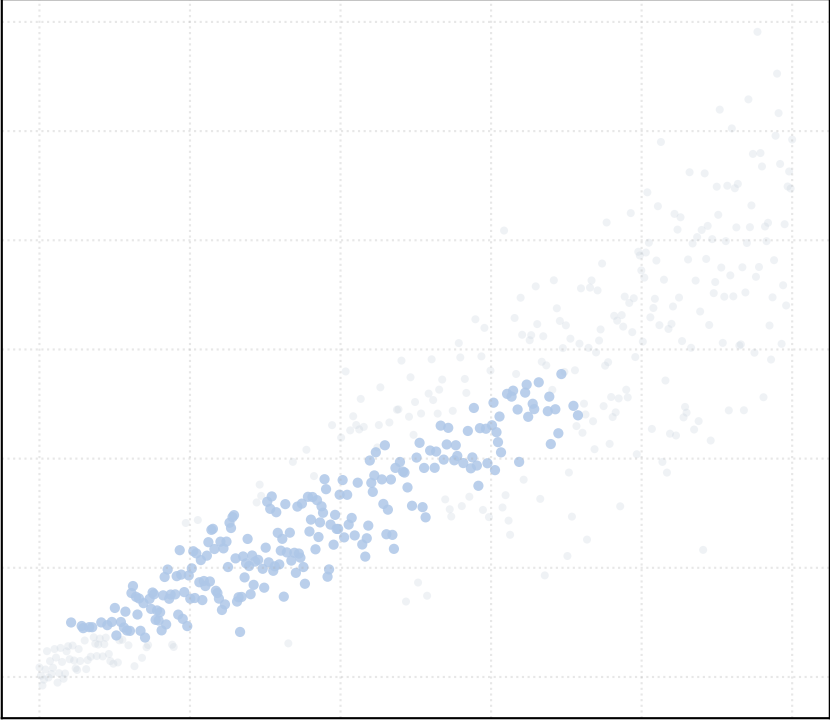
quant = 0.15 (Default) [1 Clusters]



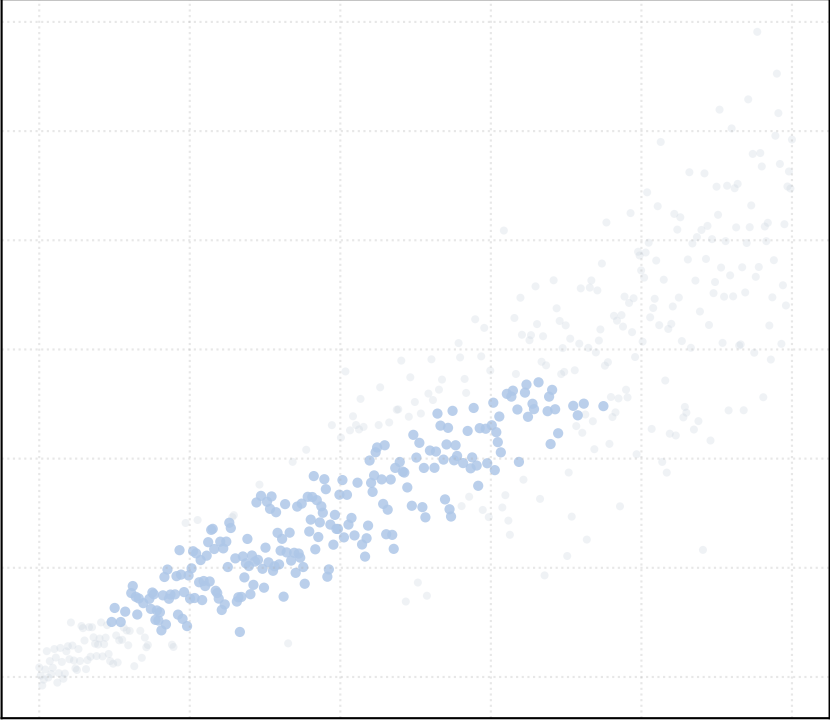
quant = 0.2 [1 Clusters]



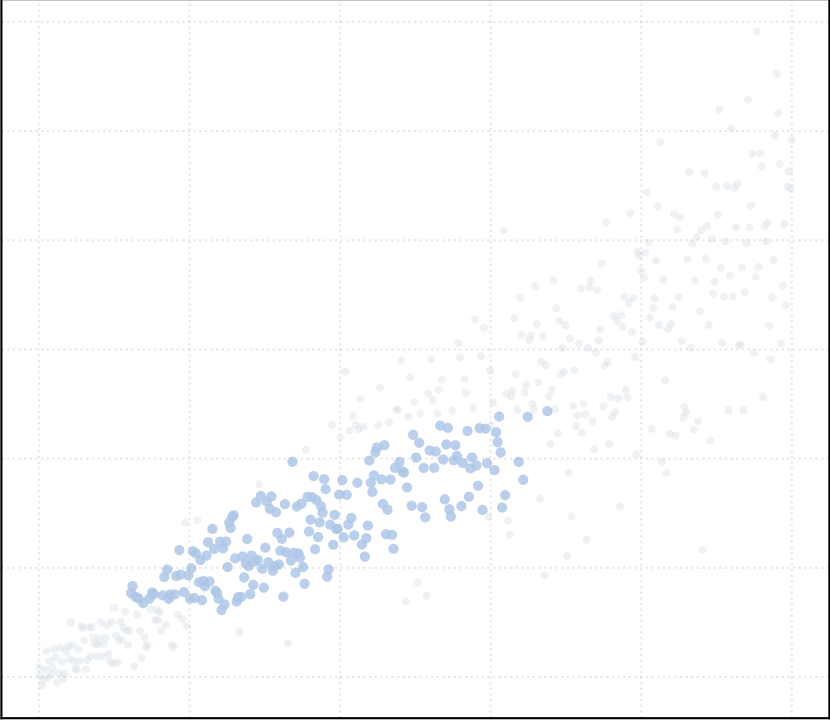
quant = 0.25 [1 Clusters]



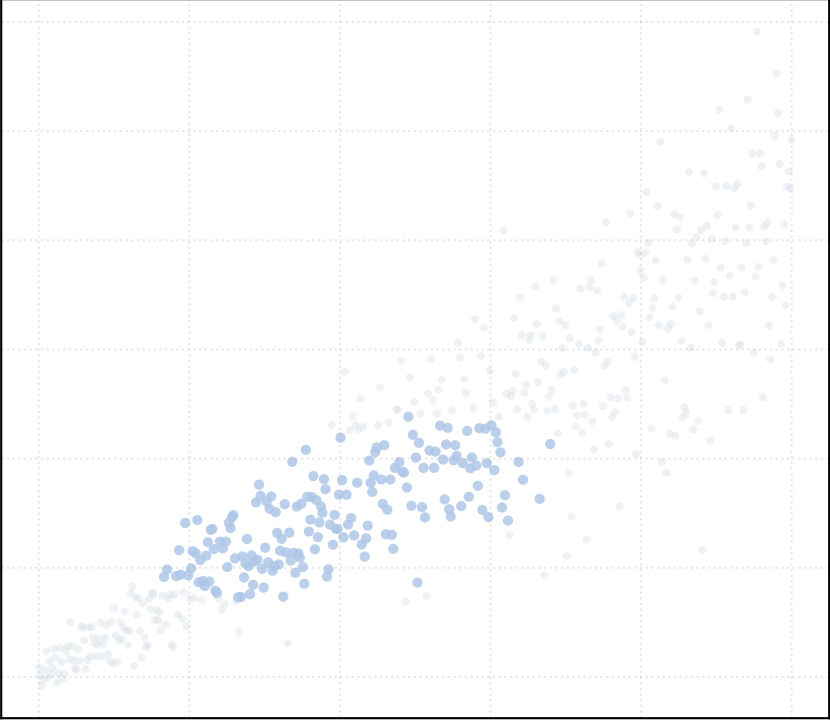
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

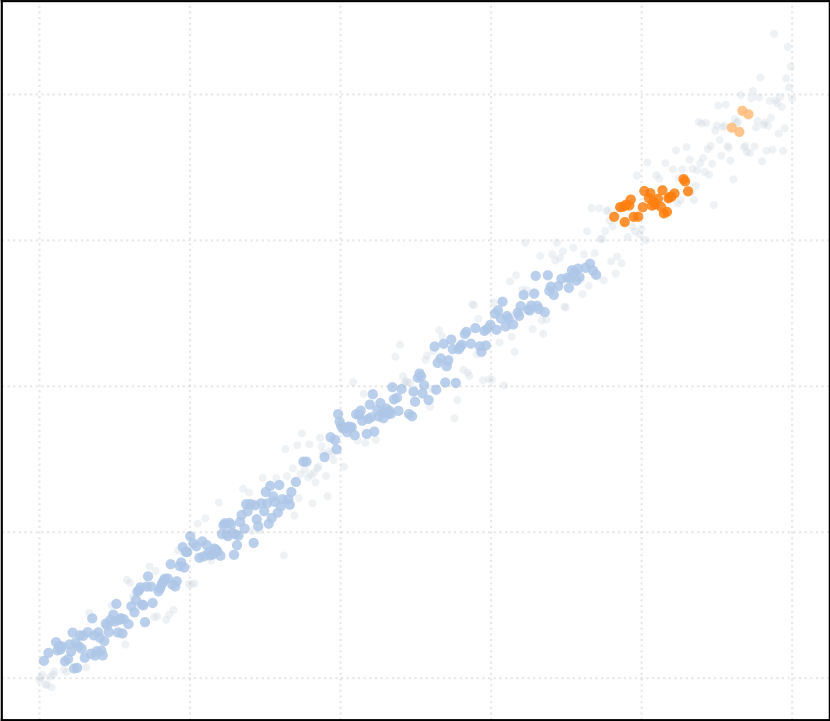


quant = 0.5 [1 Clusters]

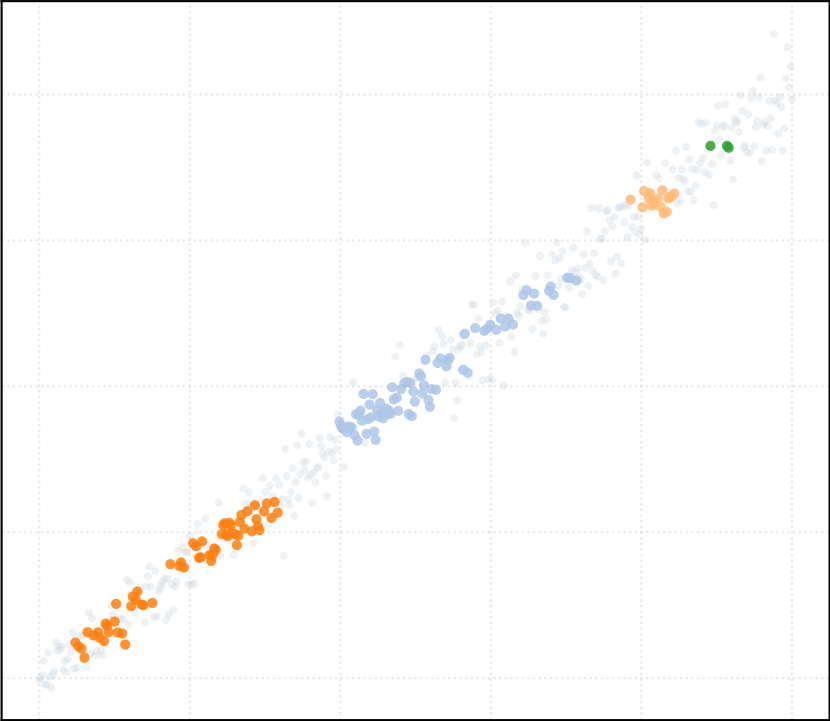


Dataset: heteroscedastic_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

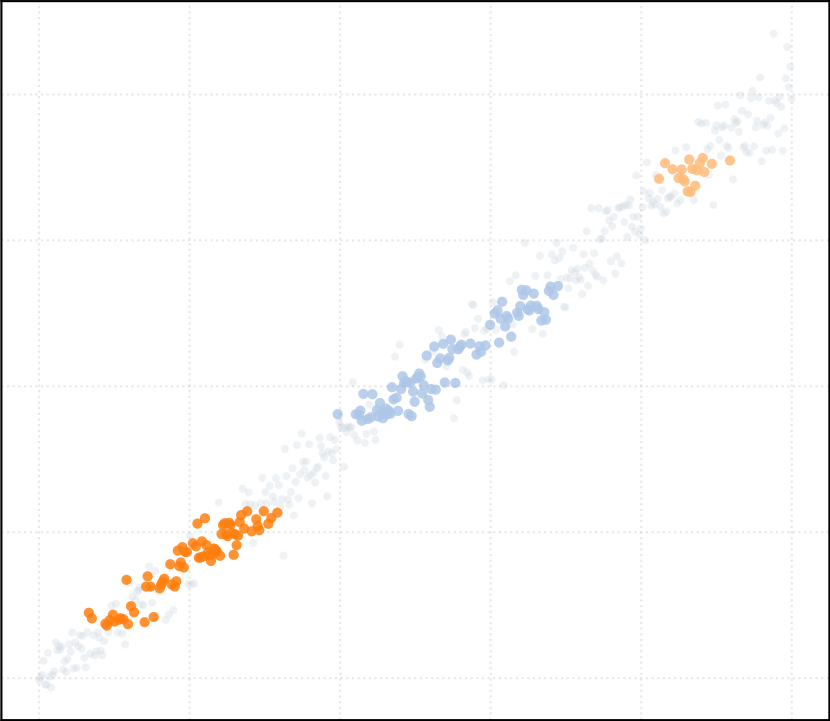
quant = 0.05 [3 Clusters]



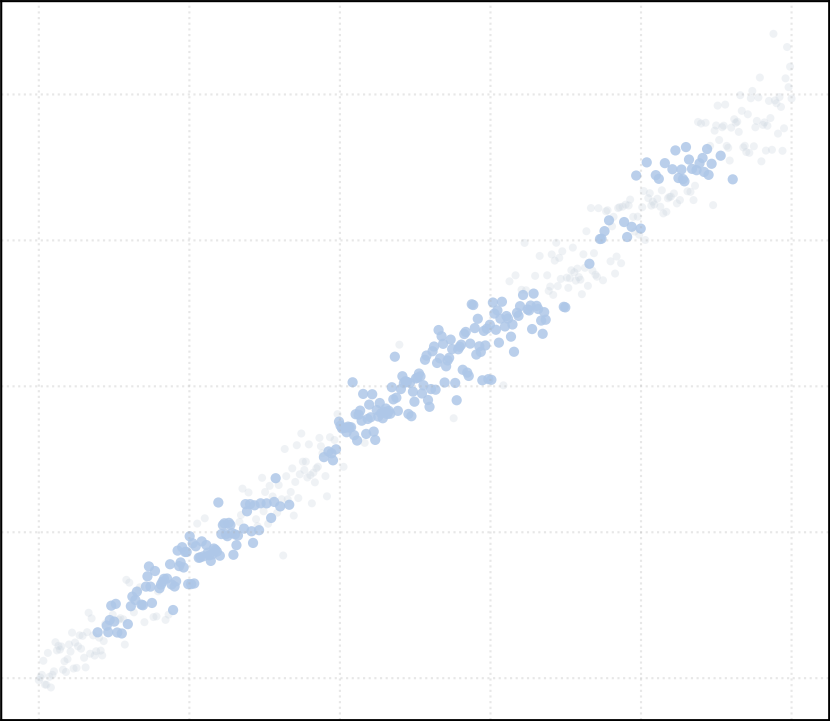
quant = 0.1 [4 Clusters]



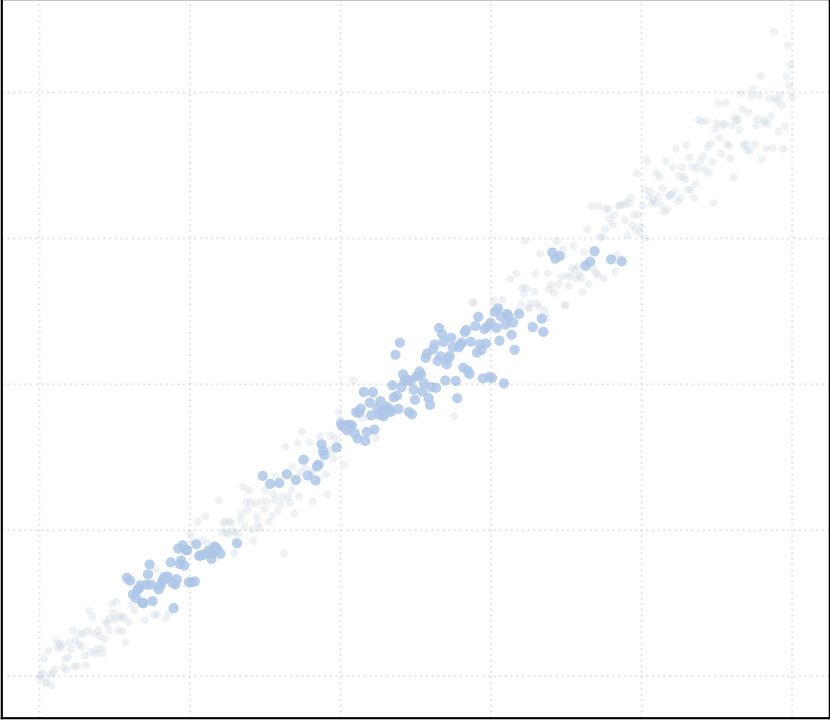
quant = 0.15 (Default) [3 Clusters]



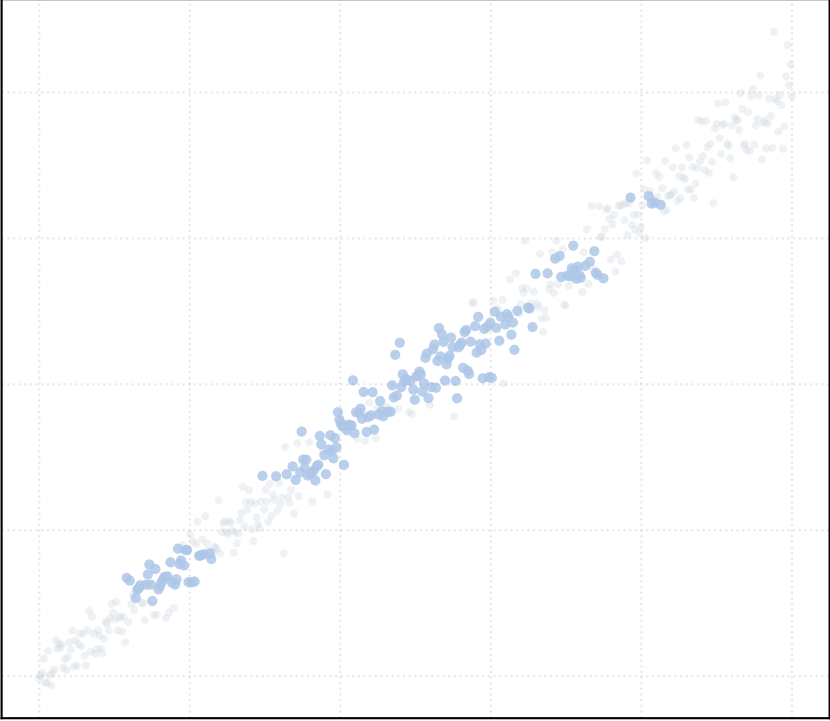
quant = 0.2 [1 Clusters]



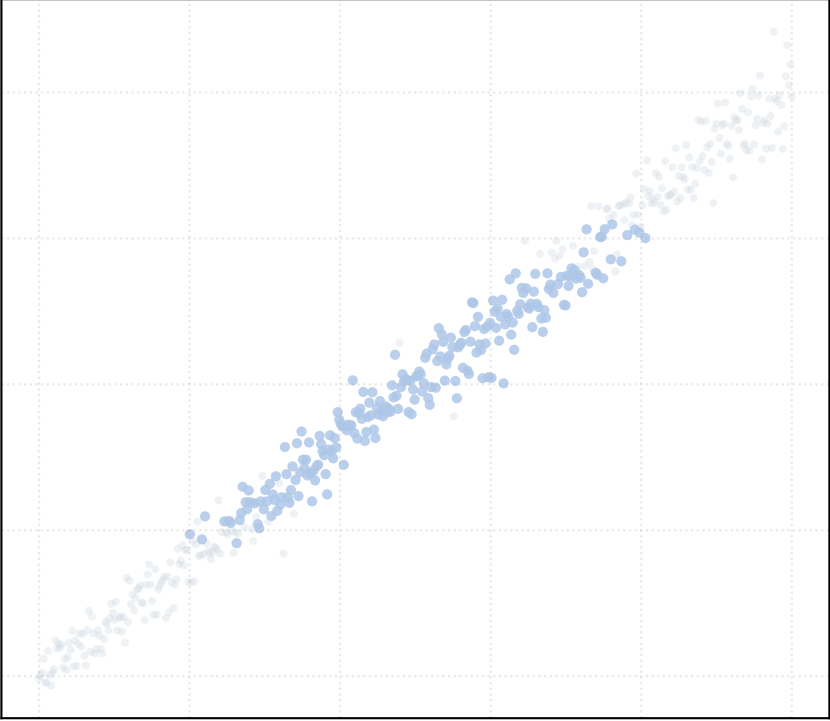
quant = 0.25 [1 Clusters]



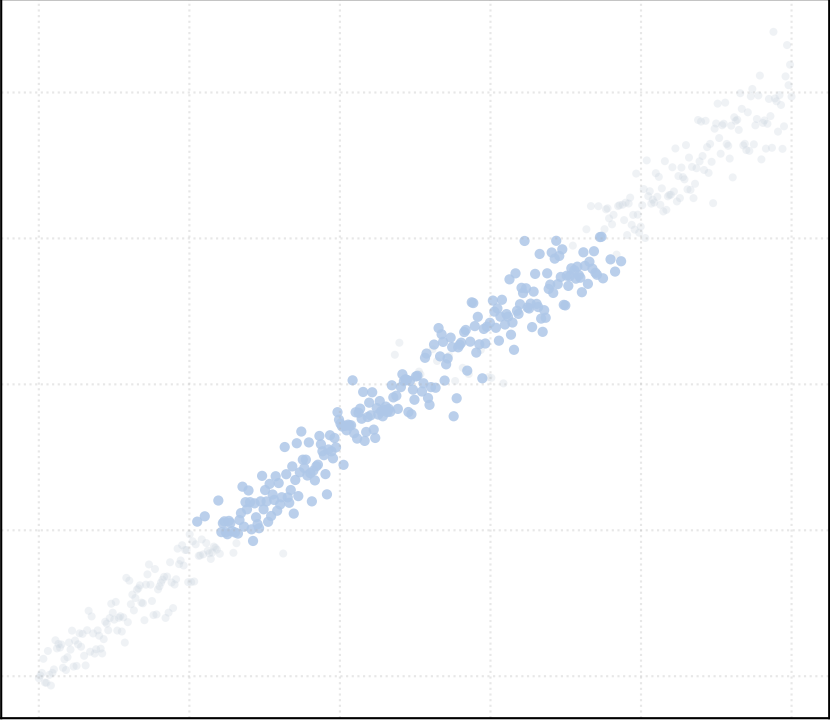
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

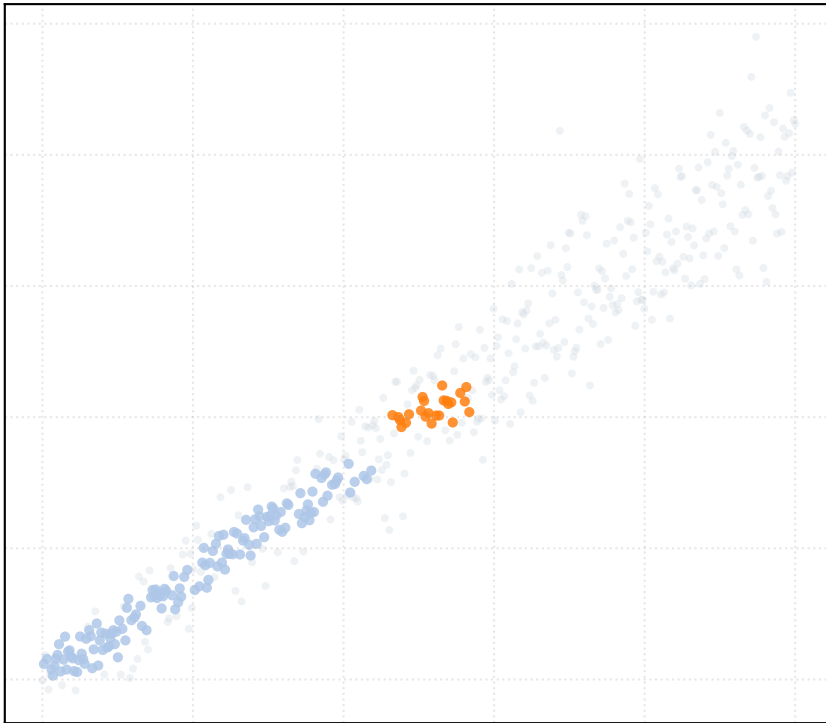


quant = 0.5 [1 Clusters]

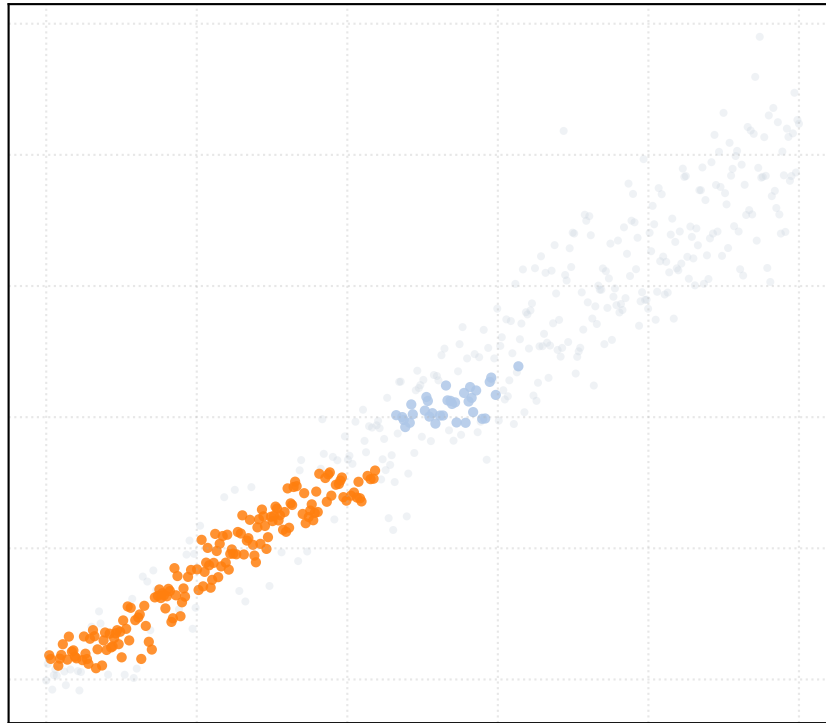


Dataset: heteroscedastic_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

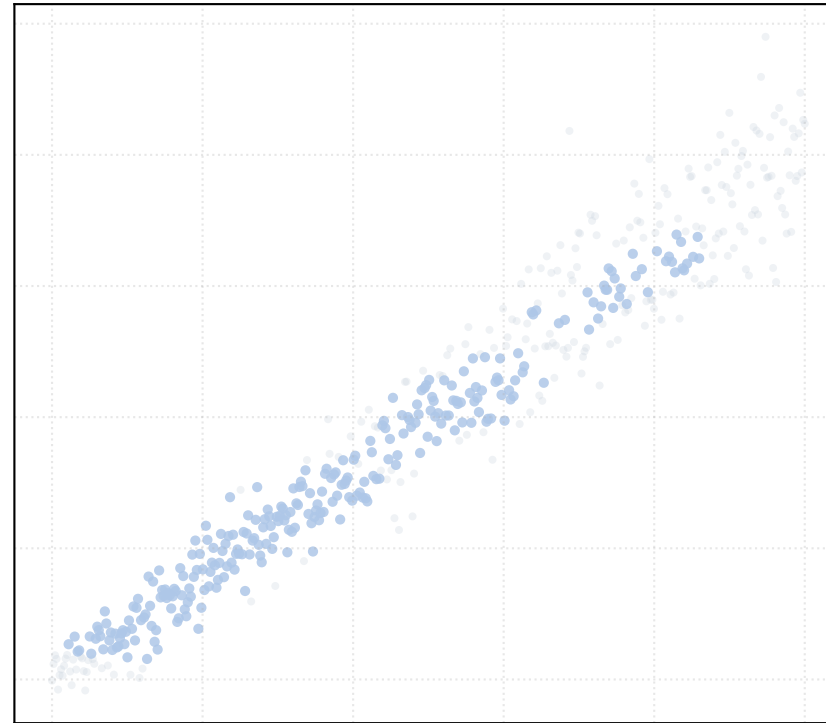
quant = 0.05 [2 Clusters]



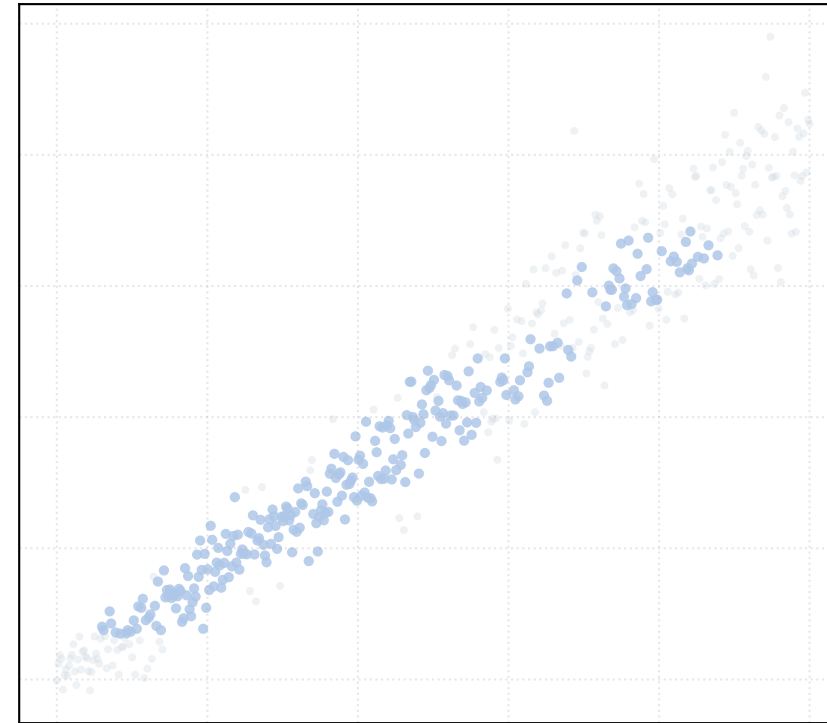
quant = 0.1 [2 Clusters]



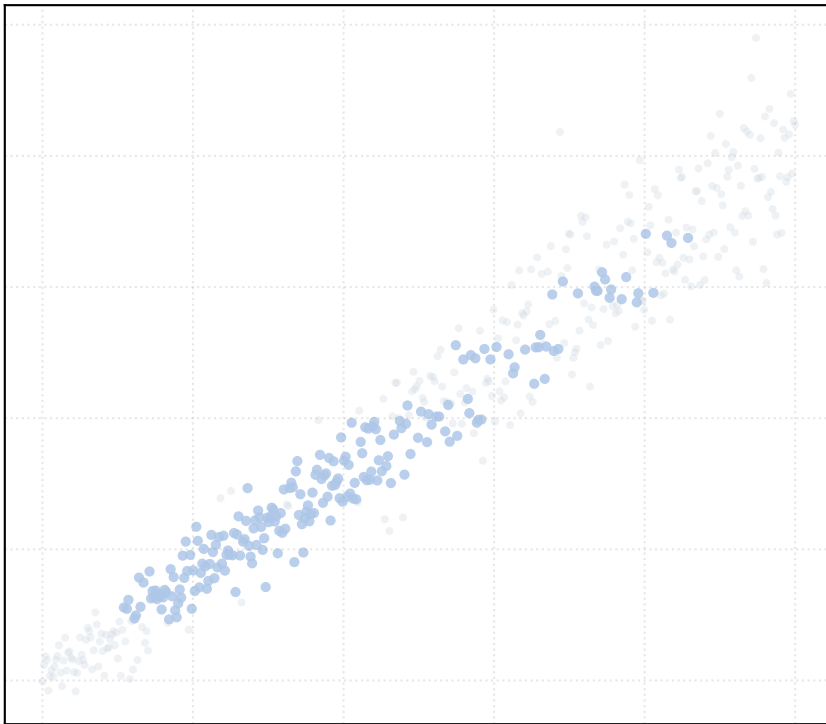
quant = 0.15 (Default) [1 Clusters]



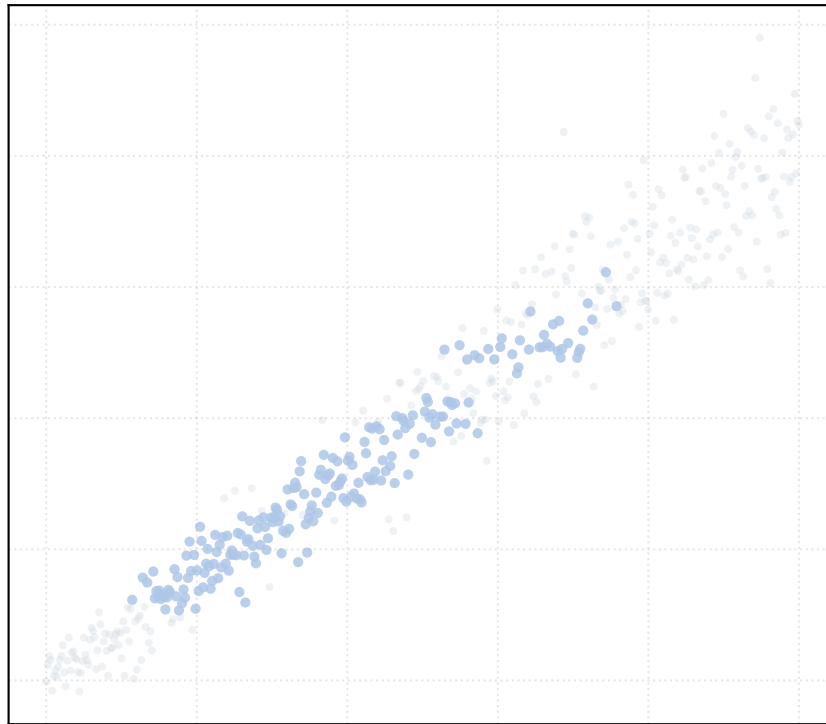
quant = 0.2 [1 Clusters]



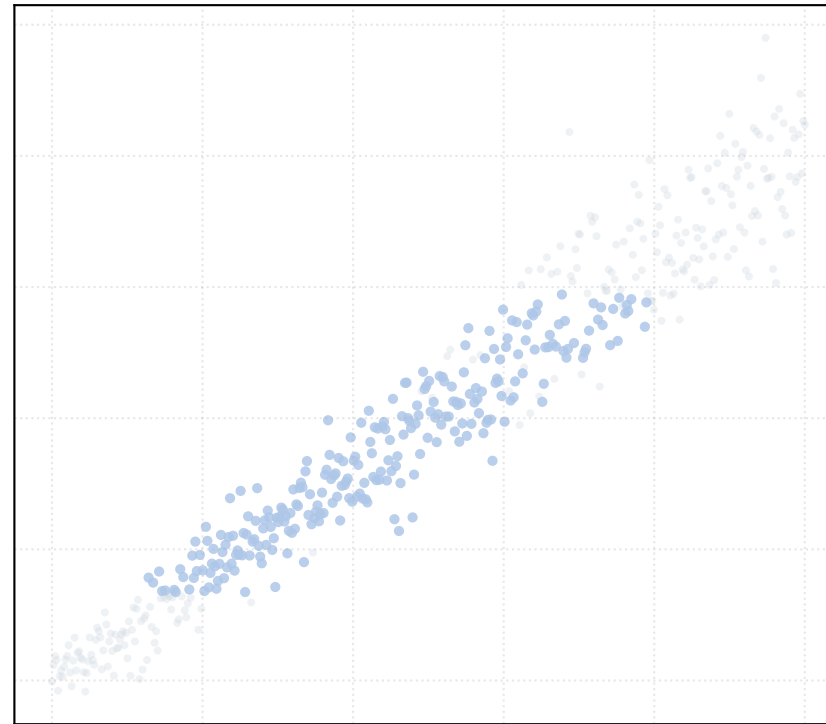
quant = 0.25 [1 Clusters]



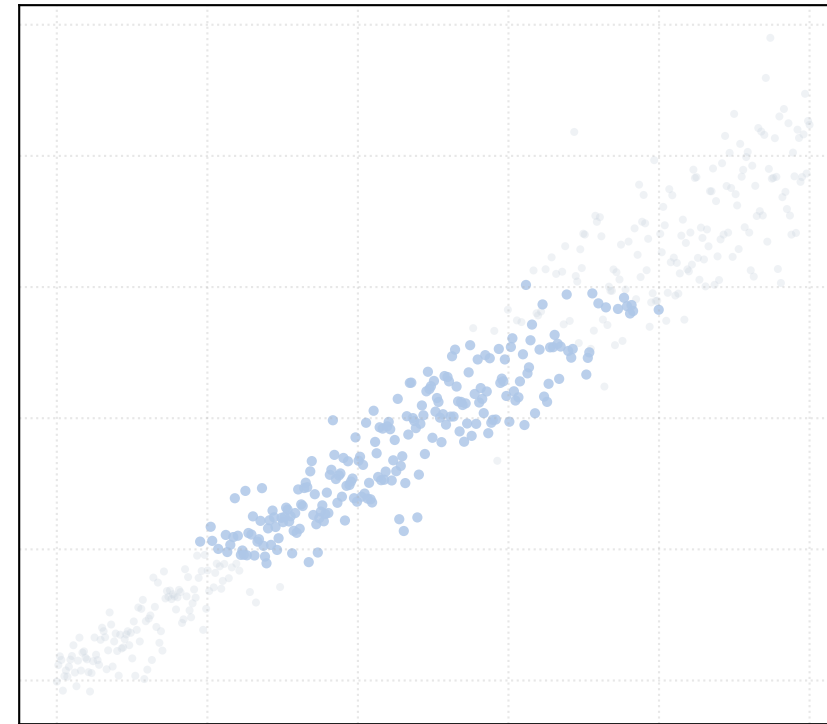
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

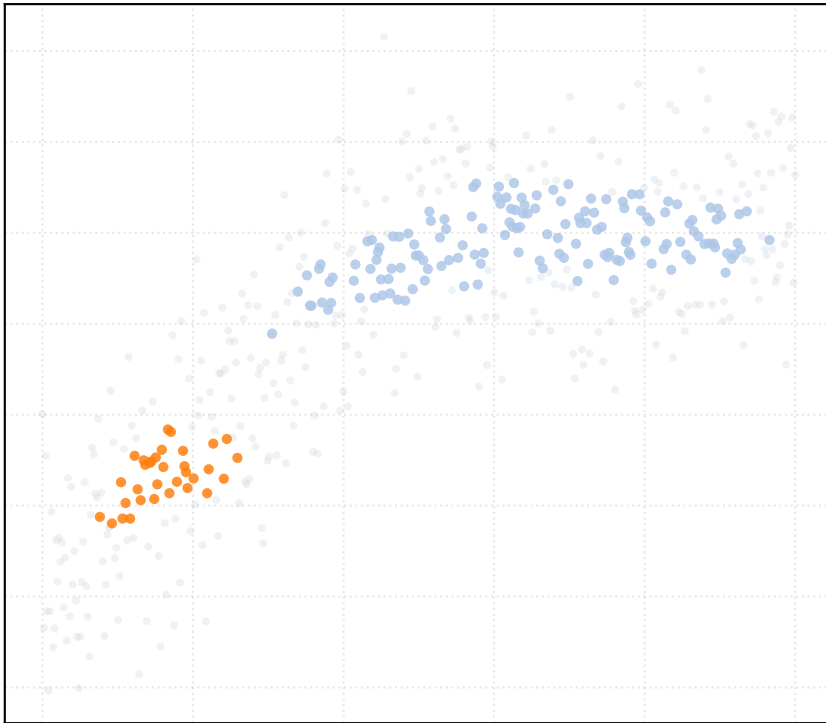


quant = 0.5 [1 Clusters]

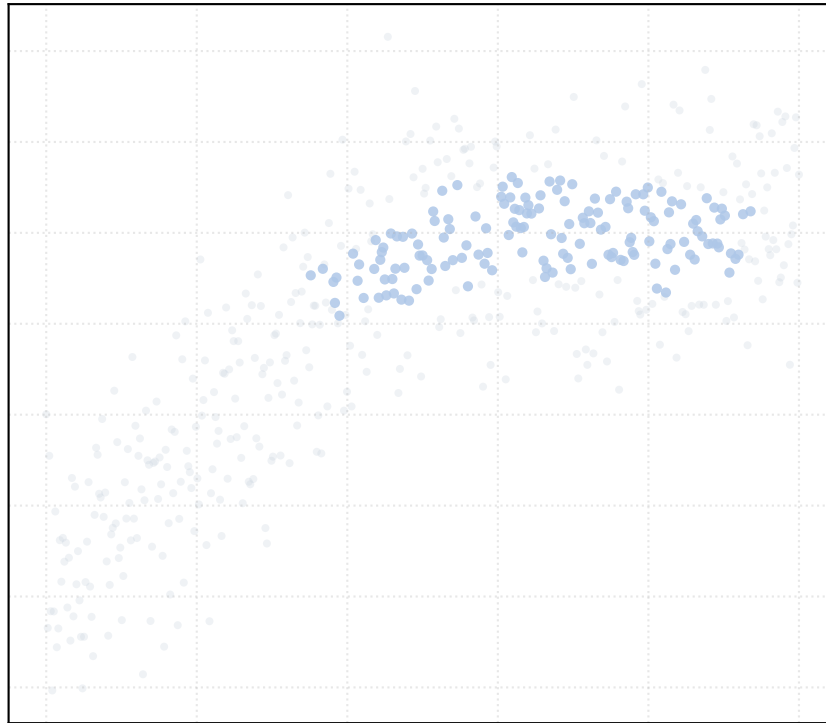


Dataset: kinked_curve_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

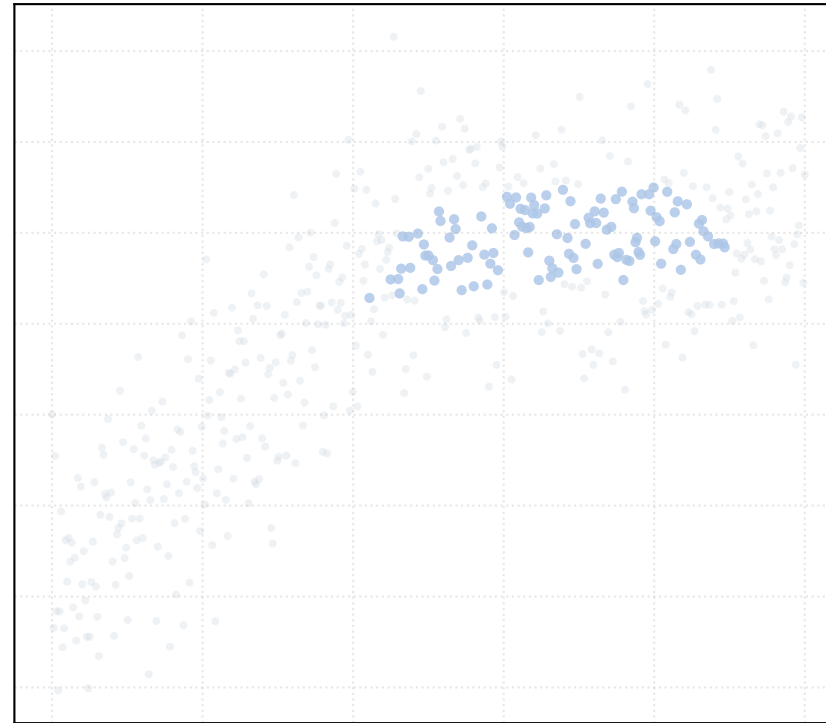
quant = 0.05 [2 Clusters]



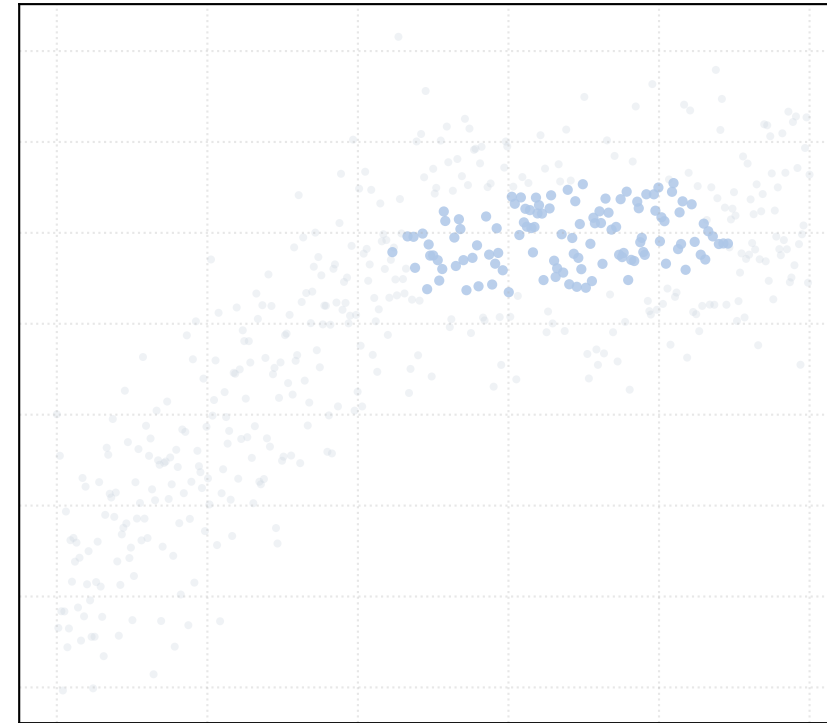
quant = 0.1 [1 Clusters]



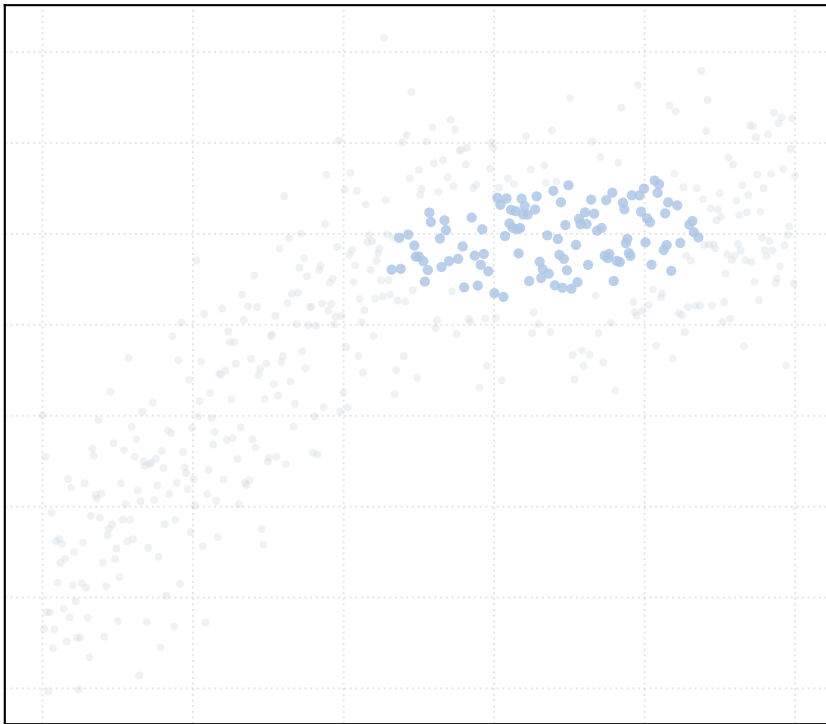
quant = 0.15 (Default) [1 Clusters]



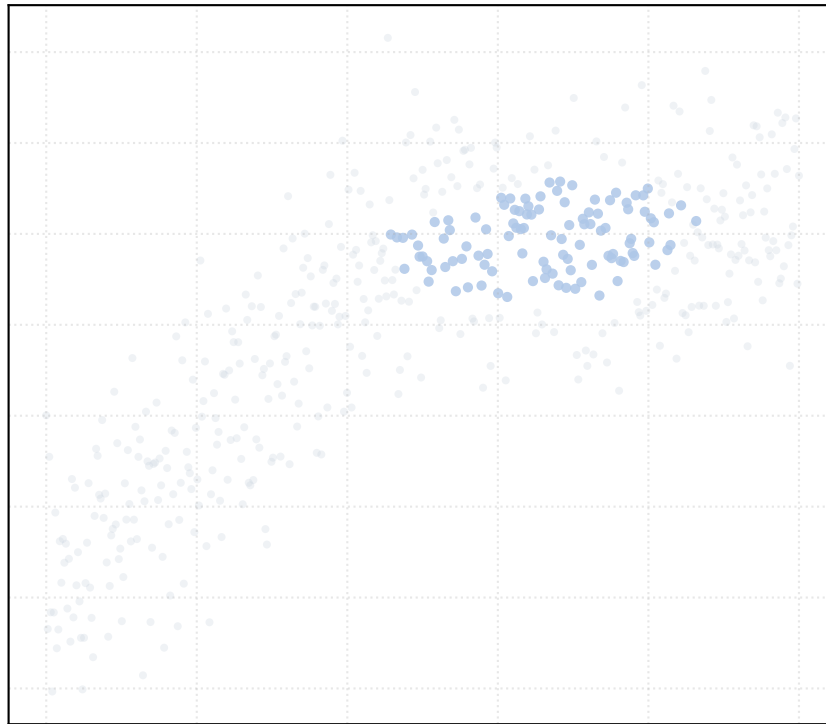
quant = 0.2 [1 Clusters]



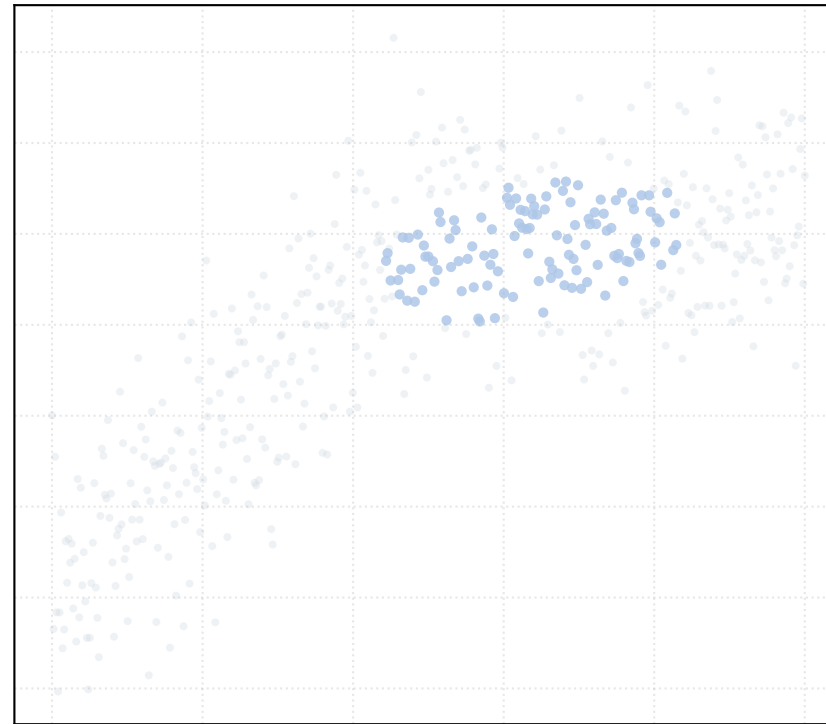
quant = 0.25 [1 Clusters]



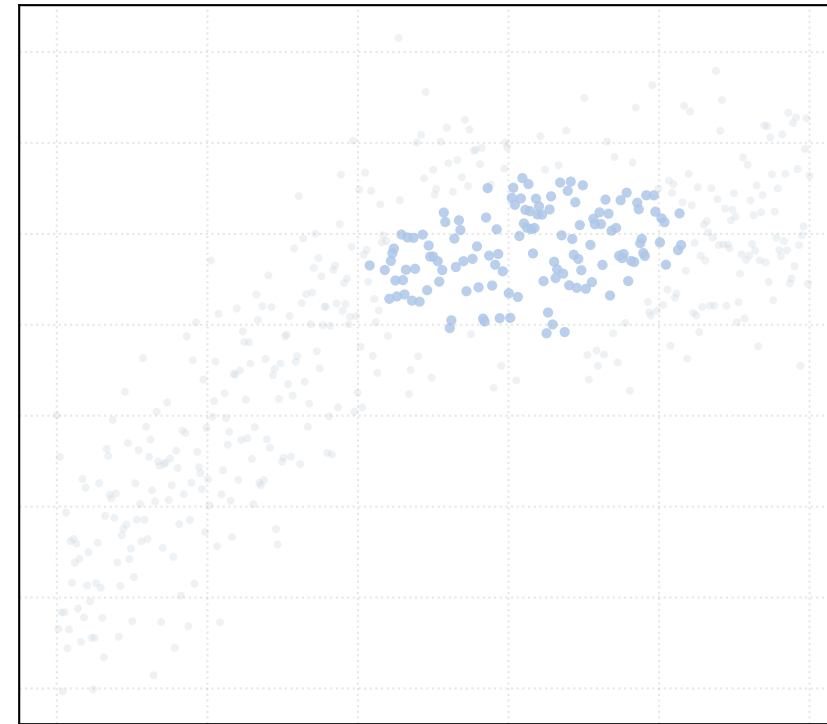
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

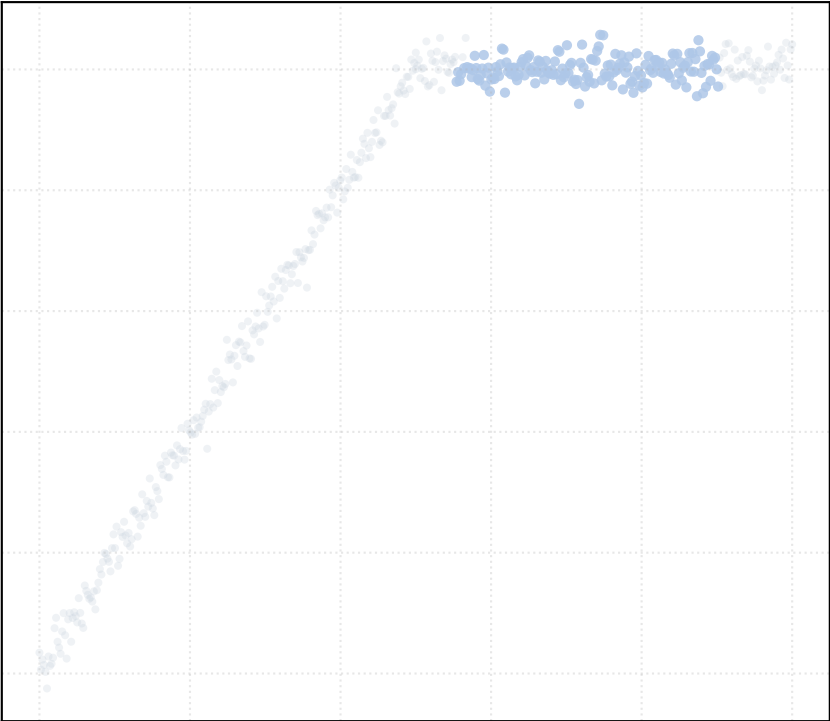


quant = 0.5 [1 Clusters]

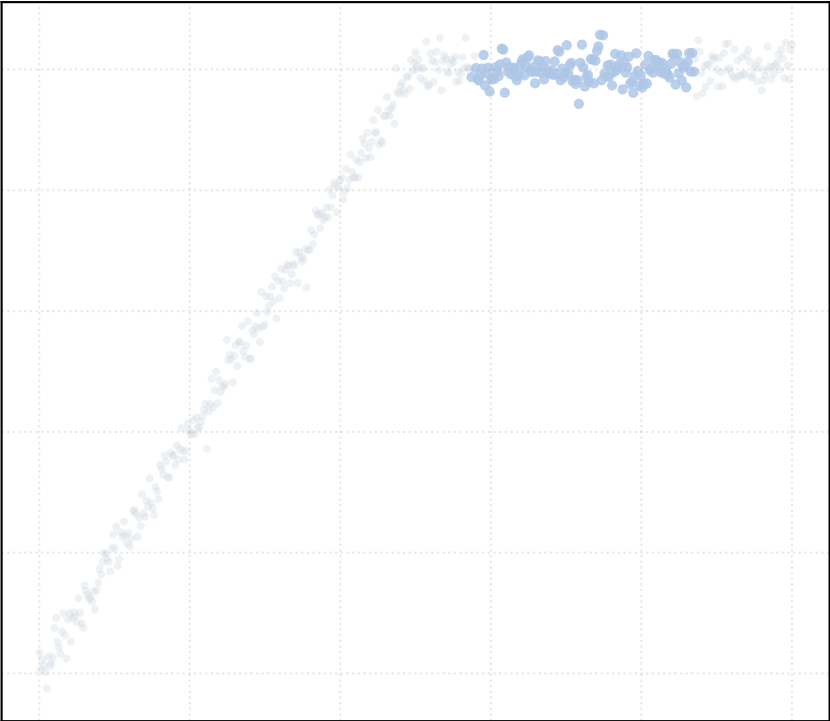


Dataset: kinked_curve_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

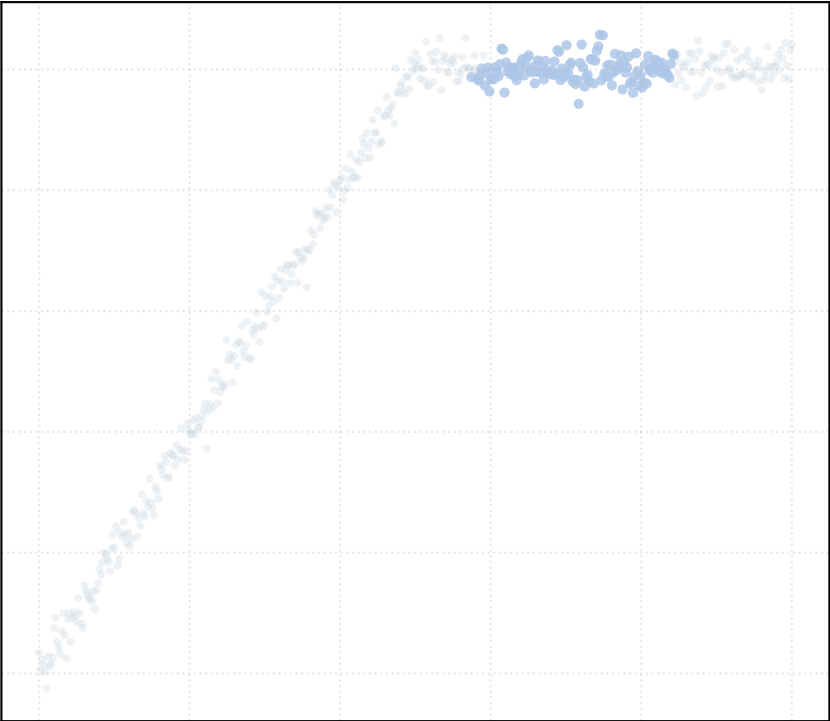
quant = 0.05 [1 Clusters]



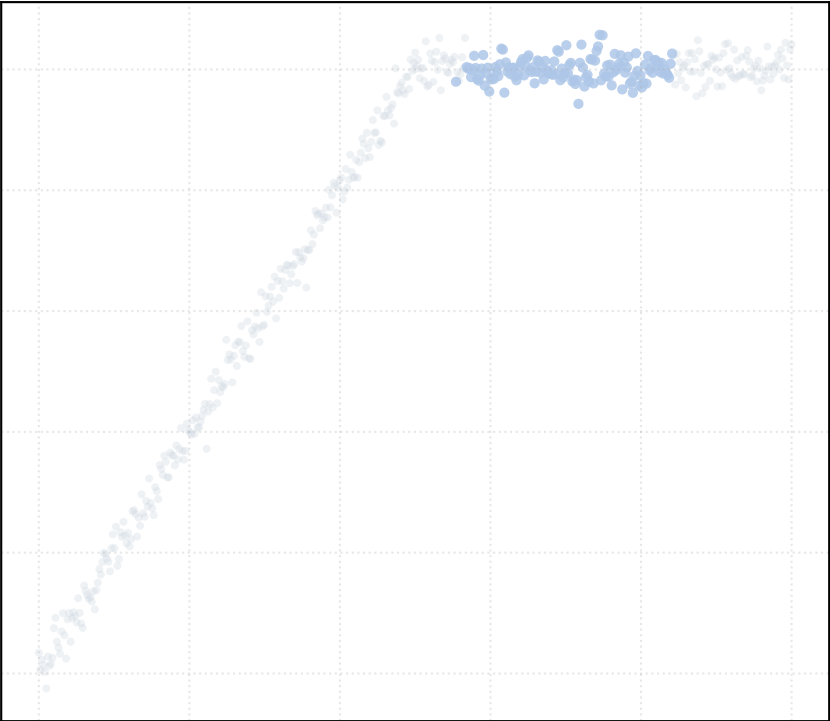
quant = 0.1 [1 Clusters]



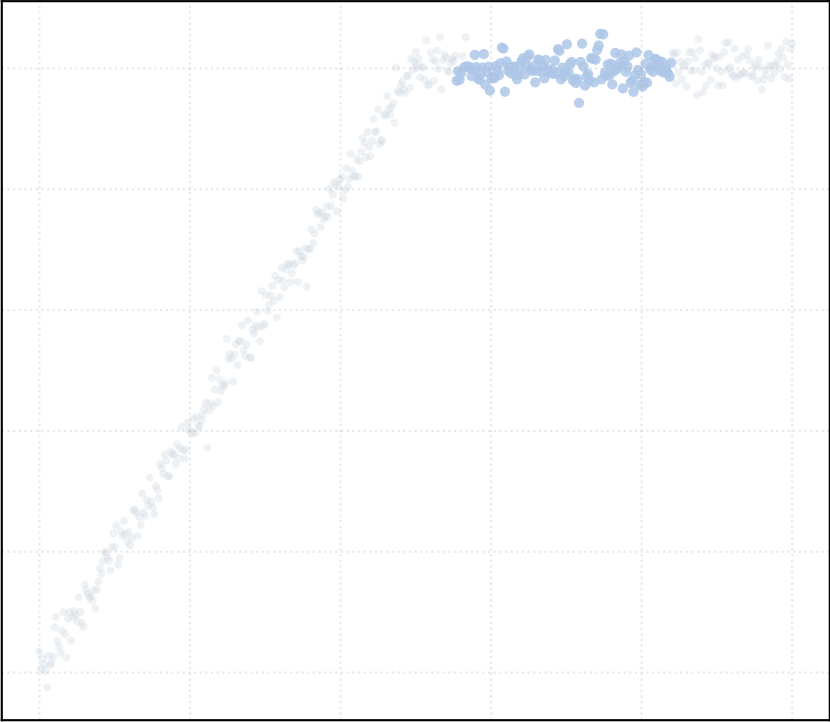
quant = 0.15 (Default) [1 Clusters]



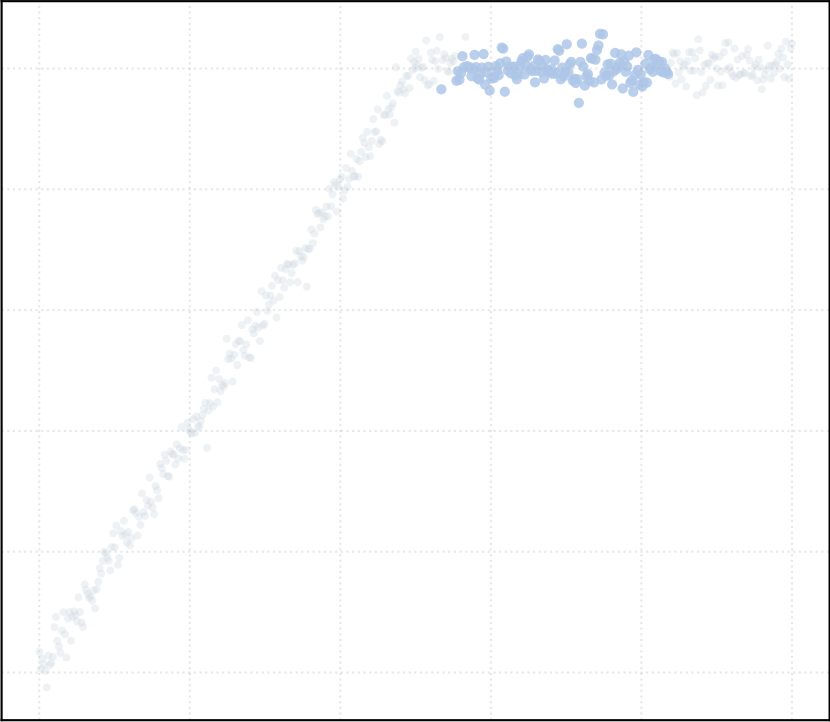
quant = 0.2 [1 Clusters]



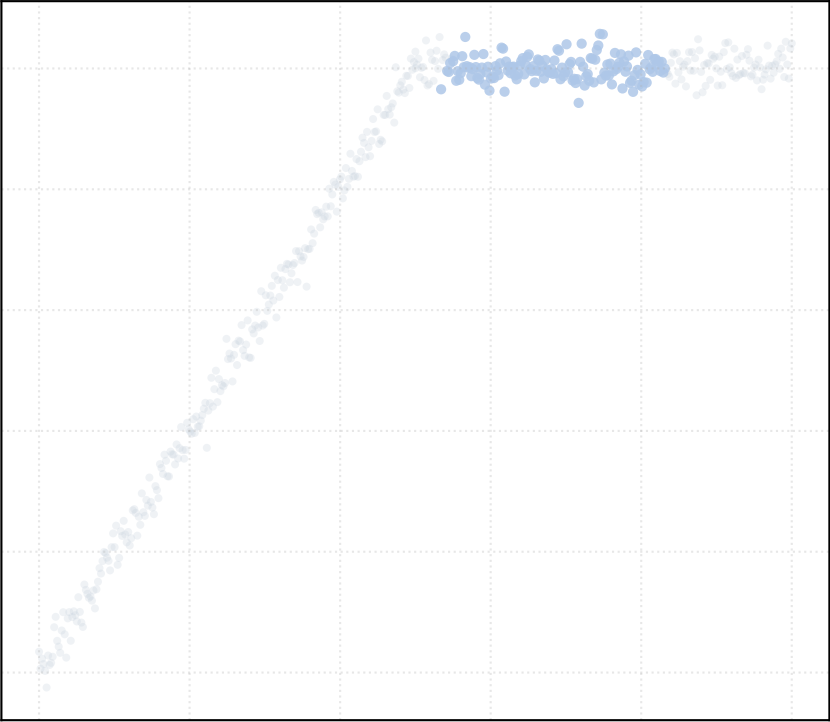
quant = 0.25 [1 Clusters]



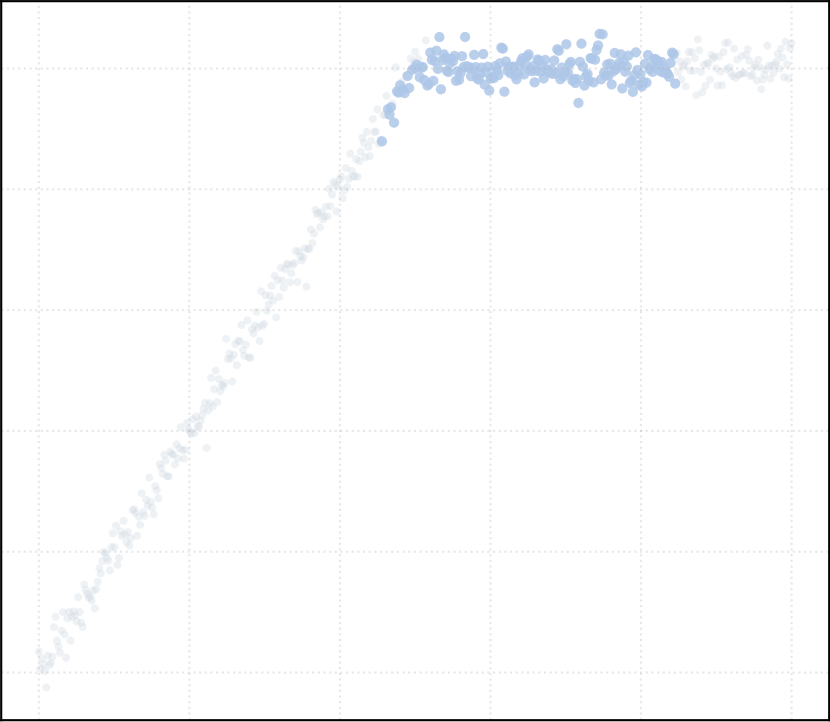
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

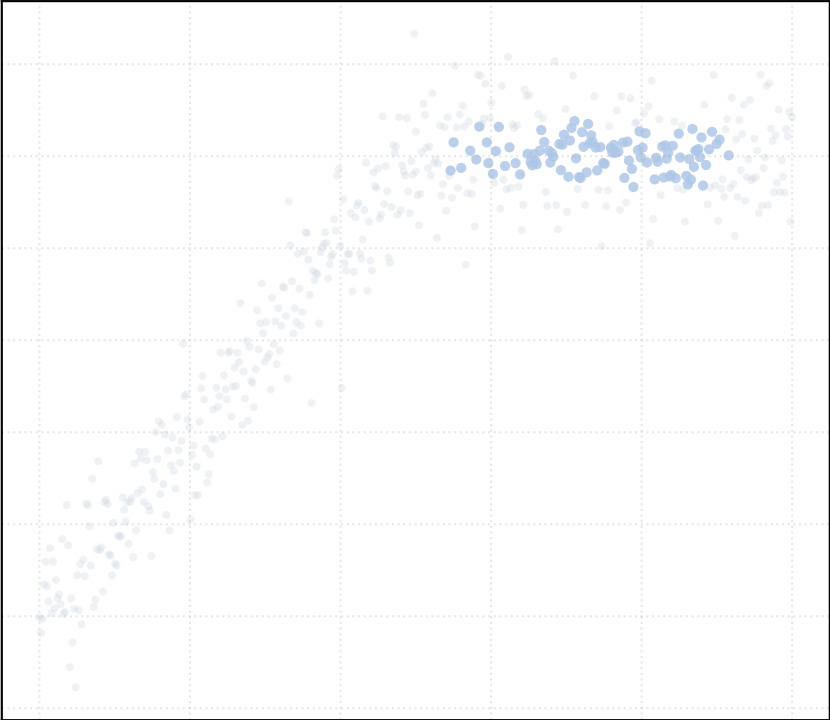


quant = 0.5 [1 Clusters]

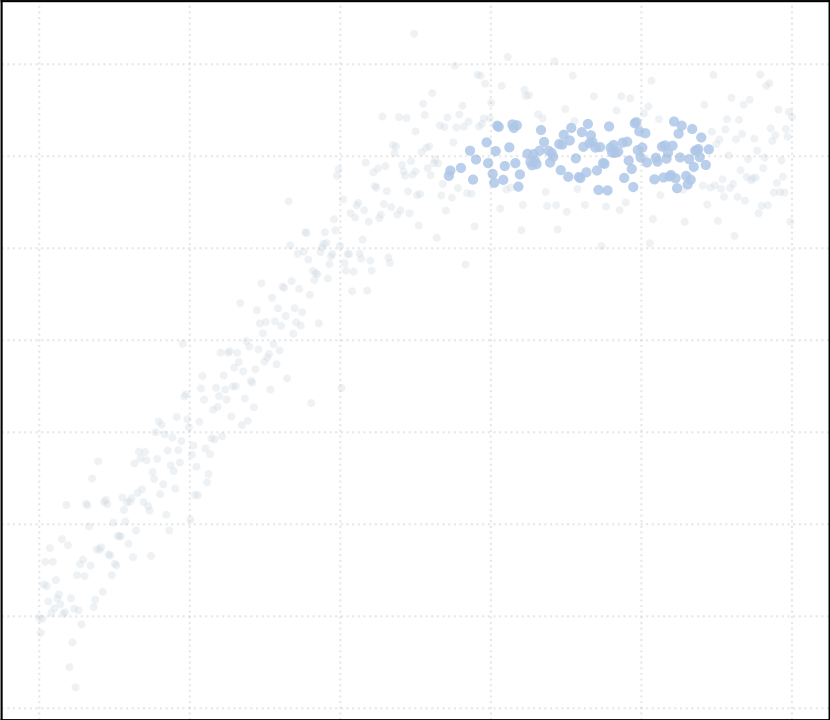


Dataset: kinked_curve_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

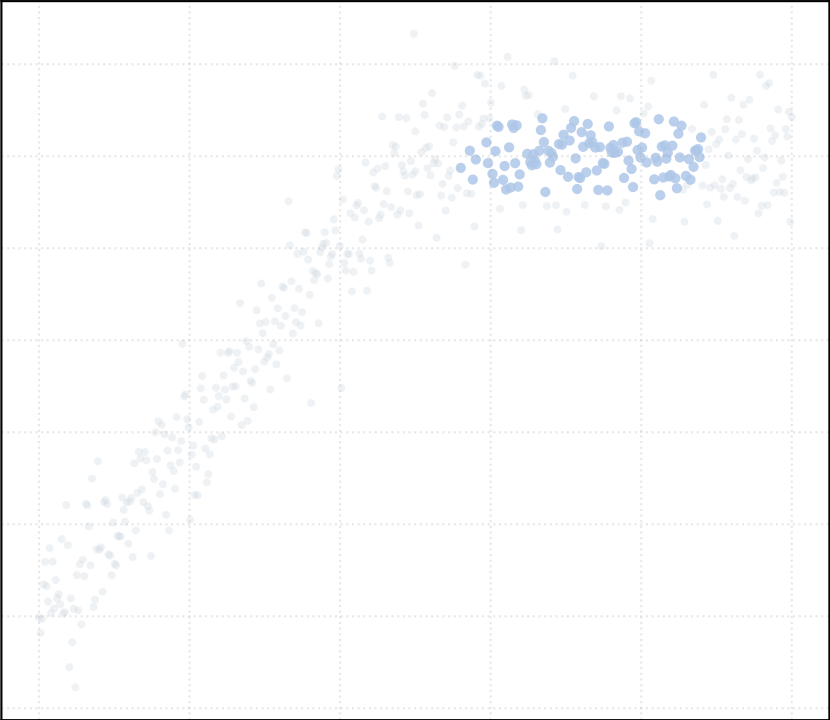
quant = 0.05 [1 Clusters]



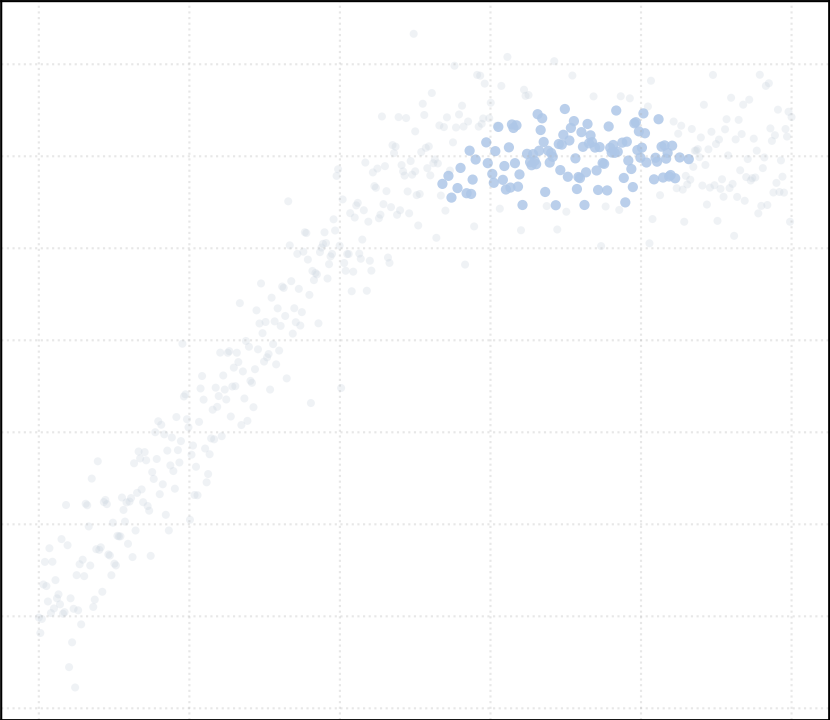
quant = 0.1 [1 Clusters]



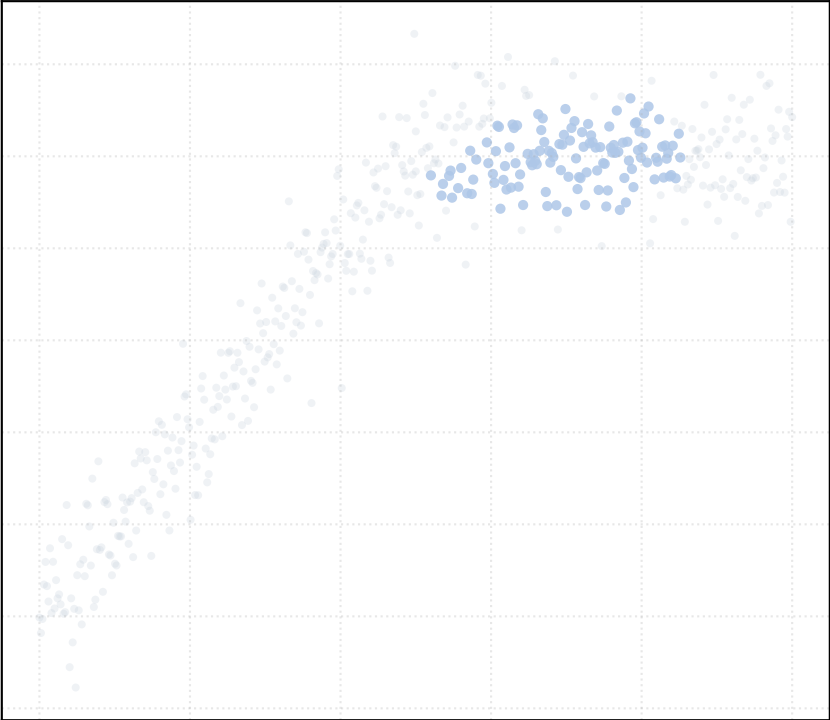
quant = 0.15 (Default) [1 Clusters]



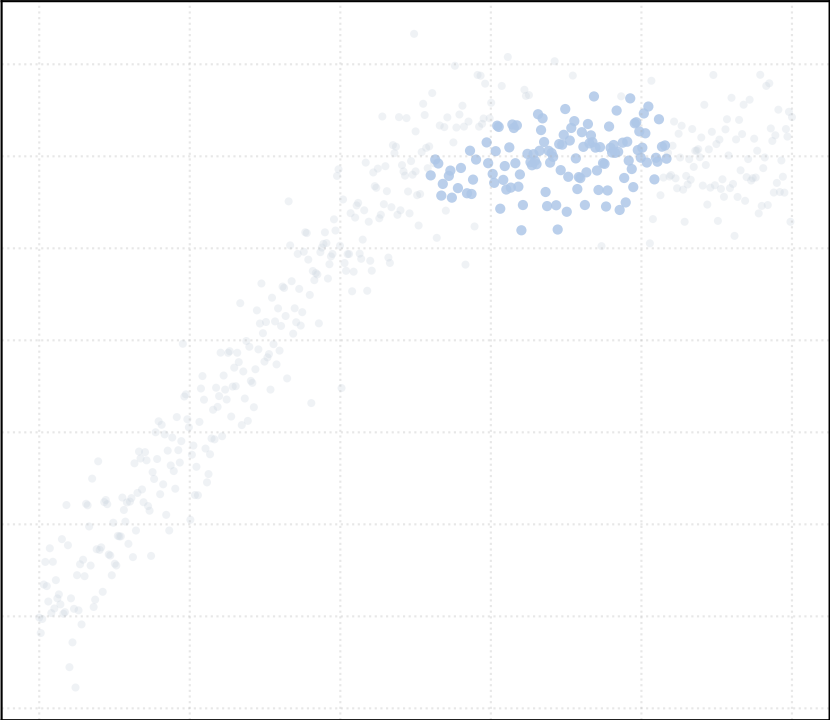
quant = 0.2 [1 Clusters]



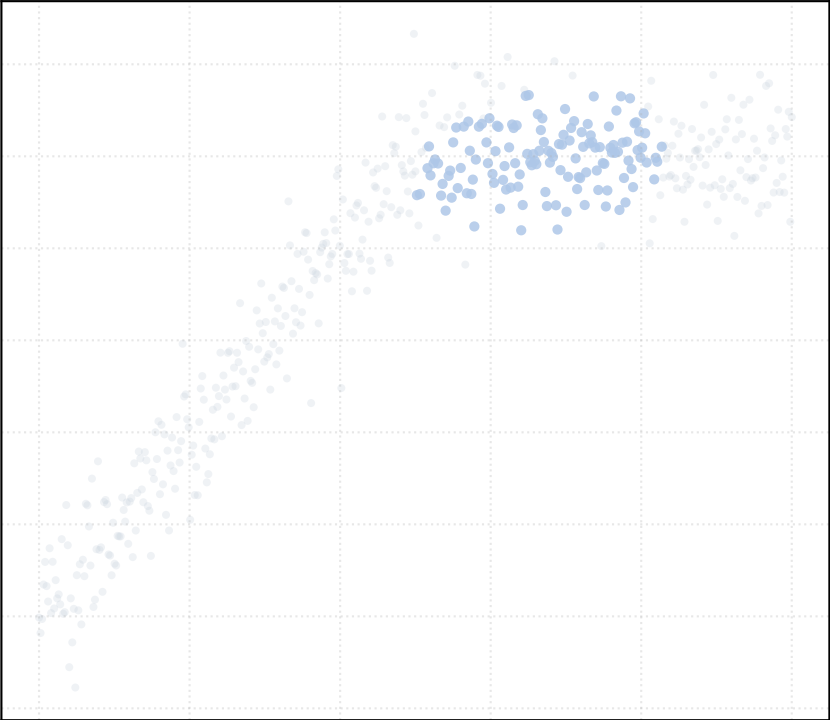
quant = 0.25 [1 Clusters]



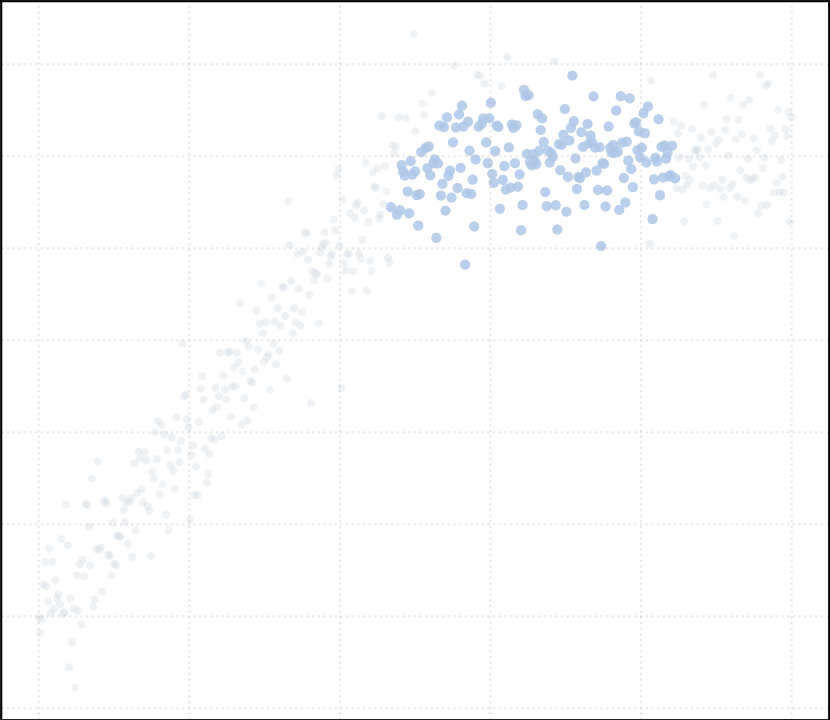
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

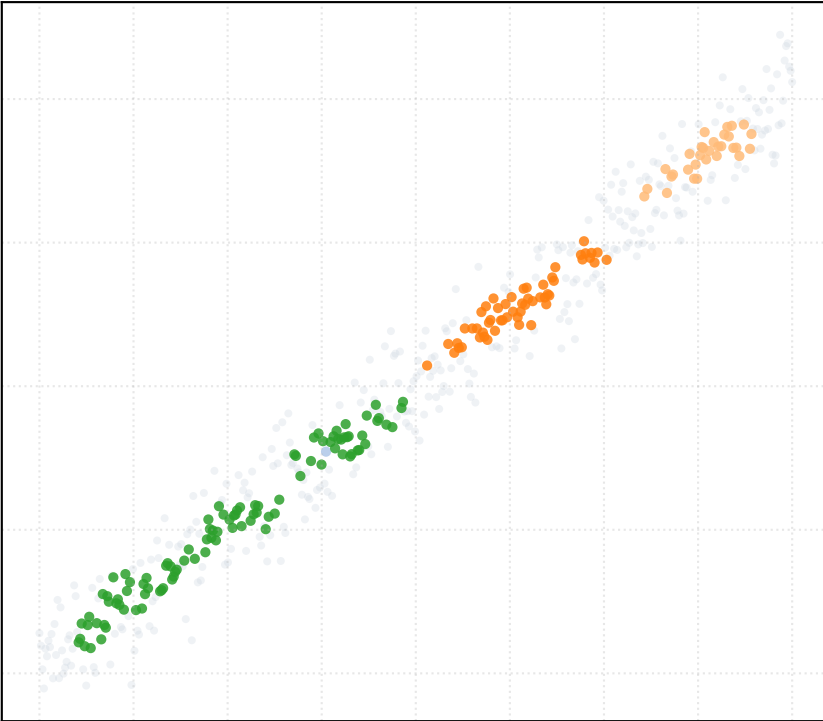


quant = 0.5 [1 Clusters]

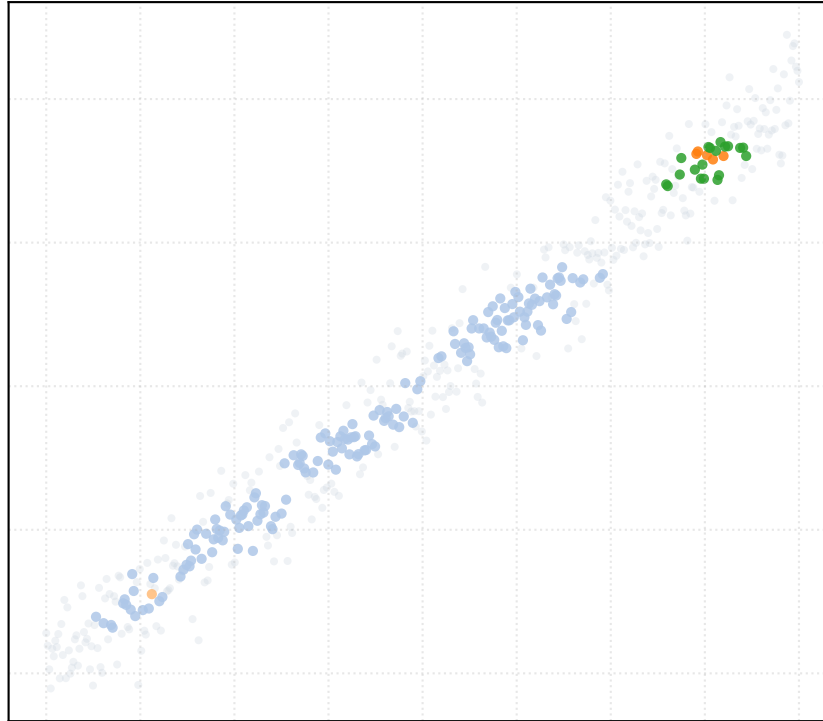


Dataset: linear_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

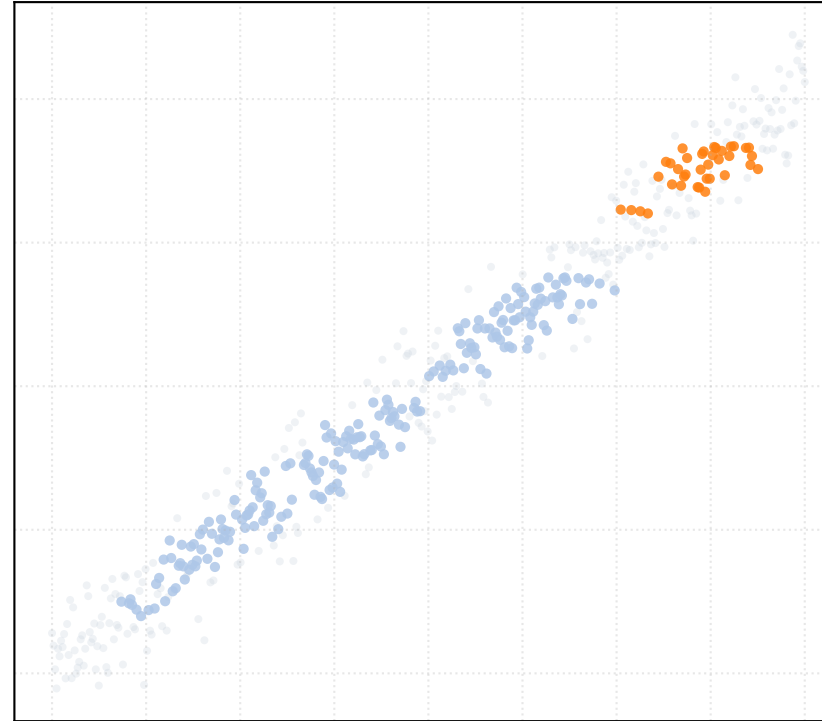
quant = 0.05 [4 Clusters]



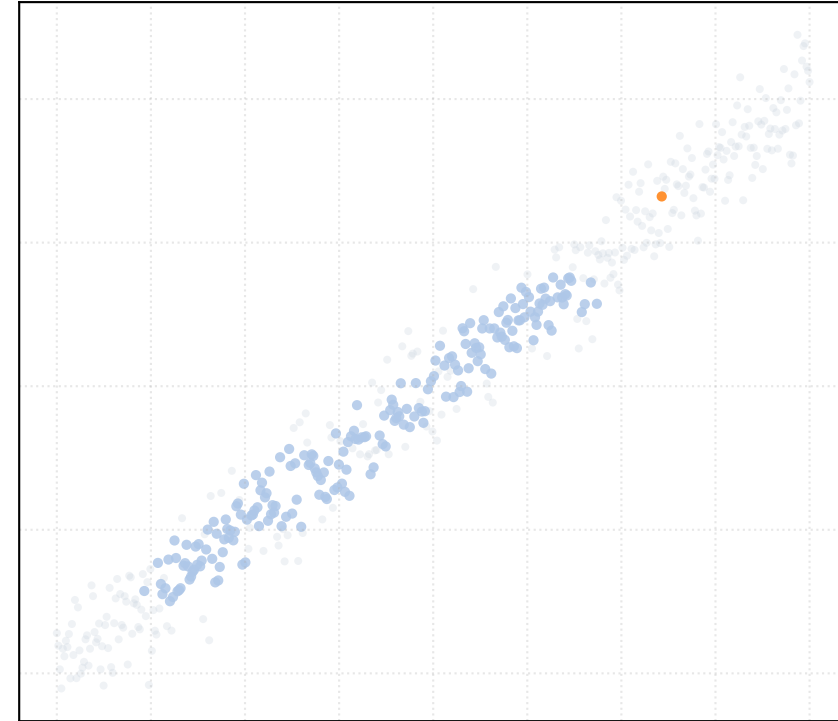
quant = 0.1 [4 Clusters]



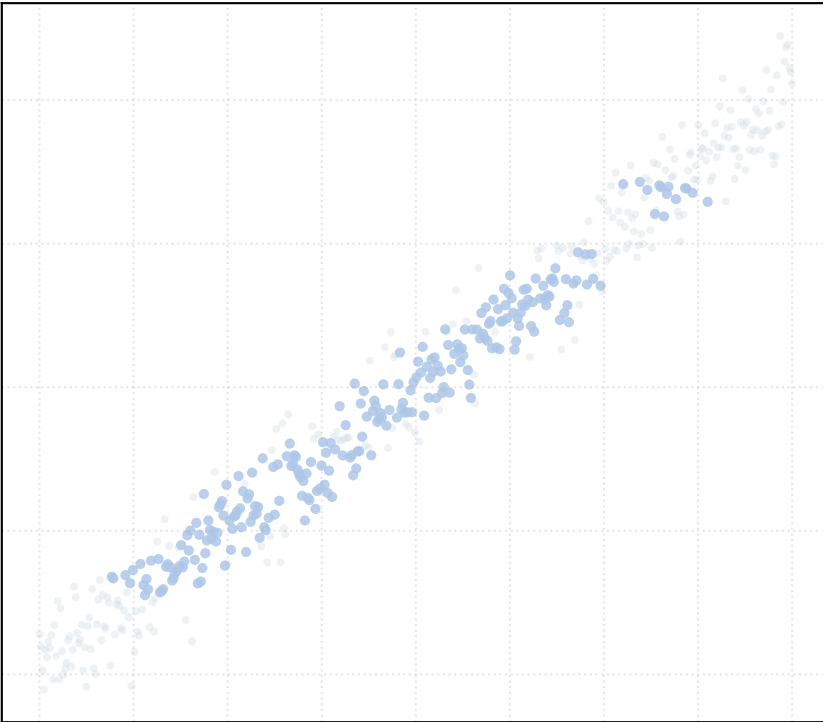
quant = 0.15 (Default) [2 Clusters]



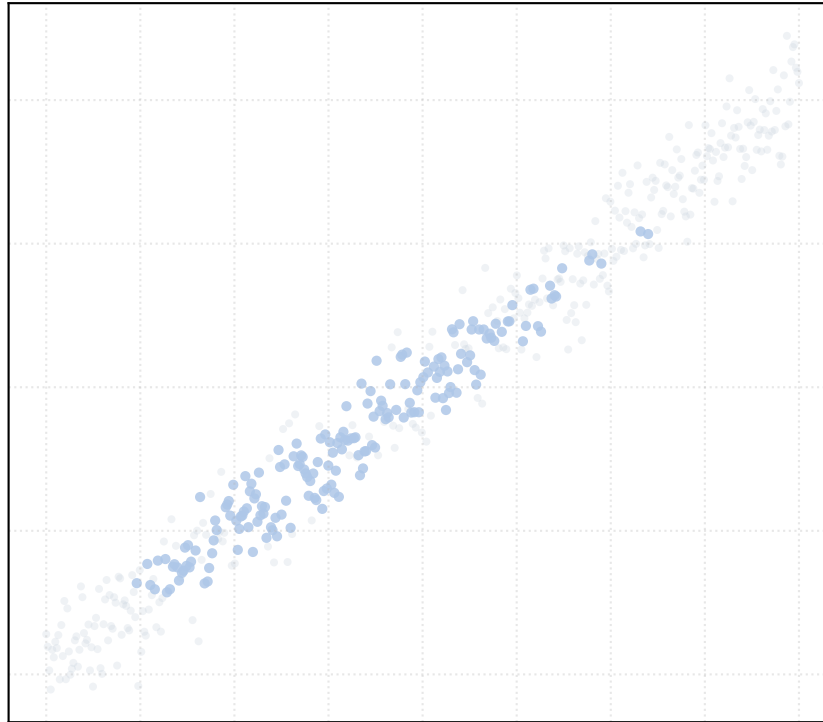
quant = 0.2 [2 Clusters]



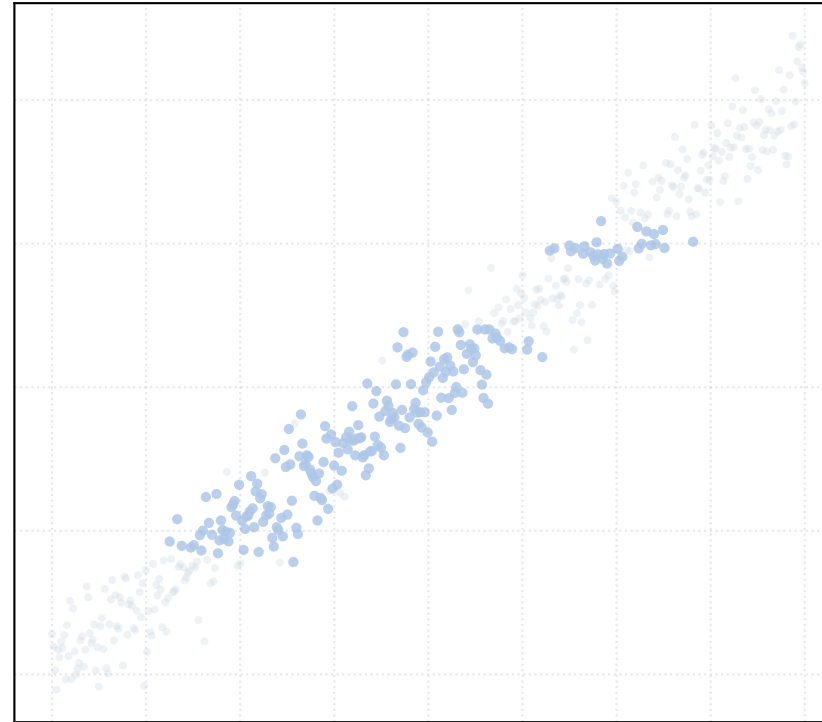
quant = 0.25 [1 Clusters]



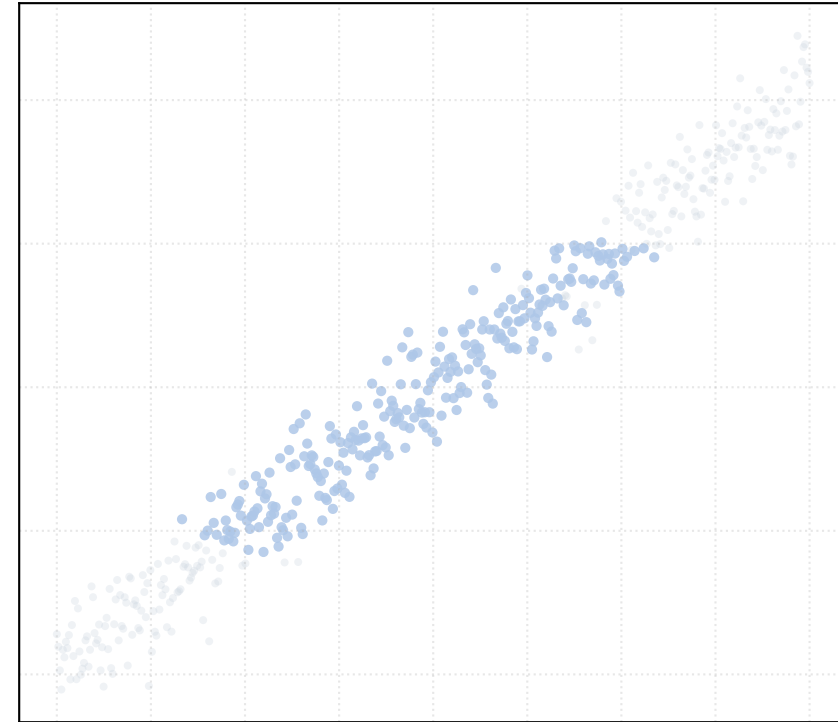
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

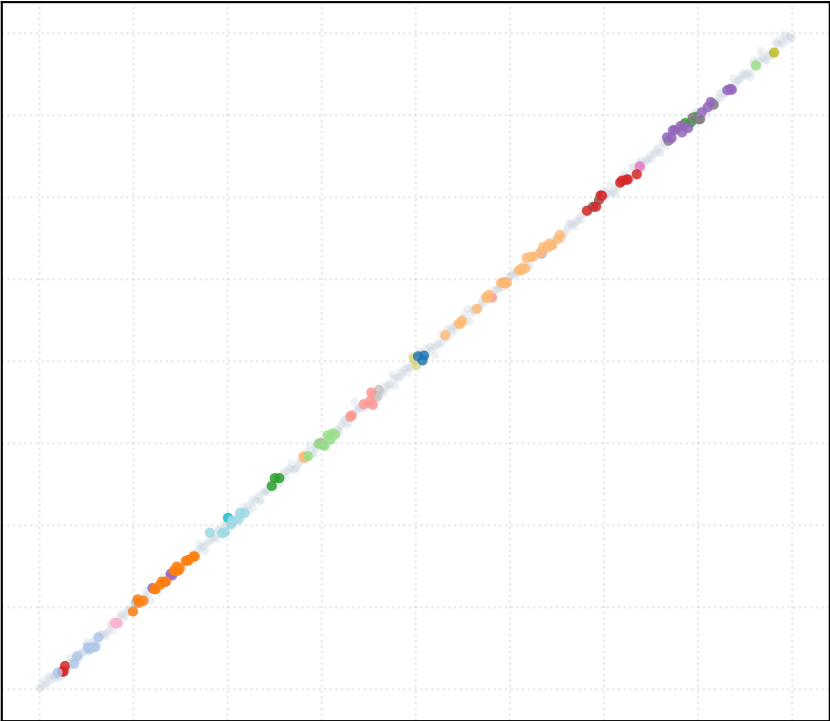


quant = 0.5 [1 Clusters]

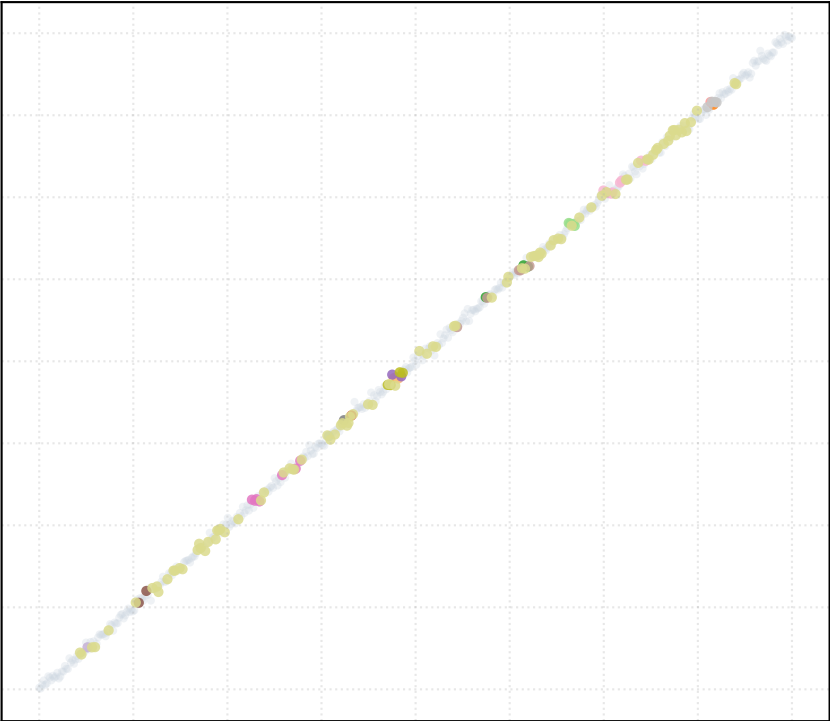


Dataset: linear_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

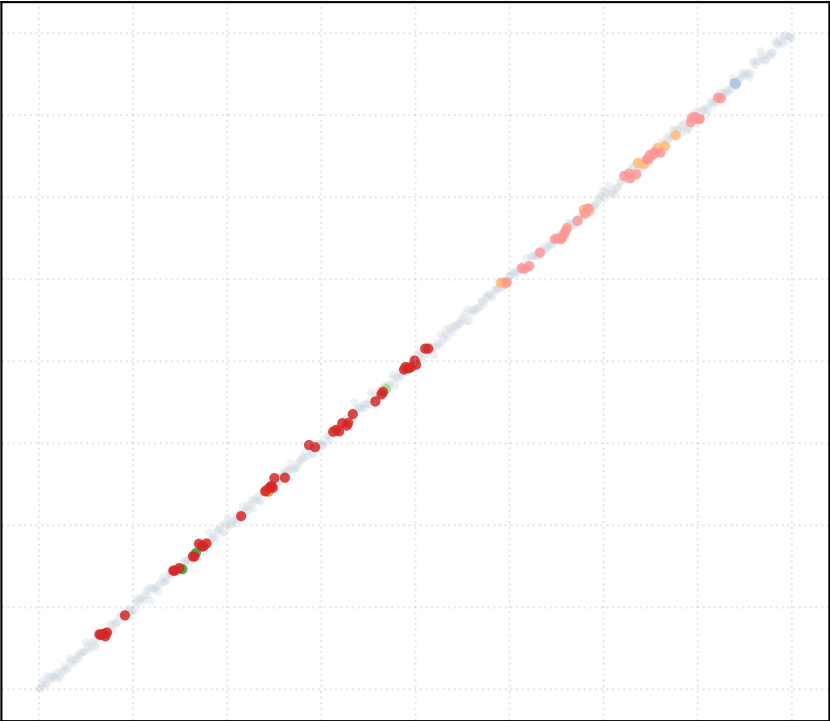
quant = 0.05 [28 Clusters]



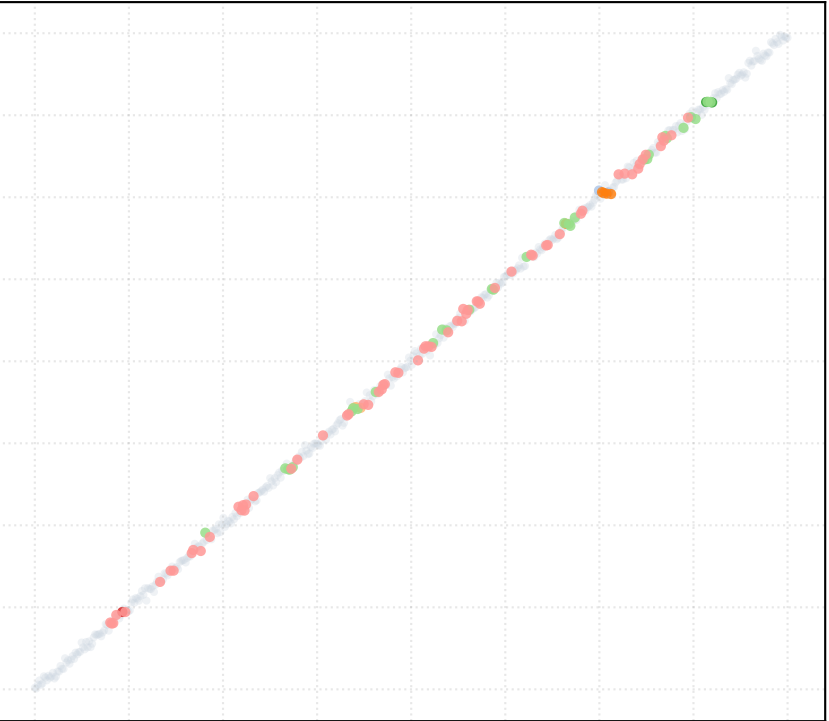
quant = 0.1 [17 Clusters]



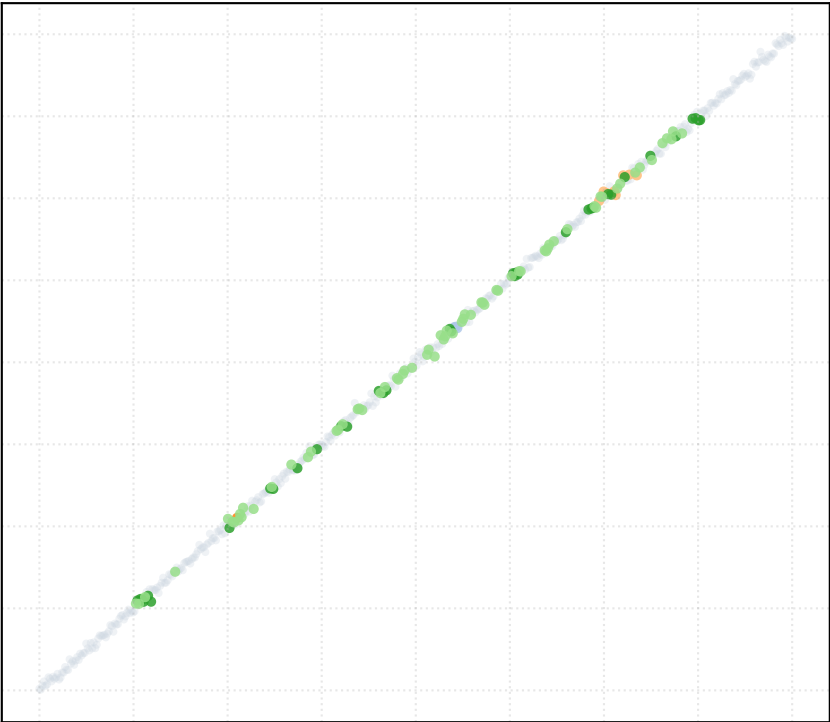
quant = 0.15 (Default) [7 Clusters]



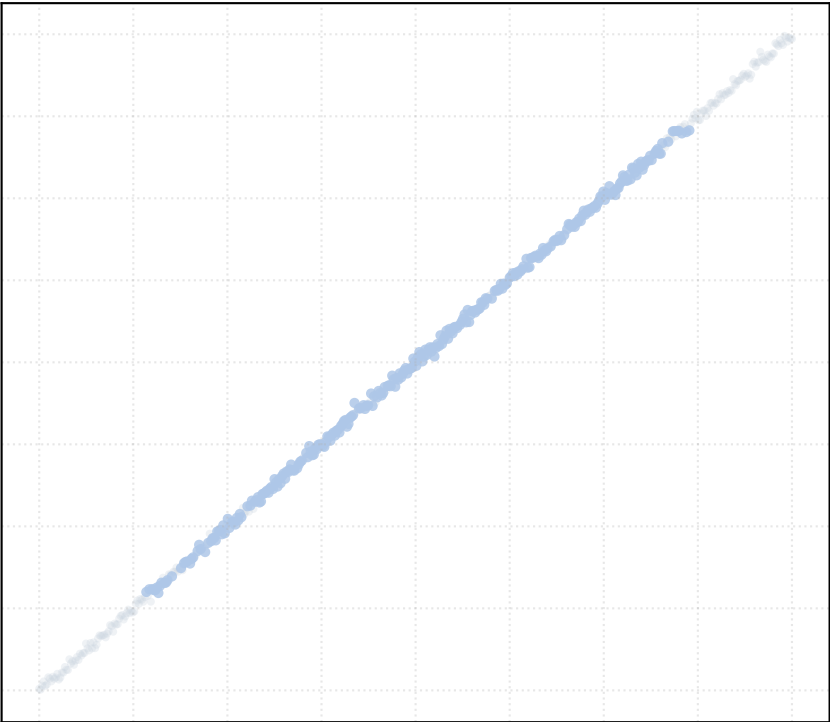
quant = 0.2 [7 Clusters]



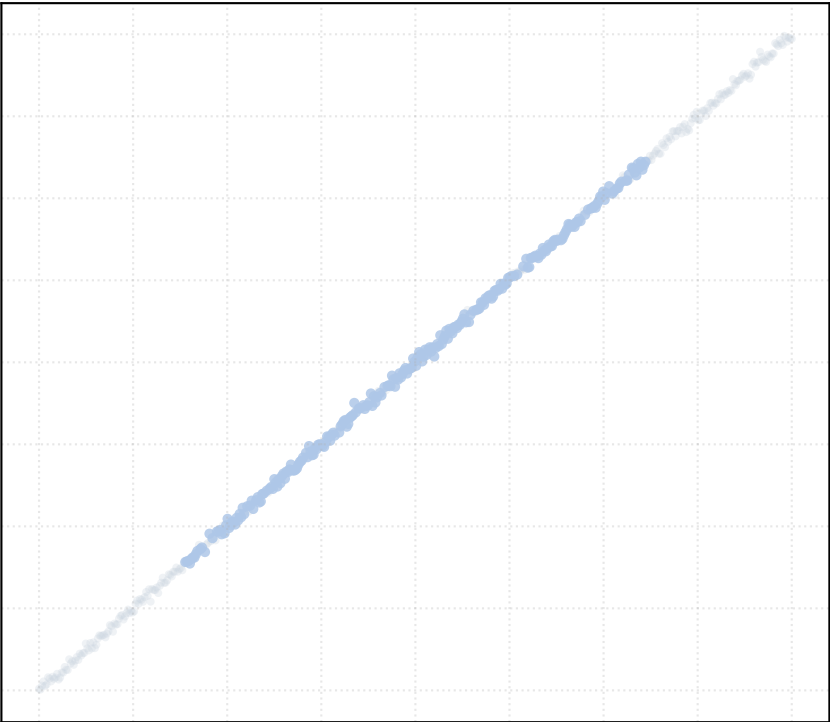
quant = 0.25 [5 Clusters]



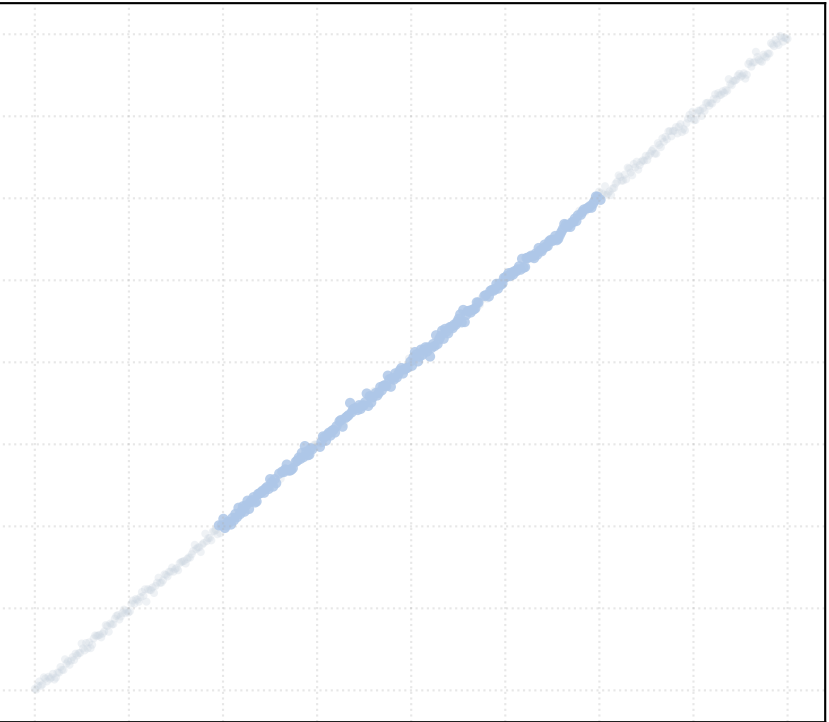
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

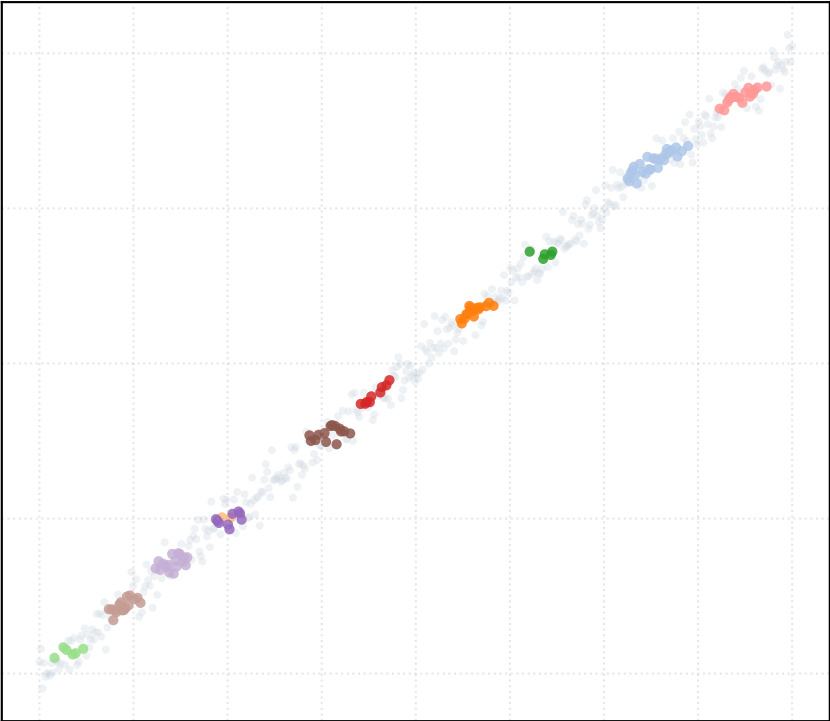


quant = 0.5 [1 Clusters]

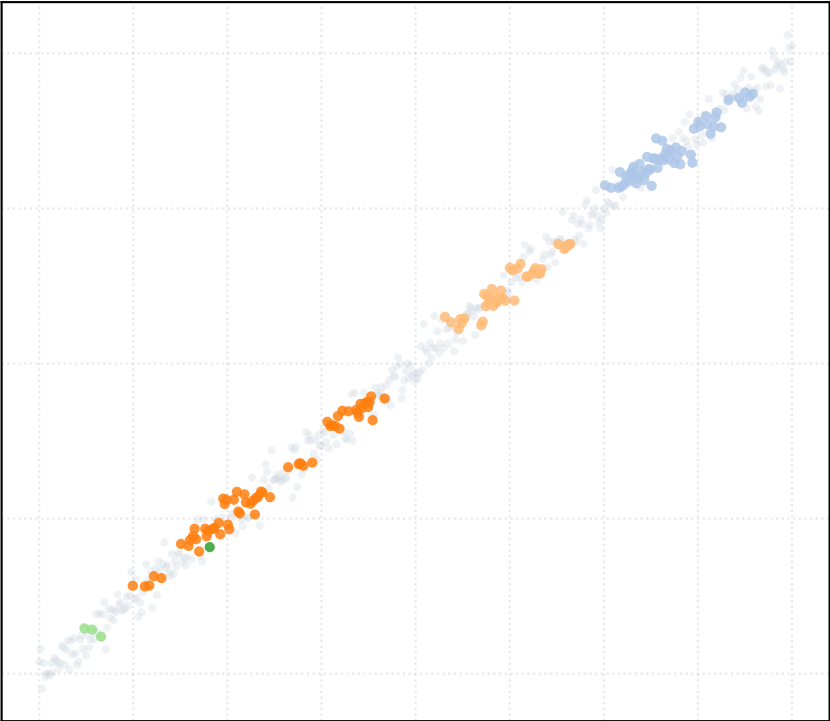


Dataset: linear_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

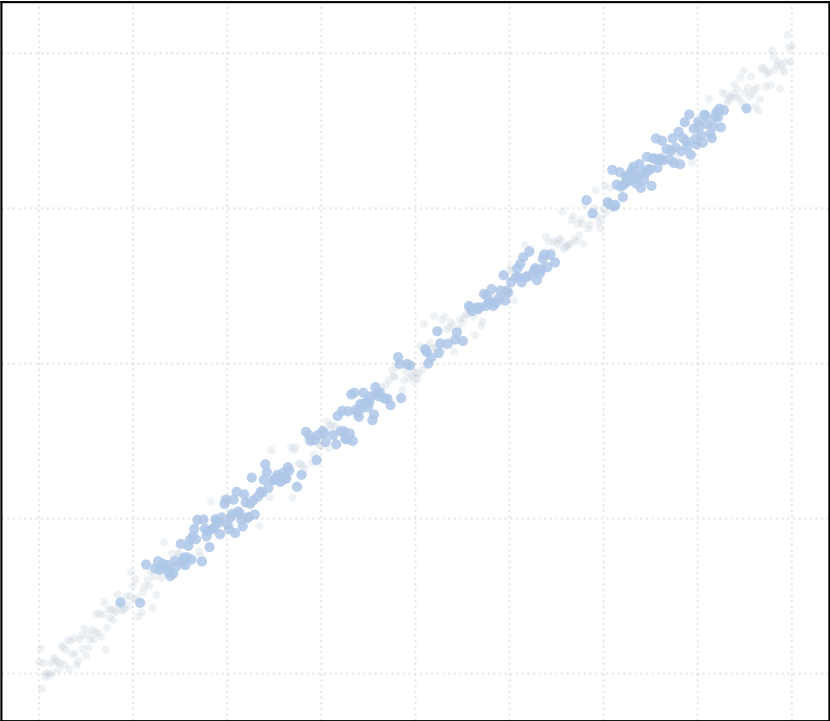
quant = 0.05 [11 Clusters]



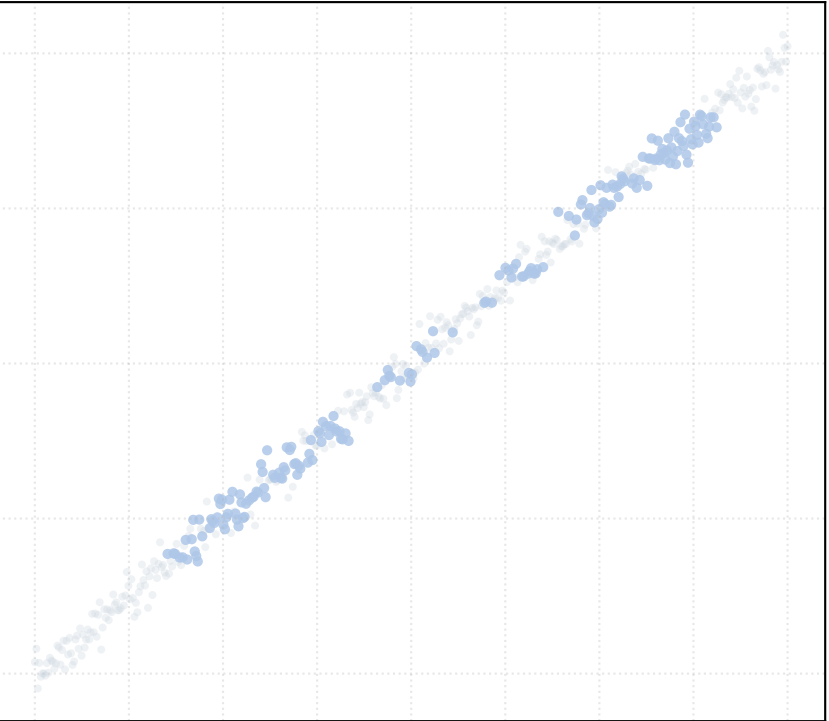
quant = 0.1 [5 Clusters]



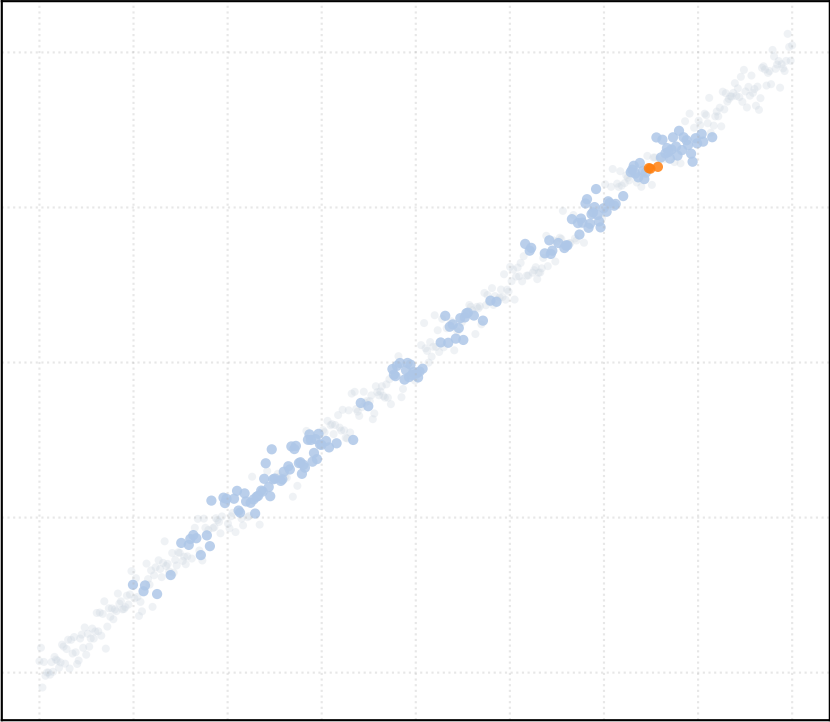
quant = 0.15 (Default) [1 Clusters]



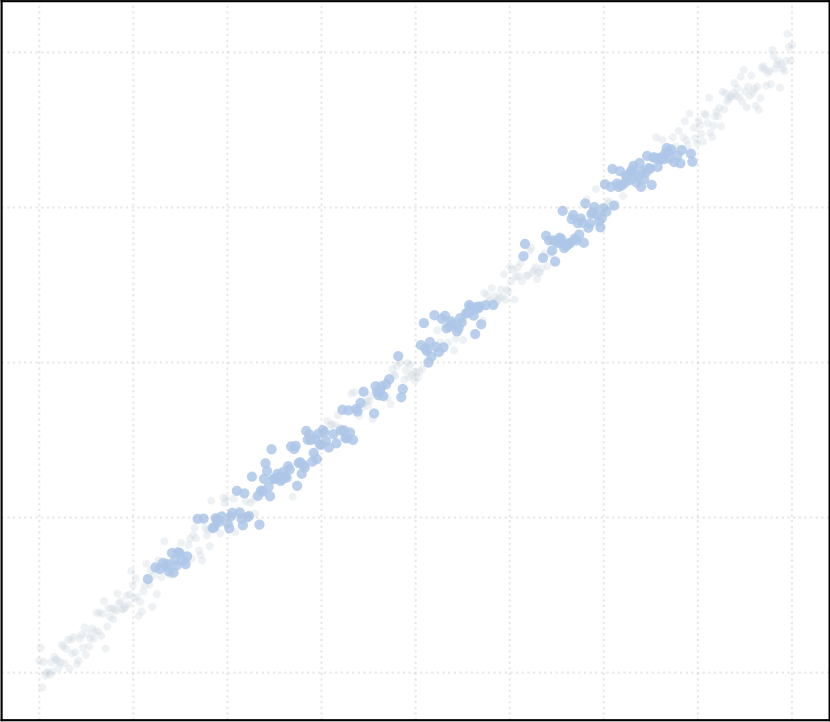
quant = 0.2 [1 Clusters]



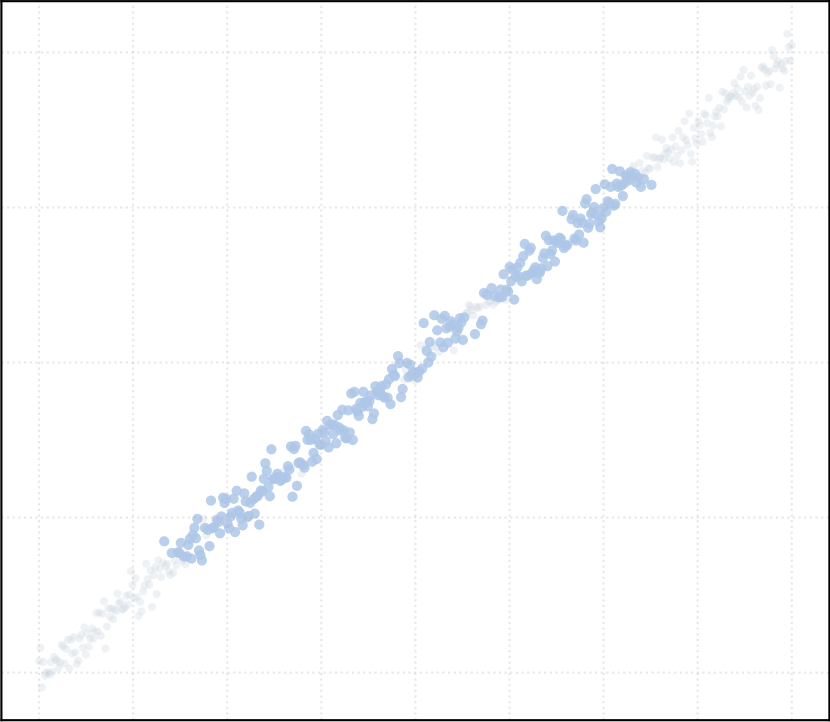
quant = 0.25 [2 Clusters]



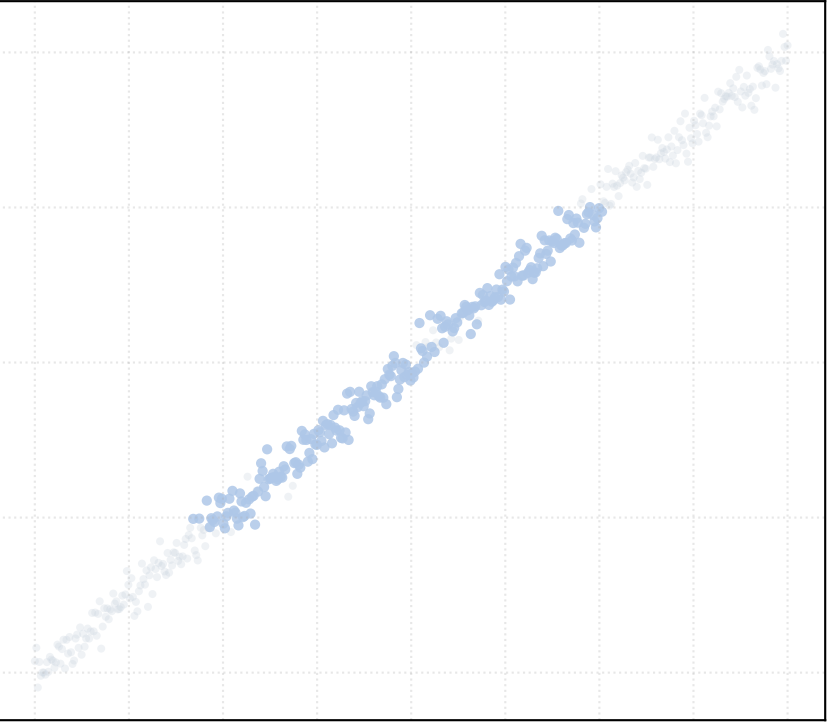
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

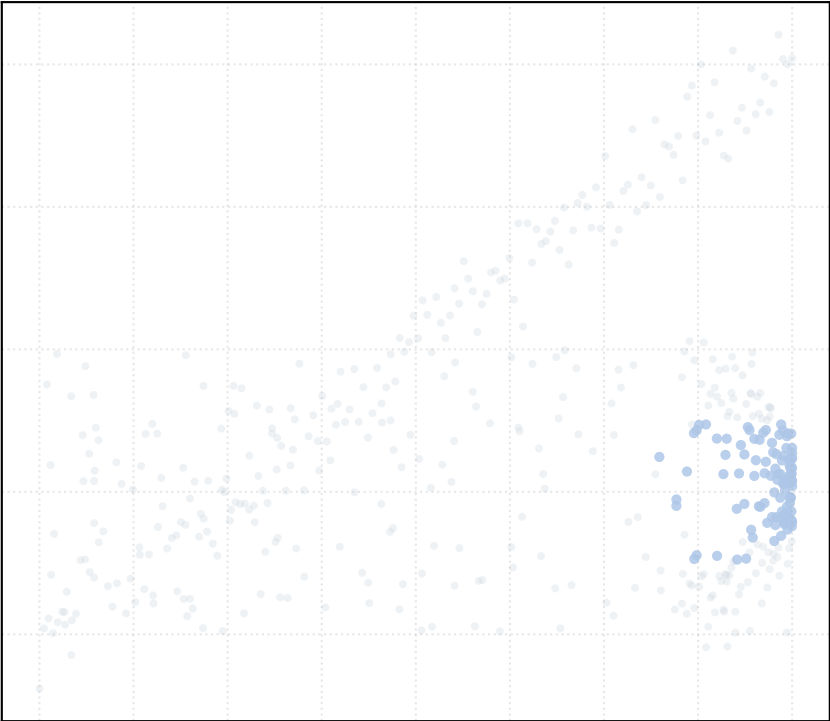


quant = 0.5 [1 Clusters]

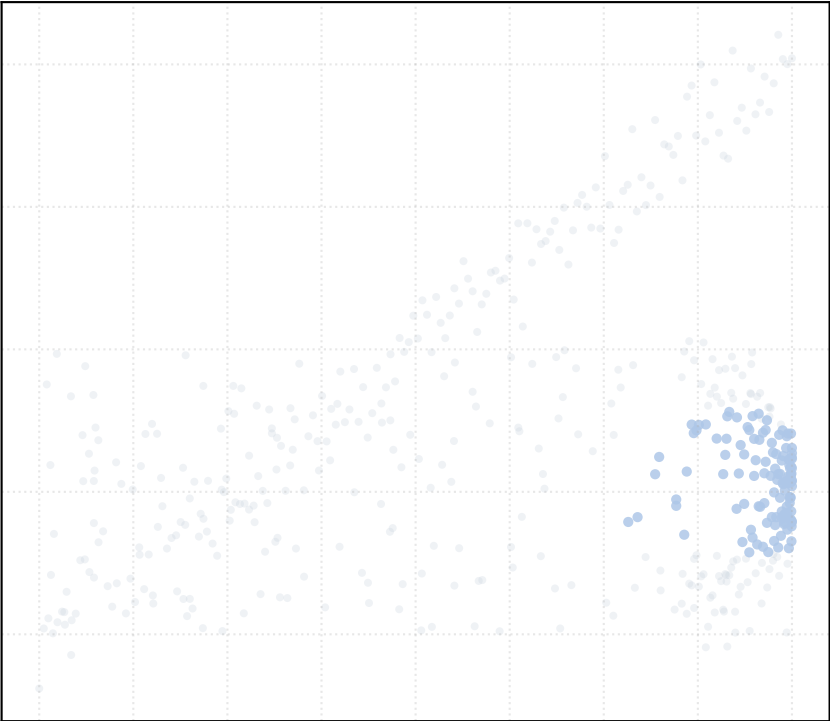


Dataset: mixed_generators_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

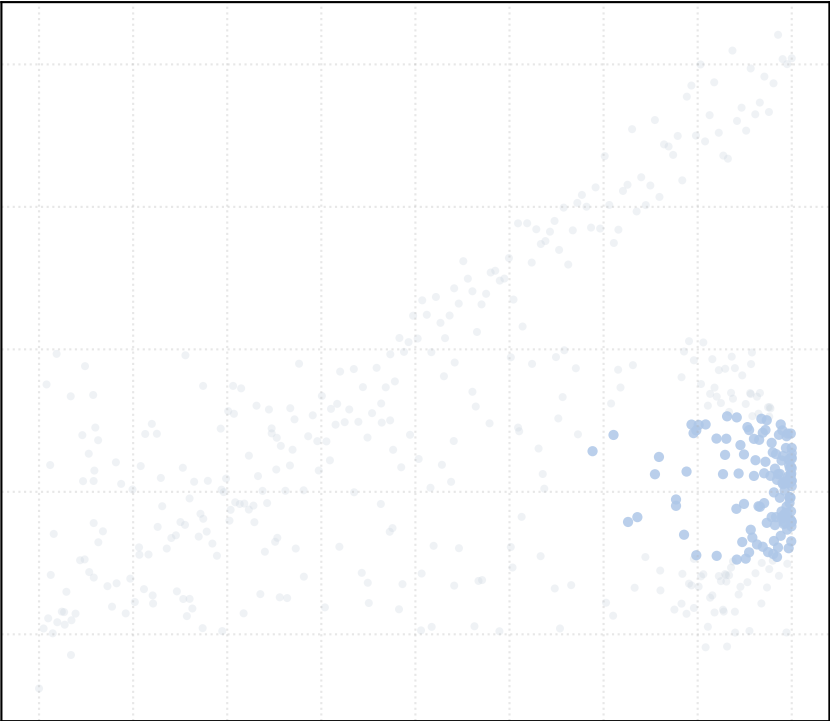
quant = 0.05 [1 Clusters]



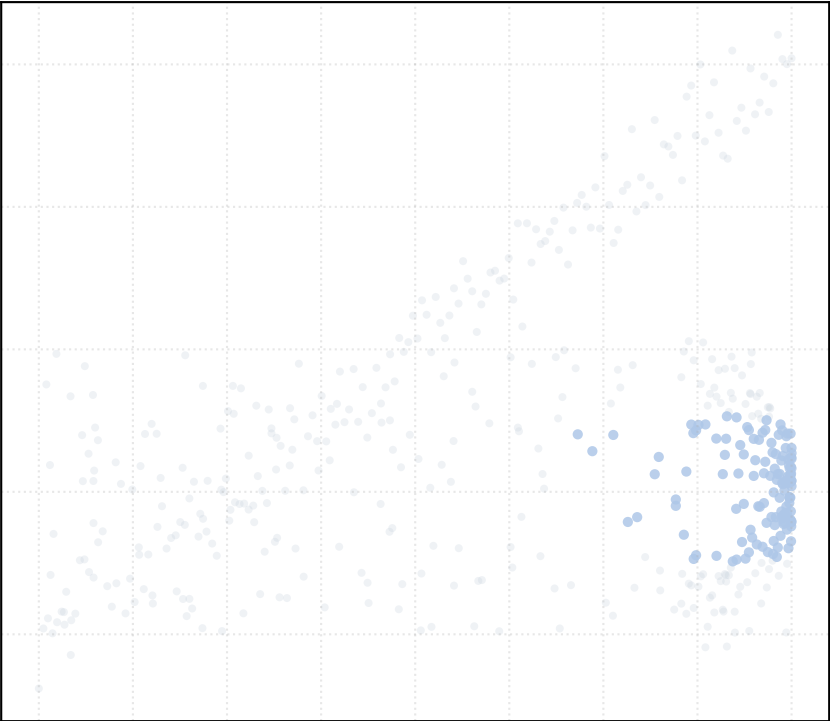
quant = 0.1 [1 Clusters]



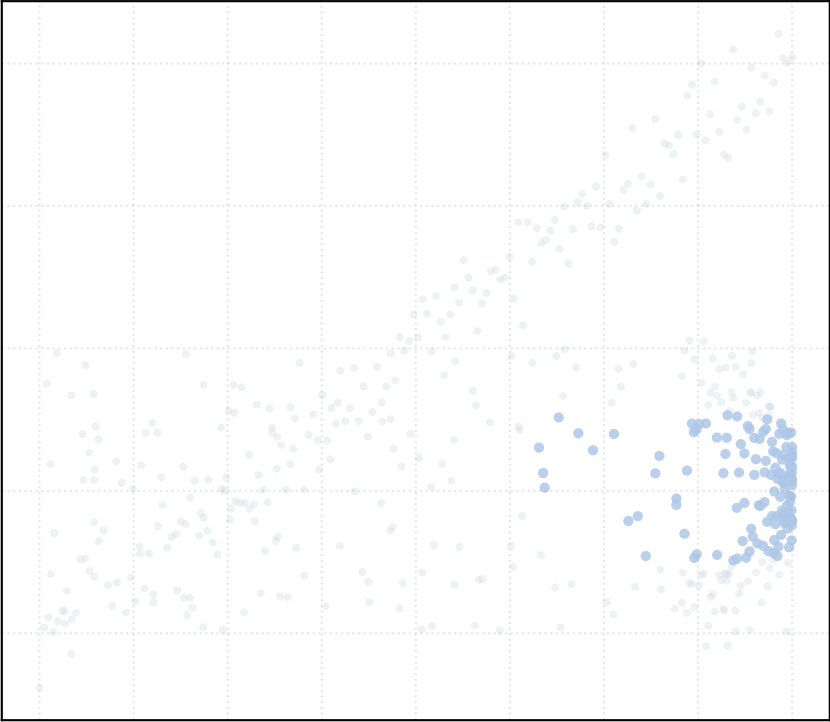
quant = 0.15 (Default) [1 Clusters]



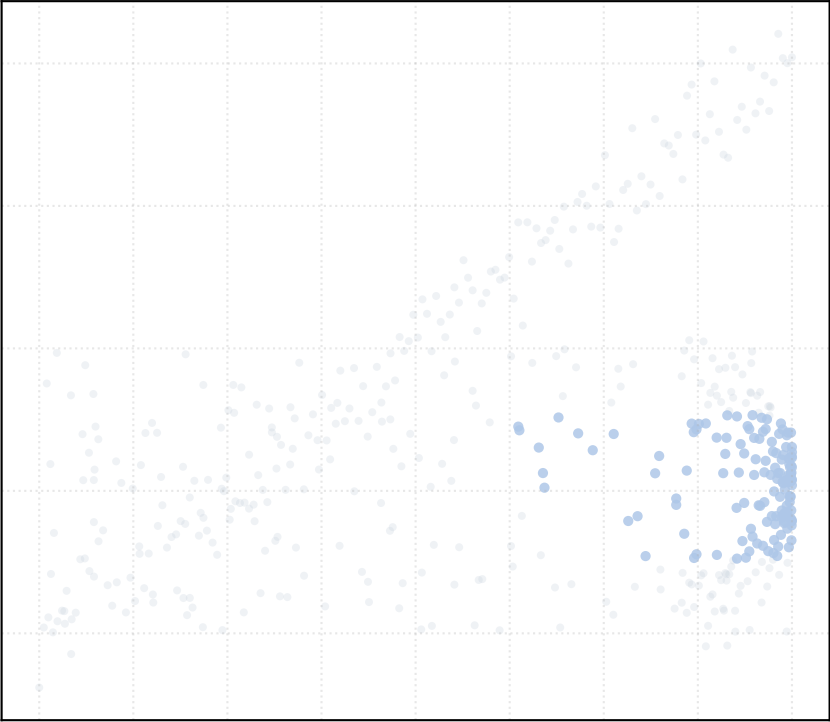
quant = 0.2 [1 Clusters]



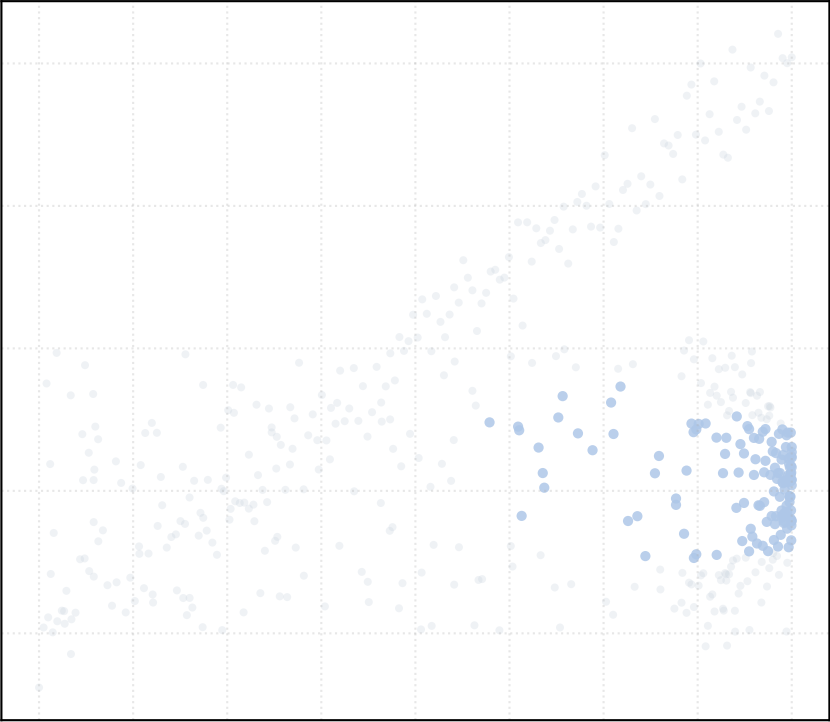
quant = 0.25 [1 Clusters]



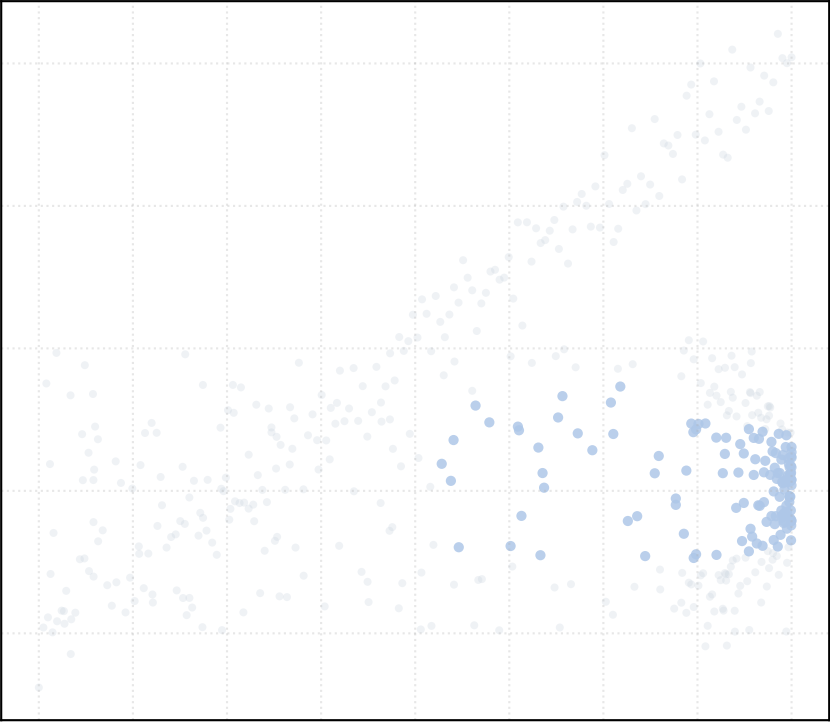
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

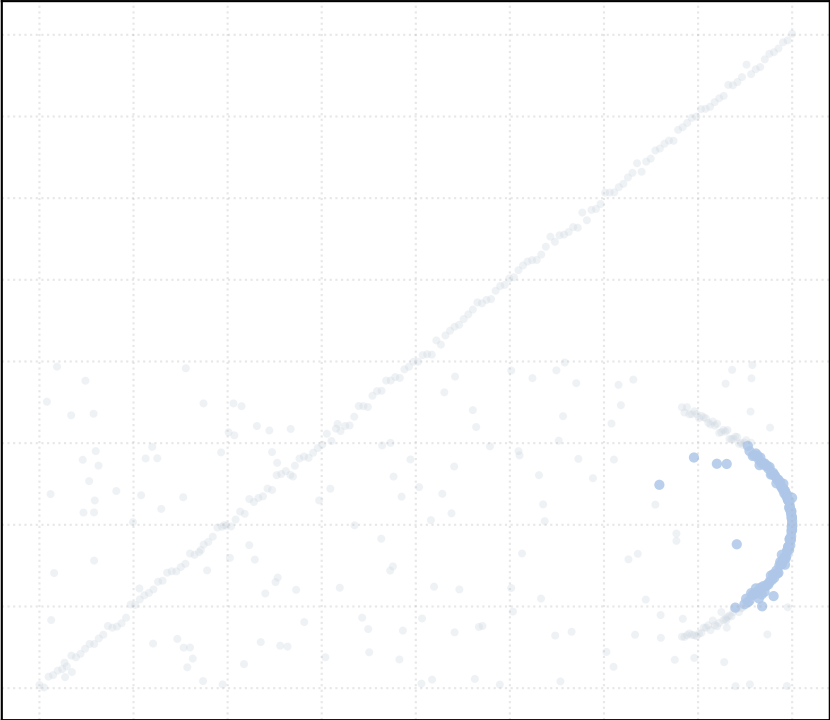


quant = 0.5 [1 Clusters]

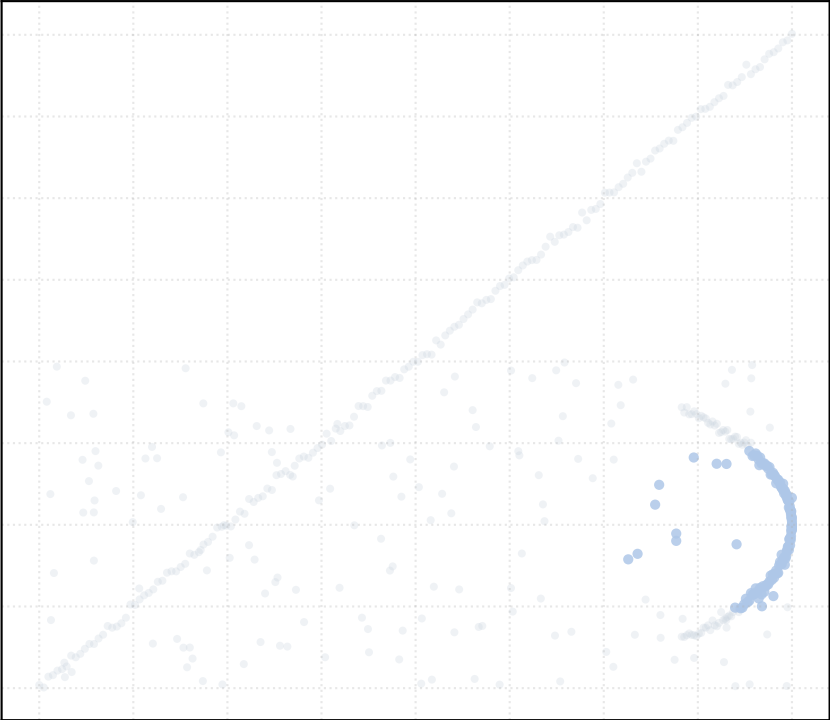


Dataset: mixed_generators_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

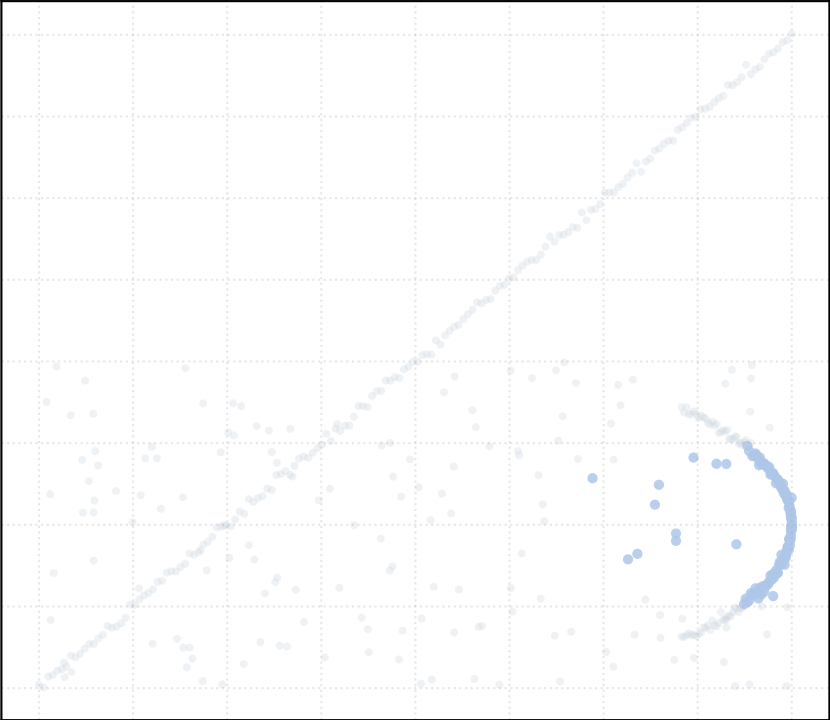
quant = 0.05 [1 Clusters]



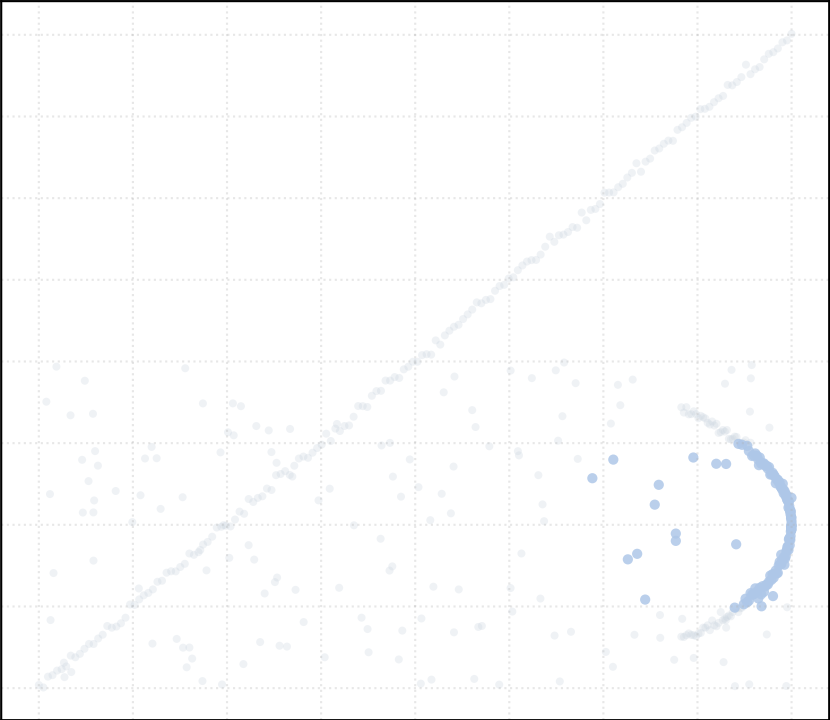
quant = 0.1 [1 Clusters]



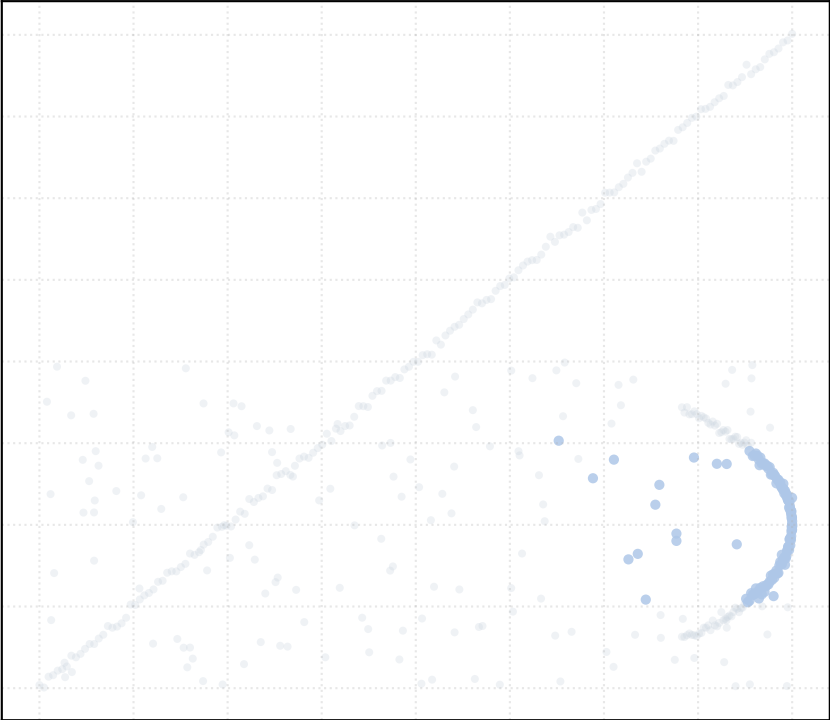
quant = 0.15 (Default) [1 Clusters]



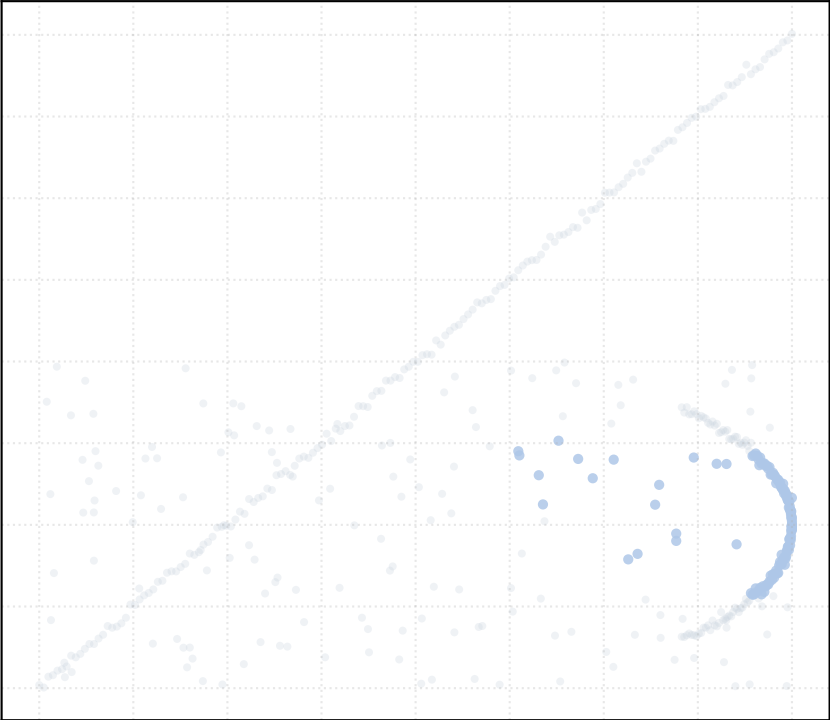
quant = 0.2 [1 Clusters]



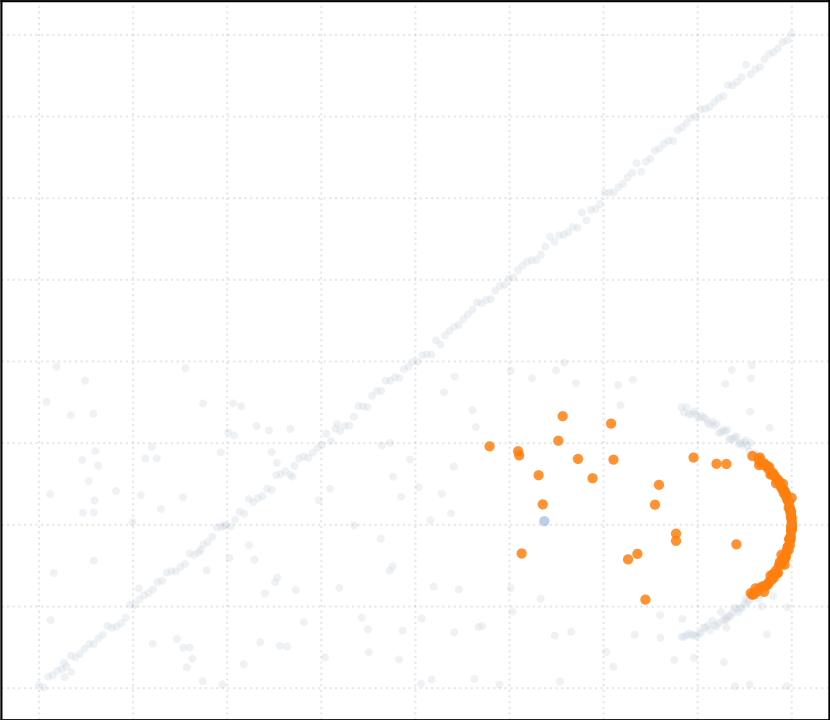
quant = 0.25 [1 Clusters]



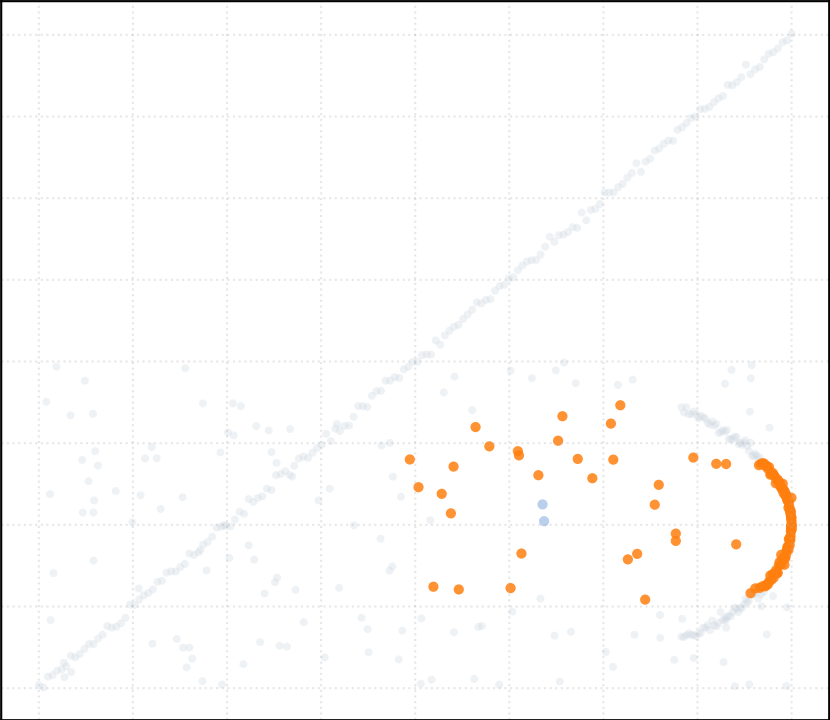
quant = 0.3 [1 Clusters]



quant = 0.4 [2 Clusters]

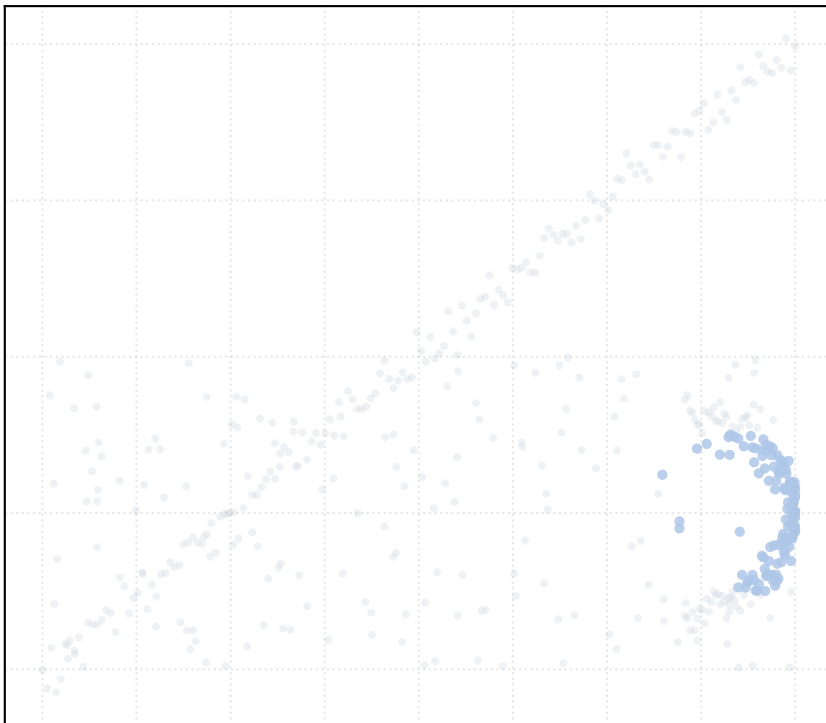


quant = 0.5 [2 Clusters]

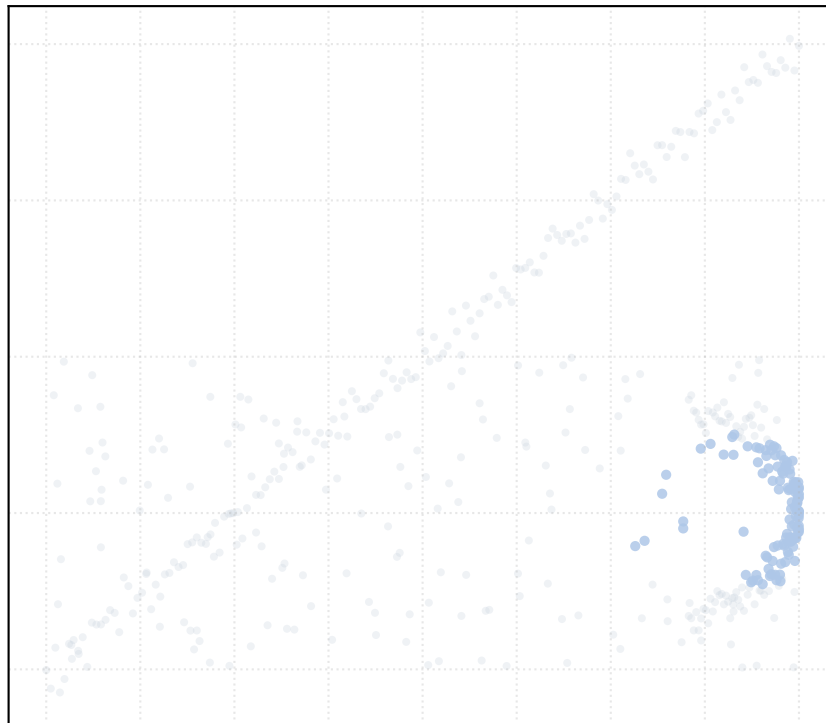


Dataset: mixed_generators_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

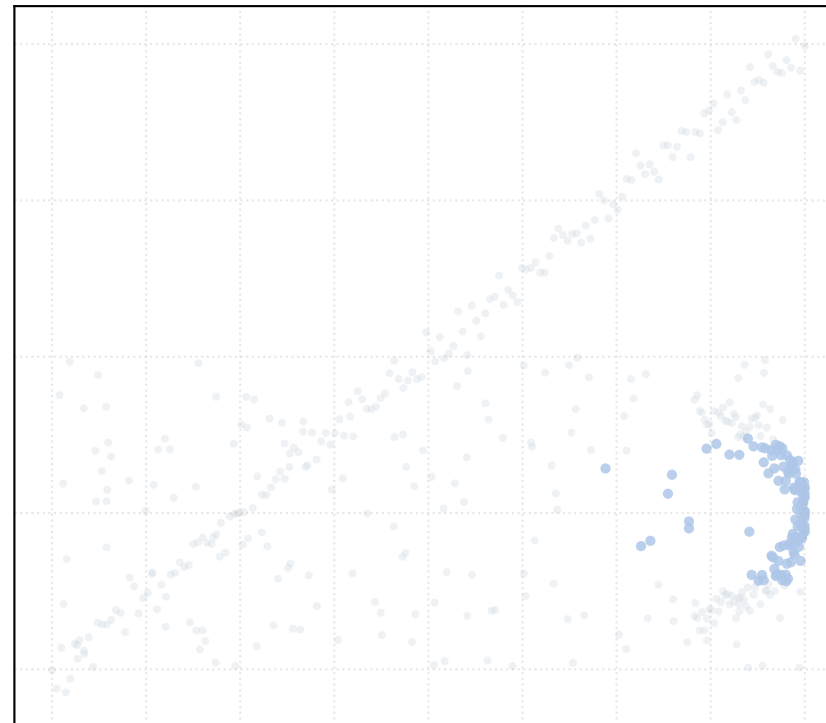
quant = 0.05 [1 Clusters]



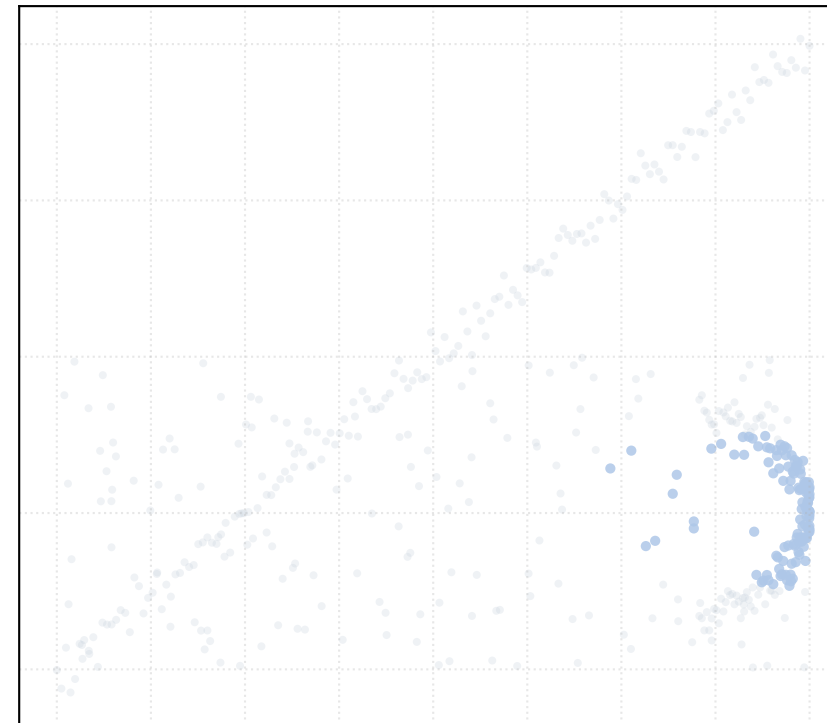
quant = 0.1 [1 Clusters]



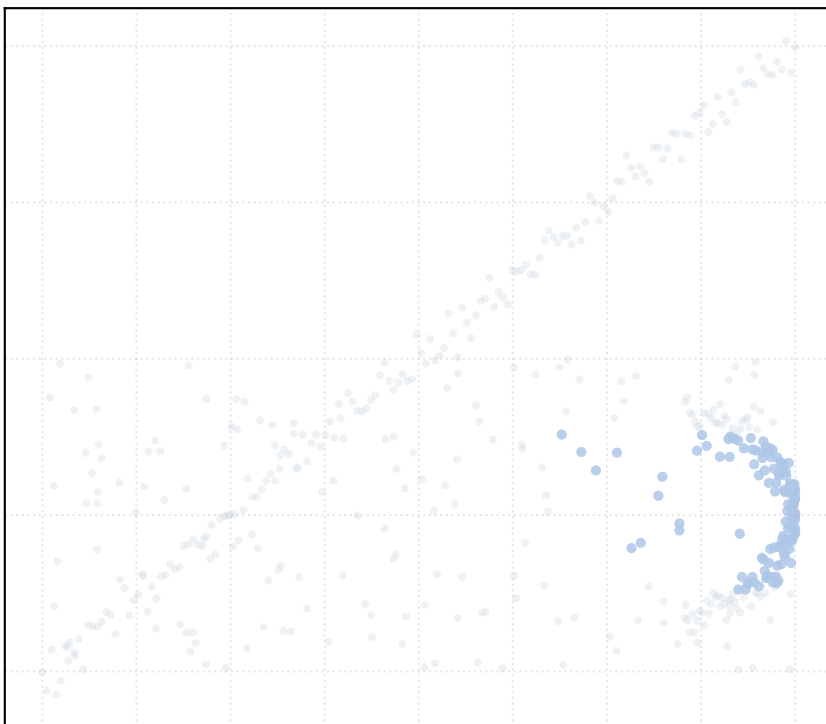
quant = 0.15 (Default) [1 Clusters]



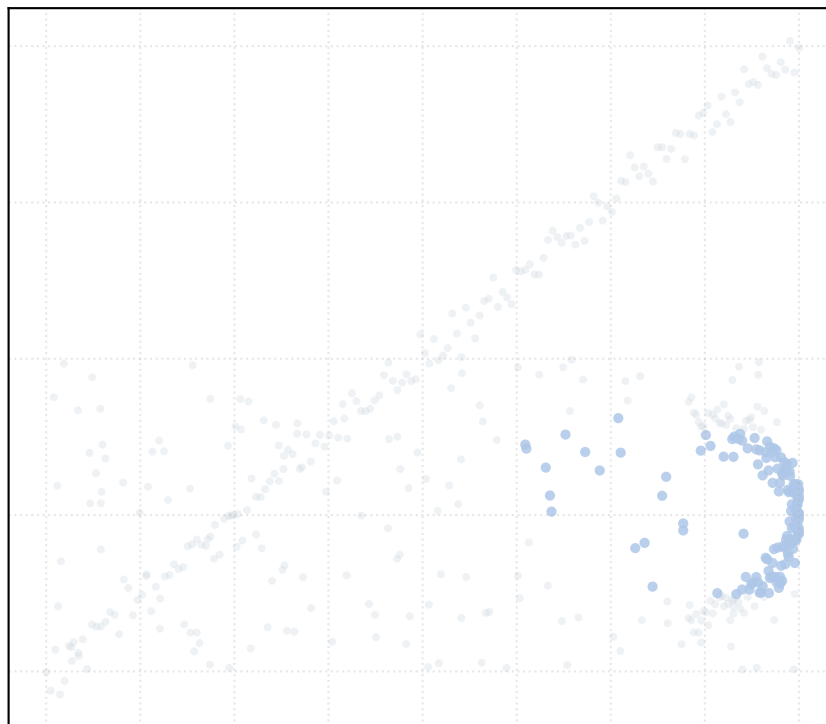
quant = 0.2 [1 Clusters]



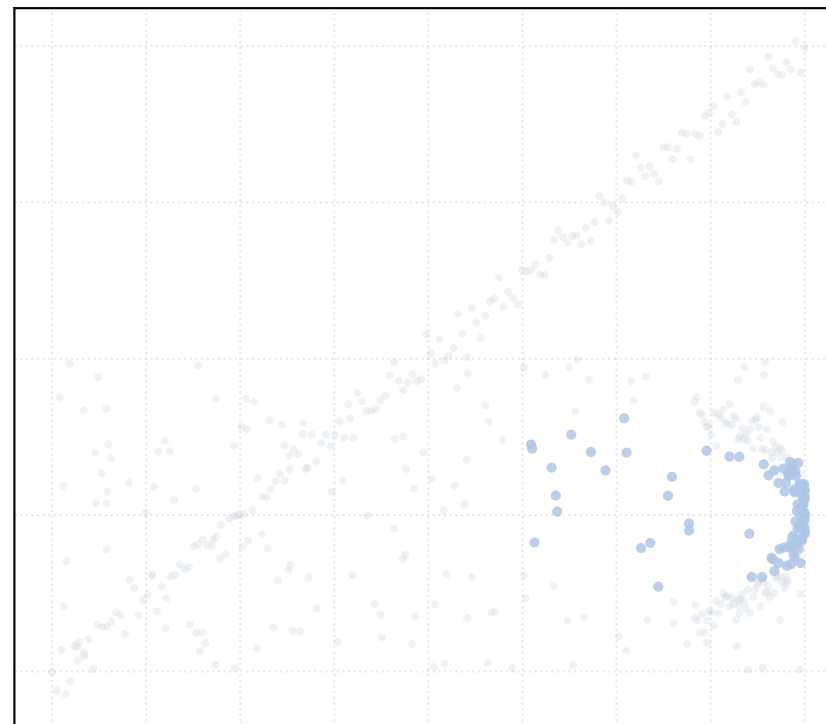
quant = 0.25 [1 Clusters]



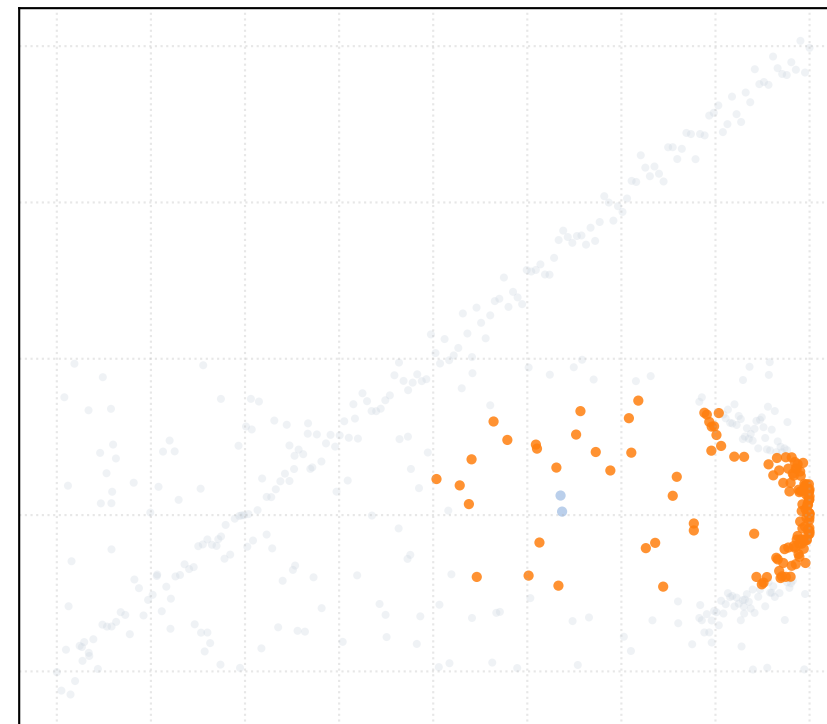
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

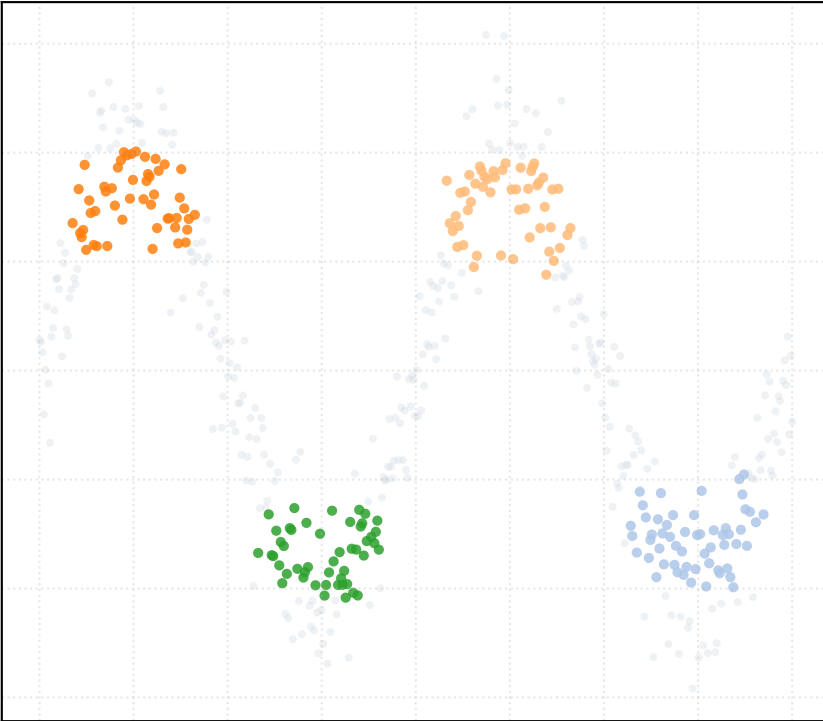


quant = 0.5 [2 Clusters]

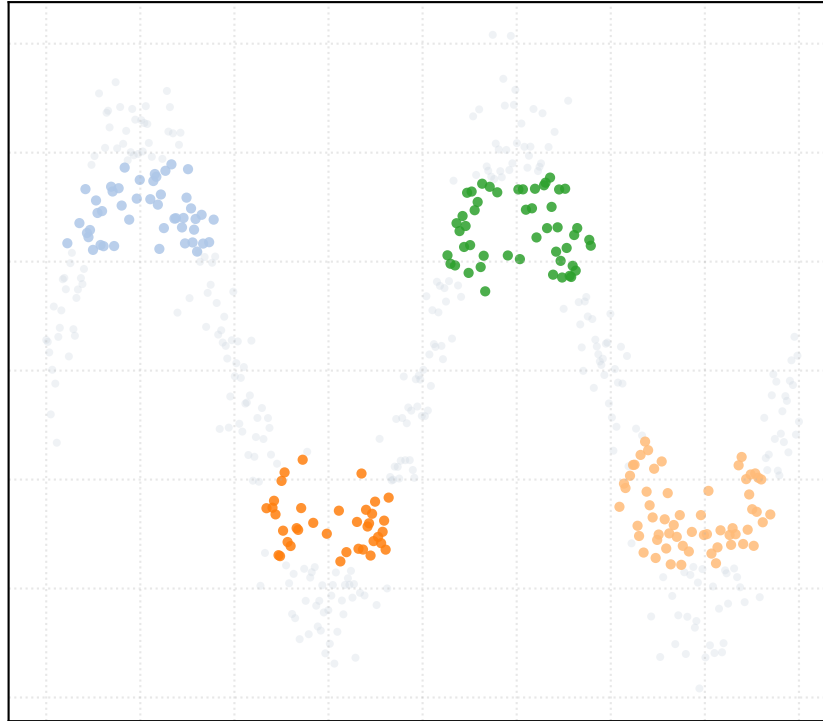


Dataset: s_curve_noise_high | Quantile Sweep (mean_to_other_vecs_dist_quant)

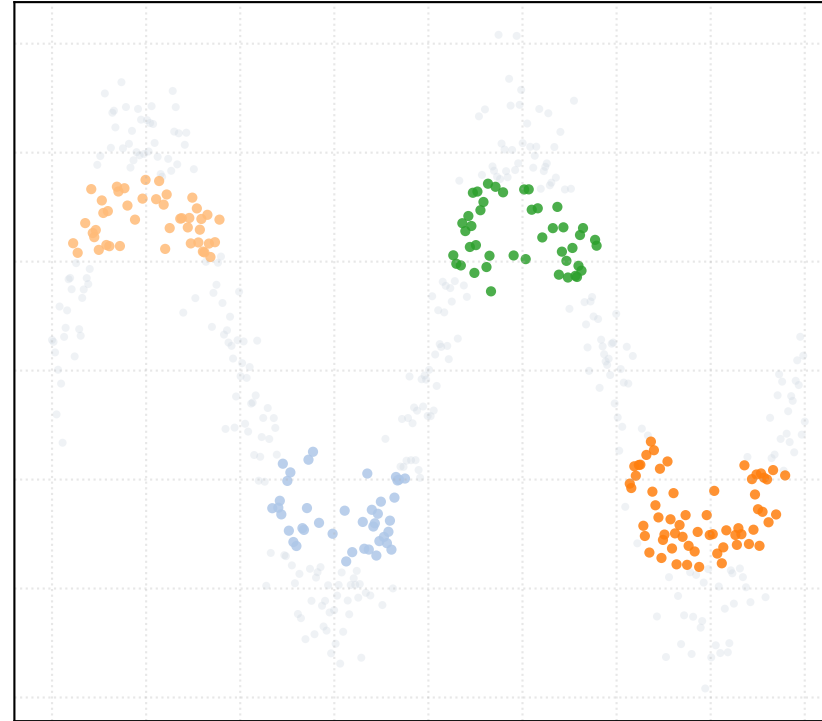
quant = 0.05 [4 Clusters]



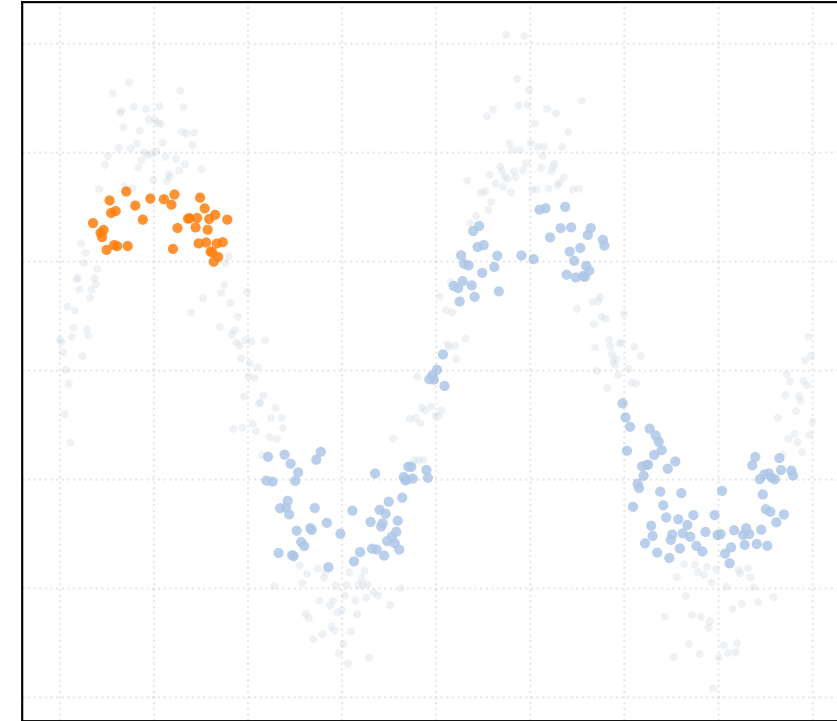
quant = 0.1 [4 Clusters]



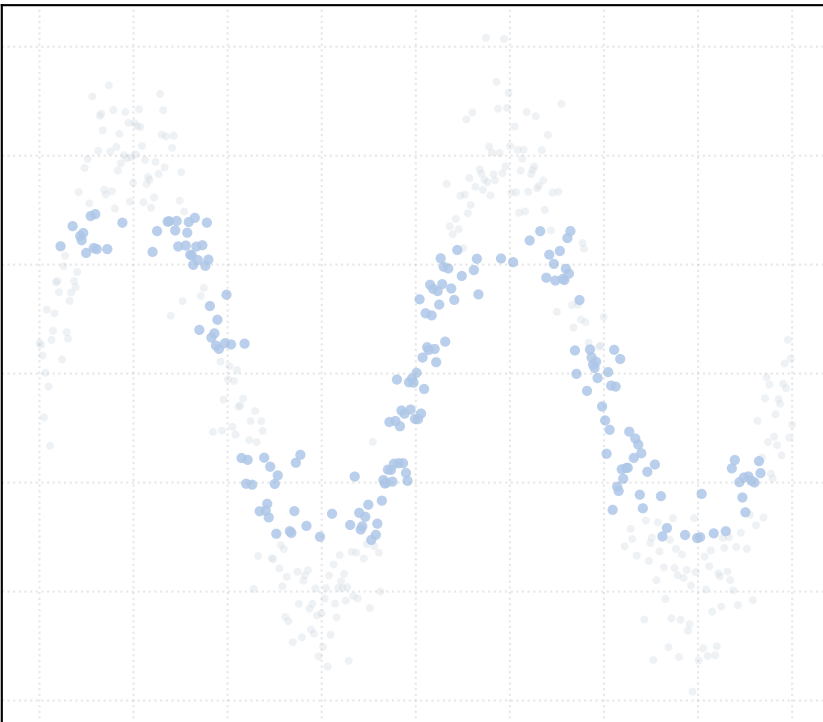
quant = 0.15 (Default) [4 Clusters]



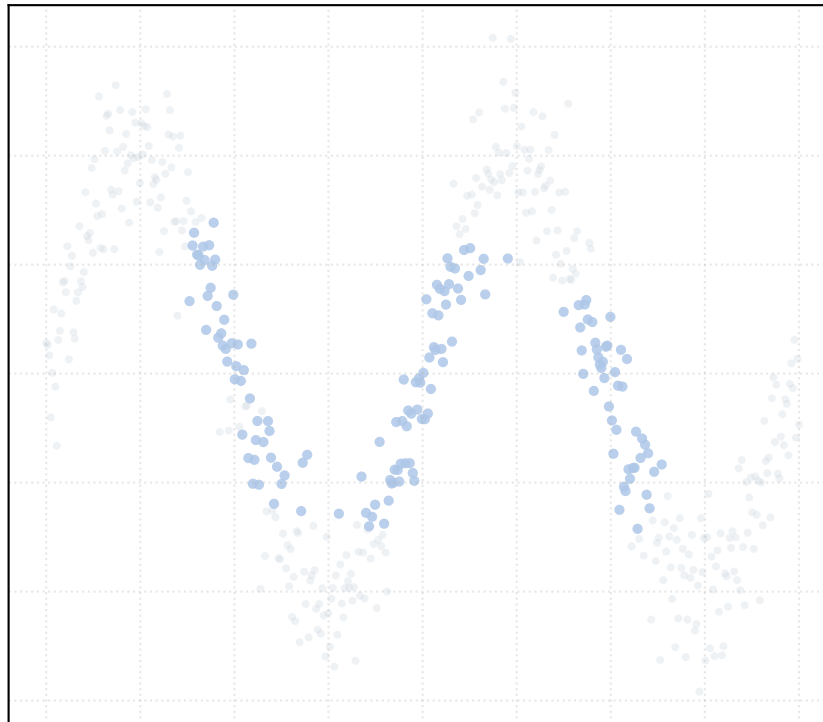
quant = 0.2 [2 Clusters]



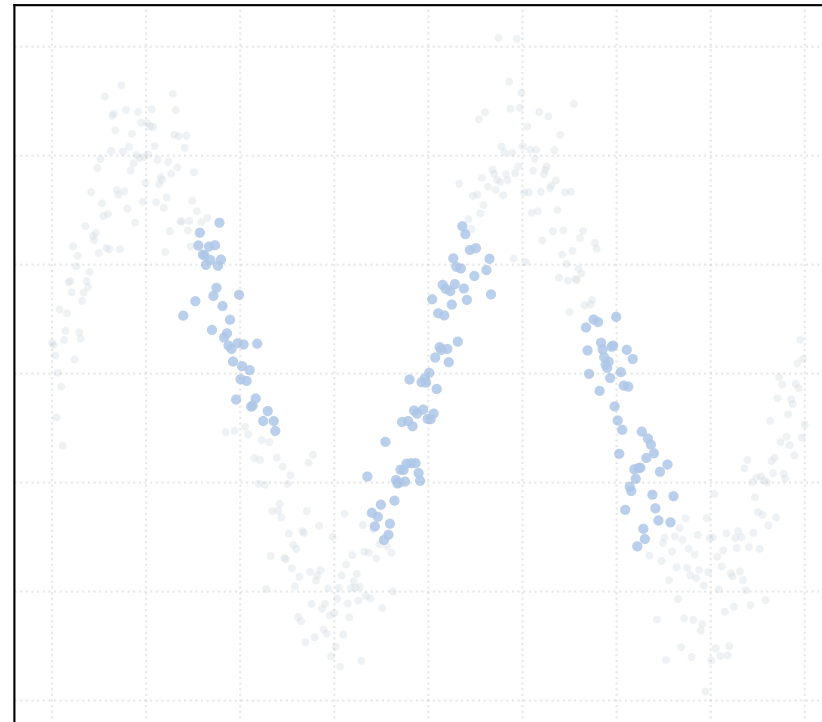
quant = 0.25 [1 Clusters]



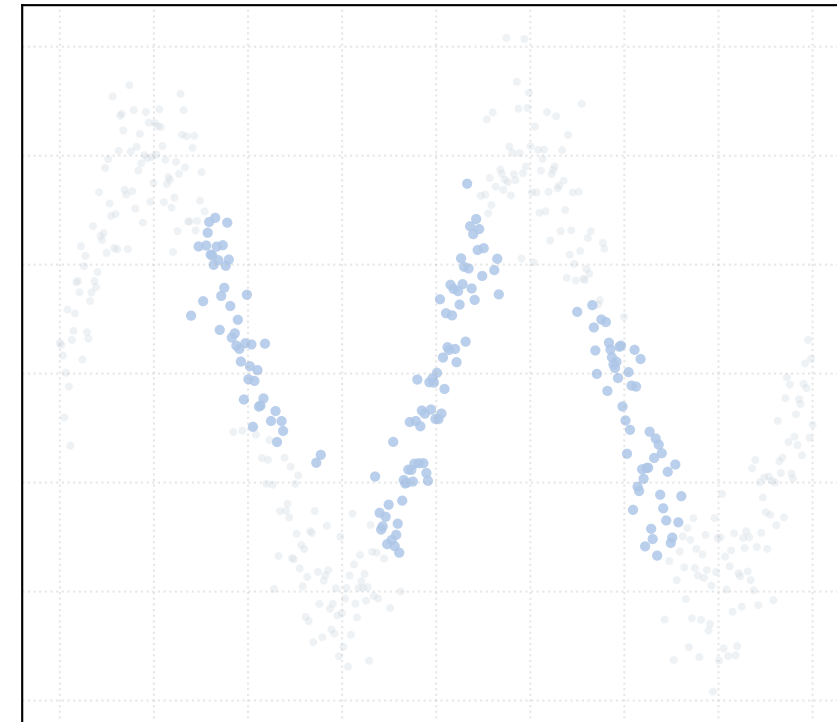
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

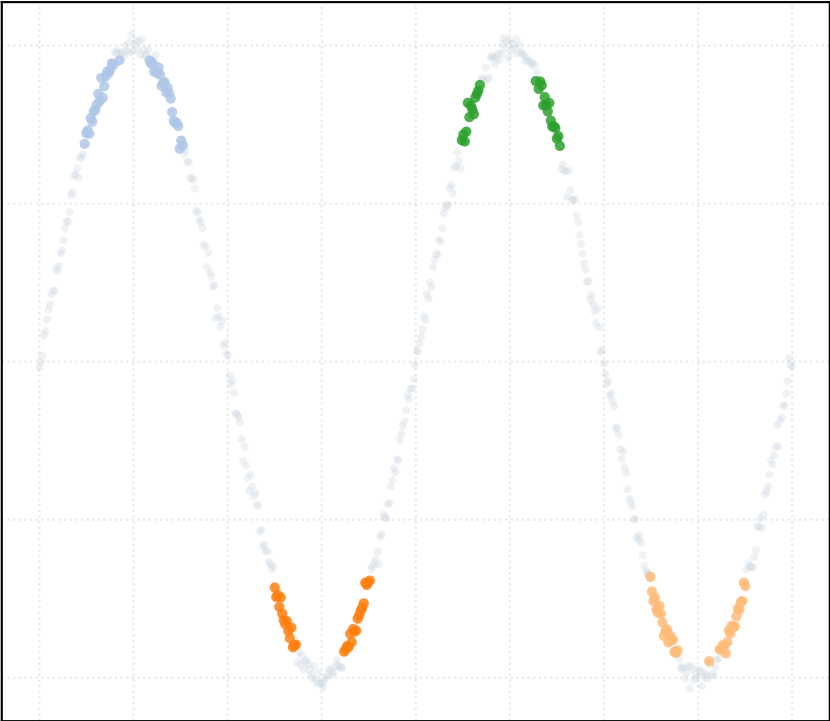


quant = 0.5 [1 Clusters]

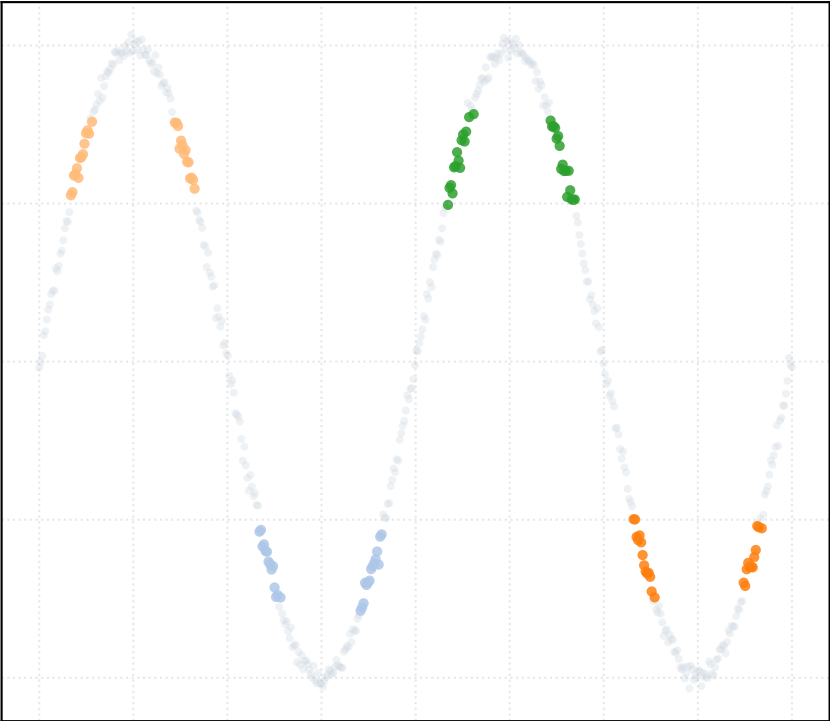


Dataset: s_curve_noise_low | Quantile Sweep (mean_to_other_vecs_dist_quant)

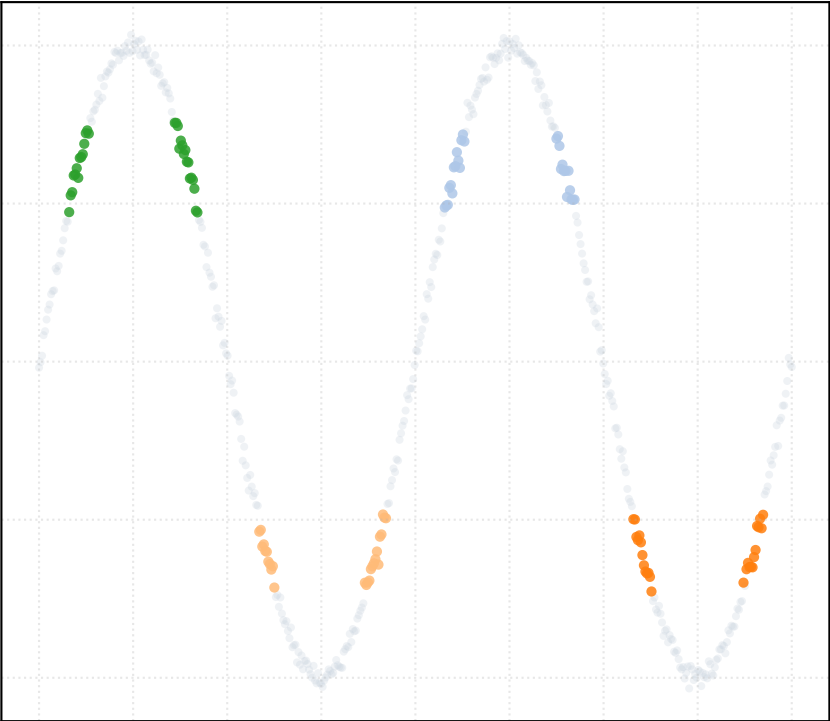
quant = 0.05 [4 Clusters]



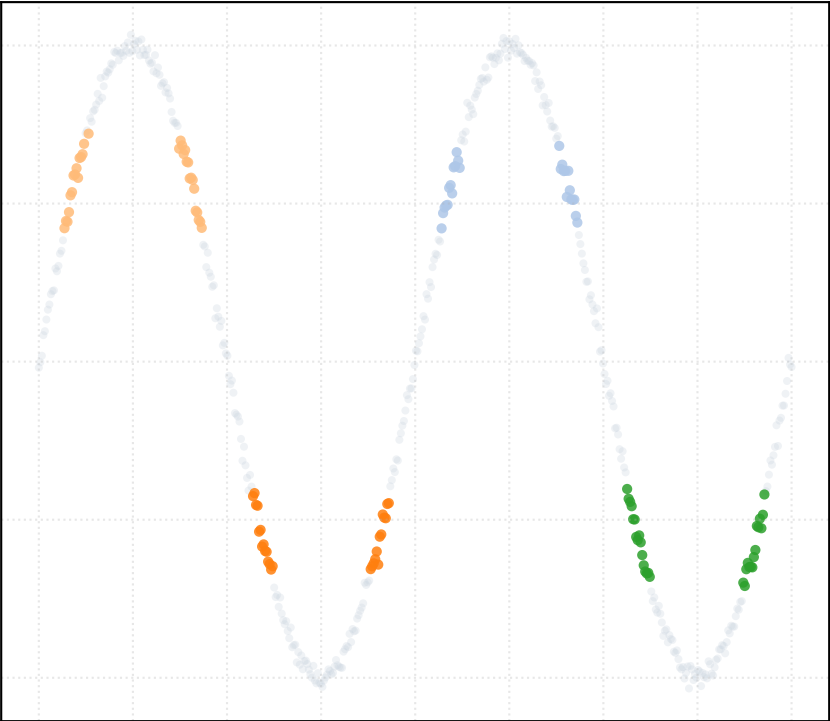
quant = 0.1 [4 Clusters]



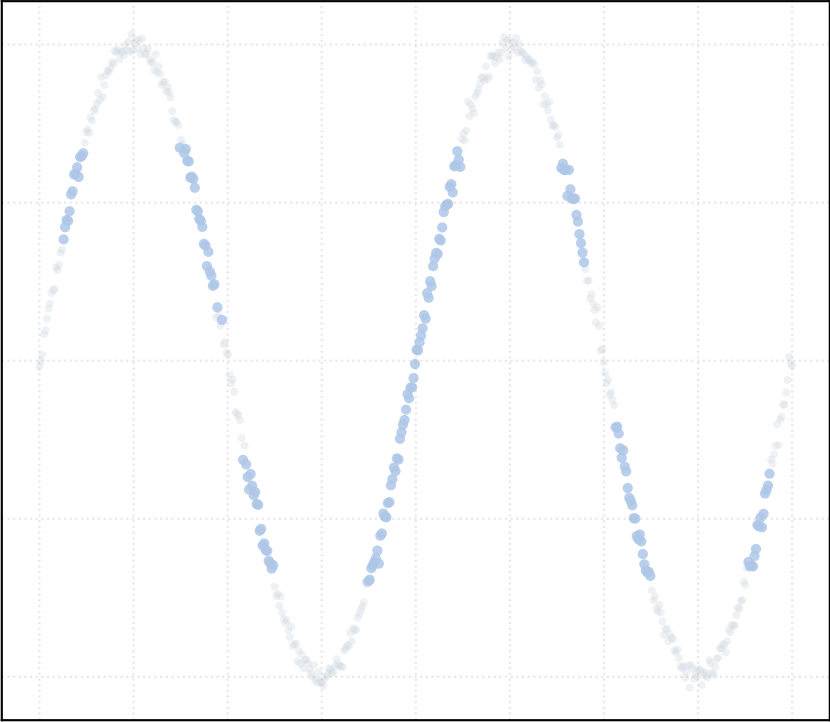
quant = 0.15 (Default) [4 Clusters]



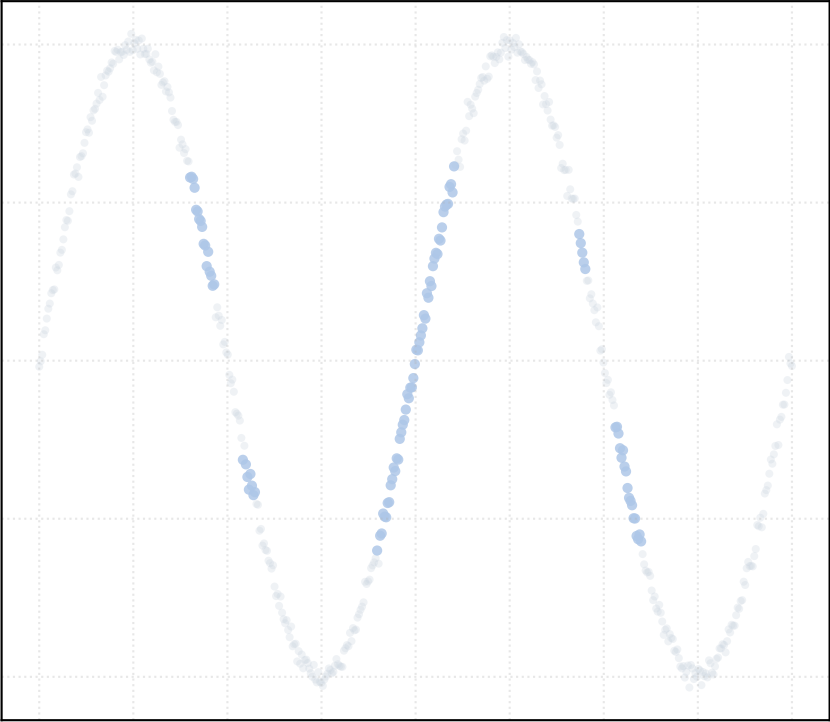
quant = 0.2 [4 Clusters]



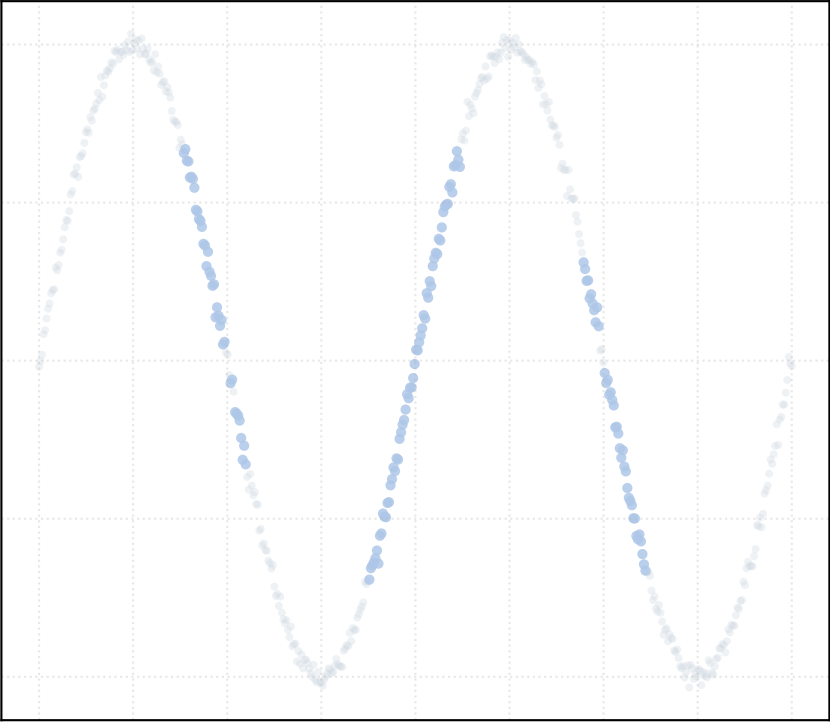
quant = 0.25 [1 Clusters]



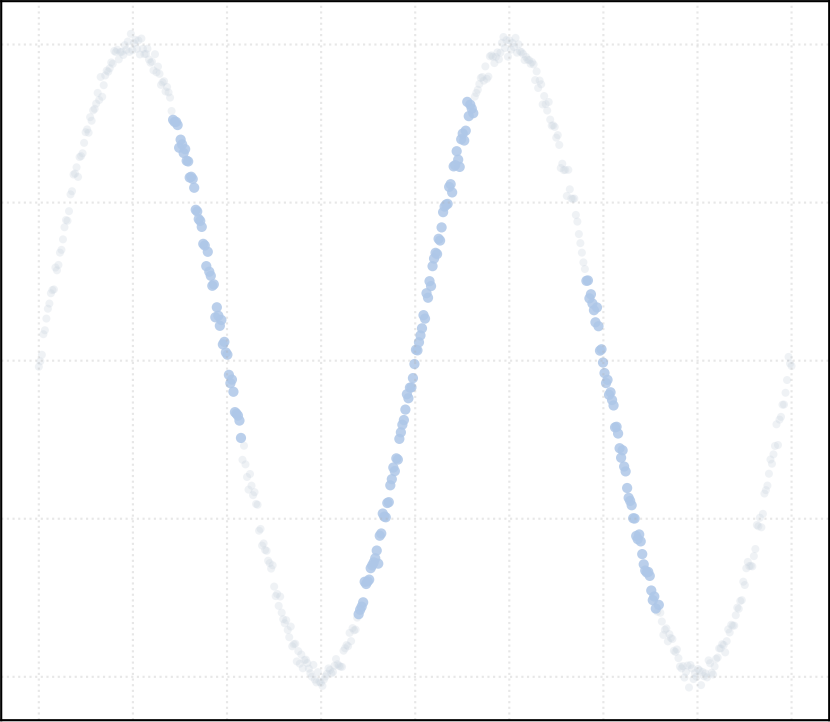
quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]

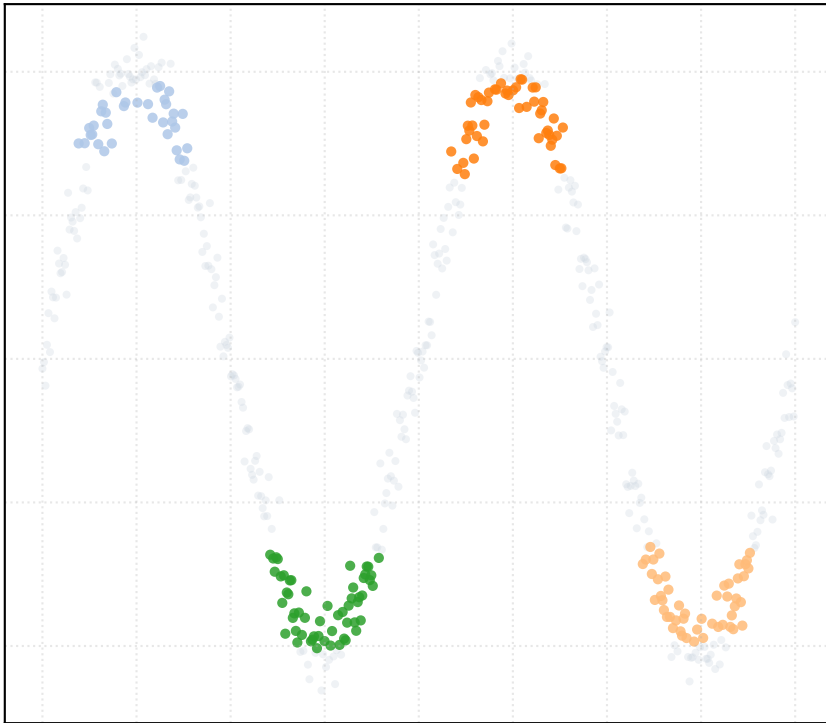


quant = 0.5 [1 Clusters]

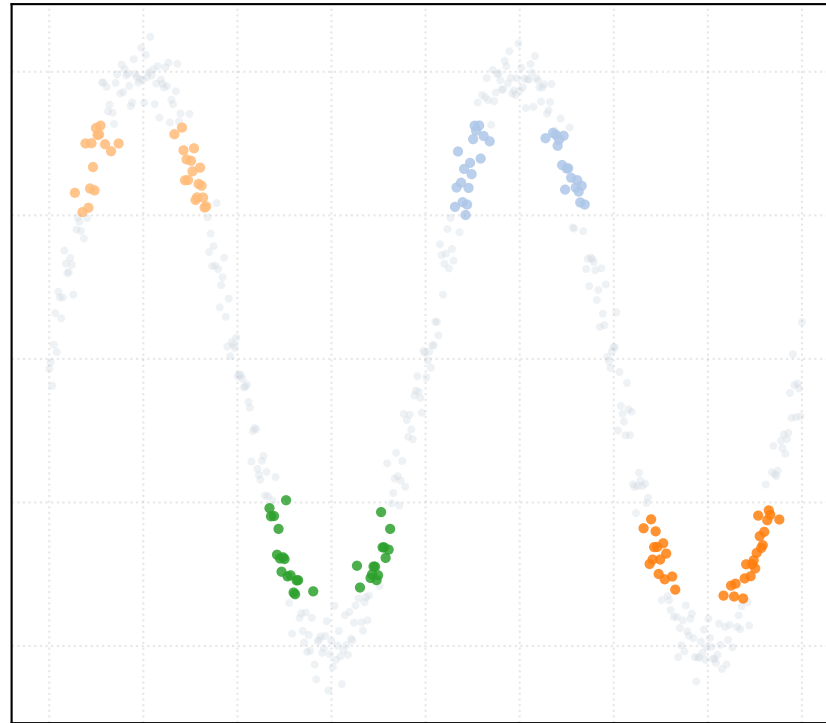


Dataset: s_curve_noise_medium | Quantile Sweep (mean_to_other_vecs_dist_quant)

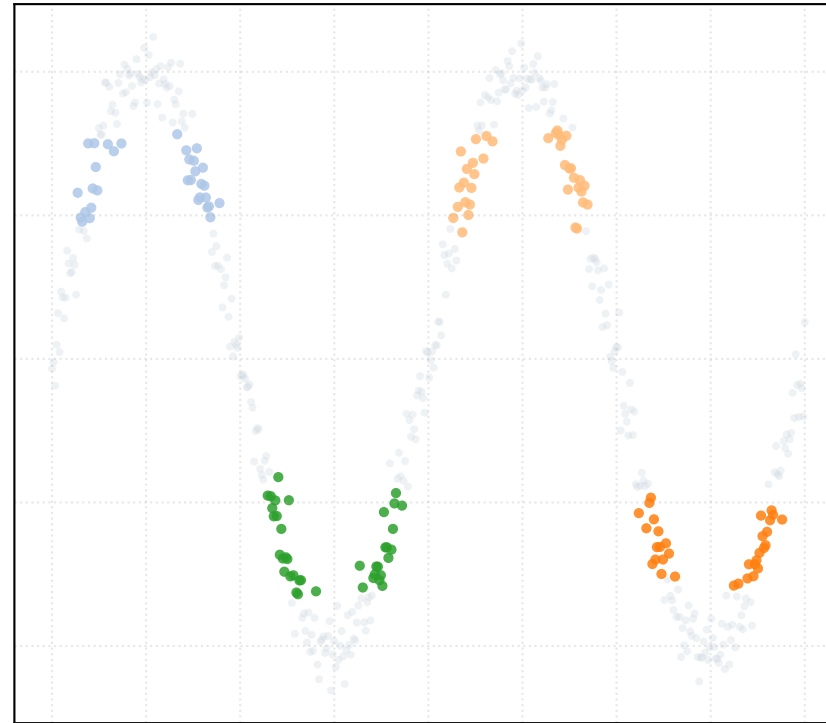
quant = 0.05 [4 Clusters]



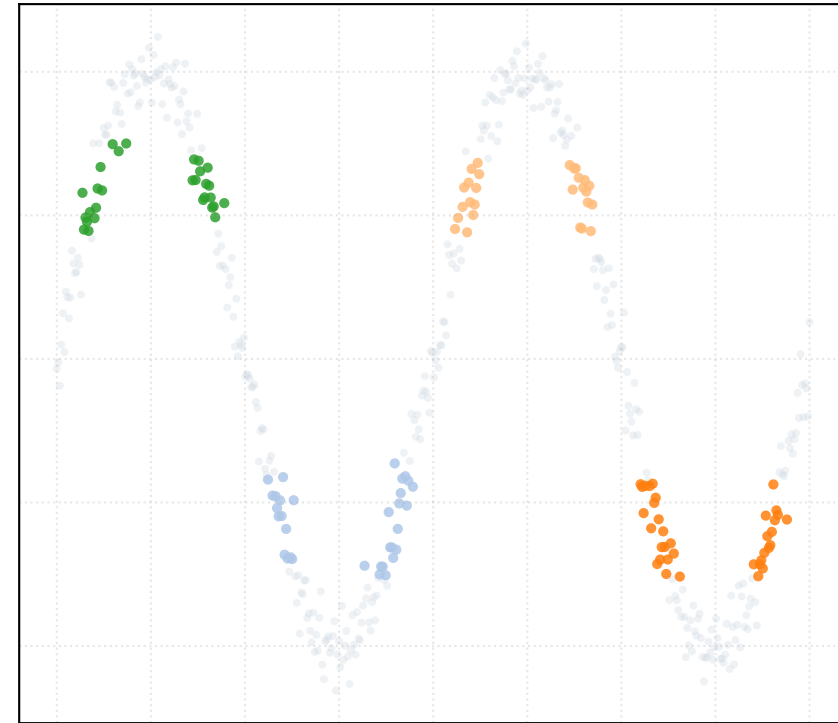
quant = 0.1 [4 Clusters]



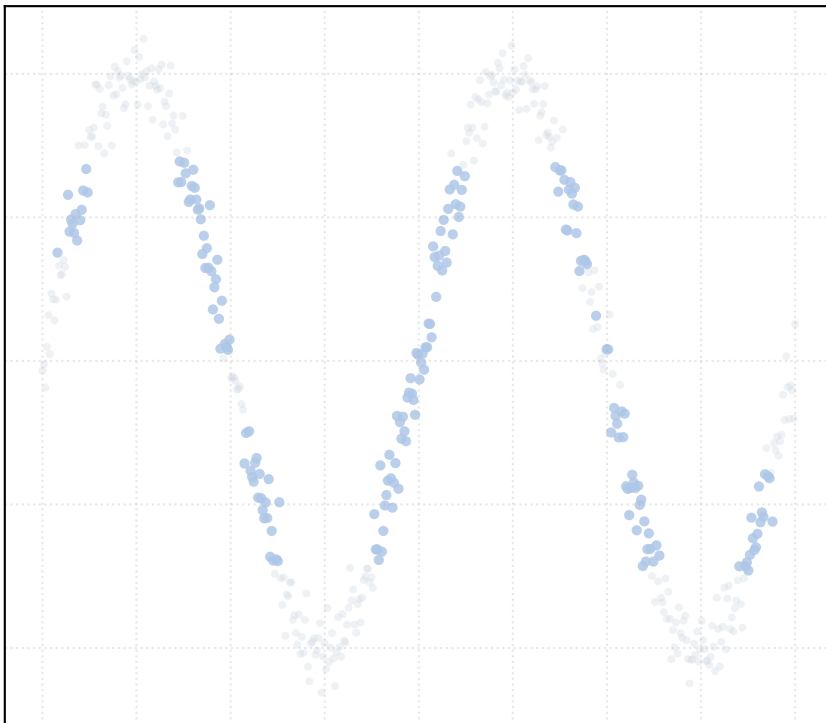
quant = 0.15 (Default) [4 Clusters]



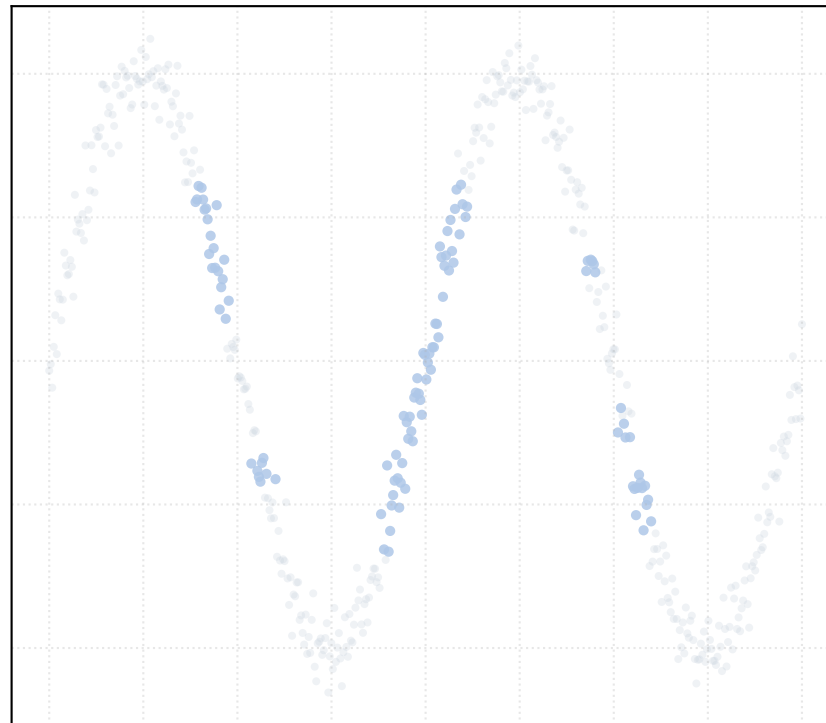
quant = 0.2 [4 Clusters]



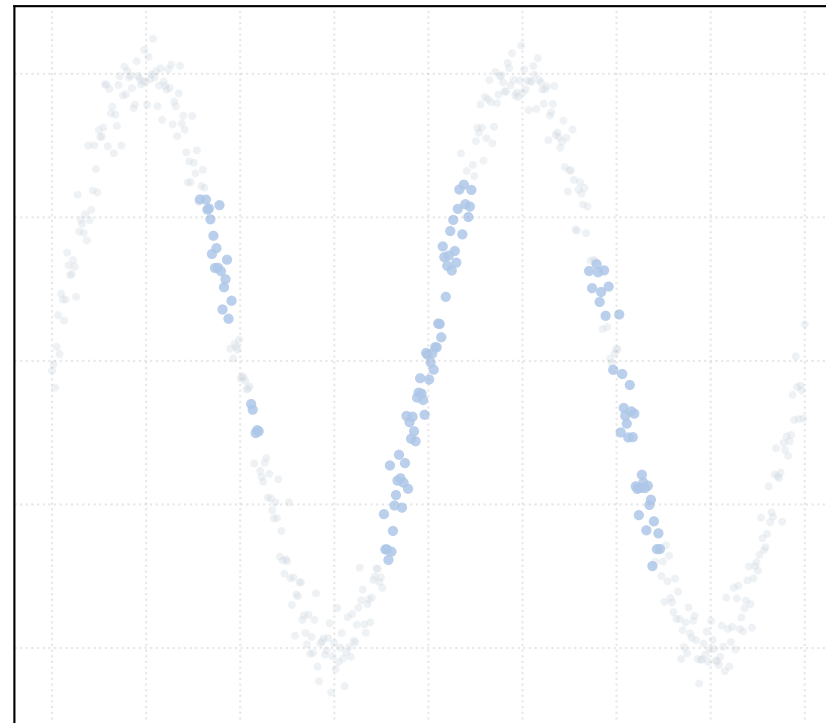
quant = 0.25 [1 Clusters]



quant = 0.3 [1 Clusters]



quant = 0.4 [1 Clusters]



quant = 0.5 [1 Clusters]

